Class 8

Mathematics

(Solutions)

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1. Rational and Irrational Numbers

Solutions

LEVEL-I

1. Correct option: (C)

Let the two numbers be a and b.

Let a=917. We have to find b.

We know that $\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b$

$\Rightarrow 131 \times 8253 = 917 \times b$

$\Rightarrow \frac{131 \times 8253}{917} = b$

$\Rightarrow b = \frac{1081143}{917}$.

$\therefore b = 1179$.

Therefore, the other number is 1179.

2. Correct option: (D)

A number is divisible by 3 if the sum of the digits is evenly divisible by 3.

Sum of digits divisible by 3

A) 5+6+1+1+1+0=14

B) 2+2+1+2+2+0+1= 10

C) 2+2+0+0+0+2+2= 8

D) 5+5+5+5+5+5= 30

Therefore, 30 is divisible by 3. Hence 555555 is divisible by 3.

3. Correct option: (C)

List out all of the prime factors for each number:

- The prime factors of 3^3 are 3, 3 and 3.
- The prime factors of 3 is 3.
- The prime factors of 4^2 are 2, 2, 2, and 2.
- The prime factors of 4 are 2 and 2.

The least common multiple is the product of all factors in the greatest number of their occurrence.

Here the common and uncommon prime factors are 2, 2, 2, 2, 3, 3, 3.

On Multiplying 2, 2, 2, 2, 3, 3, 3 to get the LCM of the given numbers.

L.C.M of $(3^3, 3, 4^2, 4) = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 = 432$.

$\therefore$ The required L.C.M least common multiple of $3^3, 3, 4^2, 4$ is 432.

4. Correct option: (A)

Observe that this number is a multiple of 3, since the sum of its digits is a multiple of 3.

Specifically, the sum of its digits is 3 x 300 = 900, which is divisible by 3.

Therefore, the greatest number among the follo Which of these is a prime number 3123,219,573, 467?

A) 3123 B) 219 C) 573 D) 467wing which divides 3333……3(300times) is 3.

5. Correct option: (B)

The divisibility test of 6 states that a number is divisible by 6 if the number is even and the sum of the digits is divisible by 3.

54672 is divisible by 2 as unit's digit is 2, and

It is also divisible by 3 because sum of digits = 5+4+6+7+2 =24 is a multiple of 3.

Hence, it is divisible by 6.

6. Correct option: (A)

A number is divisible by 5 if the last digit ends with 0 or 5.

A) 5555^4:

• Yes, 5555^4 is divisible by 5, because the number 5555 ends in the digit 5, which is a multiple of 5. Therefore, any power of 5555 will also end in the digit 5, and hence it will also be divisible by 5.

• More specifically, we can write:

5555^4 = (5500 + 55)^4

• Using the binomial theorem, we can expand this expression as:

5555^4 = 5500^4 + 4 x 5500^3 x 55 + 6 x 5500^2 x 55^2 + 4 x 5500 x 55^3 + 55^4

• All the terms in this expansion, except for the last one (55^4), are multiples of 5, because they contain a factor of 55, which is a multiple of 5. The last term, 55^4, is itself a power of 5, since 55 = 5 x 11 and 11 is not a multiple of 5. Therefore, we have:

5555^4 = 5 x (a multiple) + 5^4.

• Since the right-hand side is divisible by 5, we can conclude that 5555^4 is divisible by 5.

B) 7^{5555}

• The integer 7^{5555} is not necessarily divisible by 5.

• Use the fact that for any integer n, if n is not divisible by 5, then n^k is also not divisible by 5 for any positive integer k. In other words, if the last digit of n is not 0 or 5, then the last digit of n^k is not 0 or 5.

• The last digit of 7 is 7, which is not 0 or 5. Therefore, the last digit of 7^k is also 7 for any positive integer k. In particular, the last digit of 7^{5555} is 7, which means that 7^{5555} is not divisible by 5.

C) 4^{555}

• The integer 4^{555} is not always divisible by 5.

• To see why, we can use the fact that for any integer n, if n is not divisible by 5, then n^k is also not divisible by 5 for any positive integer k. In other words, if the last digit of n is not 0 or 5, then the last digit of n^k is not 0 or 5.

• The last digit of 4 is 4, which is not 0 or 5. Therefore, the last digit of 4^k is always 6 for any positive integer k that is greater than 1. In particular, the last digit of $4^{555}$is 6, which means that 4^{555} is not divisible by 5.

7. Correct option: (A)

Let the required numbers be x, 2x and 3x.

Then, their H.C.F. = x. So, $x = 12$.

• $x = 12$.

• $2x = 2 \times 12 = 24$.

• $3x = 3 \times 12 = 36$.

∴The numbers are 12, 24 and 36.

8. Correct option: (B)

To check if a number is divisible by 8, check if the last three digits of the number is divisible by 8.

Last 3 digits of the given numbers are:

A. 142

B. 176

C. 214

D. 462

The only number divisible by 8 is 176.

∴ 37450176 is the only number divisible by 8.

9. Correct option: (A)

i.The prime factorization of 306

$306 = 2 \times 3 \times 3 \times 17$

ii.The prime factorization of 657

$657 = 3 \times 3 \times 73$

iii. To find the HCF, multiply all the prime factors common to both numbers that is $3 \times 3 = 9$.

Therefore, GCD (306,657) = 9.

10. Correct option: (D)

On subtracting, we get.

- $2053 - 5 = 2048$
- $967 - 7 = 960$

Finding the prime factors for 2048 and 960.

- Prime factors of 2048=$\mathbf{2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2}$
- Prime factors of 960=$\mathbf{2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 5}$
- Thus, HCF of 2048 and 960= $\mathbf{2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64.}$

Therefore, GCD(2048, 960) = 64.

11. Correct option: (A)

By Euclid division lemma,

$\Rightarrow \quad 72 = 56 \times 1 + 16 \; ------ (1)$

$\Rightarrow \quad 56 = 16 \times 3 + 8 \; ------ (2)$

$\Rightarrow \quad 16 = 8 \times 2 + 0$

So HCF of 56 and 72 is 8.

From 2,

$\Rightarrow \quad 8 = 56 - (16 \times 3)$

$\Rightarrow \quad 8 = 56 - ((72 - 56) \times 3)$ (Since from 1).

$\Rightarrow \quad 8 = 56 - 72 \times 3 + 56 \times 3$

$\Rightarrow \quad 8 = 56 \times 4 - 72 \times 3$

$\Rightarrow \quad 8 = 56 \times 4 + 72(-3) = 56\,p + 72\,q.$

Therefore, the value of p = 4 and q = -3.

12. Correct option: (A)

By Euclid division lemma,

$\Rightarrow \quad 1032 = 408 \times 2 + 216 \; ------ (1)$

$\Rightarrow \quad 408 = 216 \times 1 + 192 \; ------ (2)$

$\Rightarrow \quad 216 = 192 \times 1 + 24 ------ (3)$

$\Rightarrow \quad 192 = 24 \times 8 + 0$

So HCF of 408 and 1032 is 24.

From 2,

$\Rightarrow \quad 24 = 216 - (192 \times 1).$

$\Rightarrow \quad 24 = 216 - (408 - 216 \times 1)$

$\Rightarrow \quad 24 = 1032 - 408 \times 2 - (408 - (1032 - 408 \times 2))$

$\Rightarrow \quad 24 = 1032 - 408 \times 2 - 408 + 1032 - 408 \times 2$

$\Rightarrow \quad 24 = 1032 \times 2 - 408 \times 5$

$\Rightarrow \quad 24 = -(408 \times 5 - 1032 \times 2)$

Therefore, the value of $p = 5$ and $q = 2$.

Now, $\frac{p}{q} = \frac{5}{2}$

Therefore, the required value is $\frac{5}{2}$.

13. Correct option: (B)

Highest power of 3 dividing 80! Is

$\Rightarrow \quad \frac{80}{3} + \frac{80}{3^2} + \frac{80}{3^3} + \frac{80}{3^4} = 26 + 8 + 2 + 0 = 36$

3^{36} is the highest power of 3 that divides 80!

14. Correct option: (D)

$3^4 \leq 100 < 3^5$

Hence,

$\Rightarrow \quad \frac{100}{3} + \frac{100}{3^2} + \frac{100}{3^3} + \frac{100}{3^4} = 33+11+3+1= 48$ ----(i)

$5^2 \leq 100 < 5^3$

Hence,

$\Rightarrow \quad \frac{100}{5} + \frac{100}{5^2} = 20 + 4 = 24 - - - - - - - -$ (ii)

Hence, 100! is exactly divisible by 3^{48} and 5^{24}.

Hence, the maximum power of 15 in 100! is 24.

15. Correct option: (D)

An exponent of a number, represents the number of times the number is multiplied to itself.

If 8 is multiplied by itself for n times, then, it is represented as:

$8 \times 8 \times 8 \times 8 \times \ldots\ldots \text{n times} = 8^n$

The above expression, 8^n, is said as 8 raised to the power n. Therefore, exponents are also called power or sometimes indices.

$8^6 = 8 \times 8 \times 8 \times 8 \times 8 \times 8 = 262144$

$5^6 = 5 \times 5 \times 5 \times 5 \times 5 \times 5 = 15625.$

On Subtracting we get, $8^6 - 5^6 = 262144 - 15625 = 246519$

By trial and error,

$\Rightarrow \quad \frac{246519}{91} = 2709.$

$\Rightarrow \quad \frac{246519}{49} = 5031.$

$\Rightarrow \quad \frac{246519}{129} = 1911.$

$\Rightarrow \quad 8^6 - 5^6$ is divisible by 91, 49, and 129.

16. Correct option: (B)

Numbers divisible by 5 upto 1155 = 231.

Numbers divisible by 7 upto 1155 = 165

Numbers which are divisible by both 5 and 7 (i.e., 35) upto 1155 = 33

$\therefore$ Numbers which are exactly divisible by 5 or 7 = $(231 + 165) - 33 = 363$

Numbers which are divisible by 5 and 11 both (i.e., 55) = 21

Numbers which are divisible by 7 and 11 both (i.e., 77) = 15

Numbers which are divisible by 5, 7 and 11 simultaneously = 3

$\therefore$ Numbers which are only divisible by either 5 and 11 or 7 and 11= $21 + 15 - 3 = 33$

Therefore, the number of numbers which are divisible by 5 or 7 but not by $11 = 363 - 33 = 330$.

17. Correct option: (A)

On subtracting the remainders from the given numbers,

$\Rightarrow \quad 1313 - 17 = 1296$

$\Rightarrow \quad 621 - 9 = 612$

Now, Check the divisibility rule for 9 for the numbers 1256 and 612.

If the sum of the digits are divisible by 9, then the number is divisible by 9,

$\Rightarrow \quad 1256 = 1 + 2 + 9 + 6 = 18$; 18 is divisible by 9.

$\Rightarrow \quad 612 = 6 + 1 + 2 = 9$; 9 is divisible by 9.

Therefore, total number of factor of a greatest possible number which when divides 1313 and 621, the respective remainders obtained are 17 and 9 is 9.

18. Correct option: (B)

$(2^{121}\,(2^2 - 2^1 - 2^0) \times 3^{232}(3^2 - 3^1 - 3^0))$ (Since $a^0 = 1$)

$\Rightarrow$ $(2^{121}\,(4 - 2 - 1) \times 3^{232}(9 - 3 - 1))$

$\Rightarrow$ $(2^{121}\,(1) \times 3^{232}(5))$

$\Rightarrow$ $(2^{121} \times 3^{232} \times 5)$

$\Rightarrow$ $(2^{120} \times 3^{232} \times 2 \times 5)$

$\Rightarrow$ $(2^{120} \times 3^{232} \times 10)$

Therefore, number of zeros at the end of $((2^{123} - 2^{122} - 2^{121}) \times (3^{234} - 3^{233} - 3^{232}))$ is 1.

19. Correct option: (B)

$(1^1 \times 2^2 \times 3^3 \times 4^4 \times 5^5) + (1^6 \times 2^7 \times 3^8 \times 4^9 \times 5^{10}) + (1^{11} \times 2^{12} \times 3^{13} \times 4^{14} \times 5^{15}) \ldots\ldots$
$+ (1^{96} \times 2^{97} \times 3^{98} \times 4^{99} \times 5^{100})$

$\Rightarrow$ In the first expression the number of zeros $1^1 \times 2^2 \times 3^3 \times 4^4 \times 5^5$ are 10.

$\Rightarrow$ In the second expression the number of zeros $1^6 \times 2^7 \times 3^8 \times 4^9 \times 5^{10}$ are >10.

$\Rightarrow$ In the third expression the number of zeros $1^{11} \times 2^{12} \times 3^{13} \times 4^{14} \times 5^{15}$ are >10.

$\Rightarrow$ In the fourth expression the number of zeros $1^{96} \times 2^{97} \times 3^{98} \times 4^{99} \times 5^{100}$ are >10.

Since on adding the 10, 100, 1000, 10000, 100000, will get the minimum number of zeroes.

$\Rightarrow$ $1^1 \times 2^2 \times 3^3 \times 4^4 \times 5^5 = 1 \times 4 \times 27 \times 2^5 \times 5^5 = 8640000$

$\Rightarrow$ $8 + 6 + 4 + 0 + 0 + 0 + 0 = 18$.

Therefore, the required number of zeroes for the given expression is 18.

20. Correct option: (B)

We know,

The sum of first n odd numbers is $S_n = 1 + 3 + 5 + 7 + \cdots + n = n^2$.

$S_n = \frac{(11111)^2}{11111}$.

Hence, it is divisible by 11111.

LEVEL-II

1. Correct option: (C)

A cube of a natural number so writes it as x^3 for some natural number x.

To find another natural number y such that when we shift the digits of x^3, we get y^3.

Shifting the digits of a number involves moving the last digit to the front, so write the shifted number as (100a + 10b + c), where a, b, and c are the digits of the original number.

Now, we can write the shifted number as $(10^2a + 10b + c) + 100a = 100\,y^3$. Simplifying, we get:

$\Rightarrow$ $1000a + 10b + c = 100\,y^3 - 100a$

$\Rightarrow$ $1000a + 100a = 100\,y^3 - 10b - c$

$\Rightarrow$ $1100a = 100\,y^3 - 10b - c$.

Since a is a digit, it can only take on the values 1, 2, 3, ..., 9.

Start with the case where a = 1. Then we have:

$\Rightarrow$ $1100 = 100\,y^3 - 10b - c$

Since y, b, and c are digits, y can only take on the values 1, 2, or 3.

Try each of these values and see if we get a valid solution:

If y = 1, then we have 1000 = 10b + c + 100, which simplifies to b + c = 90. There are no two digits that add up to 90, so this case doesn't work.

If y = 2, then we have 1300 = 10b + c + 800, which simplifies to b + c = 50. The only two digits that add up to 50 are 5 and 5, so we have b = 5 and c = 5. Therefore, the original number is 125 and the shifted number is 512, both of which are cubes of natural numbers.

If y = 3, then we have 2600 = 10b + c + 2700, which simplifies to b + c = 10. The only two digits that add up to 10 are 1 and 9, but we've already used 5, so this case doesn't work. Therefore, the unique 3-digit number that satisfies the conditions is 125, which is the cube of 5 and can be shifted to form 512, which is the cube of 8.

2. Correct option: (A)

⇒ The number of times he pressed the keys for one digit number = 9

⇒ The number of times he pressed the keys for two-digit number = $90 \times 2 = 180$

⇒ The number of times he pressed the keys for the three-digit number = $900 \times 3 = 2700$

⇒ The number of times he pressed the keys for the number 1000 to 1999 is $1000 \times 4 = 4000$

The total number of times the keys are pressed = $9 + 180 + 2700 + 4000 = 6889$.

Therefore, total number of times the keys are pressed is 6889.

3. Correct option: (B)

Insert the values of a and b from the given options one by one and check for the expression a5523879b.

i) a= 8 and b= 6

a5523879b = 855238796.

Check the divisibility for 8 and 6.

⇒ $\frac{855238796}{8}$ = 106904849.5

⇒ $\frac{855238796}{6}$ = 142539799.333

This mean it is not divisible by 8 and 6. So the values of a ≠ 8 and b ≠ 6.

ii) a= 7 and b= 2

a5523879b = 755238792.

Check the divisibility for 7 and 2.

⇒ $\frac{755238792}{7}$ = 107891256

⇒ $\frac{755238792}{2}$ = 377619396

This mean it is divisible by 7 and 2. So the values of a = 7 and b = 2.

iii) a= 8 and b= 1

a5523879b = 855238791.

Check the divisibility for 8 and 1.

⇒ $\frac{855238791}{8}$ = 106904848.875

⇒ $\frac{855238791}{1}$ = 855238791

This mean it is not divisible by 8 and 1. So the values of a ≠ 8 and b = 1.

Therefore, the values of the digits a and b respectively, in the number a5523879b, if it is divisible by both a and b is (7, 2).

4. Correct option: (D)

We know that a prime number is having factors 1 and itself only.

⇒ The factors of 3123 are: $3123 = 1 \times 347$

Thus, it is not a prime number.

⇒ The factors of 219 are: $219 = 1 \times 3 \times 73$

Thus, it is not a prime number.

⇒ The factors of 573 are: $573 = 1 \times 3 \times 191$

Thus, it is not a prime number.

⇒ The factors of 467 are: $467 = 1 \times 467$

Thus, it is a prime number.

Therefore, the prime number is 467.

5. Correct option: (C)

7777 - n must be a perfect square.

Now, using the option, we find that n = 33

$\Rightarrow \quad 7777 - 33 = 7744$ is perfect square.

Therefore, the minimum value of 'n' is 33.

6. Correct option: (B)

HCF of $2^4, 8^2, 16^2, 20^3 = 2^4$;

LCM of $2^4, 8^2, 16^2, 20^3 = 256 \times 125 = 32000$.

Therefore, The H.C.F. and L.C.M. of $2^4, 8^2, 16^2,$ and 20^3 are 2^4 ;32000

7. Correct option: (B)

On dividing 15192327 by 99 we get,

$\frac{15192327}{99} = 153457.84$.

The quotient is 153457 and the remainder is 84.

Therefore, the remainder is 84.

8. Correct option: (A)

A) Numbers divisible by 2: $\frac{200}{2} = 100$

B) Numbers divisible by 3: $\frac{200}{3} = 66$

C) Numbers divisible by 5: $\frac{200}{5} = 40$

counting twice

AB) Numbers divisible by $(2 \times 3 = 6)$; 6: $\frac{200}{6} = 33$

AC) Numbers divisible by $(2 \times 5 = 10)$; 10: $\frac{200}{10} = 20$

BC) Numbers divisible by $(3 \times 5 = 15)$; 15: $\frac{200}{15} = 13$

counting 3 times

ABC) Numbers divisible by $(2 \times 3 \times 5 = 30)$; 30: $\frac{200}{30} = 6$

Total of numbers $= A + B + C - AB - AC - BC + ABC$

$= 100 + 66 + 40 - 33 - 20 - 13 + 6 = 146.$

Therefore, the number of numbers lying between 1 and 200 which are divisible by either of 2,3 or 5 is 146.

9. Correct option: (C)

Let the numbers be 34 a and 34 b

According to question,

$\Rightarrow \quad 34a \times 34b = 1020 \times 34$

$\Rightarrow \quad \frac{34(a \times b)}{34} = \frac{1020}{34}$

$\Rightarrow \quad a \times b = 30$

So (a, b) can be $(1, 30), (3, 10), (5, 6)$.

$\Rightarrow$ Numbers when a and b are $(1, 30) = 34$ and 1020.

$\Rightarrow$ Numbers when a and b are $(3, 10) = 102$ and 340.

$\Rightarrow$ Numbers when a and b are $(5, 6) = 170$ and 204.

Therefore, the L.C.M. of two numbers is 1020 and their H.C.F. is 34 the possible pair of numbers is 204, 170.

10. Correct option: (A)

Let H be the HCF of two numbers N_1 and N_2.

Then, we get, $N_1 = H \times a$ and $N_2 = H \times b$

where a and b are co-primes.

Sum of the two numbers $N_1 + N_2 = H \times (a + b)$

From given question, we have H = 43 and $N_1 + N_2 = 430$

Thus, $43 \times (a + b) = 430$

$\Rightarrow a + b = 10$

So, we have to find all possible (a,b) such that a & b are coprime numbers satisfying $a + b = 10$

$\Rightarrow \quad (a, b) = (1,9), (2,8), (3,7), (4,6), (5,5).$

Therefore, the total number of 5 such possibilities.

11. Correct option: (A, B)

A) [0.5] = 0 is true. The greatest integer less than or equal to 0.5 is 0.

B) $[1 + \frac{1}{2}] = 1$ is true. $1 + \frac{1}{2} = 1.5$, and the greatest integer less than or equal to 1.5 is 1.

C) [-9.07] = -10 is false. The greatest integer less than or equal to -9.07 is -10, not -8.

D) $\frac{15}{2} = 7$ is false. $\frac{15}{2}$ is equal to 7.5, not 7.

Therefore, [0.5] = 0 and $[1 + \frac{1}{2}] = 1$ are correct.

12. Correct option: (D)

Start by simplifying the given equation:

$\Rightarrow \quad [x] + x = \frac{4}{3}[x]$

$\Rightarrow \quad [x] + x = -\frac{1}{3}[x]$

Since the greatest integer function [x] is always an integer, conclude that x must be a negative number.

Assume that $x = -a$ for some positive number a. Then :

$\Rightarrow \quad [-a] = \frac{1}{3}a$

Since [-a] is a negative integer, we can write it as $[-a] = -b$ for some positive integer b.

Substituting this into the equation above:

$\Rightarrow \quad -b = \frac{1}{3}a$

$\Rightarrow \quad a = -3b$

So, any value of x of the form $x = -a$, where $a = 3b$ for some positive integer b, will satisfy the given equation.

In other words, the possible values of x are:

$x = -3, -6, -9, -12, \ldots$

Therefore, 0 is among these possible values of x.

13. Correct option: (B,C)

The first four odd prime numbers are:

- $a = 3$
- $b = 5$
- $c = 7$
- $d = 11$

Check each of the given statements:

A) a+b+c is not odd composite

$\Rightarrow \quad a + b + c = 3 + 5 + 7 = 15$

15 is an odd number, but it is not a composite number. It is a prime number. Therefore, statement (A) is false.

B) b+c+d is prime

$\Rightarrow \quad b + c + d = 5 + 7 + 11 = 23$

23 is a prime number. Therefore, statement (B) is true.

C) abc is odd

$\Rightarrow \quad abc = 3 \times 5 \times 7 = 105$

105 is an odd number. Therefore, statement (C) is true.
D) a+c+b+d is not even composite
$\Rightarrow \quad a + c + b + d = 3 + 7 + 5 + 11 = 26$
26 is an even number, but it is not a composite number. It is a semi-prime number (i.e., a composite number with exactly two prime factors: 2 and 13).
Therefore, statement (D) is false.
Therefore, the correct statements are (B) and (C).

14. Correct option: (D)

The prime factorization of 25680 is:
$\Rightarrow \quad 25680 = 2^4 \times 3^3 \times 5 \times 7$
Use the fact that the number of divisors of a number is equal to the product of one more than each exponent in its prime factorization.
So, for 25680, the number of divisors is:
$\Rightarrow \quad (4 + 1) \times (3 + 1) \times (1 + 1) \times (1 + 1) = 5 \times 4 \times 2 \times 2 = 80$
However, subtract 2 from this number to exclude 1 and 25680 itself.
So the total number of divisors of 25680 except 1 and itself is:
$\Rightarrow \quad 80 - 2 = 78$
Therefore, the required value is 78.

15. Correct option: (A)

The prime factorization of 10500is:
$\Rightarrow \quad 10500 = 2^2 \times 3^1 \times 5^3 \times 7$
Use the fact that the number of divisors of a number is equal to the product of one more than each exponent in its prime factorization.
So, for 10500, the number of divisors is:
$\Rightarrow \quad (2 + 1) \times (1 + 1) \times (3 + 1) \times (1 + 1) = 3 \times 2 \times 4 \times 2 = 48$
So the total number of odd divisors of 10500 is equal to $(q + 1)(r + 1)(s + 1)$.
$\Rightarrow \quad (1 + 1) \times (3 + 1) \times (1 + 1) = 2 \times 4 \times 2 = 16.$
Therefore, the required value is 16.

16. Correct option: (C)

The prime factorization of 123456 is:
$\Rightarrow \quad 123456 = 2^6 \times 3^1 \times 643$
Use the fact that the number of divisors of a number is equal to the product of one more than each exponent in its prime factorization.
So, for 123456, the number of divisors is:
$\Rightarrow \quad (6 + 1) \times (1 + 1) \times (1 + 1) = 7 \times 2 \times 2 = 28$
So the total number of even divisors of 123456 is equal to $p(q + 1)(r + 1)$.
$\Rightarrow \quad 6 \times (1 + 1) \times (1 + 1) = 6 \times 2 \times 2 = 24.$
Therefore, the required value is 24.

17. Correct Matching is: a) → R. , b) →P. , c) → S. , d) → Q.

a) The prime factorization of 36 is:
$36 = 2^2 \times 3^2$
So, for 36, the number of divisors is:
$\Rightarrow \quad (2 + 1) \times (2 + 1) = 3 \times 3 = 9.$
Therefore, the total number of factors of 36 is 9.
b) The prime factorization of 240 is:
$240 = 2^4 \times 3^1 \times 5^1$
So, for 240, the number of odd divisors is:
$\Rightarrow \quad (1 + 1) \times (1 + 1) = 2 \times 2 = 4.$
Therefore, the total number of odd factors of 240 is 4.

c) The prime factorization of 270 is:

• $270 = 2^1 \times 3^3 \times 5$

The sum of all the factors is $\frac{(2^{p+1}-1)(3^{q+1}-1)(5^{r+1}-1)}{(2-1)(3-1)(5-1)}$

$$\frac{(2^{p+1}-1)(3^{q+1}-1)(5^{r+1}-1)}{(2-1)(3-1)(5-1)} = \frac{(2^{1+1}-1)(3^{3+1}-1)(5^{1+1}-1)}{(2-1)(3-1)(5-1)}$$

$$= \frac{(2^2-1)(3^4-1)(5^2-1)}{(1)(2)(4)}$$

$$= \frac{(4-1)(81-1)(25-1)}{(1)(2)(4)}$$

$$= \frac{(3)(80)(24)}{(1)(2)(4)} = 3 \times 10 \times 24 = 720.$$

Therefore, the sum of the factors of 270 is 720.

d) The prime factorization of 1520 is:

• $1520 = 2^4 \times 5^1 \times 19$

The sum of all the factors is $\frac{(2^{p+1}-1)(3^{q+1}-1)(5^{r+1}-1)}{(2-1)(3-1)(5-1)}$

$$\frac{(2^{p+1}-1)(5^{q+1}-1)(19^{r+1}-1)}{(2-1)(5-1)(19-1)} = \frac{(2^{4+1}-1)(5^{1+1}-1)(19^{1+1}-1)}{(2-1)(5-1)(19-1)}$$

$$= \frac{(2^5-1)(5^2-1)(19^2-1)}{(1)(4)(18)}$$

$$= \frac{(32-1)(25-1)(361-1)}{(1)(4)(18)}$$

$$= \frac{(31)(24)(360)}{(1)(4)(18)} = 31 \times 6 \times 20\ 3720.$$

The sum of the factors of 1520 except 1 is 3720 – 1 = 3719.

Therefore, the sum of the factors of 520 is 3719.

18. Correct Answer is: 28

$4^{16} \times 5^{25} = (2^2)^{16} \times 5^{25}$

$= 2^{32} \times 5^{25}$

$= 2^7 \times 2^{25} \times 5^{25}$

$= 2^7 \times (2 \times 5)^{25}$

$= 2^7 \times (10)^{25}$.

The total no. of zeroes are 25.

Now, $2^7 = 128$

The no. of digits are = 3.

Therefore, the number of significant digits in expansion of $4^{16} \times 5^{25}$ is 3 + 25 = 28.

19. Correct Answer is: 6

A perfect number is a positive integer that is equal to the sum of its proper divisors (excluding itself). The first few perfect numbers are 6, 28, 496, 8128, and so on.

Start by examining the first few perfect numbers and checking if they are divisible by 3:

⇒ 6 is divisible by 3, so it satisfies the condition.

⇒ 28 is not divisible by 3, so it does not satisfy the condition.

⇒ 496 is not divisible by 3, so it does not satisfy the condition.

⇒ 8128 is not divisible by 3, so it does not satisfy the condition.

Therefore, the least perfect number that is divisible by 3 is 6.

20. Correct Answer is: 2

To find which digit stands at tens place in 65x5 when it is divisible by 3, we need to apply the divisibility rule of 3.

Divisibility rule of 3: According to this rule the sum total of all the digits gives the resultant number which is divisible by 3, then the number completely is divisible by 3.
To apply this rule we need to find sum total of all the digits of number 65x5.
6 + 5 + x + 5 = Number divisible by 3
16 + x = Number divisible by 3
The nearest number to 16 that is divisible by 3 is 18
$\Rightarrow \quad 16 + x = 18$
$\Rightarrow \quad x = 18 - 16$
$\therefore x = 2.$
=> Sum of digits of number: 6 + 5 + 2 + 5 = 18
=> Number is 6525
If we replace 'x' by 1 or by 3, the number is not divisible by 3
Therefore, the least number that can replace 'x' is 2 and the number is 6525 which is divisible by 3.

STUDENT PRACTICE QUESTIONS

1. Correct option: (A)
The divisibility rule for 7 states that if a number is divisible by 7, then the result obtained by subtracting double the last digit from the remaining number should be 0 or divisible by 7.
A) 343:
- (3 x 2) = 6 and 34 – 6 = 28.
- 28 is divisible by 7.
- Hence 343 is divisible by 7.

B) $3^7 = 2187$.
- 218 - (7 x 2) = 204
- Since 204 is not divisible by 7.
- Hence 2187 is also not divisible by 7.

C) 3^{77}
- Use Fermat's Little Theorem, which states that if p is a prime number and a is any integer not divisible by p, then $a^{(p-1)}$ is congruent to 1 modulo p. In other words, $a^{(p-1)}$ - 1 is divisible by p.
- Since 7 is a prime number and 3 is not divisible by 7, we can use Fermat's Little Theorem to determine whether 3^6 - 1 is divisible by 7. We have:

3^6 - 1 = 729 - 1 = 728
- Since 728 is divisible by 7 (728 = 7 x 104), we can write:

3^6- 1 = 7k (where k is an integer)
- Multiplying both sides by 3^{71} , we get:

3^{77}- 3^{71} = 7k x 3^{71}
- Since 3^6 - 1 is divisible by 7, we know that 3^6 is congruent to 1 modulo 7. Therefore, we have: $3^{77} = (3^6)^{12}$ x 3^5
- Using the fact that 3^6 is congruent to 1 modulo 7, we can simplify this expression as follows:

$3^{77} = (3^6)^{12}$ x $3^5 = (1)^{12}$ x 243 = 243
- Therefore, we have:

3^{77} = 7 x 34 + 5
- Since the remainder when 3^{77} is divided by 7 is 5, we can conclude that 3^{77} is not divisible by 7.

D) 5^{777}

• Use Fermat's Little Theorem, which states that if p is a prime number and a is any integer not divisible by p, then $a^{(p-1)}$ is congruent to 1 modulo p. In other words, $a^{(p-1)}$ - 1 is divisible by p.

• Since 7 is a prime number and 5 is not divisible by 7, we can use Fermat's Little Theorem to determine whether 5^6- 1 is divisible by 7. We have:

5^6 - 1 = 15624.

• Since 15624 is not divisible by 7, we can conclude that 5^6- 1 is not divisible by 7. Therefore, 5^6 is not congruent to 1 modulo 7.

2. Correct option: (A)

A) 3^{444}

The integer 3^{444} is divisible by 9.

• Any integer that is divisible by 9 must also be divisible by 3.

• Further, for any positive integer k, we have: $3^k = (3^2)^{\frac{k}{2}} = (9)^{\frac{k}{2}}$

• Therefore, $3^{444} = (9)^{\frac{444}{2}} = 9^{222}$.

• Since 9 is divisible by 9, it follows that 9^{222} is also divisible by 9.

• Hence, we can conclude that 3^{444} is divisible by 9.

B) 4^{999}

No, the integer 4^{999} is not divisible by 9.

• Any integer that is divisible by 9 must also be divisible by 3. However, the prime factorization of 4^{999}

• 4^{999} is:

• $4^{999} = (2^2)^{999} = (2)^{2\times 999} = (2)^{1998}$

• Since 2 and 3 are relatively prime, it follows that $(2)^{1998}$ is not divisible by 3, and hence not divisible by 9.

• Therefore, we can conclude that 4^{999} is not divisible by 9.

C) 5^9

No, the integer 5^9 is not divisible by 9.

Any integer that is divisible by 9 must also be divisible by 3. However, the prime factorization of 5^9 is: $5^9 = 5^3 \times 5^3 \times 5^3 = 125 \times 125 \times 125$

Since 125 is not divisible by 3, it follows that 5^9 is not divisible by 3, and hence not divisible by 9.

Therefore, we can conclude that 5^9 is not divisible by 9.

3. Correct option: (A)

Divisibility rule of 11: The (sum of the odd numbered digits) - (sum of the even numbered digits) should be divisible by 11.

$\Rightarrow 137269 = (9 + 2 + 3) - (6 + 7 + 1) = \mathbf{1}4 - \mathbf{1}4 = 0$; So, 137269 is divisible by 11.

$\Rightarrow 66311 = (1 + 3 + 6) - (1 + 6) = \mathbf{1}0 - 7 = 3$; So, 66311 is not divisible by 11.

$\Rightarrow 901352 = (2 + \mathbf{3} + 0) - (5 + 1 + 9) = 5 - 15 = -1\mathbf{0}$; So, 901352 is not divisible by 11.

$\therefore$ Number which is divisible by 11 is 137269.

4. Correct option: (C)

i. The prime factorization of 1769

$1769 = 29 \times 61$

ii. The prime factorization of 657

$2378 = 2 \times 29 \times 41$

iii. To find the HCF, multiply all the prime factors common to both numbers that is 29.

Therefore, GCD(1769, 2378) = 29.

5. Correct option: (C)

By Euclid division lemma,

$\Rightarrow \quad 272 = 119 \times 2 + 34 \; ------ (1)$

$\Rightarrow \quad 119 = 34 \times 3 + 17 \; ------ (2)$

$\Rightarrow \quad 34 = 17 \times 2 + 0$

So HCF of 119 and 272 is 17.

From 2,

$\Rightarrow \quad 17 = 119 - (34 \times 3)$

$\Rightarrow \quad 17 = 119 - ((272 - 2(119)) \times 3)$ (Since from 1).

$\Rightarrow \quad 17 = 119 - 272 \times 3 + 2(119) \times 3$

$\Rightarrow \quad 17 = 119 \times 7 - 272 \times 3$

$\Rightarrow \quad 17 = 119 \times 7 + 272(-3) = 119\,p + 272\,q.$

Therefore, the value of $p = 7$ and $q = -3$.

Now, $p - q = 7 - (-3) = 7 + 3 = 10$.

Therefore, the required value is 10.

6. Correct option: (C)

Highest power of 2 dividing 100! Is

$\Rightarrow \quad \frac{100}{2} + \frac{100}{2^2} + \frac{100}{2^3} + \frac{100}{2^4} + \frac{100}{2^5} + \frac{100}{2^6} = 50 + 25 + 12 + 6 + 3 + 1 = 97$

2^{97} is the highest power of 2 that divides 100!

7. Correct option: (A)

Highest power of 2 dividing 100! Is

$\Rightarrow \quad \frac{1000}{7} + \frac{1000}{7^2} + \frac{1000}{7^3} + \frac{1000}{7^4} = 142 + 20 + 2 + 0 = 164$

7^{164} is the highest power of 7 that divides 1000!

8. Correct option: (B)

Step 1- Write the expression,

$\Rightarrow \quad 5^{72} \times 8^{27}$

$\Rightarrow \quad 5^{72} \times (2^3)^{27}$

Step 2 - Applying power rule, $(a^m)^n = a^{mn}$

$\Rightarrow \quad 5^{72} \times 2^{81}$

Step 3 - Split the power, $a^m \times a^n = a^{m+n}$

$\Rightarrow \quad 5^{72} \times 2^{72+9}$

$\Rightarrow \quad 5^{72} \times 2^{72} \times 2^9$

Step 4 - Multiplying when power is same, $a^m \times b^n = (ab)^{mn}$

$\Rightarrow \quad (5 \times 2)^{72} \times 2^9$

$\Rightarrow \quad (10)^{72} \times 2^9$

$\Rightarrow \quad 512 \times 2^9$

Number of digits are 3+72=75.

Therefore, the number of digits in the product is 75.

9. Correct option: (B)

Step 1- Write the expression,

$\Rightarrow \quad 4^{1111} \times 5^{2222}$

$\Rightarrow \quad (2^2)^{1111} \times 5^{2222}$

Step 2 - Applying power rule, $(a^m)^n = a^{mn}$

$\Rightarrow \quad (2^2)^{1111} \times 5^{2222}$

$\Rightarrow \quad 2^{2222} \times 5^{2222}$

Step 3 - Multiplying when power is same, $a^m \times b^n = (ab)^{mn}$

$\Rightarrow \quad 2^{2222} \times 5^{2222}$

$\Rightarrow \quad (2 \times 5)^{2222}$

$\Rightarrow \quad (10)^{2222}$

We know, the number of digits for $10^n = n + 1$

$\Rightarrow$ Number of digits are $(10)^{2222} = 2222 + 1 = 2223$.

Therefore, the number of digits in the product is 2223.

10. Correct option: (B)

$2^3 \times 3^4 \times 4^5 \times 5^6 + 3^5 \times 5^7 \times 7^9 \times 8^{10} + 4^5 \times 5^6 \times 6^7 \times 7^8 - 10^2 \times 15^3 \times 20^4$

$\Rightarrow$ In the first expression the number of zeros $2^3 \times 3^4 \times 4^5 \times 5^6$ are 6.

$\Rightarrow$ In the second expression the number of zeros $3^5 \times 5^7 \times 7^9 \times 8^{10}$ are 7.

$\Rightarrow$ In the third expression the number of zeros $4^5 \times 5^6 \times 6^7 \times 7^8$ are 6.

$\Rightarrow$ In the fourth expression the number of zeros $10^2 \times 15^3 \times 20^4$ are 9.

Since on adding the 10, 100, 1000, 10000, 100000, will get the minimum number of zeroes.

Therefore, the minimum zeroes for the given expression is 6, 7, 6, 9.

Therefore, the required number of zeroes for the given expression is 6.

11. Correct option: (B)

The first4 two-digit composite numbers are 10,12,14,15.

$\Rightarrow \quad a = 10, b = 12, c = 14, d = 15$

Their sum $10 + 12 + 14 + 15 = 51$

so the nearest square to this number is $64 = 8^2$

$\Rightarrow \quad 51 + e = 64$

$\Rightarrow \quad 15 + e - 15 = 64 - 15$

$\Rightarrow \quad e = 13.$

Therefore, the value of e is 13.

12. Correct option: (C)

Calculating contributions of the different powers of 5.

we have $\frac{100}{5} = 20$, $\frac{100}{5^2} = 4$.

$\Rightarrow \quad 20 + 4 = 24.$

Hence the total contributions to the power of 5 is 24 or the number 100! is divisible by 5^{24}

Let $N = 2^p 3^q 5^1$ …… then

i) Total number of divisor of N is equal to $(p + 1)(q + 1)(r + 1)$.

ii) Total number of odd divisor of N is equal to $(q + 1)(r + 1)$.

iii) Total number of even divisor of N is equal to $p\,(q + 1)(r + 1)$.

(OR) Total number of divisors - Total number of odd divisors

13. Correct option: (A)

The prime factorization of 1240 is:

$\Rightarrow \quad 1240 = 2^3 \times 5^1 \times 31$

The sum of all the factors is $\frac{(2^{p+1}-1)(5^{q+1}-1)(31^{r+1}-1)}{(2-1)(5-1)(31-1)}$

$$\frac{(2^{p+1}-1)(5^{q+1}-1)(31^{r+1}-1)}{(2-1)(5-1)(31-1)} = \frac{(2^{3+1}-1)(5^{1+1}-1)(31^{1+1}-1)}{(2-1)(5-1)(31-1)}$$

$$= \frac{(2^4-1)(5^2-1)(31^2-1)}{(2-1)(5-1)(31-1)}$$

$$= \frac{(16-1)(25-1)(961-1)}{(1)(4)(30)}$$

$$= \frac{(15)(24)(960)}{(1)(4)(30)}$$

$$= 15 \times 6 \times 32 = 2880$$

Therefore, the sum of factors of 1240 is 2880.

14. Correct option: (C)

$7056 = 84^2 = 2^4 \times 3^2 \times 7^2$

No of factors= $(4+1)(2+1)(2+1) = 45$

Product of divisor/factors = $(7056)^{\frac{45}{2}} = (84^2)^{\frac{45}{2}} = (84)^{45}$

Therefore, the product of divisors of 7056 is $(84)^{45}$.

15. Correct Matching is: a) → Q. , b) →P. , c) → T. , d) → T.

The HCF of 30, 42, 135

$\Rightarrow \quad 30 = 2 \times 3 \times 5$

$\Rightarrow \quad 42 = 2 \times 3 \times 7$

$\Rightarrow \quad 135 = 3 \times 3 \times 3 \times 5.$

Therefore, $\text{HCF}(30, 42, 135) = 3.$

The LCM of 420, 9009, 6270

$\Rightarrow \quad 420 = 2^2 \times 3 \times 5 \times 7$

$\Rightarrow \quad 9009 = 3^2 \times 7 \times 11 \times 13$

$\Rightarrow \quad 6270 = 2 \times 3 \times 5 \times 11 \times 19.$

Therefore, $\text{LCM}(420,9009,6270) = 2^2 \times 3^2 \times 5 \times 7 \times 11 \times 13 \times 19.$

$= 180180 \times 19 = 3423420.$

a) LCM of $\frac{4}{3}, \frac{4}{9}, \frac{2}{15}, \frac{36}{21}$.

H.C.F of a group of numbers is the greatest factor of the numbers.

$\text{L.C.M} = \frac{\text{L.C.M of numerators}}{\text{H.C.F of denominators}}$.

L.C.M of (4,4,2,36) = $2 \times 2 \times 3 \times 3 = 36$

H.C.F of (3, 9, 15, 21) = 3

Required L.C.M = $\frac{36}{3} = 12.$

∴ The required L.C.M is 12.

b) The smallest positive numbers which is exactly divisible by $\frac{1}{3}, \frac{1}{2}, \frac{3}{7}, \frac{4}{11}$.

H.C.F of a group of numbers is the greatest factor of the numbers.

$\text{L.C.M} = \frac{\text{L.C.M of numerators}}{\text{H.C.F of denominators}}$.

L.C.M of (1, 1, 3, 4) = $2 \times 2 \times 3 = 12$

H.C.F of (3, 2, 7, 11) = 1

Required L.C.M = $\frac{36}{3} = 12.$

∴ The required L.C.M is 12.

16. Correct Answer is: 3

The prime factorization of each of the three numbers and then multiply them together to get the prime factorization of their product:

$\Rightarrow \quad 14^{11} = 2^{11} \times 7^{11}$

$\Rightarrow \quad 7^5 = 7^5$

$\Rightarrow \quad 11^3 = 11^3$

So, the product of these three numbers is:

$\Rightarrow \quad 14^{11} \times 7^5 \times 11^3 = 2^{11} \times 7^{11} \times 7^5 \times 11^3 = 2^{11} \times 7^{16} \times 11^3$

The prime factors of this product are 2, 7, and 11.

The exponent of 2 is 11, the exponent of 7 is 16, and the exponent of 11 is 3.

Therefore, the product has a total of 3 prime factors: 2, 7, and 11.

17. Correct Answer is: 6

The HCF of 2508, 1638

$\Rightarrow \quad 2508 = 2 \times 2 \times 3 \times 11 \times 19$

$\Rightarrow \quad 1638 = 2 \times 3 \times 3 \times 7 \times 13$

Therefore, $\text{HCF}(2508, 1638) = 2 \times 3 = 6.$

18. Correct Answer is: 0

For divisibility rule of 4 last two digits must divisible by 4

Chance to be divisible

- 20
- 24
- 28

Among least is 20

∴'x' is 0.

Therefore, the least number that can replace 'x' is 0 and the number is 6702x is divisible by 4.

19. Correct Answer is: $\frac{3}{2}$

When the sum of the digits of a number is divisible by 3, the number will be divisible by 3.

Given number is 111.....1 (n times)

a) when n = 1, number is 1, not divisible by 3.

b) when n = 2, number is 11, a prime number and thus not divisible by 3.

c) when n = 3, number is 111 and 1+1+1 = 3, divisible by 3 and thus 111 is divisible by 3.

Therefore, the least possible value of n is 3 and the least possible value of $\frac{n}{2}$ is $\frac{3}{2}$.

20. Correct Answer is: 1

The consecutive odd numbers cannot have a common factor other than 1. For example, 3 and 5 do not have any common factors except 1.

Therefore, the HCF of 2 consecutive odd numbers is 1.

SURDS

LEVEL-I

1. Correct option: (A)

Let $x = 0.12333----$

Then,

$$10x = 1.2333----$$
$$100x = 12.333----$$
$$1000x = 123.3333----$$
$$-100x = 12.33333----$$

$$900x = 111$$
$$x = \frac{111}{900}$$

The correct Answer is 'A'

2. Correct option: (C)

To express the repeating decimal 0.2353535... in the form of p/q, you can use the following method:

Let x = 0.2353535...

Now, Consider 100x = 23.5353535...

Subtract the original **x** value from **100x**:

$$100x - x = 23.5353535... - 0.2353535...$$
$$\Rightarrow 99x = 23.3$$
$$\frac{23.3}{99} = \frac{233}{990}$$

The correct Answer is 'C'

3. Correct option: (A)

Given $0.2\bar{3}+0.2\bar{2}$

$$0.2\bar{3}=0.232323-----$$

$$0.2\bar{2}=0.222222----$$

Now, value of $0.2\bar{3}+0.2\bar{2}$ is equal to

$$0.2\bar{3}+0.2\bar{2}=0.232323-----+0.232323-----$$

$$0.2\bar{3}+0.2\bar{2}=0.4\bar{5}$$

The correct Answer is 'A'

4. Correct option: (C)

A rational number is a number that can be expressed as the quotient or fraction of two integers.

$\prod$ is an irrational number. It cannot be expressed as a fraction of two integers, and its decimal representation goes on forever without repeating.

e is an irrational number as well. Like π, it cannot be expressed as a fraction of two integers.

$\sqrt{144}=12$. This is a rational number because it can be expressed as a fraction of two integers (12/1).

$\sqrt{1210}$ is an irrational number. It cannot be expressed as a simple fraction of two integers.

The correct Answer is 'C'

5. Correct option: (C)

Let $x=0.712712..$

Now, subtract the non-repeating part from the repeating part.

$$1000x=712.712712...$$

$$x=0.712712...$$

Now, subtract the original x from 1000x.

$$1000x-x=712.712712...-0.712712...$$

$$999x=712$$

$$x=\frac{712}{999}$$

The correct Answer is 'C'

6. Correct option: (C)

The sum and difference of two simple quadratic surds are said to be conjugate surds to each other, and hence Conjugate of $a+\sqrt{b}$ is $a-\sqrt{b}$

The correct Answer is 'C'

7. Correct option: (B)

If $\sqrt{18-x\sqrt{2}}=\sqrt{12}-\sqrt{6}, x=?$

$$\sqrt{18-x\sqrt{2}}=\sqrt{12}-\sqrt{6}$$

Squaring on both sides implies

$$18-x\sqrt{2}=12+6-2\sqrt{72}$$

$$\Rightarrow 18-x\sqrt{2}=18-2\sqrt{72}$$

$$\therefore x\sqrt{2}=12\sqrt{2}$$

$$\Rightarrow x=12$$

The correct Answer is 'B'

8. Correct option: (B)

Square root of $7-\sqrt{48}$ is

$$\begin{aligned}&7-\sqrt{48}\\&=7-2\sqrt{12}\\&=4+3-2\sqrt{4.3}\\&=\sqrt{4^2}+\sqrt{3^2}-2\sqrt{4.3}\\&=(\sqrt{4}-\sqrt{3})^2\end{aligned}$$

Square root implies $\pm\left(2-\sqrt{3}\right)$

The correct Answer is 'B'

9. Correct option: (B)

The value of $\sqrt{12+\sqrt{12}+.....}$ is

Let $\sqrt{12+\sqrt{12}+.....}$ =y

This implies

$$\begin{aligned}&\sqrt{12+y}=y\\&\Rightarrow y^2=y+12\Rightarrow y^2-y-12=0\\&\Rightarrow y^2-4y+4y-12=0\\&y(\mathrm{y}-3)+4(\mathrm{y}-3)=0\\&\left(y-3\right)\left(y+4\right)=0\\&y=3\end{aligned}$$

The correct Answer is 'B'

10. Correct option: (C)

The positive square root of $8+2\sqrt{15}$

$$\begin{aligned}&8+2\sqrt{15}\\&=8+2\sqrt{15}\\&=5+3+2\sqrt{5.3}\\&=\sqrt{5^2}+\sqrt{3^2}+2\sqrt{5.3}\\&=(\sqrt{5}+\sqrt{3})^2\end{aligned}$$

Square root implies $\pm\left(2-\sqrt{3}\right)$

Therefore positive square root implies $\left(\sqrt{5}+\sqrt{3}\right)$

The correct Answer is 'C'

11. Correct option: ($=-1+\sqrt{100}=9$**)**

The value of $\frac{1}{1+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{3+\sqrt{4}}+.....99times$ is

$$\frac{1}{1+\sqrt{2}}\frac{1-\sqrt{2}}{1-\sqrt{2}}=\sqrt{2}-1$$

$$\frac{1}{\sqrt{2}+\sqrt{3}}\frac{\sqrt{2}-\sqrt{3}}{\sqrt{2}-\sqrt{3}}=\sqrt{3}-\sqrt{2}$$

$$\frac{1}{\sqrt{3}+\sqrt{4}}\frac{\sqrt{3}-\sqrt{4}}{\sqrt{3}-\sqrt{4}}=\sqrt{4}-\sqrt{3}$$

In the similar manner 99th term will be $\sqrt{100}-\sqrt{99}$

adding all these terms implies that $\sqrt{2}-1+\sqrt{3}-\sqrt{2}+\sqrt{4}-\sqrt{3}+\ldots.\sqrt{100}-\sqrt{99}$

$$=-1+\sqrt{100}=9$$

The correct Answer is not given in option

12. Correct option: (A)

$$\frac{5\sqrt{3}+\sqrt{7}}{4\sqrt{3}-2\sqrt{7}}=a+b\sqrt{21}=\frac{37}{10}+\frac{7\sqrt{21}}{10},a\,\&\,b=?$$

$$\Rightarrow\frac{5\sqrt{3}+\sqrt{7}}{4\sqrt{3}-2\sqrt{7}}\frac{4\sqrt{3}+2\sqrt{7}}{4\sqrt{3}+2\sqrt{7}}=a+b\sqrt{21}$$

$$\Rightarrow\frac{74+14\sqrt{21}}{48-28}=\frac{37}{10}+\frac{7\sqrt{21}}{10}=a+b\sqrt{21}$$

$$\therefore a=\frac{37}{10},b=\frac{7}{10}$$

The correct Answer is 'A'

13.Correct option: (A)

If x>0, $\sqrt{x\sqrt{x\sqrt{x\sqrt{x...\infty}}}}$

$$\sqrt{x\sqrt{x\sqrt{x\sqrt{x...\infty}}}}$$

$$=x^{\frac{1}{2}}.x^{\frac{1}{4}}.x^{\frac{1}{8}}...\infty$$

$$=x^{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+....\infty}$$

$$\therefore\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+....\infty$$

$$s_\infty=\frac{\frac{1}{2}}{1-\frac{1}{2}}=1$$

$$\Rightarrow x^1=x$$

The correct Answer is 'A'

14. Correct option: (B)

The given set of surds $\sqrt[4]{10},\sqrt[3]{6},\sqrt{3}$

$$\sqrt[4]{10},\sqrt[3]{6},\sqrt{3}$$

$$10^{\frac{1}{4}},6^{\frac{1}{3}},3^{\frac{1}{2}}$$

LCM of denominators of the exponentials is 12 and hence multiply all these exponentials with 12 implies

$$10^{\frac{1}{4}\times 12}, 6^{\frac{1}{3}\times 12}, 3^{\frac{1}{2}\times 12}$$
$$10^3, 6^4, 3^6$$
$$1000, 1296, 729$$

Therefore arranging surds in ascending order implies $\sqrt{3}, \sqrt[4]{10}, \sqrt[3]{6}$

The correct Answer is 'B'

15. Correct option: (B)

$$x = \frac{1}{2}\left(\sqrt{a} + \frac{1}{\sqrt{a}}\right) = \frac{a+1}{2\sqrt{a}}$$
$$x^2 - 1 = \frac{(a+1)^2}{4a} - 1 = \frac{(a+1)^2 - 4a}{4a} = \frac{(a-1)^2}{4a}$$
$$\sqrt{x^2-1} = \sqrt{\frac{(a-1)^2}{4a}} = \frac{a-1}{2\sqrt{a}}$$
$$x - \sqrt{x^2-1} = \frac{a+1}{2\sqrt{a}} - \frac{a-1}{2\sqrt{a}} = \frac{2}{2\sqrt{a}}$$
$$\therefore \frac{\sqrt{x^2-1}}{x-\sqrt{x^2-1}} = \frac{\frac{a-1}{2\sqrt{a}}}{\frac{2}{2\sqrt{a}}} = \frac{a-1}{2}$$

The correct Answer is 'B'

16. Correct option: (B)

If $x = \sqrt[3]{2} + \sqrt[3]{4}$ then $x^3 - 6x =$

$x = \sqrt[3]{2} + \sqrt[3]{4}$

$\Rightarrow x^3 = \left(\sqrt[3]{2} + \sqrt[3]{4}\right)^3 = 2 + 4 + 3\sqrt[3]{8}(\sqrt[3]{2} + \sqrt[3]{4})$

$\Rightarrow x^3 = 6 + \sqrt[3]{27.8}(x)$

$\Rightarrow x^3 - 6x = 6$

The correct Answer is 'B'

17. Correct option: (B)

Rationalizing factor of $\left(27^{\frac{1}{9}} - 4^{\frac{1}{6}}\right)$ **is**

$$\left(27^{\frac{1}{9}} - 4^{\frac{1}{6}}\right) = \left((27^3)^{\frac{1}{3}} - (4^2)^{\frac{1}{3}}\right)$$
$$\sqrt[3]{27^6} + \sqrt[3]{4^4} + \sqrt[3]{27^3.4^2}$$

The correct Answer is 'B'

18. Correct option: (D)

If $\frac{5-\sqrt{3}}{2+\sqrt{3}} = x + y\sqrt{3}, x \& y = ?$

$$\frac{5-\sqrt{3}}{5+\sqrt{3}} = x + y\sqrt{3}$$

$$x + y\sqrt{3} = \frac{5-\sqrt{3}}{2+\sqrt{3}}\frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{10+3-7\sqrt{3}}{1} = 13-7\sqrt{3}$$

$$x = 13, y = -7$$

The correct Answer is 'D'

19. Correct option: (C)

If $\sqrt{13-a\sqrt{10}} = \sqrt{8}+\sqrt{5}, \text{a} = ?$

$$\sqrt{13-a\sqrt{10}} = \sqrt{8}+\sqrt{5}$$

Squaring on both sides implies

$$13-a\sqrt{10} = 8+5+2\sqrt{40}$$

$$\Rightarrow 13-a\sqrt{10} = 13+4\sqrt{10}$$

$$\therefore a\sqrt{10} = -4\sqrt{10}$$

$$\Rightarrow a = -4$$

The correct Answer is 'C'

20. Correct option: (B)

If $x = 2\sqrt{2}+\sqrt{7}, \frac{1}{2}\left(x+\frac{1}{x}\right) =$

$$\frac{1}{x} = \frac{1}{2\sqrt{2}+\sqrt{7}}\frac{2\sqrt{2}-\sqrt{7}}{2\sqrt{2}-\sqrt{7}} = 2\sqrt{2}-\sqrt{7}$$

$$x+\frac{1}{x} = 2\sqrt{2}+\sqrt{7}+2\sqrt{2}-\sqrt{7} = 4\sqrt{2}$$

$$\frac{1}{2}\left(x+\frac{1}{x}\right) = \frac{1}{2}.4\sqrt{2} = 2\sqrt{2}$$

The correct Answer is 'B'

LEVEL-II

1. Correct option: (C)

$$\sqrt[4]{193+132\sqrt{2}}$$

$$\Rightarrow \sqrt[4]{(3+\sqrt{2})^4} = 3+\sqrt{2}$$

The correct Answer is 'C'

2. Correct option: (B)

$$\sqrt[4]{137-36\sqrt{14}}$$

$$\Rightarrow \sqrt[4]{(\sqrt{7}-\sqrt{2})^4} = \sqrt{7}-\sqrt{2}$$

The correct Answer is 'B'

3. Correct option: (A)

$$\sqrt{\frac{15}{2}+2\sqrt{3}+2\sqrt{6}+4\sqrt{2}}=$$

$$\sqrt{\left(\sqrt{\frac{3}{2}}+2+\sqrt{2}\right)^2}=\sqrt{\frac{3}{2}}+2+\sqrt{2}$$

The correct Answer is 'A'

4. Correct option: (A)

$$\sqrt{a+3b+4+4\sqrt{a}-4\sqrt{3b}-2\sqrt{3ab}}=$$

$$\sqrt{\left(2+\sqrt{a}-\sqrt{3b}\right)^2}=2+\sqrt{a}-\sqrt{3b}$$

The correct Answer is 'A'

5. Correct option: (A)

$$\sqrt[3]{72-32\sqrt{5}}=\sqrt[3]{(3-\sqrt{5})^3}=3-\sqrt{5}$$

The correct Answer is 'A'

6. Correct option: (C)

The value of $\sqrt[3]{38+17\sqrt{5}}=$

$$\sqrt[3]{38+17\sqrt{5}}=\sqrt[3]{(x+\sqrt{y})^3}$$

$$\therefore (x+\sqrt{y})^3=38+17\sqrt{5}$$

$$\Rightarrow x^3+y\sqrt{y}+3xy+3x^2\sqrt{y}=38+17\sqrt{5}$$

$$\Rightarrow x^3++3xy+\sqrt{y}(y+3x^2)=38+17\sqrt{5}$$

$$\Rightarrow \sqrt{y}=\sqrt{5}\Rightarrow y=5$$

$$\Rightarrow y+3x^2=17\Rightarrow 5+3x^2=17\Rightarrow x=2$$

$$\therefore x+\sqrt{y}=2+\sqrt{5}$$

The correct Answer is 'C'

7. Correct option: (A)

If $a=x+\sqrt{x^2+1}$ then x=?

$$a=x+\sqrt{x^2+1}$$

$$\Rightarrow a-x=\sqrt{x^2+1}$$

Squaring on both sides implies

$$(a-x)^2=x^2+1$$

$$\Rightarrow a^2+x^2-2ax=x^2+1$$

$$\Rightarrow a^2+2ax=1$$

$$\Rightarrow x=\frac{a^2-1}{2a}=\frac{1}{2}\left(a-\frac{1}{a}\right)$$

The correct Answer is 'A'

8. Correct option: (B)

The value of $\sqrt[3]{54\sqrt{3}+41\sqrt{5}}=$

$$\sqrt[3]{54\sqrt{3}+41\sqrt{5}}=\sqrt[3]{(\sqrt{x}+\sqrt{y})^3}$$
$$\therefore(\sqrt{x}+\sqrt{y})^3=54\sqrt{3}+41\sqrt{5}$$
$$\Rightarrow x\sqrt{x}+y\sqrt{y}+3y\sqrt{x}+3x\sqrt{y}=54\sqrt{3}+41\sqrt{5}$$
$$\Rightarrow (\mathrm{x}+3\,\mathrm{y})\sqrt{x}+(\mathrm{y}+3\,\mathrm{x})\sqrt{y}=54\sqrt{3}+41\sqrt{5}$$
$$\Rightarrow \sqrt{y}=\sqrt{5}\Rightarrow y=5$$
$$\Rightarrow y+3x=41\Rightarrow 5+3x=41\Rightarrow x=12\Rightarrow \sqrt{x}=\sqrt{12}=2\sqrt{3}$$
$$\therefore \sqrt{x}+\sqrt{y}=2\sqrt{3}+\sqrt{5}$$

The correct Answer is 'B'

9. Correct option: (C)

The value of $\sqrt[3]{34\sqrt{5}-18\sqrt{3}}=$

$$\sqrt[3]{34\sqrt{5}-18\sqrt{3}}=\sqrt[3]{(\sqrt{x}-\sqrt{y})^3}$$
$$\therefore(\sqrt{x}-\sqrt{y})^3=34\sqrt{5}-18\sqrt{3}$$
$$\Rightarrow x\sqrt{x}-y\sqrt{y}+3y\sqrt{x}-3x\sqrt{y}=34\sqrt{5}-18\sqrt{3}$$
$$\Rightarrow (\mathrm{x}-3\,\mathrm{y})\sqrt{x}-(\mathrm{y}+3\,\mathrm{x})\sqrt{y}=34\sqrt{5}-18\sqrt{3}$$
$$\Rightarrow \sqrt{y}=\sqrt{3}\Rightarrow y=3$$
$$\Rightarrow y+3x=18\Rightarrow 3+3x=18\Rightarrow x=5\Rightarrow \sqrt{x}=\sqrt{5}$$
$$\therefore \sqrt{x}-\sqrt{y}=\sqrt{5}-\sqrt{3}$$

The correct Answer is 'C'

10. Correct option: (C)

$$x=\sqrt{17}-\sqrt{16},\mathrm{y}=\sqrt{15}-\sqrt{14}\Rightarrow$$

$$x=\sqrt{17}-\sqrt{16}=0.1231$$

$$\mathrm{y}=\sqrt{15}-\sqrt{14}=0.1313$$

This implies x<y

The correct Answer is 'C'

11. Correct option: (A)

If x>0, $\sqrt{x\sqrt{x\sqrt{x\sqrt{x...\infty}}}}$

$$\sqrt{x\sqrt{x\sqrt{x\sqrt{x...\infty}}}}$$

$$= x^{\frac{1}{2}}.x^{\frac{1}{4}}.x^{\frac{1}{8}}...\infty$$

$$= x^{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+....\infty}$$

$$\therefore \frac{1}{2}+\frac{1}{4}+\frac{1}{8}+....\infty$$

$$s_\infty = \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1$$

$$\Rightarrow x^1 = x$$

The correct Answer is 'A'

12. Correct option: (B)

If $\sqrt{18+x\sqrt{2}} = \sqrt{12}+\sqrt{6}, x = ?$

$$\sqrt{18+x\sqrt{2}} = \sqrt{12}+\sqrt{6}$$

Squaring on both sides implies

$$18+x\sqrt{2} = 12+6+2\sqrt{72}$$

$$\Rightarrow 18+x\sqrt{2} = 18+2\sqrt{72}$$

$$\therefore x\sqrt{2} = 12\sqrt{2}$$

$$\Rightarrow x = 12$$

The correct Answer is 'B'

13. Correct option: (C)

If x= $\sqrt{30+\sqrt{30}+.....}$ then x

$$x = \sqrt{30+\sqrt{30}+.....}$$

$$\Rightarrow x = \sqrt{30+x}$$

$$\Rightarrow x^2 = 30+x$$

$$\Rightarrow x^2 - x - 30 = 0$$

$$\Rightarrow x^2 - 6x + 5x - 30 = 0$$

$$\Rightarrow x(x-6)+5(x-6) = 0$$

$$(x-6)(x+5) = 0$$

$$x = 6, -5$$

The correct Answer is 'C'

14. Correct option: (C)

The value of $\sqrt{1+\sqrt{7+\sqrt{1+\sqrt{7}}}+.....}$

$$\sqrt{1+\sqrt{7+\sqrt{1+\sqrt{7}}}+.....}=x$$

$$\Rightarrow \sqrt{1+\sqrt{7+x}}=x$$

$$\Rightarrow x^2=1+\sqrt{7+x}$$

$$\Rightarrow x^2-1=\sqrt{7+x}$$

$$\Rightarrow \left(x^2-1\right)^2=\left(\sqrt{7+x}\right)^2$$

$$\Rightarrow x^4-2x^2+1=7+x$$

$$\Rightarrow x^4-2x^2-x-6=0$$

After solving the value of x=2,-1.81

The correct Answer is 'C'

15. Correct option: (C)

The value of $2+\sqrt{2+\sqrt{2}+.....}$ is

Let $\sqrt{2+\sqrt{2}+.....}$ =y

This implies

$$\sqrt{2+y}=y$$

$$\Rightarrow y^2=y+2\Rightarrow y^2-y-2=0$$

$$\Rightarrow y^2-2y+y-2=0$$

$$y(y-2)+1(y-2)=0$$

$$(y-2)(y+1)=0$$

$$y=2or-1$$

$$\therefore 2+\sqrt{2+\sqrt{2}+.....}=2+y=2+2(\text{or})2-1=4\,\text{or}\,1$$

The correct Answer is 'C'

16. Correct Matching is: a-Q; b-T; c-; d-S

a) If $x=4-\sqrt{15}$ then $x^2=0.0161,\frac{1}{x^2}=61.9838,x^2-\frac{1}{x^2}=-61.96\approx-16\sqrt{15}$

The correct match is 'Q'

b) If $x=2+\sqrt{3},xy=1\Rightarrow y=\frac{1}{x}=\frac{1}{2+\sqrt{3}}\frac{2-\sqrt{3}}{2-\sqrt{3}}=2-\sqrt{3}$

Then $\frac{x}{x-2}+\frac{y}{y-2}=\frac{2+\sqrt{3}}{2+\sqrt{3}-2}+\frac{2-\sqrt{3}}{2-\sqrt{3}-2}=2$

The correct match is 'T'

c)

$$\frac{5-\sqrt{11}}{7+2\sqrt{11}}=a\sqrt{11}-b$$

$$\Rightarrow \frac{5-\sqrt{11}}{7+2\sqrt{11}}\frac{7-2\sqrt{11}}{7-2\sqrt{11}}=a\sqrt{11}-b$$

$$\Rightarrow \frac{57-17\sqrt{11}}{5}=a\sqrt{11}-b$$

There is no correct match

d)

$$x = 7 + 4\sqrt{3}, x + \frac{1}{x} = ? \Rightarrow$$

If $\frac{1}{x} = \frac{1}{7+4\sqrt{3}} \frac{7-4\sqrt{3}}{7-4\sqrt{3}} = 7 - 4\sqrt{3}$

$$x + \frac{1}{x} = 7 + 4\sqrt{3} + 7 - 4\sqrt{3} = 14$$

The correct match is 'S'

17. Correct Answer is: 27

If $x = \left(49 - 20\sqrt{6}\right)^{\frac{-1}{2}}$ then the value of $x^3 + \frac{1}{x^3} - 9\left(x^2 + \frac{1}{x^2}\right) - 7\left(x + \frac{1}{x}\right) - 18$

$$x + \frac{1}{x} = k$$

$$\Rightarrow \left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2$$

$$\therefore k^2 = x^2 + \frac{1}{x^2} + 2 \Rightarrow x^2 + \frac{1}{x^2} = k^2 - 2$$

$$\Rightarrow x^3 + \frac{1}{x^3} = k^3 - 3$$

Therefore $x^3 + \frac{1}{x^3} - 9\left(x^2 + \frac{1}{x^2}\right) - 7\left(x + \frac{1}{x}\right) - 18 =$

$$k^3 - 3 - 9(k^2 - 2) - 7\text{k} - 18$$

$$\Rightarrow k^3 - 9k^2 - 7k - 3$$

Now $x + \frac{1}{x} = k$

$$x + \frac{1}{x} = k$$

$$\Rightarrow k = \left(49 - 20\sqrt{6}\right)^{\frac{-1}{2}} + \frac{1}{\left(49 - 20\sqrt{6}\right)^{\frac{-1}{2}}} = 9.99 \approx 10$$

Now the value of $k^3 - 9k^2 - 7k - 3$ becomes

$$10^3 - 9(10)^2 - 7(10) - 3$$

$$= 1000 - 900 - 70 - 3 = 27$$

18. Correct Answer is: 7

The value of $\frac{\sqrt[3]{343} \times \sqrt[5]{16807} \times \sqrt[7]{11767}}{49}$

$$\frac{\sqrt[3]{343} \times \sqrt[5]{16807} \times \sqrt[7]{11767}}{49}$$

$$= \frac{\sqrt[3]{7^3} \times \sqrt[5]{7^5} \times \sqrt[7]{7^7}}{49}$$

$$= \frac{7 \times 7 \times 7}{49} = 7$$

19. **Correct Answer is: 6**

If $\sqrt[5]{4a^{3k+2}}\sqrt[5]{8a^{6-k}} = 2a^4$ then 'k'

$$\sqrt[5]{4a^{3k+2}}\sqrt[5]{8a^{6-k}} = 2a^4$$
$$\Rightarrow \sqrt[5]{4a^{3k+2}8a^{6-k}} = 2a^4$$
$$\Rightarrow \sqrt[5]{32a^{3k+2+6-k}} = 2a^4$$
$$\Rightarrow 2\sqrt[5]{a^{3k+2+6-k}} = 2a^4$$
$$\Rightarrow a^{3k+2+6-k} = a^{20}$$
$$\Rightarrow 2k+8 = 20 \Rightarrow k = 6$$

20. **Correct Answer is: 1**

$$\frac{\sqrt{13}-\sqrt{11}}{\sqrt{13}+\sqrt{11}} = a - b\sqrt{143}, a^2 - 143b =$$

$$\frac{\sqrt{13}-\sqrt{11}}{\sqrt{13}+\sqrt{11}} = a - b\sqrt{143}$$
$$\frac{\sqrt{13}-\sqrt{11}}{\sqrt{13}+\sqrt{11}}\frac{\sqrt{13}+\sqrt{11}}{\sqrt{13}+\sqrt{11}} = a - b\sqrt{143}$$
$$\frac{2}{24+2\sqrt{143}} = a - b\sqrt{143}$$
$$\Rightarrow a - b\sqrt{143} = \frac{1}{12+\sqrt{143}}\frac{12-\sqrt{143}}{12+\sqrt{143}}$$
$$\Rightarrow a - b\sqrt{143} = 12 - \sqrt{143}$$
$$\Rightarrow a = 12, \mathrm{b} = 1$$
$$\therefore a^2 - 143b = 12^2 - 143 = 1$$

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)

Let $x = 1.0001$.

To convert this decimal into a fraction, you need to find a common denominator and eliminate the decimal point.

$x = 1.0001$

$x \times 10,000 = 1.0001 \times 10,000$

$10,000x = 10,001$

$$x = \frac{10,001}{10,000}$$

The correct Answer is 'C'

2. Correct option: (B)

Given $0.62\bar{1}$

Let x=0.62161-----

$$1000x = 621.621.....$$
$$x = 0.621621.....$$
$$999x = 621$$

$$x = \frac{621}{999}$$
$$x = \frac{69}{111}$$
$$x = \frac{23}{37}$$

The correct Answer is 'B'

3. Correct option: (C)

An irrational number is a number that cannot be expressed as a fraction of two integers, and its decimal representation goes on forever without repeating.

$\sqrt{\frac{16}{25}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$. This is a rational number because it can be expressed as a fraction of two integers.

$\sqrt{\frac{12}{75}} = \sqrt{\frac{4}{25}} = \frac{2}{\sqrt{5}}$. This is a rational number because it can be expressed as a fraction of two integers.

$\sqrt{\frac{80}{16}} = \sqrt{5}$. This is an irrational number because it cannot be expressed as a fraction of two integers.

$\sqrt{121} = 11$. This is a rational number because it can be expressed as a fraction of two integers.

The correct Answer is 'C'

4. Correct option: (A)

Let $x = 1.565656----$

Now $100x = 156.565656...$

and x = 1.565656...

$$100x - x = 156.565656... - 1.565656...$$
$$99x = 155$$

Now, divide both sides by 99 to solve for x.

$$x = \frac{155}{99}$$

The correct Answer is 'A'

5. Correct option: (D)

The value of $\sqrt{5+2\sqrt{6}}$

$$5 + 2\sqrt{6} = 3 + 2 + 2\sqrt{3.2} = \left(\sqrt{3}\right)^2 + \left(\sqrt{2}\right)^2 + 2\sqrt{3}\sqrt{2} = \left(\sqrt{3} + \sqrt{2}\right)^2$$
$$\therefore \sqrt{5+2\sqrt{6}} = \sqrt{\left(\sqrt{3}+\sqrt{2}\right)^2} = \left(\sqrt{3}+\sqrt{2}\right)$$

The correct Answer is 'D'

6. Correct option: (D)

If $\sqrt{a+\sqrt{b}} = 2+\sqrt{3}$ then $\sqrt{a^2 - b} =$

$$\sqrt{a+\sqrt{b}} = 2+\sqrt{3}$$

Squaring on both sides

$$a+\sqrt{b}=4+3+4\sqrt{3}$$
$$\Rightarrow a+\sqrt{b}=7+4\sqrt{3}=7+\sqrt{48}$$
$$\Rightarrow a=7, b=48$$

Therefore $\sqrt{a^2-b}=\sqrt{7^2-48}=1$

The correct Answer is 'D'

7. Correct option: (A)

If $\sqrt{19-4\sqrt{x}}=\sqrt{12}-\sqrt{7}, x=?$

$$\sqrt{19-4\sqrt{x}}=\sqrt{12}-\sqrt{7}$$

Squaring on both sides implies that

$$\Rightarrow 19-4\sqrt{x}=12+7-2\sqrt{84}$$
$$\Rightarrow 19-4\sqrt{x}=19-4\sqrt{21}$$
$$\Rightarrow x=21$$

The correct Answer is 'A'

8.Correct option: (D)

Rationalizing factor of $\sqrt[3]{81}$

$$\sqrt[3]{81}$$
$$=\sqrt[3]{3\times 3\times 3\times 3}$$
$$=3\sqrt[3]{3}$$

To solve the cube root the above surd is to be multiplied with

$$3\sqrt[3]{3}\times\sqrt[3]{3^2}$$
$$=3\sqrt[3]{3\times 3^2}=9$$

To rationalize the given root $\sqrt[3]{3^2}$ is needed to multiply and hence it is the rationalizing factor

The correct Answer is 'D'

9. Correct option: (C)

The given set of surds $\sqrt[6]{2}, \sqrt[4]{2}, \sqrt[3]{2}$

$$\sqrt[6]{2}, \sqrt[4]{2}, \sqrt[3]{2}$$
$$2^{\frac{1}{6}}, 6^{\frac{1}{4}}, 2^{\frac{1}{3}}$$

LCM of denominators of the exponentials is 12 and hence multiply all these exponentials with 12 implies

$$2^{\frac{1}{6}\times 12}, 2^{\frac{1}{4}\times 12}, 2^{\frac{1}{3}\times 12}$$
$$2^2, 2^3, 2^4$$

Therefore arranging order of surds implies

$$\sqrt[6]{2}<\sqrt[4]{2}<\sqrt[3]{2}$$

The correct Answer is 'C'

10. Correct option: (C)

if $\sqrt{x}+\sqrt{x-\sqrt{1-x}}=1$ then the value of 'x' is

$$\sqrt{x}+\sqrt{x-\sqrt{1-x}}=1$$

$$\Rightarrow \sqrt{x}-1=\sqrt{x-\sqrt{1-x}}$$

Squaring on both sides implies

$$\Rightarrow x+1-2\sqrt{x}=x-\sqrt{1-x}$$

$$\Rightarrow 1-2\sqrt{x}=-\sqrt{1-x}$$

Squaring on both sides implies

$(1-2\sqrt{x})^2 = 1-x$

$\Rightarrow 1+4x-4\sqrt{x}=1-x$

$\Rightarrow 5x=4\sqrt{x}$

$\Rightarrow \sqrt{x}=\frac{5}{4} \Rightarrow x=\frac{25}{16}$

The correct Answer is 'C'

11. Correct option: (D)

If $x=5+2\sqrt{6}$ then the value of $\sqrt{x}+\frac{1}{\sqrt{x}}$

$x = 5+2\sqrt{6}$

$\Rightarrow x = 2+3+2\sqrt{2.3}$

$\Rightarrow (\sqrt{x})^2 = (\sqrt{2})^2+(\sqrt{3})^2+2\sqrt{2.3}$

$\Rightarrow (\sqrt{x})^2 = (\sqrt{2}+\sqrt{3})^2$

$\Rightarrow \sqrt{x}=\sqrt{2}+\sqrt{3}$

$\therefore \frac{1}{\sqrt{x}}=\frac{1}{\sqrt{2}+\sqrt{3}}\frac{-\sqrt{2}+\sqrt{3}}{-\sqrt{2}+\sqrt{3}}=\frac{\sqrt{3}-\sqrt{2}}{1}$

$\Rightarrow \sqrt{x}+\frac{1}{\sqrt{x}}=\sqrt{2}+\sqrt{3}+\sqrt{3}-\sqrt{2}=2\sqrt{3}$

The correct Answer is 'D'

12. Correct option: (A,B)

The value of continued function $1+\frac{1}{1+\frac{1}{1+.....}}$ **is**

Let $\frac{1}{1+\frac{1}{1+.....}}=x$

$$\therefore x=\frac{1}{1+x}$$

$$\Rightarrow x^2+x-1=0$$

$$x=\frac{-1\pm\sqrt{1+4}}{2}=\frac{-1\pm\sqrt{5}}{2}$$

$$\therefore 1+\frac{1}{1+\frac{1}{1+.....}}=1+\frac{-1\pm\sqrt{5}}{2}=\frac{1\pm\sqrt{5}}{2}$$

The correct Answer is 'A'& 'B'

13. Correct option: (B)

It is known the rationalizing factor for $\sqrt[3]{a}+\sqrt[3]{b}=\sqrt[3]{a^2}-\sqrt[3]{ab}+\sqrt[3]{b^2}$

$$\therefore 7^{\frac{1}{3}}+7^{\frac{-1}{3}}=\sqrt[3]{7}+\sqrt[3]{\frac{1}{7}}=\sqrt[3]{7^2}-\sqrt[3]{7.\frac{1}{7}}+\sqrt[3]{(\frac{1}{7})^2}=\sqrt[3]{49}-1+\sqrt[3]{\frac{1}{49}}$$

The correct Answer is 'B'

14. Correct option: (B)

It is known the rationalizing factor for $\sqrt[3]{a}-\sqrt[3]{b}=\sqrt[3]{a^2}+\sqrt[3]{ab}+\sqrt[3]{b^2}$

$$\therefore \sqrt[3]{25}+1+\sqrt[3]{\frac{1}{25}}=5^{\frac{1}{3}}-5^{\frac{-1}{3}}=\sqrt[3]{5}-\sqrt[3]{\frac{1}{5}}$$

The correct Answer is 'B'

15. Correct option: (D)

$$\sqrt[4]{3}+1=\sqrt[4]{3^2}-\sqrt[4]{3.1}+\sqrt[4]{1^2}=\sqrt[4]{9}-\sqrt[4]{3}+1$$

The correct Answer is 'D'

16. Correct Matching is: a-S; b-Q; c-R; d-T

If $x=2+\sqrt{3}$

a) $x+\frac{1}{x}=(2+\sqrt{3})+\frac{1}{(2+\sqrt{3})}=3.732+0.2679=3.999\approx 4$

The correct match is 'S'

b) $\left(x+\frac{1}{x}\right)^2=x^2+\frac{1}{x^2}+2\Rightarrow 16=x^2+\frac{1}{x^2}+2\Rightarrow x^2+\frac{1}{x^2}=14$

The correct match is 'Q'

c) $\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)\Rightarrow 64=x^3+\frac{1}{x^3}+12\Rightarrow x^3+\frac{1}{x^3}=52$

The correct match is 'R'

d) $x=2+\sqrt{3}\Rightarrow x^4=193.99,\frac{1}{x^4}=0.0051\left(x^4-\frac{1}{x^4}\right)^2=37632$

The correct match is 'T'

2. Polynomials

Solutions

LEVEL-I

1. Correct option: (C)

The power of x must be a non-negative integer in a polynomial.

C) $\frac{x^2+x^3}{x} = x + x^2$.

As the exponent of x is the whole number, it is a polynomial.

From the given options only C has a non-negative integer in powers.

Therefore, $\frac{x^2+x^3}{x}$ is a polynomial.

2. Correct option: (B)

$\frac{\sqrt{2}x^5-x^{10}+1}{x} = \sqrt{2}x^{5-1} - x^{10-1} + x^{-1} = \sqrt{2}x^4 - x^9 + x^{-1}$.

As 9 is the highest exponent above, it is also the degree of the polynomial.

3. Correct option: (C)

Expression having three terms are is trinomial.

The polynomial $x^2 + 2x + 10$ has three terms.

From the given options only C has a three terms.

Therefore, $x^2 + 2x + 10$ is a trinomial.

4. Correct option: (C)

Let $p(x)$ be any polynomial of degree greater than or equal to one and let $'a'$ be any real number.

If a polynomial p(x) is divided by $x - a$ then the remainder *is* $p(a)$.

Let $p(x) = x^3 - ax^2 + 6x - a$

The root of $x - a = 0$ *is* a.

$p(a) = (a)^3 - a(a)^2 + 6(a) - a = (a)^3 - (a)^3 + 5a = 5a$

Hence by remainder theorem, $5a$ is the remainder when $x^3 - ax^2 + 6x - a$ is divided by $x - a$.

5. Correct option: (C)

6.

$$
\begin{array}{rl}
 & 2x - 1 \\
x + 2 \;\overline{)}\; & 2x^2 + 3x + 1 \\
(-)\;\; & 2x^2 + 4x \\
\hline
 & -x + 1 \\
(+)\;\; & -x - 2 \\
\hline
 & +3
\end{array}
$$

Therefore, the remainder is 3.

6. Correct option: (B).

$f(x) = px^3 + 4x^2 + 3x - 4$

$g(x) = p + x^3 - 4$

By remainder theorem, if f(x) is divided by $(x - a)$ then the remainder is $f(a)$.

Here, when $f(x)$ *and* $g(x)$ are divided by $(x - 3)$ the remainders are $f(3)$ *and* $g(3)$ respectively.

Put $x = 3$ in both $f(x)\ and\ g(x)$, so $f(3) = g(3)$ as they are equal
$\Rightarrow p(3)^3 + 4(3)^2 + 3(3) - 4 = p + (3)^3 - 4$
$\Rightarrow 27p + 36 + 9 - 4 = 27 - 12 + p$
$\Rightarrow 27p - p = 15 - 41$
$\Rightarrow 26p = -26$
$\therefore p = -1.$

7. Correct option: (C)

Using Remainder theorem which states that if a polynomial $p(x)$ is divisible by polynomial of form $x - a$ then remainder is given by $p(a)$.
According to remainder theorem,
$R1 = f(1) = 4(1)^3 + 3(1)^2 - 12a(1) - 5 = 4 + 3 - 12a - 5 = 2 - 12a$
$R2 = g(-2) = 2(-2)^3 + a(-2)^2 - 6(-2) + 2 = -16 + 4a + 12 + 2 = 4a - 2$
$Now,$
$3 \times R1 + R2 + 28 = 0$
$3(2 - 12a) + 4a - 2 + 28 = 0$
$6 - 36a + 4a + 26 = 0$
$-32a = -32$
$a = 1.$

Therefore, Value of a is 1.

8. Correct option: (A)

Since $x^2 - 1 = (x - 1)\ is\ a\ factor\ of$
$p(x) = ax^2 + bx^3 + cx^2 + dx + e$
$\therefore$ p(x) is divisible by (x+1) and (x-1) separately
$\Rightarrow p(1) = 0\ and\ p(-1) = 0$
$p(1) = a(1)^4 + b(1)^3 + c(1)^2 + d(1) + e = 0$
$\Rightarrow a + b + c + d + e = 0 ----(i)$
Similarly, $p(-1) = a(-1)^4 + b(-1)^3 + c(-1)^2 + d(-1) + e = 0$
$\Rightarrow a - b + c - d + e = 0$
$\Rightarrow a + c + e = b + d ----(ii)$
Putting the value of a + c + e in eqn , we get
$a + b + c + d + e = 0$
$\Rightarrow a + c + e + b + d = 0$
$\Rightarrow b + d + b + d = 0$
$\Rightarrow 2(b + d) = 0$
$\Rightarrow b + d = 0 ----(iii)$
comparing equations (ii) and (iii) , we get
$a + c + e = b + d = 0$
Therefore, $a + c + e = 0.$

9. Correct option: (C)

Let , $p(x) = kx^2 + 2k^2x + m^3$
If $(x + k)$ exactly divides $p(x)$, then for $x = -k$, $p(x) = 0$
$i.e.\ p(-k) = 0.$
$p(-k) = k(-k)^2 + 2k^2(-k) + m^3$
$k^3 - 2k^3 + m^3 = 0$
$-k^3 + m^3 = 0$
$m^3 - k^3 = 0.$
[Since $(a^3 - b^3) = (a - b)(a^2 + b^2 + ab)$]
$(m^3 - k^3) = (m - k)(m^2 + k^2 + mk) = 0$

$(m-k)(m^2+k^2+mk)=0$
$(m^2+k^2+mk)=0$ (Divide both sides by $(m-k)$)
Therefore, the value of(m^2+k^2+mk) is 0.

10. Correct option: (B).

$f(x)=6x^2+mx-4$
$g(x)=nx^2-11x+3$
$f\left(\frac{1}{2}\right)=6\left(\frac{1}{2}\right)^2+m\left(\frac{1}{2}\right)-4=0$
$2m-10=0$
$m=5$
$g\left(\frac{1}{2}\right)=n\left(\frac{1}{2}\right)^2-11\left(\frac{1}{2}\right)+3=0$
$n-10=0$
$n=10$
Now,
$n^2-m^2+n-m=(10)^2-(5)^2+5-10=(10)^2-(5)^2-5=100-30=70$
Therefore, the required answer is 70.

11. Correct option: (D)

Let , $p(x)=x^3-\mathrm{k}x^2+3x+10$
If $(x-2)$ exactly divides $p(x)$, then for $x=2$, $p(x)=0$
$i.e.\ p(2)=0.$
$p(2)=(2)^3-\mathrm{k}(2)^2+3(2)+10$
$8-4\mathrm{k}+6+10=0$
$-4k=24$
$k=-6.$
Therefore, the value of k is -6.

12. Correct option: (B)

$p(x)=3x^2-x-10$
$q(x)=2x^2-x-6$
Let $(x-k)$ be a common factor of p(x) and q(x).
Therefore, $p(k)=q(k)=0$
Therefore,
$3k^2-k-10=\ 2k^2-k-6$
$k^2=4$
$k\ can\ be\ \pm 2\ but\ if\ k=-2,$
$p(-2)=12+2-10=4=0$
$Hence\ k=2$
Therefore, required common factor is $(x-2)$.

13. Correct option: (D)

$$\frac{(x^2+11x+28)}{(x^2+13x+40)}\div\frac{(x^2+6x+8)}{(x^2+11x+24)}=\frac{(x+4)(x+7)}{(x+5)(x+8)}\div\frac{(x+4)(x+2)}{(x+3)(x+8)}$$

We know that, $\frac{p}{q}\div\frac{r}{s}=\frac{p}{q}\times\frac{s}{r}$

$$\frac{(x^2+11x+28)}{(x^2+13x+40)}\div\frac{(x^2+6x+8)}{(x^2+11x+24)}=\frac{(x+4)(x+7)}{(x+5)(x+8)}\times\frac{(x+3)(x+8)}{(x+4)(x+2)}$$

$$\frac{(x^2+11x+28)}{(x^2+13x+40)}\div\frac{(x^2+6x+8)}{(x^2+11x+24)}=\frac{(x+7)(x+3)}{(x+5)(x+2)}.$$

Therefore, the required answer is $\frac{(x+7)(x+3)}{(x+5)(x+2)}$.

14. Correct option: (A)

$x^2 + 5\sqrt{3}x + 12 = 0$

$x^2 + 4\sqrt{3}x + \sqrt{3}x + 12 = 0$

$x(x + 4\sqrt{3}) + \sqrt{3}(x + 4\sqrt{3}) = 0$

$(x + 4\sqrt{3})\ (x + \sqrt{3}) = 0.$

Therefore, the required answer is $(x + 4\sqrt{3})$.

15. Correct option: (B)

We have

$216a^3 - 125 = (6a)^3 - (5)^3$

Since $[a^3 - b^3 = (a - b)(a^2 + ab + b^2)]$

$(6a)^3 - (5)^3 = (6a - 5)\{(6a)^2 + 6a \times 5 + (5)^2\}$

$(6a)^3 - (5)^3 = (6a - 5)(36a^2 + 30a + 25)$

Therefore, the required answer is $(36a^2 + 30a + 25)$.

16. Correct option: (C)

$$\frac{(5.01)^3 - (0.01)^3}{(5.01)^2 + 5.01 \times 0.01 + (0.01)^2}$$

We know that $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

$$a - b = \frac{a^3 - b^3}{(a^2 + ab + b^2)}$$

Putting $a\ =\ 5.01\ and\ b\ =\ 0.01,$

$a - b\ =\ 5.01 -\ 0.01\ =\ 5.$

Therefore, the required answer is 5.

17. Correct option: (D)

Given: $4x^2 - 9$

$$\frac{4x^2 - 9}{4} = 0$$

$$x^2 + 0x - \frac{9}{4} = 0$$

It is in the form of $x^2 + (\alpha + \beta)x + \alpha\beta$

$where\ a = 1,\ b\ =\ 0\ and\ c = -\frac{9}{4}.$

The product of roots is $\alpha\beta = \frac{c}{a}.$

$$\alpha\beta = \frac{c}{a} = -\frac{\frac{9}{4}}{1} = -\frac{9}{4}$$

Therefore, the required answer is $-\frac{9}{4}$.

18. Correct option: (C)

Let the zeroes of the cubic polynomial be $\alpha = 3, \beta = -1, \gamma = -\frac{1}{3}$

Then, $\alpha + \beta + \gamma = 3 - 1 - \frac{1}{3} = 2 - \frac{1}{3} = \frac{6-1}{3} = \frac{5}{3}$

$\alpha\beta + \beta\gamma + \gamma\alpha\ =\ (3)(-1) + (-1)\left(-\frac{1}{3}\right) + \left(-\frac{1}{3}\right)(3) = -3 + \frac{1}{3} - \frac{3}{3} = -\frac{11}{3}$

$\alpha\beta\gamma\ =\ 3 \times\ -1\ \times\ -\frac{1}{3} =\ 1.$

Now,

required cubic polynomial $= x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma$

$$= x^3\ -\left(\frac{5}{3}\right)x^2 + \left(-\frac{11}{3}\right)x -\ 1$$

$$= 3x^3 - 5\,x^2 - 11x\ -\ 3.$$

$So, 3x^3 - 5\,x^2 - 11x\ -\ 3$ is the required cubic polynomial which satisfy the given conditions.

19. Correct option: (D).

To find the value of $\alpha^2 - \beta^2$, where α and β are the zeros of the polynomial $x^2 - \sqrt{2}x$, Use the difference of squares formula.

The difference of squares formula states that $a^2 - b^2 = (a + b)(a - b)$.

In this case, α and β are the zeros of the polynomial, so $\alpha^2 - \beta^2$ can be expressed as (α + β)(α - β).

From the quadratic equation $x^2 - \sqrt{2}x$, factor out an x:

$x(x - \sqrt{2}) = 0$.

This implies that either $x = 0 \, or \, x - \sqrt{2} = 0$.

From x = 0, we get one root: α = 0.

From $x - \sqrt{2} = 0$, we get the other root: $\beta = \sqrt{2}$.

Now, substitute these values into (α + β)(α - β):

$(0 + \sqrt{2})(0 - \sqrt{2}) = (\sqrt{2})(-\sqrt{2}) = -2$.

Therefore, $\alpha^2 - \beta^2$ is equal to -2.

20. Correct option: (B).

To find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$ for the polynomial $x^2 - x - 6$, we need to determine the values of α and β, which are the roots of the polynomial.

Factor the polynomial as follows: $x^2 - x - 6 = (x - 3)(x + 2)$

Setting each factor equal to zero,

$x - 3 = 0 \; --> \; x = 3$

$x + 2 = 0 \; --> \; x = -2$

Therefore, the roots of the polynomial are α = 3 and β = -2.

Now, calculate $\frac{1}{\alpha} + \frac{1}{\beta}$:

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{1}{3} + \frac{1}{-2} = \frac{2-3}{3 \times 2} = \frac{-1}{6}$$

Therefore, $\frac{1}{\alpha} + \frac{1}{\beta}$ is equal to $\frac{-1}{6}$.

LEVEL -II

1. Correct option: (B)

$\frac{x^{100} - 3x^2 + x^6}{x^2} = x^{100-2} - 3x^{2-2} + x^{6-2} = x^{98} - 3x^0 + x^4$.

As 98 is the highest exponent above, it is also the degree of the polynomial.

2. Correct option: (B)

A binomial is a polynomial that has only 2 terms.

A) $x^2 + x + 1$ has 3 terms.

B) $x^{10} - \frac{1}{3}x^8$ has 2 terms.

C) $x^5 - \sqrt{2}x + 3$ has 3 terms.

D) x^{100} has 1 terms.

Therefore, $x^{10} - \frac{1}{3}x^8$ has 2 terms, that represents Binomial.

3. Correct option: (C)

A cubic polynomial is a polynomial with degree 3.

Cubic polynomial refers to the degree of the term with exponent value equal to 3.

Therefore, the degree of the term $x^3 - 3x$ with an exponent value equal to 3 is a cubic polynomial.

4. Correct option: (B)

$$
\begin{array}{rl}
 & 2x^3 + 2x^2 + 3x + \frac{1}{2} \\
4x + 2 \;\big|\; & 8x^4 + 12x^3 + 16x^2 + 8x + 10 \\
(-)\; & 8x^4 + 4x^3 \\
\hline
 & 8x^3 + 16x^2 + 8x \\
(-)\; & 8x^3 + 4x^2 \\
\hline
 & 12x^2 + 8x \\
(-)\; & 12x^2 + 6x + 10 \\
\hline
 & 2x + 10 \\
(-)\; & 2x + 1 \\
\hline
 & 9
\end{array}
$$

Therefore, the remainder is 9.

5. Correct option: (A)

$(x + \sqrt{\frac{5}{3}})(x - \sqrt{\frac{5}{3}}) = (x^2 - \frac{5}{3})$ is a factor.

On dividing $P(x) = 3x^4 + 6x^3 - 2x^2 - 10x - 5$ by $g(x) = \left(x^2 - \frac{5}{3}\right)$ then the quotient is $q(x) = 3x^2 + 6x + 3$.

On Factorising q(x),

$3x^2 + 6x + 3 = 3x^2 + 3x + 3x + 3 = 3x(x + 1) + 3(x + 1) = (x + 1)(3x + 3)$

$(x + 1) = 0$ $\quad$ $(3x + 3) = 0.$

$x = -1$ $\quad$ $3x = -3.$

$x = -1.$

Therefore, values of x = -1.

6. Correct option: (B)

$3x^2 - x - 10 = 3x^2 + 5x - 6x - 10 = x(3x + 5) - 2(3x + 5) = (3x + 5)(x - 2)$

$2x^2 - x - 6 = 2x^2 + 3x - 4x - 6 = x(2x + 3) - 2(2x + 3) = (2x + 3)(x - 2)$

Therefore, the common factor of $3x^2 - x - 10$, $2x^2 - x - 6$ is $x - 2$.

7. Correct option: (A)

$p(x) = x^4 - y^4$ *and zero of* $x - y$ *is* $x = y$.

$p(y) = y^4 - y^4 = 0.$

Therefore, remainder when $x^4 - y^4$ is divided by $x - y$ *is* 0.

8. Correct option: (A)

$x^4 + px^3 + qx^2 - 1 = (x^2 + x - 1)(x^2 + tx + 1)$

Note that coeff. of x^2 and constant have been chosen to suit the product.

$x^4 + px^3 + qx^2 - 1 = x^4 + (t + 1)x^3 + tx^2 + (t - 1)x - 1.$

Comparing the coefficients $t - 1 = 0, t = 1.$

$p = t + 1 = 1 + 1 = 2$

$q = t = 1$

Therefore, the values of p, q are 2, 1.

9. Correct option: (A)

Substitute $\frac{1}{2}$ in first option and check, $p(x) = 4x^2 - 8x + 3$

$p\left(\frac{1}{2}\right) = 4\left(\frac{1}{2}\right)^2 - 8\left(\frac{1}{2}\right) + 3 = 1 - 4 + 3 = 3 - 3 = 0$

Substitute $\frac{3}{2}$ in first option and check,

$p(x) = 4x^2 - 8x + 3$

$p\left(\frac{3}{2}\right) = 4\left(\frac{3}{2}\right)^2 - 8\left(\frac{3}{2}\right) + 3 = 9 - 12 + 3 = 12 - 12 = 0$

Therefore, the zeroes of quadratic polynomial $\frac{1}{2}, \frac{3}{2}$ *is* $4x^2 - 8x + 3$.

10. Correct option: (B)

$$100 - 9x^2 = (10)^2 - (3x)^2 = (10 + 3x)(10 - 3x).$$

11. Correct option: (B,C)

$2x^{99} + 3x^{98} + 2x^{97} + 3x^{96} + \cdots \ldots \ldots + 2x + 3 = 0$------ equation (i)

Taking $2x + 3$ as common,

$(2x + 3)\ (x^{99} + x^{98} + x^{97} + x^{96} + \ldots\ldots\ldots + x + 1) = 0$

Since $(x^{99} + x^{98} + x^{97} + x^{96} + \cdots \ldots \ldots + x + 1)$*is not equal to* 0.

Therefore, only $(2x + 3)$ *or* $x = \frac{-3}{2}$ has real root.

Hence, number of real roots of equation (i) is 1.

12. Correct option: (A, B)

$(x - \sqrt{2})\ (x + \sqrt{2}) = (x^2 - 2)$ is a factor.

On dividing $P(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$ by $g(x) = (x^2 - 2)$ then the quotient is $q(x) = 2x^2 - 3x + 1$.

On Factorising q(x),

$2x^2 - 3x + 1 = 2x^2 - 2x - x + 1 = 2x(x - 1) - 1(x - 1) = (2x - 1)(x - 1)$

$(2x - 1) = 0 \qquad (x - 1) = 0.$

$x = \frac{1}{2} \qquad x = 1.$

Therefore, values of x = $\frac{1}{2}, 1$.

13. Correct option: (A, B)

$$\alpha + \beta = -\frac{b}{a} = \frac{-(-20)}{1} = 20,$$

$\alpha\beta = \frac{c}{a} = \frac{(-30)}{1} = -30.$

14. Correct option: (B)

$R_1 = p(4) = a(4)^3 + 3(4)^2 - 3 = 64a + 45.$

$R_2 = p(4) = 2(4)^3 - 5(4) + a = 108 + a$

$R_1 = R_2$

$64a + 45 = 108 + a$

$64a + 45 - 108 - a = 0.$

$63a - 63 = 0.$

$a = 1.$

Therefore, the value of a is 1.

15. Correct option: (A)

$R_1 = p(4) = a(4)^3 + 3(4)^2 - 3 = 64a + 45.$

$R_2 = p(4) = 2(4)^3 - 5(4) + a = 108 + a$

$R_1 + R_2 = 0$

$64a + 45 + 108 + a = 0$

$65a + 153 = 0.$

$63\,a\, = -153.$

$a\, = -\frac{153}{65}.$

Therefore, the value of a is $-\frac{153}{65}$.

16. Correct option: (D)

$R_1 = p(4) = a(4)^3 + 3(4)^2 - 3 = 64a + 45.$

$R_2 = p(4) = 2(4)^3 - 5(4) + a = 108 + a$

$2R_1\, -\, R_2\, =\, 0.$

$2(64a + 45)\, -\, (108 + a) =\, 0.$

$128a + 90\, -\, 108 - a =\, 0.$

$127a -\, 18 =\, 0.$

$127a = -\, 18$

$a = \frac{-18}{127}$

Therefore, the value of a is $-\frac{18}{127}$.

17. Correct Matchin is: $a.\rightarrow$ S., $b.\rightarrow$ R., $c.\rightarrow$ T., $d.\rightarrow$ S.

a. If x^{11} is divisible by x +1, then remainder is

$\Rightarrow$ $p(-1) =\, (-1)^{11} =\, -1.$ (Since the power is an odd number so its $(-1)^{11} =\, -1$)

b. If the polynomial $f(x)$ is such that $f(-3) = 0,$, then the factor of $f(x)$ *is* $x + 3.$

c. One of the factors of $x^2 + x\, - 6 = (x - 2)\,(x + 3).$

d. If $(\,x - 2)$ is a factor of $x^2 + 3px\, - 2p$ then the value of p is

$\Rightarrow$ $f(2) =\, (2)^2 + 3p(2)\, - 2p =\, 4 + 6p - 2p = 4 + 4p$

$\Rightarrow$ $4 + 4p = 0.$

$\Rightarrow$ $p =\, -1.$

18. Correct Matching is: $a.\rightarrow$ T., $b.\rightarrow$ P., $c.\rightarrow$ T., $d.\rightarrow$ R.

a. If 2 is the root of $x^2 + px\, + 26 = 0,$ then the value of p is

$\Rightarrow$ $f(2) = 0,$

$\Rightarrow$ $(2)^2 + p(2) + 26 = 0,$

$\Rightarrow$ $4 + 2p + 26 = 0,$

$\Rightarrow$ $2p + 30 = 0.$

$\Rightarrow$ $p =\, -15.$

b. If 5 is root of $x^2 - (k - 1)x\, + 10 = 0,$ then the value of k is

$\Rightarrow$ $f(5) = 0$

$\Rightarrow$ $(5)^2 - (k - 1)(5) + 10 = 0$

$\Rightarrow$ $25 - 5k + 5 + 10 = 0,$

$\Rightarrow$ $-5k = -40.$

$\Rightarrow$ $k =\, 8.$

c. Zeroes of polynomial $4x^2 -\, 9$ is

$\Rightarrow$ $4x^2 -\, 9 = 0$

$\Rightarrow$ $4x^2 =\, 9$

$\Rightarrow$ $x^2 = \frac{9}{4}$

$\Rightarrow$ $x = \pm\frac{3}{2}$

d. Degree of $(x - a)\,(x - b)(x - c)\,(x - d)$ is 4.

The above expression $(x - x\,)$ would be zero so the product would be zero.

consider X as a variable and a,b,c,d as constants then the degree would be 4.

19. Correct Answer is: $x^2\,(a + b + c) + x(2a + 2b) + (a + 2c)$.

To simplify the expression, let's combine like terms:

$(ax^2 + bx + c) + (ax + bx^2 + c) + (a + bx + cx^2)$

Rearranging the terms:

$(ax^2 + bx^2 + cx^2) + (ax + bx + bx) + (a + c + c)$

Combining like terms:

$(ax^2 + bx^2 + cx^2) + (2ax + 2bx) + (a + 2c)$

Now, let's simplify further:

$x^2\,(a + b + c) + x(2a + 2b) + (a + 2c)$

Therefore, the simplified expression is:

$x^2\,(a + b + c) + x(2a + 2b) + (a + 2c)$.

20. Correct Answer is: 5

Vieta's formulas state that for a quadratic equation of the form $ax^2 + bx + c = 0$with roots α and β, the following relationships hold: $\alpha + \beta = -\frac{b}{a}$

$$\alpha\beta = \frac{c}{a}$$

In this case, $a = 1, b = -15, and\ c = 3$.

Using Vieta's formulas: $\alpha + \beta = -\frac{(-15)}{1} = 15$

$\alpha\beta = \frac{3}{1} = 3$

Now, calculate $\frac{1}{\alpha} + \frac{1}{\beta}$:

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{(\alpha + \beta)}{\alpha\beta} = \frac{15}{3} = 5$$

Therefore, $\frac{1}{\alpha} + \frac{1}{\beta}$ is equal to 5.

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)

$x - x^5 + 1$.

As 5 is the highest exponent above, it is also the degree of the polynomial.

2. Correct option: (D)

The polynomial $4x^2 + 5x + 3$ has three terms.

Each term is separated by a plus sign (+) or a minus sign (-), and the number of terms is equal to the number of distinct algebraic terms in the polynomial.

In this case, there are three distinct terms: $4x^2,\ 5x\ , and\ 3$.

Therefore, the number of terms in the polynomial is 3.

3. Correct option: (B)

The polynomials with degree 4 is known as biquadratic polynomial.

The polynomial $x^4 - 30x^3 + 3x^2$ has a degree 4.

From the given options only B has a degree 4.

Therefore, $x^4 - 30x^3 + 3x^2$ is a biquadratic polynomial.

4. Correct option: (C)

Let $p(x) = 4 + x^3 - x + 2x^2$

$p(-2) = 4 + (-2)^3 - (-2) + 2(-2)^2$

$p(-2) = 4 - 8 + 2 + 8 = 6.$

$p(-2) = 6.$

Therefore, the value of the polynomial at x=-2 is 6.

5. Correct option: (C)

The remainder = $f(1)$ by remainder theorem.

$\therefore Remainder = 1 + 1^2 + 1^3 + 1^4 + - - + 1^{2019}$

$\therefore Remainder = 1 + 1 + 1 + 1 \ldots . + 1(2019\ times) = 2019.$

6. Correct option: (C)

$f(x) = x^{19} + x^{17} + x^{13} + x^{11} + x^7 + x^5 + x^3$

Put $x^2 = -1$,

$f(x) = (x^2)^9 \times x + (x^2)^8 \times x + (x^2)^6 \times x + (x^2)^5 \times x + (x^2)^2 \times x + (x^2) \times x$

$f(x) = (-1)^9 \times x + (-1)^8 \times x + (-1)^6 \times x + (-1)^5 \times x + (-1)^2 \times x + (-1) \times x$

$f(x) = -x + x + x - x + x - x$

$f(x) = -x.$

Therefore, $f(x) = -x$.

7. Correct option: (D)

$$
\begin{array}{l}
x^2 + 2x + 1 \overline{)\ 3x^2 + x^2 + 2x + 5\ (\ 3x - 5} \\
\quad\ 3x^3 + 6x^2 + 3x \\
\quad\ - \quad - \quad - \\
\hline
\quad -5x^2 - x + 5 \\
\quad -5x^2 - 10x - 5 \\
\quad\ + \quad + \quad + \\
\hline
\quad\quad\ 9x + 10
\end{array}
$$

Therefore, the remainder is $9x + 10$.

8. Correct option: (D)

$$
\begin{array}{rl}
 & \quad -x^2 - 2 \\
\hline
-x^2 + 2 & \overline{)\ x^4 + 0x^3 + 0x^2 - 5x + 6} \\
(-) & x^4 + 0x^3 - 2x^2 \\
\hline
 & \quad 0x^3 + 2x^2 - 5x + 6 \\
(-) & \quad 0x^3 + 2x^2 + 0x - 4 \\
\hline
 & \quad\quad -5x + 10
\end{array}
$$

Therefore, the quotient $= -x^2 - 2$, Remainder $= -5x + 10$.

9. Correct option: (C)

$f(x) = 3x^3 + ax^2 + 3x + 5$

$g(x) = a - 2x + x^2 + 4x^3$

By remainder theorem, if f(x) is divided by $(x - a)$ then the remainder is $f(a)$.

Here, when $f(x)\ and\ g(x)$ are divided by $(x - 2)$ the remainders are $f(2)\ and\ g(2)$ respectively.

Put $x = 2$ in both $f(x)\ and\ g(x)$, so $f(2) = g(2)$ as they are equal

$\Rightarrow 3(2)^3 + a(2)^2 + 3(2) + 5 = a - 2(2) + (2)^2 + 4(2)^3$

$\Rightarrow 24 + 4a + 6 + 5 = a - 4 + 4 + 32$

$\Rightarrow 4a + 35 = a + 32$

$\Rightarrow 3a = -3$

$\therefore a = -1.$

Now, $2 + a = 2 + (-1) = 1$.
Therefore, the value of $2 + a$ is 1.

10. Correct option: (D).

Let $f(x) = x^3 - ax^2 - 13x + b$
Given that, $(x - 1)$ *and* $(x + 3)$ are factors of $\boldsymbol{f(x)}$
According to the factor theorem, if $\boldsymbol{(x - a)}$ is a factor of $\boldsymbol{f(x)}$, $\boldsymbol{then\ f(a) = 0}$.
Therefore, $f(1) = 0$ *and* $f(-3) = 0$
$\Rightarrow f(1) = 1 - a - 13 + b = 0$
$\Rightarrow b - a = 12 \ldots . (i)$
And,
$\Rightarrow f(-3) = -27 - 9a + 39 + b = 0$
$\Rightarrow b - 9a = 12$
$\therefore\ b = 12 + 9a$
Substituting the value of b in (i), we get:
$12 + 9a - a = 12$
$\Rightarrow 8a = 0$
$\therefore\ a = 0$
$b = 12 + 9a$
$\therefore\ b = 12 + 9(0) = 12$
Hence, the value of a is 0 and that of b is 12, $a + b = 0 + 12 = 12$.

11. Correct option: (B)

Let , $p(x) = x^4 - 2x^3 + 3x^2 - nx + 3$
If $(x - 3)$ exactly divides $p(x)$, then for $x = 3$, $p(x) = 0$
$i.e.\ p(3) = 0$.
$p(3) = (3)^4 - 2(3)^3 + 3(3)^2 - n(3) + 3$
$81 - 54 + 27 - 3n + 3 = 0$
$57 - 3n = 0$
$n = 19$.
Therefore, the value of n is 19.

12. Correct option: (C)

$P(x) = kx^3 + 9x^2 + 4x - 8$
$g(x) = x + 3$
$x + 3 = 0$
$x = -3$
$Then\ p(-3) = remainder$
As p(x) divided by x+3 leaves the remainder -20
$\Rightarrow p(-3) = Remainder$
$-27k + 81 - 12 - 8 = -20$
$-27k + 61 = -20$
$-27k = -81$
$k = \dfrac{-81}{-27}$
$k = 3$
Therefore, the value of k is 3.

13. Correct option: (C)

$x^2 + 3x - 4 = 0$
$x^2 + 4x - x - 4 = 0$
$x(x + 4) - 1(x + 4) = 0$

$(x+4)(x-1)=0$
Therefore, the factors are $(x+4)(x-1)$.

14. Correct option: (D)

$4x^2-13x+10=0$
$4x^2-5x-8x+10=0$
$x(4x-5)-2(4x-5)=0$
$(4x-5)(x-2)=0$
Therefore, the factors are $(4x-5)(x-2)$.

15. Correct option: (B)

$81-16x^2=(9)^2-(4x)^2$
$Since\ (a^2-b^2)=(a+b)(a-b)$
$(9)^2-(4x)^2=(9+4x)(9-4x)$
Therefore, the required answer is $(9+4x)(9-4x)$.

16. Correct option: (A)

$x^2+4\sqrt{2}x+6=0$
$x^2+3\sqrt{2}x+\sqrt{2}x+6=0$
$x(x+3\sqrt{2})+\sqrt{2}(x+3\sqrt{2})=0$
$(x+3\sqrt{2})\,(x+\sqrt{2})=0.$
Therefore, the required answer is $\left(x+3\sqrt{2}\right)$.

17. Correct option: (D)

$(28)^3+(-15)^3+(-13)^3$
$\because\ (28)+(-15)+(-13)=0$
$Using\ identity, if\ x+y+z=0, then\ x^3+y^3+z^3=3xyz.$
$(28)^3+(-15)^3+(-13)^3=3\times(28)\times(-15)\times(-13)=16380$
Therefore, the required answer is 16380.

18. Correct option: (D)

$x^2-2x-8=0$
It is in the form of $x^2+(\alpha+\beta)x+\alpha\beta$
$where\ a=1, b=-2\ and\ c=-8.$
The product of roots is $\alpha\beta=\frac{c}{a}$.
$\alpha\beta=\frac{c}{a}=-\frac{8}{1}=-8$
Therefore, the required answer is -8.

19. Correct option: (A)

Let the zeroes of the cubic polynomial be $\alpha=1,\beta=2,\gamma=-3$
Then, $\alpha+\beta+\gamma=1+2-3=3-3=0$
$\alpha\beta+\beta\gamma+\gamma\alpha=(1)(2)+(2)(-3)+(-3)(1)=2-6-3=-7$ and
$\alpha\beta\gamma=1\times2\times-3=-6.$
Now, required cubic polynomial $=x^3-(\alpha+\beta+\gamma)x^2+(\alpha\beta+\beta\gamma+\gamma\alpha)x-\alpha\beta\gamma$
$=x^3-(0)x^2+(-7)x-6$
$=x^3-7x-6.$
So, x^3-7x-6 is the required cubic polynomial which satisfy the given conditions.

20. Correct option: (D)

The sum of the roots $=0+\sqrt{2}=\sqrt{2}$
The product of roots $=0\times\sqrt{2}=0$
$Quadratic\ Polynomial=x^2-(sum\ of\ zeroes)x+(product\ of\ zeroes)\ldots(1)$
Substitute these two values in above mentioned formula.

$\Rightarrow polynomial = x^2 - (\sqrt{2})x + (0)$

$\Rightarrow polynomial = x^2 - \sqrt{2}\,x$

Therefore, the required answer $x^2 - \sqrt{2}\,x$.

21. Correct option: (C)

Vieta's formulas state that for a quadratic equation of the form $ax^2 + bx + c = 0$ with roots α and β, the following relationships hold:

$\alpha + \beta = -\frac{b}{a}$

$\alpha\beta = \frac{c}{a}$

In this case, the polynomial is $x^2 + 4x + 3, so\ we\ have\ a = 1, b = 4, and\ c = 3.$

Using Vieta's formulas:

$\alpha + \beta = \frac{-b}{a} = \frac{-4}{1} = -4$

$\alpha\beta = \frac{c}{a} = \frac{3}{1} = 3$

Now, we want to find $\alpha^2 + \beta^2$. We can express $\alpha^2 + \beta^2$ in terms of (α + β) and αβ using the following identity:

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2 \propto \beta$

Substituting the values obtained from Vieta's formulas:

$\alpha^2 + \beta^2 = (-4)^2 - 2(3) = 16 - 6 = 10$

Therefore, $\alpha^2 + \beta^2$ is equal to 10.

22. Correct option: (C)

In this case, the polynomial is $x^2 + 8x + 6, so\ we\ have\ a = 1, b = 8, and\ c = 6.$

Using Vieta's formulas:

$\alpha + \beta = \frac{-b}{a} = \frac{-8}{1} = -8.$

$\alpha\beta = \frac{c}{a} = \frac{6}{1} = 6.$

$(\alpha + \beta)^2 = (-8)^2$

$\alpha^2 + \beta^2 + 2\alpha\beta = 64.$

$\alpha^2 + \beta^2 + 2(6) = 64$

$x^2 + \beta^2 = 64 - 12$

$\alpha^2 + \beta^2 = 52$

Now,

$(\alpha - \beta)^2 = \alpha^2 + \beta^2 - 2\alpha\beta$

$\alpha - \beta = \sqrt{a^2 + \beta^2 - 2\alpha\beta}$

$\alpha - \beta = \sqrt{52 - 2(6)} = \sqrt{52 - 12} = \sqrt{40}$

$\alpha - \beta = \sqrt{40}$

Therefore, the required answer is $\sqrt{40}$.

23. Correct option: (C)

To find the sum of the cubes of the roots of the quadratic equation $2x^2 - 4x +$ 5(represented as $\alpha^3 + \beta^3$), use the following relationship based on Vieta's formulas:

$\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2)$ ·

From Vieta's formulas, we know that:

$\alpha + \beta - \frac{(-4)}{2} = 2$

$\alpha\beta = \frac{5}{2}$

Now, to find α^2 and β^2:

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2 \propto \beta = 2^2 - 2\left(\frac{5}{2}\right) = 4 - 5 = -1$

Substituting these values into the equation:
$\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 + \beta^2 - \alpha\beta) \cdot$
$\alpha^3 + \beta^3 = (2)\left((-1) - \left(\frac{5}{2}\right)\right) = 2\left(-1 - \frac{5}{2}\right) = 2\left(\frac{-2-5}{2}\right) = -7$
$\alpha^3 + \beta^3 = -7$
Therefore, $\alpha^3 + \beta^3\ is\ -7.$

24. Correct option: (C)

To find the difference of the cubes of the roots of the quadratic equation $x^2 - 11x + 30$ (represented as $\alpha^3 - \beta^3$), use the following relationship based on Vieta's formulas:
$\alpha^3 - \beta^3 = (\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2) \cdot$
From Vieta's formulas, we know that:
$\alpha + \beta - \frac{(-11)}{1} = 11$
$\alpha\beta = \frac{30}{1} = 30$
Now, to find α^2 and β^2:
$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2 \propto \beta = (11)^2 - 2(30) = 121 - 60 = 61.$
$(\alpha - \beta)^2 = \alpha^2 + \beta^2 - 2\alpha\beta$
$\alpha - \beta = \sqrt{a^2 + \beta^2 - 2\alpha\beta}$
$\alpha - \beta = \sqrt{61 - 2(30)} = \sqrt{61 - 60} = \sqrt{1}$
$\alpha - \beta = 1.$
Substituting these values into the equation:
$\alpha^3 - \beta^3 = (\alpha - \beta)(\alpha^2 + \beta^2 + \alpha\beta) \cdot$
$\alpha^3 - \beta^3 = (1)\big((61) + 30\big) = 91$
$\alpha^3 - \beta^3 = 91.$
Therefore, $\alpha^3 - \beta^3$ is 91.

25. Correct option: (B)

To find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ for the polynomial $x^2 + 2x + 1$, first need to determine the values of α and β, the roots of the polynomial.
Factor the polynomial as follows: $x^2 + 2x + 1 = (x + 1)(x + 1)$
Setting each factor equal to zero:
$x + 1 = 0 \ \ --> \ \ x = -1$
Therefore, both roots of the polynomial are $\alpha = \beta = -1.$
Now, we can calculate $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$:
$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{1}{(1)^2} + \frac{1}{(1)^2} = \frac{1}{1} + \frac{1}{1} = 1 + 1 = 2.$
Therefore, $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ is equal to 2.

26. Correct option: (D)

The degree of zero polynomial is not defined since it corresponds to any value which when put in the variable and solved makes the equation 0.
The polynomial 0 has no terms at all and is called a zero polynomial.
Because the zero polynomial has no non-zero terms, the zero polynomial has no degree.

27. Correct option: (B)

$(x + 2)$ is a factor of $x^2 + ax + 2b$
$(x + 2) = 0.$
$x = -2.$
$x^2 + ax + 2b = (-2)^2 + a(-2) + 2b = 4 - 2a + 2b = a - b = 2.$

On adding,
$a - b = 2.$
$a + b = 4$
$2a \quad = 6$
$2a = 6$
$a = 3.$
$a + b = 4$
$3 + b = 4$
$b = 1.$
Therefore, the values of a, b is 3, 1.

28. Correct option: (B)

Substitute $-\frac{7}{5}$ in $5x^2 - 17x - 14.$

$5x^2 - 17x - 14 = 5\left(-\frac{7}{5}\right)^2 - 17\left(-\frac{7}{5}\right) - 14 = 5\left(-\frac{49}{25}\right) + 17\left(\frac{7}{5}\right) - 14$.

$5x^2 - 17x - 14 = \left(\frac{49}{5}\right) + \left(\frac{119}{5}\right) - 14 = \frac{49+119-70}{5} = 0.$

Therefore, one of zeroes of polynomial $5x^2 - 17x - 14$ is $-\frac{7}{5}$

29. Correct option: (D)

$\alpha + \beta = 3$
$\alpha\beta = 2$ then $\alpha = \frac{2}{\beta}.$

Substitute $\frac{2}{\beta}$ $for\ \alpha$ in $\alpha + \beta = 3$

$$\frac{2}{\beta} + \beta = 3$$
$$\frac{2 + \beta^2}{\beta} = 3$$
$2 + \beta^2 = 3\,\beta$
$\beta^2 - 3\,\beta + 2 = 0$
$\beta^2 - \beta - 2\,\beta + 2 = 0$
$\beta(\beta - 1) - 2(\beta - 1) = 0$
$(\beta - 2)(\beta - 1) = 0$
Therefore, $\beta = 1, 2.$
Therefore, if $\alpha = 1,$ then $\beta = 2.$ Or if $\alpha = 2,$ then $\beta = 1.$
Now, Substitute $\alpha = 1,$ and $\beta = 2$ in $\left(1 + \frac{\alpha}{\beta}\right)\left(1 + \frac{\beta}{\alpha}\right)$

$\left(1 + \frac{\alpha}{\beta}\right)\left(1 + \frac{\beta}{\alpha}\right) = \left(1 + \frac{1}{2}\right)\left(1 + \frac{2}{1}\right) = \left(\frac{2+1}{2}\right)(1 + 2) = \left(\frac{3}{2}\right) \times 3 = \frac{9}{2}.$

30. Correct option: (B, D)

Substitute $m = 2, n = 1,$ in
$2x^{3m+2n} + 4x^{5m+2n} + 2x^{m+7n} = 2x^{3(2)+2(1)} + 4x^{5(2)+2(1)} + 2x^{(2)+7(1)} = 2x^8 + 4x^{12} + 2x^9.$
Therefore, the highest degree is 12.

31. Correct option: (B)

Since polynomial with degree 4 is called bi-quadratic.

32. Correct option: (A, B, C)

$p(x) = x^n - y^n\ and\ zero\ of\ x - y\ is\ x = y.$
$p(y) = y^n - y^n = 0.$
$Therefore,$ n is any natural, even, and odd number.

33. Correct option: (A, B)

$p(x) = x^3 + ax^2 + bx - 12$ (1)
one of the factor is $x - 2$

$x - 2 = 0$
$x = 2.$
putting $x = 2$ in equation (1)
$p(2) = (2)^3 + a(2)^2 + b(2) - 12 = 0$
$8 + 4a + 2b - 12 = 0.$
$4a + 2b - 4 = 0$ (2)
now, one of the factor is $x + 3$
$x + 3 = 0$
$x = -3.$
putting $x = -3$ in equation (1)
$p(-3) = (-3)^3 + a(-3)^2 + b(-3) - 12 = 0$
$-27 + 9a - 3b - 12 = 0.$
$9a - 3b - 39 = 0$ (3)
Multiply $4a + 2b - 4 = 0$ by 3 and $9a - 3b - 39 = 0$ by 2.
$12a + 6b - 12 = 0$ (4)
$18a - 6b - 78 = 0$ (5)
now on adding (4) and (5)
$12a + 6b - 12 = 0$
$18a - 6b - 78 = 0$
$30a - 90 = 0$
~~$30a = 90$~~
$a = 3.$
so, substitute $a = 3$ in $4a + 2b - 4 = 0$
$4(3) + 2b - 4 = 0$
$b = -4.$
Hence the value of 'a=3' and 'b= -4'.

34. Correct option: (A, B, C)

$(x + 1)(x + 2)(x + 3) = (x^2 + 2x + x + 2)(x + 3) = (x^2 + 3x + 2)(x + 3)$
$= x^3 + 3x^2 + 3x^2 + 9x + 2x + 6$
$= x^3 + 6x^2 + 11x + 6.$

35. Correct option: (A, B, C)

$x^3 - x = 0.$
$x(x^2 - 1) = 0.$
$x = 0$ $x^2 - 1 = 0$
$x = \pm 1.$

36. Correct option: (A, B, D)

Substitute 1 , 2, and -3 in $p(x) = x^3 - 7x + 6.$
$p(1) = (1)^3 - 7(1) + 6 = 7 - 7 = 0.$
$p(2) = (2)^3 - 7(2) + 6 = 14 - 14 = 0.$
$p(-3) = (-3)^3 - 7(-3) + 6 = 27 - 27 = 0.$
Therefore, zeroes of a polynomial $x^3 - 7x + 6$ are 1, 2, -3.

37. Correct option: (B)

$\alpha + \beta = -\frac{b}{a} = \frac{1}{2}, where\ b = -1\ and\ a = 2.$
$\alpha\beta = \frac{c}{a} = -1$
$\frac{c}{2} = -1, \quad c = -2.$
Now quadratic expression is written as 2
On substituting the values of a, b, c is $2x^2 - x - 2.$

Therefore, the quadratic polynomial is $2x^2 - x - 2$.

38. Correct option: (A)

Put $x = 1$ in $f(x) = x^5 - x^4 + x^3 - x^2 + x - 1$

$f(1) = (1)^5 - (1)^4 + (1)^3 - (1)^2 + (1) - 1 = 0$

$f(1) = 0$.

Therefore, $(x - 1)$ is a factor of $f(x)$.

39. Correct option: (B)

Put $x = -1$ in $g(x) = x^3 + x^2 + x + 1$

$g(-1) = (-1)^3 + (-1)^2 + (-1) + 1 = -1 + 1 - 1 + 1 = 0$

$g(-1) = 0$.

Therefore, $(x + 1)$ is a factor of $g(x)$.

40. Correct option: (A, C)

Put $x = 1$ in $f(x) = x^3 - 7x + 6$.

$f(1) = (1)^3 - 7(1) + 6 = 0$

$f(1) = 0$.

Therefore, $(x - 1)$ is a factor of $f(x)$.

Put $x = 2$ in $f(x) = x^3 - 7x + 6$.

$f(2) = (2)^3 - 7(2) + 6 = 14 - 14 = 0$

$f(2) = 0$.

Therefore, $(x - 2)$ is a factor of $f(x)$.

41. Correct option: (B)

Put $x = 1$ in $(3x - 1)^7 = a_7x^7 + a_6x^6 + \cdots + a_1x + a_0$

$(3(1) - 1)^7 = a_7(1)^7 + a_6(1)^6 + \cdots + a_1(1) + a_0$

$(3 - 1)^7 = a_7 + a_6 + \cdots + a_1 + a_0$

$(2)^7 = a_7 + a_6 + \cdots + a_1 + a_0$

$128 = a_7 + a_6 + \cdots + a_1 + a_0$

Therefore, $a_0 + a_1 + a_2 + ---------- + a_6 + a_7$ is 128.

Passage – III

If α, β are zeroes of a quadratic polynomial $ax^2 + bx + c$, then

$ax^2 + bx + c = a(x - \alpha)(x - \beta)$

42. Correct option: (B)

$(2a - \sqrt{3})(2a - \sqrt{3}) = (2a)^2 - 2a\sqrt{3} - 2a\sqrt{3} - (\sqrt{3})^2 = 4a^2 - 4\sqrt{3}a + 3.$

43. Correct option: (A)

$(x - 5)(x + 3) = (x^2 + 3x - 5x - 15) = x^2 - 2x - 15.$

44. Correct option: (C)

$$\frac{5.01 \times 5.01 \times 5.01 - 0.01 \times 0.01 \times 0.01}{5.01 \times 5.01 + 5.01 \times 0.01 + 0.01 \times 0.01} = \frac{125.7515}{25.1001 + 0.0501 + 0.0001} = \frac{125.7515}{25.1503} = 5.$$

45. Correct Matching is: *a.*→ R., S., *b.*→ Q., S., *c.*→Q., T., *d.*→ R., T.

a. Factors of $x^2 + 6x - 27 = 0$ is $(x - 3)(x + 9)$.

b. Factors of $x^2 + 19x + 90 = 0$ is $(x + 10)(x + 9)$.

c. Factors of $x^2 + 30x + 200 = 0$ is $(x + 10)(x + 20)$.

d. Factors of $x^2 + 17x - 60 = 0$ is $(x - 3)(x + 20)$.

46. Correct Answer is: 0

Put $x = 1$ in $f(x) = x^5 - x^4 + x^3 - x^2 + x - 1$

$f(1) = (1)^5 - (1)^4 + (1)^3 - (1)^2 + (1) - 1 = 0$

$f(1) = 0$.

Therefore, $(x - 1)$ is a factor of $f(x)$.

47. Correct Answer is: 2

$(2)^2 + a(2) + b = (2)^2 + c(2) + d$

$4 + 2a + b = 4 + 2c + d$

$2a - 2c = b - d$

$2(a - c) = b - d$

$$\frac{b-d}{c-a} = 2.$$

48. Correct Answer is: 3

First, let's simplify the left side of the equation: $\frac{8m^2-6mn}{2m} = 4m - kn$

We can simplify the numerator: $8m^2 - 6mn = 4m \times 2m - 2mkn$

$8m^2 - 6mn = 8m^2 - (2mn)k$

Now, let's equate the coefficients of like terms on both sides of the equation:

$-6mn = -(2mn)k$

To solve for k, we divide both sides of the equation by -2: k = 3

Therefore, k is equal to 3.

49. Correct Answer is: $a = 3$.

Given, $f(x) = x^3 + ax^2 + bx - 12$

Let us assume $x - 2 = 0$ ***and*** $x + 3 = 0$

$\Rightarrow x = 2$ ***and*** $x = -3$

Also given that, $(x - 2)$ ***and*** $(x + 3)$ are factors of $f(x)$

$\therefore$ By factor theorem, $f(2) = 0$ ***and*** $f(-3) = 0$

Now, $f(2) = 0$

$\Rightarrow 23 + a(2)2 + b(2) - 12 = 0$

$\Rightarrow 8 + 4a + 2b - 12 = 0$

$\Rightarrow 4a + 2b - 4 = 0$

Dividing te equation by 2

$\Rightarrow 2a + b - 2 = 0 \ldots (i)$

And, $f(-3) = 0$

$\Rightarrow (-3)3 + a(-3)2 + b(-3) - 12 = 0$

$\Rightarrow -27 + 9a - 3b - 12 = 0$

$\Rightarrow 9a - 3b - 39 = 0$

Dividing the equation by 3

$\Rightarrow 3a - b = 13 \ldots (ii)$

Now, adding equation (i) and equation (ii) we get, $(2a + b) + (3a - b) = 2 + 13$

$\Rightarrow 2a + 3a + b - b = 15$

$\Rightarrow 5a = 15$

$\Rightarrow a = \frac{\mathbf{15}}{\mathbf{5}} = 3$

Substituting this value of a in equation (i)

$\Rightarrow 2a + b = 2$

$\Rightarrow 2(3) + b = 2$

$\Rightarrow 6 + b = 2$

$\Rightarrow b = 2 - 6$

$\Rightarrow b = -4$

Hence, $a = 3$ ***and*** $b = -4$

50. Correct Answer is: 18,7.

$(x - 18)(x - 7) = (x^2 - 18x - 7x + 126) = x^2 - 25x + 126.$

The factors are $(x - 18)(x - 7)$.

Therefore, the value of a is 18, 7

3. Pair of Linear Equations in Two Variable

Solutions

LEVEL-I

1. Correct option: (B)

The given equation is

$$6-3(2x-3)=5(6-x)-4x$$
$$\Rightarrow 6-6x+9=30-5x-4x$$
$$\Rightarrow -6x+9x=30-15 \Rightarrow x=5$$

2. Correct option: (B)

$$\frac{2x+3}{3x+4}=\frac{2x-3}{3x-2}$$
$$\Rightarrow (2x+3)(3x-2)=(2x-3)(3x+4)$$
$$\Rightarrow 6x^2+5x-6=6x^2-x-12$$
$$\Rightarrow 6x=-6 \Rightarrow x=-1$$

3. Correct option: (D)

$$2+\frac{2x-3}{2x+3}=\frac{3x+4}{x+2}$$
$$\Rightarrow \frac{4x+6+2x-3}{2x+3}=\frac{3x+4}{x+2}$$
$$\Rightarrow \frac{6x+3}{2x+3}=\frac{3x+4}{x+2} \Rightarrow (6x+3)(x+2)=(3x+4)(2x+3)$$
$$\Rightarrow 6x^2+15x+6=6x^2+17x+12$$
$$\Rightarrow -2x=6 \Rightarrow x=-3$$

4. Correct option: (A)

$$x+\frac{1}{x}=5$$

Squaring on both sides implies

$$x^2+\frac{1}{x^2}+2(x)\left(\frac{1}{x}\right)=25$$
$$\Rightarrow x^2+\frac{1}{x^2}=25-2 \Rightarrow x^2+\frac{1}{x^2}=23$$

5. Correct option: (C)

$$\frac{x}{2}-\frac{1}{5}=\frac{x}{3}+\frac{1}{4}$$

Simplifying the above expression implies

$$\frac{x}{2}-\frac{x}{3}=\frac{1}{5}+\frac{1}{4}$$
$$\Rightarrow \frac{x}{6}=\frac{9}{20} \Rightarrow x=\frac{27}{10}$$

6. Correct option: (A)

Sol. Let the number of students be 'x'

If each student was given by two chocolates implies total number of chocolates are with students will be equal to 2x

After distributing accordingly number of chocolates left with teacher is 7

This entire hypothesis can be represented in the form of equation as follows

$$2x+7=45$$
$$\Rightarrow 2x=38$$
$$\Rightarrow x=19$$

7. Correct option: (A)

Sol. Let the two numbers 'x' and 'y'

Given that their sum and sum of their squares is 15 and 113 respectively

This can be represented in the form of equation as follows

$$x+y=15$$
$$x^2+y^2=113$$

Considering 1st equation and squaring it on both sides implies

$$(x+y)^2=15^2$$
$$\Rightarrow x^2+y^2+2xy=225$$
$$\Rightarrow 113+2xy=225$$
$$\Rightarrow 2xy=112\Rightarrow xy=56$$
$$\therefore (x-y)^2=(x+y)^2-4xy=225-224=1$$
$$\Rightarrow x-y=1$$

Now we have an equation pair with only 'x' and 'y'

$$x+y=15$$
$$x-y=1$$

Adding both of them yields the value of 'x' as 8

Substitute the value of 'x' in any of the equation pair yields the value of 'y=7'

8. Correct option: (B)

Let the sides of perpendicular of a right angle be 3x and 4 respectively

The perimeter of the right angle triangle is given by 96cm. Formula for the perimeter of a right angle triangle is given below

$$p=a+b+\sqrt{a^2+b^2}$$

Here a, b are the perpendicular sides of the right angle triangles respectively.

Substitute the values of perpendicular sides of right angle triangle in above formula implies

$$p=a+b+\sqrt{a^2+b^2}$$
$$\Rightarrow 96=3x+4x+\sqrt{(3x)^2+(4x)^2}$$
$$\Rightarrow x=8cm$$

One perpendicular side of a right angle triangle will be 3x=24cm

Another perpendicular side of a right angle triangle will be 4x=32cm

If two perpendicular sides of the right angle triangle is known the other side can be calculated by using Pythagarous theorem

According to it $d^2=\sqrt{p_1^2+p_2^2}$

Here d is diagonal and p_1, p_2 are perpendicular sides of right angle triangle respectively

$$d^2=\sqrt{24^2+32^2}\Rightarrow d=40cm$$

The largest side of the triangle is 40cm

9. Correct option: (D)

$$x+y=3$$
$$xy=2$$

$$(x-y)^2=(x+y)^2-4xy$$
$$\Rightarrow (x-y)^2=(3)^2-4(2)=1$$
$$\Rightarrow x-y=1$$

Solving $x+y, x-y$ together implies

$$x+y=3$$
$$x-y=1$$

Adding both the equations implies $x=2$ this yield $y=1$

$$\Rightarrow x^3-y^3=2^3-1^3=7$$

10. Correct option: (A)

$$4x+3y=14$$
$$9x-5y=55$$

To solve this linear equation pair it is essential to find the values of x & y.
In order to do that either of 'x' or 'y' term should be eliminated.
Let's eliminate y term, multiply 1st equation with '5' and multiply second equation with '3' implies

$$20x+15y=70$$
$$27x-15y=165$$

Adding above two equations implies

$$47x=235$$
$$\Rightarrow x=\frac{235}{47}=5$$

Now substitute value of 'x' in any of the equation will yields the value of 'y'

$$4(5)+3y=14$$
$$\Rightarrow 3y=14-20$$
$$\Rightarrow y=\frac{-6}{3}=-2$$

Solution of given equation pair is (5,-2)

11. Correct option: (A)

$$3^{x+y}=27$$
$$3^{x-y}=243$$

$$3^{x+y}=27=3^3$$
$$\Rightarrow x+y=3(\because ifa^m=a^n \Rightarrow m=n)$$

$$3^{x-y}=243=3^5$$
$$\Rightarrow x-y=5(\because ifa^m=a^n \Rightarrow m=n)$$

Adding both the equations implies

$$2x=8\Rightarrow x=4$$

12. **Correct option: (B)**

$$\frac{1}{2x}-\frac{1}{y}=-1$$
$$\frac{1}{x}+\frac{1}{2y}=8.$$

After simplifying above two equations implies

$$y-2x=-2xy$$
$$2y+x=16xy$$

In order to solve above equation pair multiply 1st equation with '1' and 2nd equation with '2' and add the resultant equations implies

$$5y=30xy$$
$$\Rightarrow 6x=1$$
$$\therefore x=\frac{1}{6}$$

Substitute the value of 'x' in any of the above equation, implies

$$\Rightarrow y-2\left(\frac{1}{6}\right)=-2\left(\frac{1}{6}\right)y$$
$$\Rightarrow y+\frac{y}{3}=\frac{1}{3}$$
$$\Rightarrow \frac{4y}{3}=\frac{1}{3}\Rightarrow y=\frac{1}{4}$$

13. **Correct option: (B)**

Let 'x' and 'y' be the two numbers and according to given data

$$x+y=40$$
$$x-y=6$$

Adding both equations implies

$$2x=46$$
$$x=\frac{46}{2}=23$$

Substituting value of 'x' in any of above equations implies $23+y=40\Rightarrow y=17$

14. **Correct option: (A)**

Let the fraction be $\frac{x}{y}$

It is given that both the numerator and the denominator are decreased by one, it is Equal to $\frac{1}{2}$ this implies the resultant fraction is equals to $\frac{x-1}{y-1}=\frac{1}{2}$

Also it is given that if the numerator is increased by 2 it is equal to $\frac{x+2}{y}=\frac{2}{3}$

Simplifying these two equations implies that

$$2x-2=y-1, 3x+6=2y$$
$$2x-y=1 \rightarrow \times 2$$
$$3x-2y=-6 \rightarrow \times 1$$
$$4x-2y=2$$
$$-3x-+2y=-+6$$
$$x=8$$
$$y=15$$

Therefore the resultant fraction $\frac{x}{y}=\frac{8}{15}$

15. **Correct option: (D)**

The sum of three angles in a triangle is 180 degrees $x+3x+y=180^0$

Given the relation between three angles is $3y-5x=30^\circ$

Solving these two together implies that

$$4x+y=180^0 \rightarrow \times 5$$
$$-5x+3y=30^0 \rightarrow \times 4$$
$$20x+5y=900$$
$$-20x+12y=120$$
$$17y=1020^0$$
$$y=60^0$$
$$4x+60^0=120^0$$
$$4x=120^0$$
$$x=30^0$$
$$3x=90^0$$
$$30^0, 90^0, 60^0$$

Since one angle of triangle is 90 degrees and hence it is an right angled triangle

16. **Correct option: (B)**

Given the ratio of ages between 'A' and 'B' is $A:B=5:3$

Let the Age be 'x'

After 6 years their respective ages with ratio is $5x+6:3x+6=7:5$

$$\frac{5x+6}{3x+6}=\frac{7}{5}$$
$$25x+30=21x+42$$
$$4x=42-30$$
$$4x=12, x=3$$
$$\therefore A=15, B=9$$

17. **Correct option: (A)**

5 pencils and 7 pens together cost Rs. 50 this implies that $5p_e+7p=50$

7 pencils and 5 pens together cost Rs.46 this implies that $7p_e+5p=46$

The sum of costs of 1 pencil and 1 pen is $p+p_e$

Solving these two together implies that

$$5p_e + 7p = 50 \rightarrow \times 5$$
$$7p_e + 5p = 46 \rightarrow \times 7$$
$$25p_e + 35p = 250$$
$$-49p_e - 35p = -322$$
$$-24p_e = -172$$
$$p_e = 3$$
$$5(3) + 7p = 50$$
$$7p = 35, p = 5$$
$$p + p_e = 8$$

18. **Correct option: (D)**

$$2x - y = 5 \rightarrow \times 2$$
$$3x + 2y = 11 \rightarrow \times 1$$
$$4x - 2y = 10$$
$$3x + 2y = 11$$
$$7x = 21$$
$$x = 3$$
$$6 - y = 5 \Rightarrow y = 1$$
$$2x + 3y \Rightarrow 2(3) + 3(1) = 9$$

19. **Correct option: (C)**

The condition for pair of equations is coincident implies $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Given pair of equations

$$3x + 4y + 2 = 0$$
$$9x + 12y + k = 0$$
$$\frac{\cancel{3}}{\cancel{4}} = \frac{\cancel{4}}{\cancel{12}} = \frac{2}{k}$$
$$\frac{1}{3} = \frac{2}{k} \Rightarrow k = 6$$

20. **Correct option: (134)**

Let length and Breadth be l, b respectively and the Area be A=l∗b.

Given that the area of a rectangle gets reduced by 80 sq. units if its length is reduced by 5 units and the breadth increased by 2 units, this implies that

$$(l-5)(b+2) = lb - 80$$

Simplifying implies that

$$lb + 2l - 5b - 10 = lb - 80$$
$$2l - 5b = -70$$

If we increase the length by 10 units and decrease the breadth by 5 units, the area will be increased by 50 sq. units this implies that

$$(l+10)(b-5) = lb + 50$$

Simplifying implies that

$$lb - 5l + 10b - 50 = lb + 50$$
$$-5l + 10b = 100$$

$$-5l+10b=100 \rightarrow \times 1$$
$$2l-5b=-70 \rightarrow \times 2$$
$$-5l+10b=100$$
$$4l-10b=-140$$
$$-l=-40$$
$$l=40$$
$$2(40)-5b=-70$$
$$80+70=5b$$
$$b=30$$
$$2(l+b)=140$$
$$2(35+37)=134$$

LEVEL - II

1. Correct option: (C)

Given

$$\frac{4p+9q}{p}=\frac{5q}{p-q}$$

Cross- multiply

$$(4p+9q)(p-q)=(5q)(p)$$
$$4p^2-4pq+9pq-9q^2=5pq$$
$$4p^2+5pq-5pq=9q^2$$
$$4p^2=9q^2$$
$$\frac{p^2}{q^2}=\frac{9}{4}$$
$$\frac{p}{q}=\sqrt{\frac{9}{4}}$$
$$\frac{p}{q}=\frac{3}{2}$$

Hence, the value of $\frac{p}{q}$ is $\frac{3}{2}$.

2. Correct option: (B)

Given that x, y and $y-\frac{1}{x}$ are not equal to zero

So

$$\frac{x-\frac{1}{y}}{y-\frac{1}{x}}=\frac{\frac{xy-1}{y}}{\frac{xy-1}{x}}$$
$$\frac{x-\frac{1}{y}}{y-\frac{1}{x}}=\frac{x}{y}$$

Hence, the value of $\frac{x-\frac{1}{y}}{y-\frac{1}{x}}$ is $\frac{x}{y}$.

3. Correct option: (B)

Let x be the number of coins of 25 paise and y be the number of coins of 50 paise.

Given that there are 47 coins in total. So

$$x + y = 47$$

Also worth of coins is Rs. 14.50 = 1450 paise

Then

$$25x + 50y = 1450$$

Consider

$$x + y = 47$$

$$y = 47 - x$$

Put value of y in equation $25x + 50y = 1450$

$$25x + 50y = 1450$$

$$25x + 50(47 - x) = 1450$$

$$25x + 2350 - 50x = 1450$$

$$2350 - 1450 = 25x$$

$$25x = 900$$

$$x = \frac{900}{25}$$

$$x = 36$$

Then y is

$$y = 47 - x$$

$$y = 47 - 36$$

$$y = 11$$

Hence, there are 36 coins of 25 paise and 11 coins of 50 paise.

4. Correct option: (C)

Given

$$10x + 3y = 75$$

$$6x - 5y = 11$$

Solve equations to get value of x and y

For this consider

$$6x - 5y = 11$$

$$x = \frac{11 + 5y}{6}$$

Put the value of x in equation $10x + 3y = 75$

$$10x + 3y = 75$$

$$10\left(\frac{11 + 5y}{6}\right) + 3y = 75$$

$$\frac{110 + 50y}{6} + 3y = 75$$

$$\frac{110 + 50y + 18y}{6} = 75$$

$$110 + 68y = 450$$

$$68y = 450 - 110$$
$$68y = 340$$
$$y = \frac{340}{68}$$
$$y = 5$$

Then value of x is

$$x = \frac{11 + 5y}{6}$$
$$x = \frac{11 + 5(5)}{6}$$
$$x = \frac{11 + 25}{6}$$
$$x = \frac{36}{6}$$
$$x = 6$$

Now $x + y = 6 + 5$

$$x + y = 11$$

Hence, value of x + y is 11.

5. Correct option: (C)

Let the numerator of fraction be N

Given that denominator is 1 more than the numerator. So, denominator is N+1

Also, if 1 is subtracted from the numerator and denominator, the fraction becomes equal to $1/2$

$$\frac{N-1}{(N+1)-1} = \frac{1}{2}$$
$$\frac{N-1}{N} = \frac{1}{2}$$
$$2(N-1) = N$$
$$2N - 2 = N$$
$$2N - N = 2$$
$$N = 2$$

Numerator is 2

Then denominator is 2 + 1 = 3

Hence, the fraction is $\frac{2}{3}$

6. Correct option: (B)

Let the number of goats be x and the number of hens be y.

Goats have 4 legs and hens have 2 legs, which are $4x + 2y$

Given that legs are 10 more than twice the number of heads, which can be formulated as

$$4x + 2y = 2(x + y) + 12$$
$$4x + 2y = 2x + 2y + 12$$
$$2x = 12$$
$$x = 6$$

Hence, the number of goats is 6.

7. Correct option: (B)

Given that (1, 4) is the point of intersection of the lines $2x + by = 6$ and $3y = 8 + ax$

Therefore, the value x=1 and y=4 satisfy the set of equations

$$2x + by = 6$$

$$3y = 8 + ax$$

So

$$2(1) + b(4) = 6$$

$$2 + 4b = 6$$

$$4b = 4$$

$$b = 1$$

and

$$3y = 8 + ax$$

$$3(4) = 8 + a(1)$$

$$12 = 8 + a$$

$$a = 4$$

Now

$$a - b = 4 - 1 = 3$$

Hence, value of a - b is 3.

8. Correct option: (D)

Given (a + b, a - b) is the solution of equation $3x + 2y = 20$ and $4x - 5y = 42$

So put value of x as (a + b) and y as (a - b) in above equation

$$3x + 2y = 20$$

$$3(a + b) + 2(a - b) = 20$$

$$3a + 3b + 2a - 2b = 20$$

$$5a + b = 20$$

And

$$4x - 5y = 42$$

$$4(a + b) - 5(a - b) = 42$$

$$4a + 4b - 5a + 5b = 42$$

$$9b - a = 42$$

$$a = 9b - 42$$

Put the value of a in equation 5a + b = 20

$$5a + b = 20$$

$$5(9b - 42) + b = 20$$

$$45b - 210 + b = 20$$

$$46b = 210 + 20$$

$$46b = 230$$

$$b = 5$$

Hence, the value of b is 5

9. Correct option: (B)

Let N be numerator and D be denominator

If 1 is added to the numerator and 1 is subtracted from denominator, the fraction becomes equal to 1

$$\frac{N+1}{D-1}=1$$
$$N+1=D-1$$
$$N=D-2$$

Also, if 1 is added to denominator, then fraction becomes equal to $1/2$

$$\frac{N}{D+1}=\frac{1}{2}$$
$$2N=D+1$$

Put the value of N in above equation

$$2N=D+1$$
$$2(D-2)=D+1$$
$$2D-4=D+1$$
$$2D-D=4+1$$
$$D=5$$

Then numerator is

$$2N=D+1$$
$$2N=5+1$$
$$2N=6$$
$$N=3$$

The fraction is $3/8$

Hence, sum of numerator and denominator is 3+5=8

10. Correct option: (A)

Given $kx+2y=5$

Or $kx+2y-5=0$

And $3x+y=1$

Or $3x+y-1=0$

Comparing with general form of equation

$$a_1x+b_1y+c_1=0$$
$$a_1=k, b_1=2, c_1=-5$$

And

$$a_2x+b_2y+c_2=0$$
$$a_2=3, b_2=1, c_2=-1$$

Now for the system of equation to have zero solution

$$\frac{a_1}{a_2}=\frac{b_1}{b_2}\neq\frac{c_1}{c_2}$$
$$\frac{k}{3}=\frac{2}{1}\neq\frac{-5}{-1}$$

Clearly

$$\frac{b_1}{b_2}\neq\frac{c_1}{c_2}$$

Then the given system of equation will have no solution, if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\frac{k}{3} = \frac{2}{1}$$

$$k = 6$$

Hence, the given system of equations will have zero solution, if k=6

11. Correct option: (D)

Given

$$\frac{ax}{b} - \frac{by}{a} = a + b$$

or

$$a^2x - b^2y = ab(a + b)$$

or

$$a^2x - b^2y - ab(a + b) = 0$$

And

$$ax - by = 2ab$$

$$ax - by - 2ab = 0$$

Comparing with general form of equation

$$a_1x + b_1y + c_1 = 0$$

$$a_1 = a^2, b_1 = -b^2, c_1 = -ab(a + b)$$

And

$$a_2x + b_2y + c_2 = 0$$

$$a_2 = a, b_2 = -b, c_2 = -2ab$$

Formula for cross- multiplication is

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

$$\frac{x}{(-b^2)(-2ab) - (-b)[-ab(a + b)]} = \frac{y}{[-ab(a + b)](a) - (-2ab)(a^2)} = \frac{1}{a^2(-b) - (a)(-b^2)}$$

$$\frac{x}{2ab^3 - a^2b^2 - ab^3} = \frac{y}{-a^3b - a^2b^2 + 2a^3b} = \frac{1}{-a^2b + ab^2}$$

$$\frac{x}{ab^3 - a^2b^2} = \frac{y}{a^3b - a^2b^2} = \frac{1}{ab^2 - a^2b}$$

$$\frac{x}{ab^2(b - a)} = \frac{y}{-a^2b(b - a)} = \frac{1}{ab(b - a)}$$

$$\frac{x}{ab^2(b - a)} = \frac{1}{ab(b - a)}$$

$$x = b$$

$$\frac{y}{-a^2b(b - a)} = \frac{1}{ab(b - a)}$$

$$y = -a$$

Hence, the value of x are b and y are -a

12. Correct option: (B)

Given $3 < x < 6$

$2 < y < 4$

Multiply 2 in equation $3 < x < 6$ and 3 in equation $2 < y < 4$

$6 < 2x < 12$

$6 < 2y < 12$

Then $6 < \frac{2x}{2y} < 12$

$3 < \frac{x}{y} < 6$

Hence, the greatest value of $\frac{x}{y}$ is 6

13. Correct option: (A, C)

Given

A work can be finished in 18 days

A and B can do a work in 10 days

$$\frac{10}{A} + \frac{10}{B} = 1$$

$$10B + 10A = AB$$

Now, A alone can finish a work in 12 days and B alone can finish a work in 6 days

$$\frac{12}{A} + \frac{6}{B} = 1$$

$$12B + 6A = AB$$

Comparing equation $10\mathrm{B} + 10\mathrm{A} = \mathrm{AB}$

$12\mathrm{B} + 6\mathrm{A} = \mathrm{AB}$

$12\mathrm{B} + 6\mathrm{A} = 10\mathrm{B} + 10\mathrm{A}$

$12B - 10B = 10A - 6A$

$2B = 4A$

$B = 2A$

Put value of B in equation

$$\frac{10}{\mathrm{A}} + \frac{10}{\mathrm{B}} = 1$$

$$\frac{10}{\mathrm{A}} + \frac{10}{2\mathrm{A}} = 1$$

$$\frac{20 + 10}{2\mathrm{A}} = 1$$

$$\frac{30}{2A} = 1$$

$$2A = 30$$

$$A = 15$$

Then value of B is $\mathrm{B} = 2\mathrm{A}$

$\mathrm{B} = 2(15)$

$B = 30$

Hence, the time required for A is 15 days and time required for B is 30 days.

14. Correct option: (A)

Given

$$\frac{1}{2(2x+3y)}+\frac{12}{7(3x-2y)}=\frac{1}{2}$$

$$\frac{7}{2x+3y}+\frac{4}{3x-2y}=2$$

Let

$$\frac{1}{(2x+3y)}=X$$

And

$$\frac{1}{3x-2y}=Y$$

Then

$$\frac{1}{2}X+\frac{12}{7}Y=\frac{1}{2}$$

$$7X+24Y=7$$

and

$$7X+4Y=2$$

Subtracting above equation, we get

$$7X-7X+24Y-4Y=7-2$$

$$20Y=5$$

$$Y=\frac{1}{4}$$

Putting value of Y in equation

$$7X+4Y=2$$

$$7X+4\left(\frac{1}{4}\right)=2$$

$$7X+1=2$$

$$7X=1$$

$$X=\frac{1}{7}$$

Now

$$\frac{1}{(2x+3y)}=X$$

$$\frac{1}{(2x+3y)}=\frac{1}{7}$$

$$2x+3y=7$$

and

$$\frac{1}{3x-2y}=Y$$

$$\frac{1}{3x-2y}=\frac{1}{4}$$

$$3x-2y=4$$

Solve above equation to get value of x and y

$2x + 3y = 7$..... (i)

$3x - 2y = 4$..... (ii)

Multiply equation (i) by 2 and equation (ii) by 3

$$4x + 6y = 14$$
$$9x - 6y = 12$$

Adding above equations

$$4x + 9x + 6y - 6y = 14 + 12$$
$$13x = 26$$
$$x = 2$$

Then y is

$$9(2) - 6y = 12$$
$$18 - 6y = 12$$
$$18 - 12 = 6y$$
$$6y = 6$$
$$y = 1$$

Hence value of x is 2 and value of y is 1.

15. Correct option: (C)

x = 2

y = 1

x + y = 2 +1 =3

Hence, value of x + y is 3

16. Correct option: (B)

x = 2

y = 1

x – y = 2 – 1 =1

Hence, the value of x – y is 1

17. Correct Matching is: a-P, b- R, c- S d- Q

A) $2x + 3y + 4 = 0;\ 5x + 6y + 9 = 0$	P) $\left[-1, \frac{-2}{3}\right]$
B) $x + 3y = 1;\ 2x + 6y = 2$	T) $\left(1, \frac{2}{3}\right)$
C) The solution set for inconsistent system of equations	S) Null set
D) $y = mx;\ m \neq 0$ passes through	Q) (0, 0)

Explanation:

A) Given

$$2x + 3y + 4 = 0$$
$$5x + 6y + 9 = 0$$

Multiply 2 in equation $2x + 3y + 4 = 0$

$$2(2x + 3y + 4) = 0$$
$$4x + 6y + 8 = 0$$

Subtract this equation by $5x + 6y + 9 = 0$

$$5x + 6y + 9 - (4x + 6y + 8) = 0$$
$$5x - 4x + 6y - 6y + 9 - 8 = 0$$
$$x + 1 = 0$$
$$x = -1$$

Put value of x in equation

$$2x + 3y + 4 = 0$$
$$2(-1) + 3y + 4 = 0$$
$$-2 + 3y + 4 = 0$$
$$3y + 2 = 0$$
$$3y = -2$$
$$y = \frac{-2}{3}$$

Hence the value of x is -1 and y is $\frac{-2}{3}$.

B) Given

$$x + 3y = 1$$
$$2x + 6y = 2$$

or

$$x + 3y = 1$$

x = 0

$$0 + 3y = 1$$
$$y = \frac{1}{3}$$

y = 0

$$x + 0 = 1$$
$$x = 1$$

C) The solution set for inconsistent system of equations is null set.

D) $y = mx;\ m \neq 0$ passes through (0 ,0).

18. Correct Answer is: 4

Given

$$6x + 3y = 7xy$$
$$3x + 9y = 11xy$$

Solve the equation to get value of x and y

Divide both sides of equation by xy

$$\frac{6}{y} + \frac{3}{x} = 7$$
$$\frac{3}{y} + \frac{9}{x} = 11$$

Multiply above equation by 2

$$\frac{6}{y} + \frac{18}{x} = 22$$

Subtract this equation by $\frac{6}{y}+\frac{3}{x}=7$

$$\frac{6}{y}+\frac{18}{x}-\left(\frac{6}{y}+\frac{3}{x}\right)=22-7$$

$$\frac{6}{y}+\frac{18}{x}-\frac{6}{y}-\frac{3}{x}=15$$

$$\frac{18-3}{x}=15$$

$$15x=15$$

$$x=1$$

Put the value of x in $\frac{3}{y}+\frac{9}{x}=11$

$$\frac{3}{y}+\frac{9}{1}=11$$

$$\frac{3}{y}+9=11$$

$$\frac{3}{y}=2$$

$$y=\frac{3}{2}$$

Thus, value of x+2y is

$$x+2y=1+2\left(\frac{3}{2}\right)$$

$$x+2y=1+3$$

$$x+2y=4$$

Hence, the value of x+2y is 4

19. Correct Answer is: 6

Let the number be x

Given that three times of a number is increased by 6 which gives results as 24

$$3x+6=24$$

$$3x=16$$

$$x=6$$

Hence, the number is 6.

20. Correct Answer is: 2

Let the two-digit number be 10x + y

Reverse of two-digit number is 10y + x

Given the sum of two-digit number and reverse of it is 66

$$10x+y+10y+x=66$$

$$11x+11y=66$$

$$x+y=6$$

Also given that difference of digit is 2 $x-y=2$

Add equation $x+y=6$

And $x-y=2$

$$x+y+x-y=8$$

$$2x = 8$$
$$x = 4$$

then value of y is $x - y = 2$

$$4 - y = 2$$
$$y = 4 - 2$$
$$y = 2$$

Hence, two digits number are 42 and 24.

Therefore, there are two such numbers

STUDENT PRACTICE QUESTIONS

1. Correct option: (D)

$$0.16(5x-2) = 0.4x + 7$$
$$\Rightarrow 0.8x - 0.32 = 0.4x + 7$$
$$\Rightarrow 0.4x = 7.32 \Rightarrow x = \frac{7.32}{0.4} = 18.3$$

2. Correct option: (A)

$$\frac{2x+1}{3x-2} = \frac{9}{10}$$
$$\Rightarrow 10(2x+1) = 9(3x-2)$$
$$\Rightarrow 20x + 10 = 27x - 18$$
$$\Rightarrow 7x = 28 \Rightarrow x = \frac{28}{7} = 4$$

3. Correct option: (A)

$$x - \left(\frac{2x+8}{3}\right) = \frac{1}{4}\left(x - \frac{2-x}{6}\right) - 3$$
$$\Rightarrow \frac{3x - 2x - 8}{3} = \frac{6x - 2 + x}{24} - 3$$
$$\Rightarrow \frac{x-8}{3} = \frac{7x - 2 - 72}{24} \Rightarrow 8x - 64 = 7x - 74$$
$$\Rightarrow x = 10$$

4. Correct option: (A)

$$x + \frac{1}{x} = 3$$

Squaring on both sides implies

$$x^2 + \frac{1}{x^2} + 2(x)\left(\frac{1}{x}\right) = 9$$
$$\Rightarrow x^2 + \frac{1}{x^2} = 9 - 2 \Rightarrow x^2 + \frac{1}{x^2} = 7$$

Again Squaring on both sides on $x^2 + \frac{1}{x^2} = 7$ implies

$$x^4 + \frac{1}{x^4} + 2\left(x^2\right)\left(\frac{1}{x^2}\right) = 49$$
$$\Rightarrow x^4 + \frac{1}{x^4} = 49 - 2 \Rightarrow x^4 + \frac{1}{x^4} = 47$$

Again Squaring on both sides on $x^4+\frac{1}{x^4}=47$ implies

$$x^8+\frac{1}{x^8}+2\left(x^4\right)\left(\frac{1}{x^4}\right)=2209$$

$$\Rightarrow x^8+\frac{1}{x^8}=2209-2\Rightarrow x^8+\frac{1}{x^8}=2207$$

5. Correct option: (B).

$$\frac{x}{4}+\frac{x}{6}=x-7$$

$$\frac{3x+2x}{12}=x-7$$

$$\Rightarrow\frac{5x}{12}=x-7\Rightarrow 5x=12x-84\Rightarrow 7x=84$$

$$\Rightarrow x=\frac{84}{7}=12$$

6. Correct option: (C)

Sol. Let the two numbers be 3x and 8x respectively

Given that sum of those two numbers is 165

$$\Rightarrow 3x+8x=165$$

$$\Rightarrow 11x=165$$

$$\Rightarrow x=15$$

Therefore the numbers pair is $3\times15=45, 8\times15=120$ respectively

7. Correct option: (D)

Sol. Let the two consecutive even numbers be 'x' and 'x+2' respectively

Given that two-fifth of the smaller exceeds two Eleventh of the large by 4

This can be represented in the form of equation as follows

$$\frac{2x}{5}-\frac{2x+4}{11}=4$$

$$\Rightarrow\frac{22x-10x-20}{55}=4\Rightarrow 12x-20=220$$

$$\Rightarrow 12x=240\Rightarrow x=20$$

Therefore another number is 'x+2=22'.

8. Correct option: (B)

$$2y+\frac{7}{x}=3$$

$$\Rightarrow 2xy+7=3x$$

$$y-\frac{3}{x}=8$$

$$\Rightarrow xy-3=8x$$

Adding both the equations implies

$$11x-4=3xy$$

$$\Rightarrow y=\frac{11x-4}{3x}$$

Substitute the value of 'x' in any of the above equation pair implies that

$$2\left(\frac{11x-4}{3x}\right)+\frac{7}{x}=3$$
$$\Rightarrow 22x-8+21=9x$$
$$\Rightarrow 22x-9x=-13$$
$$\Rightarrow x=-1$$

Substituting the value of 'x' in value of 'y' implies

$$y=\frac{11x-4}{3x}$$
$$y=\frac{11(-1)-4}{3(-1)}=5$$

9. **Correct option: (D)**.

$$\frac{x}{a}+\frac{y}{b}=2$$
$$ax-by=a^2-b^2$$

$$\Rightarrow \frac{x}{a}+\frac{y}{b}-2=0$$
$$\Rightarrow bx+ay-2ab=0$$
$$\Rightarrow bx+ay=2ab$$

$$ax-by=a^2-b^2$$
$$bx+ay=2ab$$

In order to solve above equation pair multiply equation 1 with 'a' and equation 2 with 'b' implies

$$(ax-by)a=(a^2-b^2)a$$
$$(bx+ay)b=2ab^2$$

Simplifying further implies that

$$a^2x-aby=(a^2-b^2)a$$
$$b^2x+aby=2ab^2$$

Adding both the equations implies that

$$\left(a^2+b^2\right)x=\left(a^2-b^2\right)a+2ab^2$$
$$\Rightarrow x=\frac{\left(a^2-b^2\right)a+2ab^2}{\left(a^2+b^2\right)}=\frac{a\left(a^2+b^2\right)}{\left(a^2+b^2\right)}=a$$

10. **Correct option: (A)**

Cost of 5 pens and 7 notebooks is Rs. 82 this implies $5P+7N=82$

Also the cost of 4 pens and 4 notebooks is Rs 52 $4P+4N=52$

Asked to find the cost of 2 pens and 3 notebooks implies that $2P+3N$

$$5P+7N=82$$
$$4P+4N=52$$
$$P+N=13 \rightarrow \times 7$$
$$5P+7N=82 \rightarrow \times 1$$
$$7P+7N=91$$
$$_5P+_7N=_8$$
$$2P=9$$
$$P=4.5$$
$$N=8.5$$
$$2P+3N=2(4.5)+3(8.5)$$
$$9+25.5$$
$$34.5$$

11. Correct option: (B, C)

Given $3x + 4y = 25$

And $5x - 6y = -9$

Solve the equation to get value of x and y

Consider $3x + 4y = 25$

$$x = \frac{25 - 4y}{3}$$

Put the value of x in equation $5x - 6y = -9$

$$5x - 6y = -9$$
$$5\left(\frac{25 - 4y}{3}\right) - 6y = -9$$
$$\frac{125 - 20y}{3} - 6y = -9$$
$$\frac{125 - 20y - 18y}{3} = -9$$
$$125 - 38y = -27$$
$$125 + 27 = 38y$$
$$38y = 152$$
$$y = \frac{152}{38}$$
$$y = 4$$

Then value of x is $x = \frac{25-4y}{3}$

$$x = \frac{25 - 4(4)}{3}$$
$$x = \frac{25 - 16}{3}$$
$$x = \frac{9}{3}$$
$$x = 3$$

Hence the value of x is 3 and value of y is 4

Also, 4y=16

12. Correct option: (A,B)

Given

$\sqrt{3}x - \sqrt{2}y = \sqrt{3}$.....(i)

$\sqrt{5}x + \sqrt{3}y = \sqrt{2}$..... (ii)

Solve the equations to get value of x and y

Multiply equation (i) by $\sqrt{3}$ and equation (ii) by $\sqrt{2}$

$$3x - \sqrt{6}y = 3$$
$$\sqrt{10}x + \sqrt{6}y = 2$$

Adding above equation

$$3x - \sqrt{6}y + \sqrt{10}x + \sqrt{6}y = 5$$
$$x\left(\sqrt{10} + 3\right) = 5$$
$$x = \frac{5}{\sqrt{10} + 3}$$
$$\mathrm{x} = \frac{5}{\sqrt{10} + 3} \times \frac{\sqrt{10} - 3}{\sqrt{10} - 3}$$
$$x = \frac{5\left(\sqrt{10} - 3\right)}{10 - 3\sqrt{10} + 3\sqrt{10} - 9}$$
$$x = 5\left(\sqrt{10} - 3\right)$$

Put value of x in equation

$$\sqrt{3}\mathrm{x} - \sqrt{2}\mathrm{y} = \sqrt{3}$$
$$\sqrt{3}\left[5\left(\sqrt{10} - 3\right)\right] - \sqrt{2}\mathrm{y} = \sqrt{3}$$
$$5\sqrt{30} - 15\sqrt{3} - \sqrt{2}y = \sqrt{3}$$
$$\sqrt{2}y = 5\sqrt{30} - 15\sqrt{3} - \sqrt{3}$$
$$y = \frac{5\sqrt{30} - 16\sqrt{3}}{\sqrt{2}}$$
$$y = \frac{5\sqrt{30}}{\sqrt{2}} - \frac{16\sqrt{3}}{\sqrt{2}}$$
$$y = 5\sqrt{15} - 8\sqrt{6}$$

Hence, the value of x is $5\left(\sqrt{10} - 3\right)$ and y is $5\sqrt{15} - 8\sqrt{6}$.

13. Correct option: (B)

Given $2x - ky + 3 = 0$ and $4x + 6y - 5 = 0$ are parallel lines

Comparing with general form of equation

$$a_1x + b_1y + c_1 = 0$$
$$a_1 = 2, b_1 = -k, c_1 = 3$$
$$a_2x + b_2y + c_2 = 0$$
$$a_2 = 4, b_2 = 6, c_2 = -5$$

Condition for parallel lines is

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\frac{2}{4} = \frac{-k}{6} \neq \frac{3}{-5}$$

Taking

$$\frac{2}{4} = \frac{-k}{6}$$
$$-4k = 12$$
$$k = -3$$

Hence, the value of k is -3.

14. Correct option: (D)

Given that lines $3x + y = 5$ and $4x + (k-3)y + 7 = 0$ are intersecting

Comparing with general form of equation

$$a_1x + b_1y + c_1 = 0$$
$$a_1 = 3, b_1 = 1, c_1 = -5$$
$$a_2x + b_2y + c_2 = 0$$
$$a_2 = 4, b_2 = (k-3), c_2 = 7$$

Condition for intersecting lines is

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$
$$\frac{3}{4} = \frac{1}{k-3}$$
$$3k - 9 = 4$$
$$3k = 13$$
$$k = \frac{13}{3}$$

Hence, the value of k is $13/3$.

15. Correct Matching is: a-R, b-S, c-Q, d-T

A) $\sqrt{3}x + \sqrt{5}y = \sqrt{3}; \sqrt{5}x + \sqrt{3}y = \sqrt{3}$ R) $\frac{\sqrt{15}-3}{2}, \frac{\sqrt{15}-3}{2}$

B) $2^x + 3^y = 17; 2^{x+2} - 3^{y+1} = 5$ S) (2, 3)

C) $\frac{2}{x} + \frac{3}{y} = 13; \frac{5}{x} - \frac{4}{y} = -2 \; where \; x \neq 0, y \neq 0$ Q) $\frac{1}{2}, \frac{1}{3}$

D) $\frac{5}{x-1} + \frac{1}{y-2} = 2; \frac{6}{x-1} - \frac{3}{y-2} = 1$ T) (4, 5)

Explanation

A) Given

$$\sqrt{3}x + \sqrt{5}y = \sqrt{3}$$
$$\sqrt{5}x + \sqrt{3}y = \sqrt{3}$$

Multiply above equations by $\sqrt{3}$

$$3x + \sqrt{15}y = 3$$
$$\sqrt{15}x + 3y = 3$$

Consider

$$\sqrt{15}x + 3y = 3$$
$$y = \frac{3 - \sqrt{15}x}{3}$$

Put the value of y in equation

$$3x + \sqrt{15}y = 3$$
$$3x + \sqrt{15}\left(\frac{3 - \sqrt{15}x}{3}\right) = 3$$
$$3x + \frac{3\sqrt{15} - 15x}{3} = 3$$
$$9x + 3\sqrt{15} - 15x = 9$$
$$3\sqrt{15} - 6x = 9$$
$$6x = 3\sqrt{15} - 9$$
$$2x = \sqrt{15} - 3$$
$$x = \frac{\sqrt{15} - 3}{2}$$

Then value of y is

$$y = \frac{3 - \sqrt{15}x}{3}$$
$$y = \frac{3 - \sqrt{15}\left(\frac{\sqrt{15} - 3}{2}\right)}{3}$$
$$y = \frac{3 - \left(\frac{15 - 3\sqrt{15}}{2}\right)}{3}$$
$$y = \frac{6 - 15 + 3\sqrt{15}}{6}$$
$$y = \frac{3\sqrt{15} - 9}{6}$$
$$y = \frac{\sqrt{15} - 3}{2}$$

Hence the value of x and y are $\frac{\sqrt{15}-3}{2}$, $\frac{\sqrt{15}-3}{2}$

B) Given

$$2^x + 3^y = 17$$
$$2^{x+2} - 3^{y+1} = 5$$

Let $2^x = a$ and $3^y = b$

$$a + b = 17$$
$$2^x 2^2 - 3^y 3^1 = 5$$
$$4a - 3b = 5$$

Solve the equations to get value of a and b

$$a + b = 17$$
$$4a - 3b = 5$$

Consider

$$a + b = 17$$
$$a = 17 - b$$

Put the value of a in equation

$$4a - 3b = 5$$
$$4(17 - b) - 3b = 5$$
$$68 - 4b - 3b = 5$$
$$7b = 68 - 5$$
$$7b = 63$$
$$b = 9$$

Value of a is

$$a = 17 - b$$
$$a = 17 - 9$$
$$a = 8$$

Now $2^x = a$

$$2^x = 8$$
$$2^x = 2^3$$
$$x = 3$$

and $3^y = b$

$$3^y = 9$$
$$3^y = 3^2$$
$$y = 2$$

Hence, the value of x is 3 and value of y is 2

C) Given

$$\frac{2}{x} + \frac{3}{y} = 13$$
$$\frac{5}{x} - \frac{4}{y} = -2$$

Let $\frac{1}{x} = a$ and $\frac{1}{y} = b$

$2a + 3b = 13$…… (i)

$5a - 4b = -2$…… (ii)

Multiply equation (i) by 4 and (ii) by 3

$$8a + 12b = 52$$
$$15a - 12b = -6$$

Add above equations

$$8a + 15a + 12b - 12b = 52 - 6$$
$$23a = 46$$
$$a = 2$$

Put the value of a in equation

$$2a + 3b = 13$$
$$2(2) + 3b = 13$$
$$3b = 13 - 4$$

$$3b = 9$$

$$b = 3$$

Now $\frac{1}{x} = a$

$$\frac{1}{x} = 2$$

$$x = \frac{1}{2}$$

and $\frac{1}{y} = b$

$$\frac{1}{y} = b$$

$$y = \frac{1}{3}$$

D) Given

$$\frac{5}{x-1} + \frac{1}{y-2} = 2$$

$$\frac{6}{x-1} - \frac{3}{y-2} = 1$$

Let $\frac{1}{x-1} = a$ and $\frac{1}{y-2} = b$

$5a + b = 2$…… (i)

$6a - 3b = 1$…… (ii)

Multiply equation (i) by 3

$$15a + 3b = 6$$

Add this equation by equation (ii)

$$15a + 6a + 3b - 3b = 6 + 1$$

$$21a = 7$$

$$a = \frac{1}{3}$$

Put value of x in equation $6a - 3b = 1$

$$6\left(\frac{1}{3}\right) - 3b = 1$$

$$2 - 3b = 1$$

$$2 - 1 = 3b$$

$$3b = 1$$

$$b = \frac{1}{3}$$

Now $\frac{1}{x-1} = a$

$$\frac{1}{x-1} = a$$

$$\frac{1}{x-1} = \frac{1}{3}$$

$$3 = x - 1$$

$$x = 3 + 1$$

$$x = 4$$

and $\frac{1}{y-2} = b$

$$\frac{1}{y-2} = b$$
$$\frac{1}{y-2} = \frac{1}{3}$$
$$3 = y - 2$$
$$y = 5$$

Hence, the value of x is 4 and value of y is 5

16. Correct Answer is: 2

Given

$$x + y = 1$$
$$kx + y = 2$$

Comparing with general form of equation

$$a_1x + b_1y + c_1 = 0$$
$$a_1 = 1, b_1 = 1, c_1 = -1$$
$$a_2x + b_2y + c_2 = 0$$
$$a_2 = k, b_2 = 1, c_2 = -2$$

Condition for no solution is

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$
$$\frac{1}{k} = \frac{1}{1} \neq \frac{-1}{-2}$$

Consider

$$\frac{1}{k} = \frac{1}{1}$$
$$k = 1$$

Then value of k + 1 is

$$k + 1 = 1 + 1$$
$$k + 1 = 2$$

Hence, value of k + 1 is given by 2

17. Correct Answer is: 3

Let the number of bees be x and number of flowers be y

Given that if one bee sits on each flower, then one bee will be left and if two bees sit on each flower, one flower will be left.

Then

$$x - y = 1$$
$$y - \frac{x}{2} = 1$$

Solve equations to get value of x and y

Consider

$$x - y = 1$$
$$x = y + 1$$

Put value of x in equation

$$y - \frac{x}{2} = 1$$
$$y - \frac{y+1}{2} = 1$$
$$\frac{2y - (y+1)}{2} = 1$$
$$2y - y - 1 = 2$$
$$y = 3$$

Then value of x is

$$x = y + 1$$
$$x = 3 + 1$$
$$x = 4$$

Hence, the number of bees is 4 and number of flowers is 3.

18. Correct Answer is: 0

Given

$$\frac{5x-3}{4} + \frac{y+1}{5} = 2x + y$$
$$\frac{x+11}{6} - \frac{3y-4x}{4} = 2x + 3$$

Solve the equations to get value of x and y

Consider

$$\frac{x+11}{6} - \frac{3y-4x}{4} = 2x + 3$$
$$\frac{4(x+11) - 6(3y-4x)}{24} = 2x + 3$$
$$\frac{4x + 44 - 18y + 24x}{24} = 2x + 3$$
$$28x - 18y + 44 = 24(2x + 3)$$
$$28x - 18y = 48x + 72 - 44$$
$$-20x - 18y = 28$$
$$20x = -28 - 18y$$
$$x = \frac{-28 - 18y}{20}$$

Put the value of x in equation

$$\frac{5x-3}{4} + \frac{y+1}{5} = 2x + y$$
$$\frac{5\left(\frac{-28-18y}{20}\right) - 3}{4} + \frac{y+1}{5} = 2\left(\frac{-28-18y}{20}\right) + y$$
$$\frac{\left(\frac{-28-18y}{4}\right) - 3}{4} + \frac{y+1}{5} = \left(\frac{-28-18y}{10}\right) + y$$
$$\frac{-28 - 18y - 12}{16} + \frac{y+1}{5} = \left(\frac{-28 - 18y + 10y}{10}\right)$$

$$\frac{-40-18y}{16}+\frac{y+1}{5}=\frac{-28-8y}{10}$$
$$\frac{5(-40-18y)+16(y+1)}{80}=\frac{-28-8y}{10}$$
$$\frac{-240-90y+16y+16}{8}=-28-8y$$
$$-224-74y=-224-8y$$
$$0=74y-8y$$
$$0=66y$$
$$y=0$$

Hence, value of y is 0

19. Correct Answer is: 2

Given $2x+3y=34$

$$1+\frac{x}{y}=\frac{13}{8}$$

Solve the equations to get value of x and y

Consider $2x+3y=34$

$$x=\frac{34-3y}{2}$$

Put the value of x in equation

$$1+\frac{x}{y}=\frac{13}{8}$$
$$1+\frac{\left(\frac{34-3y}{2}\right)}{y}=\frac{13}{8}$$
$$1+\frac{34-3y}{2y}=\frac{13}{8}$$
$$\frac{2y+34-3y}{2y}=\frac{13}{8}$$
$$8(34-y)=13(2y)$$
$$272-8y=26y$$
$$34y=272$$
$$y=8$$

Then value of x is $x=\frac{34-3y}{2}$

$$x=\frac{34-3(8)}{2}$$
$$x=\frac{34-24}{2}$$
$$x=\frac{10}{2}$$
$$x=5$$

Thus, value of 2x – y is $2x-y=2(5)-8$

$$2x-y=10-8$$
$$2x-y=2$$

Hence, value of 2x – y is 2.

20. Correct Answer is:

Given $x+\frac{4}{y}=4$

$y+\frac{6}{x}=6$

Solve the equation to get value of x and y

Consider $x+\frac{4}{y}=4$

$xy+4=4y$

$4y-xy=4$

$y(4-x)=4$

$y=\frac{4}{4-x}$

Substitute value of y in equation $y+\frac{6}{x}=6$

$\frac{4}{4-x}+\frac{6}{x}=6$

$\frac{4x+6(4-x)}{x(4-x)}=6$

$4x+24-6x=6x(4-x)$

$24-2x=24x-6x^2$

$6x^2-24x-2x+24=0$

$6x^2-26x+24=0$

$3x^2-13x+12=0$

$3x^2-9x-4x+12=0$

$3x(x-3)-4(x-3)=0$

$(x-3)(3x-4)=0$

$x=3, x=\frac{4}{3}$

Value of y $y=\frac{4}{4-x}$

When x =3 $y=\frac{4}{4-3}$

$y=4$

when x = 4/3 $y=\frac{4}{4-\left(\frac{4}{3}\right)}$

$y=\frac{12}{12-4}$

$y=\frac{12}{8}$

$y=\frac{3}{2}$

Now, x + y is x = 3 and y =4

$x+y=3+4=7$

And x = 4/3 and y =3/2

$x+y=\frac{4}{3}+\frac{3}{2}$

$x+y=\frac{8+9}{6}$

$x+y=\frac{17}{6}$

4. Quadratic Equation

Solutions

LEVEL- 1

1. Correct option: (C)

To find the roots of given equation $6x^2 - 5x + 5 = 0$, will use the method of splitting terms. In splitting the middle terms, split middle term in such a way that splitted terms gives the product of first and last term and sum of it give the middle term.

So,

$$6x^2 - 5x + 1 = 0$$
$$6x^2 - 2x - 3x + 1 = 0$$
$$2x(3x - 1) - 1(3x - 1) = 0$$
$$(3x - 1)(2x - 1) = 0$$
$$x = \frac{1}{3}, x = \frac{1}{2}$$

Hence, roots of equation $6x^2 - 5x + 1 = 0$ are $1/2$ or $1/3$.

2. Correct option: (B)

Discriminant of quadratic equation $ax^2 + bx + c = 0$ is given by

$$D = b^2 - 4ac$$

So, for equation $x^2 + 6x + 36 = 0$ discriminant is

$$D = 6^2 - 4(1)(36)$$
$$D = 36 - 144$$
$$D = -108$$

where a=1, b=6 and c=36

Hence discriminant of equation $x^2 + 6x + 36 = 0$ is -108.

3. Correct option: (D)

Let x and (x+1) be the two consecutive terms.

Given that sum of their squares is 613. So,

$$(x)^2 + (x + 1)^2 = 613$$
$$x^2 + x^2 + 2x + 1 = 613$$
$$2x^2 + 2x - 612 = 0$$
$$x^2 + x - 306 = 0$$
$$x^2 + 18x - 17x - 306 = 0$$
$$x(x + 18) - 17(x + 18) = 0$$
$$(x + 18)(x - 17) = 0$$
$$x = -18, x = 17$$

As per the question value of x cannot be negative as given that the two numbers are positive.

So, choose value of x as 17 and hence value of x+1 be 18.

4. Correct option: (C)

Given that equation $\mathbf{12x^2 + mx + 5 = 0}$ has roots in ratio 3:2. So let roots be 3a and 2a.

Now sum and product of quadratic equation is given by

$$\alpha + \beta = \frac{-b}{a}$$

And $\alpha\beta = \frac{c}{a}$

Here a=12, b=m, c=5, $\alpha = 3a, \beta = 2a$

$$\alpha + \beta = \frac{-b}{a}$$

$$3a + 2a = \frac{-m}{12}$$
$$5a = \frac{-m}{12}$$
$$\alpha\beta = \frac{c}{a}$$
$$3a \times 2a = \frac{5}{12}$$
$$6a^2 = \frac{5}{12}$$
$$a^2 = \frac{5}{72}$$
$$a = \sqrt{\frac{5}{72}}$$
$$a = \frac{\sqrt{5}}{6\sqrt{2}}$$

Put the value of a in equation

$$5a = \frac{-m}{12}$$
$$5\frac{\sqrt{5}}{6\sqrt{2}} = \frac{-m}{12}$$
$$m = \pm 5\sqrt{10}$$

Hence, the value of m is $\pm 5\sqrt{10}$.

5. Correct option: (D)

If roots of quadratic equation are equal then its discriminant is equal to zero.

$$D = 0$$
$$b^2 - 4ac = 0$$

Then in equation $x^2 + px + 4 = 0$**,** a=1, b=p and c=4

$$p^2 - 4(1)(4) = 0$$
$$p^2 - 16 = 0$$
$$p^2 = 16$$
$$p = 4$$

Hence the value of p is 4.

6. Correct option: (B)

If roots of quadratic equation are equal then its discriminant is equal to zero.

$$D = 0$$
$$b^2 - 4ac = 0$$

Then in equation $(3m + 1)x^2 + 2(m + 1)x + m = 0$, a= 3m+1, b=2(m+1) and c=m

$$[2(m + 1)]^2 - 4(3m + 1)(m) = 0$$
$$4(m^2 + 2m + 1) - 4(3m^2 + m) = 0$$
$$4m^2 + 8m + 4 - 12m^2 - 4m = 0$$
$$-8m^2 + 4m + 4 = 0$$
$$2m^2 - m - 1 = 0$$
$$2m^2 - 2m + m - 1 = 0$$
$$2m(m - 1) + 1(m - 1) = 0$$
$$(2m + 1)(m - 1) = 0$$
$$m = -1/2, m = 1$$

Hence, the roots of given equation are -1/2 and 1.

7. Correct option: (B)

To find one root of equation, simplify it

$$(q-r)x^2+(r-p)x+(p-q)=0$$
$$qx^2-rx^2+rx-px+p-q=0$$
$$q(x^2-1)+r(x-x^2)+p(1-x)=0$$
$$q(x+1)(x-1)+rx(1-x)+p(1-x)=0$$
$$-q(x+1)+rx+p=0$$
$$-qx-q+rx+p=0$$
$$p-q=x(q-r)$$
$$x=\frac{p-q}{q-r}$$

Also, the sum of roots of quadratic equation is given by

$$\alpha+\beta=\frac{-b}{a}$$
$$\alpha=\frac{p-q}{q-r}$$
$$b=r-p \text{ and } a=q-r$$

So, β is

$$\frac{p-q}{q-r}+\beta=\frac{-(r-p)}{q-r}$$
$$\beta=-\frac{p-q}{q-r}-\frac{r-p}{q-r}$$
$$\beta=\frac{-p+q-r+p}{q-r}$$
$$\beta=\frac{q-r}{q-r}$$
$$\beta=1$$

Hence one root of equation is 1.

8. Correct option: (A)

Given that $\alpha \text{ and } \beta$ are roots of equation $x^2+2x+3=0$

So, sum and product of roots of equation is $\alpha+\beta=\frac{-2}{1}$

And $\alpha\beta=\frac{3}{1}$

Now, using identity of $(a+b)^2$

$$(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta$$

Putting values

$$(-2)^2=\alpha^2+\beta^2+2(3)$$
$$4=\alpha^2+\beta^2+6$$
$$\alpha^2+\beta^2=-2$$

9. Correct option: (C)

Given that $\alpha \text{ and } \beta$ are roots of equation $ax^2+bx+c=0$

So put value of x as α

$$a\alpha^2+b\alpha+c=0$$
$$\alpha(a\alpha+b)=-c$$
$$a\alpha+b=\frac{-c}{\alpha}$$

Also put value of x as β

$$a\beta^2+b\beta+c=0$$
$$\beta(a\beta+b)=-c$$
$$a\beta+b=\frac{-c}{\beta}$$

Now

$$(a\alpha + b)^{-2} + (a\beta + b)^{-2} = \left(\frac{-c}{\alpha}\right)^{-2} + \left(\frac{-c}{\beta}\right)^{-2}$$

$$(a\alpha + b)^{-2} + (a\beta + b)^{-2} = \frac{\alpha^2 + \beta^2}{c^2}$$

Also, by equation $ax^2 + bx + c = 0$ sum and product are given by

$$\alpha + \beta = \frac{-b}{a} \text{ and } \alpha\beta = \frac{c}{a}$$

Using identity of $(a + b)^2 = a^2 + b^2 + 2ab$

$$(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$$

$$\left(\frac{-b}{a}\right)^2 = \alpha^2 + \beta^2 + 2\left(\frac{c}{a}\right)$$

$$\alpha^2 + \beta^2 = \frac{b^2}{a^2} - \frac{2c}{a}$$

$$\alpha^2 + \beta^2 = \frac{b^2 - 2ac}{a^2}$$

Put the value of $\alpha^2 + \beta^2$ in expression

$$(a\alpha + b)^{-2} + (a\beta + b)^{-2} = \frac{\alpha^2 + \beta^2}{c^2}$$

$$(a\alpha + b)^{-2} + (a\beta + b)^{-2} = \frac{\frac{b^2 - 2ac}{a^2}}{c^2}$$

$$(a\alpha + b)^{-2} + (a\beta + b)^{-2} = \frac{b^2 - 2ac}{a^2c^2}$$

Hence, the value of $(a\alpha + b)^{-2} + (a\beta + b)^{-2}$ is $\frac{b^2-2ac}{a^2c^2}$

10. Correct option: (B)

Given that $\alpha \text{ and } \beta$ are roots of equation $8x^2 - 3x + 27 = 0$

Then sum and product of equation is given by

$$\alpha + \beta = \frac{3}{8}$$

$$\alpha\beta = \frac{27}{8}$$

Now

$$\left(\frac{\alpha^2}{\beta}\right)^{\frac{1}{3}} + \left(\frac{\beta^2}{\alpha}\right)^{\frac{1}{3}} = \frac{\alpha^{\frac{2}{3}}}{\beta^{\frac{1}{3}}} + \frac{\beta^{\frac{2}{3}}}{\alpha^{\frac{1}{3}}}$$

$$\left(\frac{\alpha^2}{\beta}\right)^{\frac{1}{3}} + \left(\frac{\beta^2}{\alpha}\right)^{\frac{1}{3}} = \frac{\alpha + \beta}{(\alpha\beta)^{\frac{1}{3}}}$$

$$\left(\frac{\alpha^2}{\beta}\right)^{\frac{1}{3}} + \left(\frac{\beta^2}{\alpha}\right)^{\frac{1}{3}} = \frac{\frac{3}{8}}{\left(\frac{27}{8}\right)^{\frac{1}{3}}}$$

$$\left(\frac{\alpha^2}{\beta}\right)^{\frac{1}{3}} + \left(\frac{\beta^2}{\alpha}\right)^{\frac{1}{3}} = \frac{\frac{3}{8}}{\left[\left(\frac{3}{2}\right)^3\right]^{\frac{1}{3}}}$$

$$\left(\frac{\alpha^2}{\beta}\right)^{\frac{1}{3}} + \left(\frac{\beta^2}{\alpha}\right)^{\frac{1}{3}} = \frac{\frac{3}{8}}{\frac{3}{2}}$$

$$\left(\frac{\alpha^2}{\beta}\right)^{\frac{1}{3}} + \left(\frac{\beta^2}{\alpha}\right)^{\frac{1}{3}} = \frac{3 \times 2}{8 \times 3}$$

$$\left(\frac{\alpha^2}{\beta}\right)^{\frac{1}{3}} + \left(\frac{\beta^2}{\alpha}\right)^{\frac{1}{3}} = \frac{1}{4}$$

Hence, the value of $\left(\frac{\alpha^2}{\beta}\right)^{\frac{1}{3}} + \left(\frac{\beta^2}{\alpha}\right)^{\frac{1}{3}}$ is $\frac{1}{4}$.

11. Correct option: (A)

Given that α *and* β are roots of equation $ax^2 + bx + c = 0$

Then sum and product of equation is given by

$$\alpha + \beta = \frac{-b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Then

$$\alpha^5\beta^8 + \alpha^8\beta^5 = \alpha^5\beta^5(\alpha^3 + \beta^3)$$

Using identity of $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$

$$(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$$

$$\left(\frac{-b}{a}\right)^3 = \alpha^3 + \beta^3 + 3\left(\frac{-b}{a}\right)\left(\frac{c}{a}\right)$$

$$\alpha^3 + \beta^3 = \frac{-b^3}{a^3} + \frac{3bc}{a^2}$$

$$\alpha^{\mathbf{3}} + \beta^{\mathbf{3}} = \frac{-b^3 + 3abc}{a^3}$$

Now put value of $\alpha^3 + \beta^3$ in expression

$$\alpha^5\beta^8 + \alpha^8\beta^5 = \alpha^5\beta^5(\alpha^3 + \beta^3)$$

$$\alpha^5\beta^8 + \alpha^8\beta^5 = \left(\frac{c}{a}\right)^5 \times \frac{(-b^3 + 3abc)}{a^3}$$

$$\alpha^5\beta^8 + \alpha^8\beta^5 = \frac{c^5(3abc - b^3)}{a^8}$$

Hence, the value of $\alpha^5\beta^8 + \alpha^8\beta^5$ is $\frac{c^5(3abc-b^3)}{a^8}$.

12. Correct option: (C)

Given that α *and* β are roots of equation $x^2 + kx + 2 = 0$

Then sum and product of equation is given by

$$\alpha + \beta = \frac{-k}{1}$$

$$\alpha\beta = \frac{2}{1}$$

Also $$\alpha - \beta = 1$$

Solving equation to get value of α *and* β

$$\alpha\beta = 2$$

$$\alpha = \frac{2}{\beta}$$

$$\alpha - \beta = 1$$

$$\frac{2}{\beta} - \beta = 1$$

$$\beta^2 + \beta - 2 = 0$$

$$\beta^2 - \beta + 2\beta - 2 = 0$$

$$\beta(\beta - 1) + 2(\beta - 1) = 0$$

$$(\beta - 1)(\beta + 2) = 0$$
$$\beta = 1, \beta = -2$$
$$\alpha - \beta = 1$$
$$\alpha = 2, \alpha = -1$$

Now
$$\alpha + \beta = -k$$
$$\alpha = 2, \beta = 1$$
$$-k = 2 + 1$$
$$k = -3$$

And
$$\alpha = -1, \beta = -2$$
$$-k = -1 + (-2)$$
$$k = 3$$

Hence, the value of k is± 3

13. Correct option: (C)

Given that the product of roots of equation $x^2 - 3kx + 2e^{2\log k} - 1 = 0$ is 7

Product of roots is given by

$$\alpha\beta = \frac{c}{a} = \frac{2e^{2\,log\,k} - 1}{1}$$
$$2e^{2\,log\,k} - 1 = 7$$
$$2e^{2\log k} = 8$$
$$e^{2\log k} = 4$$
$$e^{log\,k^2} = 4$$

Taking logarithm on both sides

$$\log k^2 = \log 4$$
$$k^2 = 4$$
$$k = \pm 2$$

Since input value of logarithm cannot be negative, so

$$k = 2$$

Sum of roots of equation $x^2 - 3kx + 2e^{2\log k} - 1 = 0$ is

$$\alpha + \beta = \frac{-(-3k)}{1}$$
$$\alpha + \beta = 3(2)$$
$$\alpha + \beta = 6$$

14. Correct option: (C)

Roots of equation are given

They are $(2\sqrt{3} - 5)\ and (-2\sqrt{3} - 5)$

Sum of roots is
$$\alpha + \beta = 2\sqrt{3} - 5 + (-2\sqrt{3} - 5)$$
$$\alpha + \beta = -10$$

Product of roots is
$$\alpha\beta = (\mathbf{2}\sqrt{3} - 5)(-2\sqrt{3} - 5)$$
$$\alpha\beta = -12 - 10\sqrt{3} + 10\sqrt{3} + 25$$
$$\alpha\beta = 13$$

To find equation of given roots, put value in given form

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$
$$x^2 - (-10)x + 13 = 0$$
$$x^2 + 10x + 13 = 0$$

Hence, the equation is $x^2 + 10x + 13 = 0$

15. Correct option: (B)

Let

$$x = 2 - \frac{1}{2 - \frac{1}{2 - \ldots\ldots}}$$

$$x = 2 - \frac{1}{x}$$

$$x^2 = 2x - 1$$

$$x^2 - 2x + 1 = 0$$

$$(x-1)^2 = 0$$

$$x = 1$$

Hence, the value of expression $x = 2 - \frac{1}{2 - \frac{1}{2 - \ldots\ldots}}$ is 1

16. Correct option: (A)

If roots of quadratic equation are equal then its discriminant is equal to zero.

$$D = 0$$

$$b^2 - 4ac = 0$$

In equation $x^2 - 2(5 + 2k)x + 3(7 + 10k) = 0$, a =1, b = -2(5+2k) and c=3(7+10k)

Now

$$[-2(5 + 2k)]^2 - 4(1)[3(7 + 10k)] = 0$$

$$4(4k^2 + 25 + 20k) - 12(7 + 10k) = 0$$

$$16k^2 + 100 + 80k - 84 - 120k = 0$$

$$16k^2 - 40k + 16 = 0$$

$$2k^2 - 5k + 2 = 0$$

$$2k^2 - 4k - k + 2 = 0$$

$$2k(k-2) - 1(k-2) = 0$$

$$(k-2)(2k-1) = 0$$

$$k = 2, k = \frac{1}{2}$$

Hence, value of k is $1, 1/2$.

17. Correct option: (C)

To find value of x, put $x^{\frac{1}{3}}$ as p in equation $x^{\frac{2}{3}} + x^{\frac{1}{3}} - 2 = 0$

$$p^2 + p - 2 = 0$$

$$p^2 + 2p - p - 2 = 0$$

$$p(p+2) - 1(p+2) = 0$$

$$(p+2)(p-1) = 0$$

$$p = 1, p = -2$$

Now if

p = 1

$$x^{\frac{1}{3}} = 1$$

$$x = 1$$

And if p = 8

$$x^{\frac{1}{3}} = -2$$

$$x = -8$$

Hence, the solution for equation $x^{\frac{2}{3}} + x^{\frac{1}{3}} - 2 = 0$ are 1 and -8.

18. Correct option: (A)

Given

$$2^x + 27(2)^{-x} = 12$$
$$2^x + \frac{27}{2^x} = 12$$
$$2^{2x} - 12(2^x) + 27 = 0$$

Put $2^x = p$

$$p^2 - 12p + 27 = 0$$
$$p^2 - 3p - 9p + 27 = 0$$
$$p(p-3) - 9(p-3) = 0$$
$$(p-3)(p-9) = 0$$
$$p = 3, p = 9$$

Now if

p = 3

$$2^x = p$$
$$2^x = 3$$
$$x = \log_2 3$$

And if

p = 9

$$2^x = p$$
$$2^x = 9$$
$$x = \log_2 9$$

Hence, value of x is $\log_2 3$ and $\log_2 9$.

19. Correct option: (D)

Let the two consecutive terms be x and x+1

Given that reciprocal of terms is equal to $\frac{23}{132}$, which are given by

$$\frac{1}{x} + \frac{1}{x+1} = \frac{23}{132}$$

Simplify above equation

$$\frac{x+1+x}{x(x+1)} = \frac{23}{132}$$
$$\frac{2x+1}{x^2+x} = \frac{23}{132}$$
$$264x + 132 = 23x^2 + 23x$$
$$23x^2 - 241x - 132 = 0$$
$$23x^2 - 253x + 12x - 132 = 0$$
$$23x(x-11) + 12(x-11) = 0$$
$$(23x+12)(x-11) = 0$$
$$x = \frac{-12}{23}, x = 11$$

Since number cannot be negative, so taking positive value of x which is 11.

Another number is x+1=11+1=12

Hence the two consecutive numbers are 11 and 12.

20. Correct option: (B)

Given

Sum of roots

$$\alpha + \beta = 12$$

And sum of cubes of roots

$$\alpha^3 + \beta^3 = 468$$

To find the equation of roots, consider identity

$$(\alpha+\beta)^3 = \alpha^3+\beta^3+3\alpha\beta(\alpha+\beta)$$
$$(12)^3 = 468+3\alpha\beta(12)$$
$$(12)^2 = 39+3\alpha\beta$$
$$3\alpha\beta = 144-39$$
$$3\alpha\beta = 105$$
$$\alpha\beta = 35$$

To find equation of given roots, put value in given form

$$x^2-(\alpha+\beta)x+\alpha\beta = 0$$
$$x^2-(12)x+35 = 0$$

Thus, $x^2-(12)x+35=0$ is required equation.

LEVEL – II

1. Correct option: (B)

For an equation to be quadratic, the standard form is given by

$$ax^2+bx+c=0$$

where a, b and c are real numbers.

Also, the degree of quadratic equation is 2.

Consider

$$\mathrm{x}^2+\frac{1}{\mathrm{x}}=0$$
$$x^3+1=0$$

Since, degree is 3

Hence it is not a quadratic equation.

Again

$$x^2+3=0$$

Degree is 2

Hence, it is quadratic equation

Also

$$\mathrm{x}^2-\mathrm{x}^{\frac{2}{3}}+1=0$$

Comparing with the standard form

$$ax^2+bx+c=0$$

Value of b cannot be determined.

Hence, equation is not quadratic.

Now, consider

$$\mathrm{x}+\frac{1}{\mathrm{x}}=0$$
$$x^2+1=0$$

Degree is 2

Hence, it is quadratic equation

2. Correct option: (B)

To find the roots of equation $3a^2x^2+8abx+4b^2=0$

Use quadratic formula

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

Comparing equation $3a^2x^2+8abx+4b^2=0$, value of $a=3a^2$, $b=8ab$ and $c=4b^2$

Putting value in formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$x = \frac{-(8ab) \pm \sqrt{(8ab)^2 - 4(3a^2)4(b^2)}}{2(3a^2)}$$
$$x = \frac{-8ab \pm \sqrt{64a^2b^2 - 48a^2b^2}}{6a^2}$$
$$x = \frac{-8ab \pm \sqrt{16a^2b^2}}{6a^2}$$
$$x = \frac{-8ab \pm 4ab}{6a^2}$$

So

$$x = \frac{-8ab + 4ab}{6a^2}$$
$$x = \frac{-4ab}{6a^2}$$
$$x = \frac{-2b}{3a}$$

Also

$$x = \frac{-8ab - 4ab}{6a^2}$$
$$x = \frac{-12ab}{6a^2}$$
$$x = \frac{-2b}{a}$$

Hence, the value of x is $\frac{-2b}{3a}$

3. Correct option: (D)

To find the roots of equation $25x^2 + 30x + 7 = 0$

Use quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Comparing equation $25x^2 + 30x + 7 = 0$, value of a =25, b =30 and c =7

Putting value in formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$x = \frac{-(30) \pm \sqrt{(30)^2 - 4(25)(7)}}{2(25)}$$
$$x = \frac{-(30) \pm \sqrt{900 - 700}}{50}$$
$$x = \frac{-(30) \pm \sqrt{200}}{50}$$
$$x = \frac{-(30) \pm 10\sqrt{2}}{50}$$
$$x = \frac{-3 \pm \sqrt{2}}{5}$$

Hence, value of x is $\frac{-3-\sqrt{2}}{5}$.

4. Correct option: (A)

Consider equation

$$2^{2x+3} - 57 = 65(2^x - 1)$$

Simplify $2^{2x}(2^3) - 57 = 65(2^x - 1)$

$2^{2x}(8) - 57 = 65(2^x - 1$

To find the value of x in given equation $2^{2x}(8) - 57 = 65(2^x - 1)$, put value of 2^x as p. Then

$$8p^2 - 57 = 65(p - 1)$$
$$8p^2 - 57 = 65p - 65$$
$$8p^2 - 65p - 57 + 65 = 0$$
$$8p^2 - 65p + 8 = 0$$
$$8p^2 - 64p - p + 8 = 0$$
$$8p(p - 8) - 1(p - 8) = 0$$
$$(p - 8)(8p - 1) = 0$$
$$p = 8, p = \frac{1}{8}$$

Now if

p=8

$$2^x = 8$$
$$2^x = 2^3$$
$$x = 3$$

Hence, value of x is 3.

5. Correct option: (B)

Given $\alpha + \beta = 2$

$\alpha\beta = 3$

Using identity $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$

Then

$$(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$$
$$(2)^3 = \alpha^{\mathbf{3}} + \beta^{\mathbf{3}} + 3(3)(2)$$
$$8 = \alpha^{\mathbf{3}} + \beta^{\mathbf{3}} + 18$$
$$\alpha^3 + \beta^3 = -10$$

Hence, value of $\alpha^3 + \beta^3$ is -10.

6. Correct option: (B)

Given, roots of equation are $12x^2 + mx + 5 = 0$ are in ratio of 5:4

So $\frac{\alpha}{\beta} = \frac{5}{4}$

Also sum and product of equation $12x^2 + mx + 5 = 0$ is

$$\alpha + \beta = \frac{-m}{12}$$
$$\alpha\beta = \frac{5}{12}$$

To get value of m, solve above equation

$$\frac{\alpha}{\beta} = \frac{5}{4}$$
$$\alpha = \frac{5\beta}{4}$$
$$\alpha\beta = \frac{5}{12}$$
$$\frac{5\beta}{4}\beta = \frac{5}{12}$$
$$\beta^2 = \frac{1}{3}$$
$$\beta = \pm\frac{1}{\sqrt{3}}$$

Now

$$\alpha + \beta = \frac{-m}{12}$$
$$\frac{5\beta}{4} + \beta = \frac{-m}{12}$$
$$\frac{9\beta}{4} = \frac{-m}{12}$$
$$9\beta = \frac{-m}{3}$$
$$27\beta = -m$$
$$m = \frac{-27}{\pm\sqrt{3}}$$
$$m = \pm 9\sqrt{3}$$

Hence, value of m is $\pm 9\sqrt{3}$.

7. Correct option: (C)

Given that one root is double another root of equation

$$x^2 - 10x + 3(7 + 10k) = 0$$

So

$$\alpha = a \; and \; \beta = 2a$$

Also sum and product of equation is

$$\alpha + \beta = \frac{-10}{1}$$
$$\alpha + \beta = -10$$

and

$$\alpha\beta = \frac{3(7 + 10k)}{1}$$
$$\alpha\beta = 21 + 30k$$

Now

$$\alpha + \beta = -10$$
$$a + 2a = -10$$
$$3a = -10$$
$$a = \frac{-10}{3}$$

and

$$\alpha\beta = 21 + 30k$$
$$(a)(2a) = 21 + 30k$$
$$2a^2 = 21 + 30k$$

Putting value of a

$$2\left(\frac{-10}{3}\right)^2 = 21 + 30k$$
$$\frac{200}{9} = 21 + 30k$$
$$200 = 189 + 270k$$
$$11 = 270k$$
$$k = \frac{11}{270}$$

The value of k is $\frac{11}{270}$.

8. Correct option: (D)

Given that perimeter of rectangle is 28 and area is 40

Formula for perimeter and area of rectangle is given by

$$2[length(l) + breadth(b)] = 28$$
$$lenght(l) \times breadth(b) = 40$$
$$l = \frac{40}{b}$$
$$2(l + b) = 28$$
$$2(\frac{40}{b} + b) = 28$$
$$\frac{40 + b^2}{b} = 14$$
$$40 + b^2 = 14b$$
$$b^2 - 14b + 40 = 0$$
$$b^2 - 10b - 4b + 40 = 0$$
$$b(b - 10) - 4(b - 10) = 0$$
$$(b - 10)(b - 4) = 0$$
$$b = 10, b = 4$$

When b = 10

$$l = \frac{40}{b}$$
$$l = \frac{40}{10}$$
$$l = 4$$

When b = 4

$$l = \frac{40}{b}$$
$$l = \frac{40}{4}$$
$$l = 10$$

9. Correct option: (D)

Given

The base of a triangle 4cm longer than its altitude

Height = h

Base, b = h+4

Also, the area of triangle is 48cm^2

$$A = \frac{1}{2} base \times height$$
$$A = \frac{1}{2} bh$$
$$A = \frac{1}{2}(h + 4)(h)$$
$$48 = \frac{1}{2}(h^2 + 4h)$$
$$h^2 + 4h - 96 = 0$$
$$h^2 + 12h - 8h - 96 = 0$$
$$h(h + 12) - 8(h + 12) = 0$$
$$(h + 12)(h - 8) = 0$$
$$h = -12, h = 8$$

Since, height cannot be negative. So $h = 8$

Now base is

$$b = h + 4$$
$$b = 8 + 4$$
$$b = 12$$

Hence, base of triangle is 12cm and height of triangle is 8cm.

10. Correct option: (D)

Given

α, β are roots of equation $x^2 + x + 2 = 0$

Sum and product of equation is

$$\alpha + \beta = \frac{-1}{1}$$
$$\alpha + \beta = -1$$

and

$$\alpha\beta = \frac{2}{1}$$
$$\alpha\beta = 2$$

Using identity $(a + b)^2 = a^2 + b^2 + 2ab$

$$(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$$
$$(-1)^2 = \alpha^2 + \beta^2 + 2(2)$$
$$\alpha^2 + \beta^2 = 1 - 4$$
$$\alpha^2 + \beta^2 = -3$$

Hence, $\alpha^2 + \beta^2 = -3$

11. Correct option: (A, C)

Given

$$6\left(x^2 + \frac{1}{x^2}\right) - 25\left(x - \frac{1}{x}\right) + 12 = 0$$

Put $\left(x - \frac{1}{x}\right)$ *as* y, in above equation

$$y = x - \frac{1}{x}$$

Squaring both sides

$$y^2 = \left(x - \frac{1}{x}\right)^2$$
$$y^2 = x^2 + \left(\frac{1}{x}\right)^2 - 2x\frac{1}{x}$$
$$y^2 = x^2 + \frac{1}{x^2} - 2$$
$$x^2 + \frac{1}{x^2} = y^2 + 2$$

Now

$$6\left(x^2 + \frac{1}{x^2}\right) - 25\left(x - \frac{1}{x}\right) + 12 = 0$$
$$6(y^2 + 2) - 25(y) + 12 = 0$$
$$6y^2 + 12 - 25y + 12 = 0$$
$$6y^2 - 25y + 24 = 0$$
$$6y^2 - 16y - 9y + 24 = 0$$
$$2y(3y - 8) - 3(3y - 8) = 0$$
$$(3y - 8)(2y - 3) = 0$$
$$y = \frac{8}{3}, y = \frac{3}{2}$$

When $y = \frac{8}{3}$

$$y = x - \frac{1}{x}$$
$$\frac{8}{3} = x - \frac{1}{x}$$
$$3x^2 - 3 = 8x$$

$$3x^2 - 8x - 3 = 0$$
$$3x^2 - 9x + x - 3 = 0$$
$$3x(x-3) + 1(x-3)$$
$$(x-3)(3x+1) = 0$$
$$x = 3, \qquad x = \frac{-1}{3}$$

and when $y = \frac{3}{2}$

$$y = x - \frac{1}{x}$$
$$\frac{3}{2} = x - \frac{1}{x}$$
$$2x^2 - 2 = 3x$$
$$2x^2 - 3x - 2 = 0$$
$$2x^2 - 4x + x - 2 = 0$$
$$2x(x-2) + 1(x-2) = 0$$
$$(x-2)(2x+1) = 0$$
$$x = 2, \qquad x = \frac{-1}{2}$$

Hence, value of x is 2 and 3.

12. Correct option: (A)

Consider

$$\frac{1}{k}(5x^2 - 4x + 38) + (-4x^2 - 2x - 1) = 0$$

Simplify equation

$$(5x^2 - 4x + 38) + (-4kx^2 - 2kx - k) = 0$$
$$(5x^2 - 4x + 38) - 4kx^2 - 2kx - k = 0$$
$$x^2(5 - 4k) - x(4 + 2k) + (38 - k) = 0$$

Sum and product of equation is

$$\alpha + \beta = \frac{-b}{a}$$
$$\alpha + \beta = \frac{-[-(4+2k)]}{5-4k}$$

and

$$\alpha\beta = \frac{c}{a}$$
$$\alpha\beta = \frac{38 - k}{5 - 4k}$$

Given that product of roots of equation is -5

Then

$$\alpha\beta = -5$$
$$\alpha\beta = \frac{38 - k}{5 - 4k}$$

Comparing equation

$$-5 = \frac{38 - k}{5 - 4k}$$
$$-25 + 20k = 38 - k$$
$$21k = 63$$
$$k = 3$$

Hence value of k is 3.

13. Correct option: (C)

Given that α, β are roots of equation $ax^2 + bx + c = 0$

So, sum and product of equation is

$$\alpha + \beta = \frac{-b}{a}$$

and

$$\alpha\beta = \frac{c}{a}$$

Now

$$\frac{\alpha^3 + \beta^3}{\alpha^{-3} + \beta^{-3}} = \frac{\alpha^3 + \beta^3}{\frac{1}{\alpha^3} + \frac{1}{\beta^3}}$$

$$\frac{\alpha^3 + \beta^3}{\alpha^{-3} + \beta^{-3}} = \frac{\alpha^3 + \beta^3}{\frac{\beta^3 + \alpha^3}{\alpha^3\beta^3}}$$

$$\frac{\alpha^3 + \beta^3}{\alpha^{-3} + \beta^{-3}} = \frac{\alpha^3\beta^3(\alpha^3 + \beta^3)}{\alpha^3 + \beta^3}$$

$$\frac{\alpha^3 + \beta^3}{\alpha^{-3} + \beta^{-3}} = \alpha^3\beta^3$$

$$\frac{\alpha^3 + \beta^3}{\alpha^{-3} + \beta^{-3}} = (\alpha\beta)^3$$

$$\frac{\alpha^3 + \beta^3}{\alpha^{-3} + \beta^{-3}} = \left(\frac{c}{a}\right)^3$$

Hence, value of $\frac{\alpha^3+\beta^3}{\alpha^{-3}+\beta^{-3}}$ is $\frac{c^3}{a^3}$.

14. Correct option: (A)

Given that α, β are roots of equation $ax^2 + bx + c = 0$

So, sum and product of equation is $\alpha + \beta = \frac{-b}{a}$

And $\alpha\beta = \frac{c}{a}$

Now $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3+\beta^3}{\alpha\beta}$

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)}{\alpha\beta}$$

Put value

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\left(\frac{-b}{a}\right)^3 - 3\left(\frac{c}{a}\right)\left(\frac{-b}{a}\right)}{\frac{c}{a}}$$

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\frac{-b^3}{a^3} + \frac{3bc}{a^2}}{\frac{c}{a}}$$

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\frac{-b^3 + 3bac}{a^3}}{\frac{c}{a}}$$

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{a(3abc - b^3)}{a^3c}$$

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{(3abc - b^3)}{a^2c}$$

Hence value of $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$ is $\frac{(3abc-b^3)}{a^2c}$.

15. Correct option: (B)

Given that α, β are roots of equation $ax^2 + bx + c = 0$

So, sum and product of equation is $\alpha + \beta = \frac{-b}{a}$

And $\alpha\beta = \frac{c}{a}$

Now

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\beta^2 + \alpha^2}{\alpha^2\beta^2}$$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha^2\beta^2}$$

Put value of roots

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\left(\frac{-b}{a}\right)^2 - 2\left(\frac{c}{a}\right)}{\left(\frac{c}{a}\right)^2}$$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\frac{b^2}{a^2} - 2\left(\frac{c}{a}\right)}{\frac{c^2}{a^2}}$$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\frac{b^2 - 2ac}{a^2}}{\frac{c^2}{a^2}}$$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{b^2 - 2ac}{c^2}$$

Hence, value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ is $\frac{b^2 - 2ac}{c^2}$.

16. Correct Matching is: (a-Q,b-R,c-P,d-S)

17. Correct Answer is: (1)

18. Correct Answer is: (9)

19. Correct Answer is: (3)

20. Correct Answer is: (2)

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

To find the roots of given equation $\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$, use splitting the middle term method.

In this split the middle term into product of first term and last term and sum of which gives the middle term itself.

So,

$$\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$$

$$\sqrt{3}x^2 + 4x + 6x - 8\sqrt{3} = 0$$

$$\sqrt{3}x^2 + 6x + 4x - 8\sqrt{3} = 0$$

$$\sqrt{3}x\left(x + 2\sqrt{3}\right) - 4\left(x + 2\sqrt{3}\right) = 0$$

$$\left(x + 2\sqrt{3}\right)\left(\sqrt{3}x - 4\right) = 0$$

$$x = -2\sqrt{3}, x = \frac{4}{\sqrt{3}}$$

Hence, the roots of equation $\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$ are $-2\sqrt{3}$ *and* $\frac{4}{\sqrt{3}}$.

2. Correct option: (B)

Given that one root of equation $\boldsymbol{x^2 - 5x + k = 0}$ is 2.

So put value of x as 2

$$x^2 - 5x + k = 0$$
$$(2)^2 - 5(2) + k = 0$$
$$4 - 10 + k = 0$$
$$k = 6$$

Then equation becomes

$$x^2 - 5x + 6 = 0$$

Split middle term

$$x^2 - 2x - 3x + 6 = 0$$
$$x(x - 2) - 3(x - 2) = 0$$
$$(x - 2)(x - 3) = 0$$
$$x = 2, x = 3$$

Hence the root of equation is 3.

3. Correct option: (A)

To find the value of x, simplify the given equation

$$\frac{3}{x+2} - \frac{2}{x-1} = 5$$
$$3(x - 1) - 2(x + 2) = 5(x + 2)(x - 1)$$
$$3x - 3 - 2x - 4 = 5(x^2 - x + 2x - 2)$$
$$x - 7 = 5(x^2 + x - 2)$$
$$x - 7 = 5x^2 + 5x - 10$$
$$5x^2 + 4x - 3 = 0$$

Apply quadratic formula to find the roots of following quadratic equation

$$5x^2 + 4x - 3 = 0$$

Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Here a=5, b=4, c=-3

$$x = \frac{-4 \pm \sqrt{(4)^2 - 4(5)(-3)}}{2(2)}$$
$$x = \frac{-4 \pm \sqrt{16 + 60}}{10}$$
$$x = \frac{-4 \pm \sqrt{76}}{10}$$
$$x = \frac{-4 \pm 2\sqrt{19}}{10}$$
$$x = \frac{-2 \pm \sqrt{19}}{5}$$

Hence, vale of x is $\frac{-2}{5} \pm \frac{\sqrt{19}}{5}$.

4. Correct option: (A)

Roots of equation are given

They are 3 and 5

Sum of roots is $\alpha + \beta = 3 + 5$

Product of roots is $\alpha\beta = 3 \times 5$

$$\alpha\beta = 15$$

To find equation of given roots, put value in given form

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$
$$x^2 - 8x + 15 = 0$$

Hence, the required equation is $\boldsymbol{x^2 - 8x + 15 = 0}$.

5. Correct option: (D)

Given that α *and* β are roots of equation $ax^2 + bx + c = 0$

So put value of x as α

$$a\alpha^2 + b\alpha + c = 0$$
$$\alpha(a\alpha + b) = -c$$
$$a\alpha + b = \frac{-c}{\alpha}$$

Also put value of x as β

$$a\beta^2 + b\beta + c = 0$$
$$\beta(a\beta + b) = -c$$
$$a\beta + b = \frac{-c}{\beta}$$

Now

$$(a\alpha + b)^{-3} + (a\beta + b)^{-3} = \left(\frac{-c}{\alpha}\right)^{-3} + \left(\frac{-c}{\beta}\right)^{3}$$
$$(a\alpha + b)^{-3} + (a\beta + b)^{-3} = \frac{\alpha^3 + \beta^3}{c^3}$$

Also, by equation $ax^2 + bx + c = 0$ sum and product are given by

$$\alpha + \beta = \frac{-b}{a} \text{ and } \alpha\beta = \frac{c}{a}$$

Using identity of $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$

$$(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$$
$$\left(\frac{-b}{a}\right)^3 = \alpha^3 + \beta^3 + 3\left(\frac{-b}{a}\right)\left(\frac{c}{a}\right)$$
$$\alpha^3 + \beta^3 = \frac{-b^3}{a^3} + \frac{3bc}{a^2}$$
$$\boldsymbol{\alpha^3 + \beta^3 = \frac{-b^3 + 3abc}{a^3}}$$

Put the value of $\alpha^3 + \beta^3$ in expression

$$(a\alpha + b)^{-3} + (a\beta + b)^{-3} = \frac{\alpha^3 + \beta^3}{c^3}$$
$$(a\alpha + b)^{-3} + (a\beta + b)^{-3} = \frac{\frac{-b^3 + 3abc}{a^3}}{c^3}$$
$$(a\alpha + b)^{-3} + (a\beta + b)^{-3} = \frac{-b^3 + 3abc}{a^3c^3}$$

Hence, the value of $(a\alpha + b)^{-3} + (a\beta + b)^{-3}$ is $\frac{-b^3+3abc}{a^3c^3}$

6. Correct option: (A)

Given that α *and* β are roots of equation $ax^2 + bx + c = 0$

So put value of x as α

$$a\alpha^2 + b\alpha + c = 0$$
$$\alpha(a\alpha + b) = -c$$
$$a\alpha + b = \frac{-c}{\alpha}$$

Also put value of x as β

$$a\beta^2 + b\beta + c = 0$$
$$\beta(a\beta + b) = -c$$

$$a\beta + b = \frac{-c}{\beta}$$

Now

$$\left(\frac{\alpha}{a\beta+b}\right)^3 - \left(\frac{\beta}{a\alpha+b}\right)^3 = \left(\frac{\alpha}{\frac{-c}{\beta}}\right)^3 - \left(\frac{\beta}{\frac{-c}{\alpha}}\right)^3$$

$$\left(\frac{\alpha}{a\beta+b}\right)^3 - \left(\frac{\beta}{a\alpha+b}\right)^3 = \frac{-\alpha^3\beta^3}{c^3} - \frac{(-\alpha^3\beta^3)}{c^3}$$

$$\left(\frac{\alpha}{a\beta+b}\right)^3 - \left(\frac{\beta}{a\alpha+b}\right)^3 = \frac{-\alpha^3\beta^3}{c^3} + \frac{(\alpha^3\beta^3)}{c^3}$$

$$\left(\frac{\alpha}{a\beta+b}\right)^3 - \left(\frac{\beta}{a\alpha+b}\right)^3 = 0$$

Hence, the value of $\left(\frac{\alpha}{a\beta+b}\right)^3 - \left(\frac{\beta}{a\alpha+b}\right)^3$ is 0.

7. Correct option: (C)

If roots of quadratic equation are equal then its discriminant is equal to zero.

$$D = 0$$

$$b^2 - 4ac = 0$$

Here, in equation $9x^2 + kx + 1 = 0$, a = 9, b = k and c =1

$$(k)^2 - 4(9)(1) = 0$$

$$k^2 - 36 = 0$$

$$k^2 = 36$$

$$k = \pm 6$$

Hence, the value of k is ± 6.

8. Correct option: (C)

9. Correct option: (B)

Given

$$7^{1+x} + 7^{1-x} = 50$$

$$\mathbf{7.7^x + 7.7^{-x} = 50}$$

$$\mathbf{7.7^x + \frac{7}{7^x} = 50}$$

$$\mathbf{7.7^{2x} - 50.7^x + 7 = 0}$$

Put $7^x = p$

$$7p^2 - 50p + 7 = 0$$

$$7p^2 - 49p - p + 7 = 0$$

$$7p(p-7) - 1(p-7) = 0$$

$$(p-7)(7p-1) = 0$$

$$p = 7, p = \frac{1}{7}$$

Now for p = 7

$$7^x = 7^1$$

$$x = 1$$

And p = 1/7

$$7^x = \frac{1}{7}$$

$$7^x = 7^{-1}$$

$$x = -1$$

Hence, value of x is ± 1.

10. Correct option: (A)

Given that α, β are roots of equation $ax^2 + bx + c = 0$

So, sum and product of equation is

$$\alpha + \beta = \frac{-b}{a}$$

and

$$\alpha\beta = \frac{c}{a}$$

Now

$$\sqrt{\left(\frac{-b}{a}\right)^2 - \frac{4c}{a}} = \sqrt{(\alpha+\beta)^2 - 4(\alpha\beta)}$$

$$\sqrt{\left(\frac{-b}{a}\right)^2 - \frac{4c}{a}} = \sqrt{(\alpha)^2 + \beta^2 + 2\alpha\beta - 4(\alpha\beta)}$$

$$\sqrt{\left(\frac{-b}{a}\right)^2 - \frac{4c}{a}} = \sqrt{(\alpha)^2 + \beta^2 - 2(\alpha\beta)}$$

$$\sqrt{\left(\frac{-b}{a}\right)^2 - \frac{4c}{a}} = \sqrt{(\alpha-\beta)^2}$$

$$\sqrt{\left(\frac{-b}{a}\right)^2 - \frac{4c}{a}} = |\alpha - \beta|$$

11. Correct option: (C)

Given

$\alpha\beta$ are roots of $x^2 + x + 1 = 0$

So, sum and product of equation is

$$\alpha + \beta = \frac{-1}{1}$$

$$\alpha + \beta = -1$$

And

$$\alpha\beta = \frac{1}{1}$$

$$\alpha\beta = 1$$

Using identity $(a+b)^2 = a^2 + b^2 + 2ab$

$$(\alpha+\beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$$

$$(-1)^2 = \alpha^2 + \beta^2 + 2(1)$$

$$\alpha^2 + \beta^2 = 1 - 2$$

$$\alpha^2 + \beta^2 = -\mathbf{1}$$

Consider

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 1$$

Take LHS

$$\frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{-1}{1}$$

$$\frac{\alpha^2 + \beta^2}{\alpha\beta} = -1$$

Hence, $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = -1$

12. Correct option: (B)

Given that α, β are roots of equation $x^2 + 7x + 12 = 0$

So, sum and product of equation is

$$\alpha + \beta = \frac{-7}{1}$$
$$\alpha + \beta = -7$$

and

$$\alpha\beta = \frac{12}{1}$$
$$\alpha\beta = 12$$

Now

$$(\alpha + \beta)^2 = (-7)^2$$
$$(\alpha + \beta)^2 = 49$$

and

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$
$$(\alpha - \beta)^2 = 49 - 4(12)$$
$$(\alpha - \beta)^2 = 49 - 48$$
$$(\alpha - \beta)^2 = 1$$

Equation having roots $(\alpha + \beta)^2$ and $(\alpha - \beta)^2$ is given by

$$\boldsymbol{x^2 - \left[(\alpha + \beta)^2 + (\alpha - \beta)^2\right]x + (\alpha - \beta)^2(\alpha + \beta)^2 = 0}$$
$$x^2 - [49 + 1]x + (49)(1) = 0$$
$$x^2 - 50x + 49 = 0$$

Hence, $x^2 - 50x + 49 = 0$ is required equation.

13. Correct option: (A)

Given

$$3p^2 = 5p + 2$$

and

$$3q^2 = 5q + 2$$

So

$$3p^2 = 5p + 2$$
$$3p^2 - 5p - 2 = 0$$
$$3p^2 - 6p + p - 2 = 0$$
$$3p(p - 2) + 1(p - 1) = 0$$
$$(p - 2)(3p + 1) = 0$$
$$p = 2, p = \frac{-1}{3}$$

Similarly

$$3q^2 = 5q + 2$$
$$3q^2 - 5q - 2 = 0$$
$$3q^2 - 6q + q - 2 = 0$$
$$3q(q - 2) + 1(q - 1) = 0$$
$$(q - 2)(3q + 1) = 0$$
$$q = 2, q = \frac{-1}{3}$$

The roots are $3p - 2q$ and $3q - 2p$

Now

$$\alpha = 3p - 2q$$
$$\alpha = 3(2) - 2\left(\frac{-1}{3}\right)$$

$$\alpha = 6 + \frac{2}{3}$$
$$\alpha = \frac{20}{3}$$

and

$$\beta = 3q - 2p$$
$$\beta = 3\left(\frac{-1}{3}\right) - 2(2)$$
$$\beta = -1 - 4$$
$$\beta = -5$$

So

$$\alpha + \beta = \frac{20}{3} + (-5)$$
$$\alpha + \beta = \frac{5}{3}$$

And

$$\alpha\beta = \left(\frac{20}{3}\right)(-5)$$
$$\alpha\beta = \frac{-100}{3}$$

Standard form of quadratic equation is

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$
$$x^2 - \frac{5}{3}x + \left(\frac{-100}{3}\right) = 0$$
$$3x^2 - 5x - 100 = 0$$

Hence, $3x^2 - 5x - 100 = 0$ is required equation.

14. Correct option: (B)

Given that α, β are roots of equation $ax^2 + bx + c = 0$

So, sum and product of equation is

$$\alpha + \beta = \frac{-b}{a}$$

and

$$\alpha\beta = \frac{c}{a}$$

Standard form of quadratic equation is

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$
$$x^2 - \left[\left(\frac{-b}{a}\right) + \frac{c}{a}\right]x + \left[\left(\frac{-b}{a}\right)\left(\frac{c}{a}\right)\right] = 0$$
$$x^2 - \frac{c-b}{a}x - \frac{bc}{a^2} = 0$$
$$a^2x^2 - a(c-b)x - bc = 0$$
$$a^2x^2 + a(b-c)x - bc = 0$$

Hence, $a^2x^2 + a(b-c)x - bc = 0$ is required equation.

15. Correct Matching is: (a-P,b-T,c_Q,d-P)

16. Correct Answer is: (1)

17. Correct Answer is: (3)

5. Progressions

Solutions

LEVEL-I

1. Correct option: (B) 17

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$a_n = a + (n-1)d$

where a is the first term of the AP and d is the common difference.

Given, an AP has terms 20, 25, 30, … ,100

We see, the first term of the AP is $a = 20$ and the common difference is

$d = 25 - 20$

$d = 5$

Now, the number of terms in the AP is:

$a_n = a + (n-1)d$

$100 = 20 + (n-1)5$

$80 = (n-1)5$

$16 = n - 1$

$n = 17$

Therefore, if the terms of the AP are 20, 25, 30, … ,100, then the number of terms in the AP is 17.

2. Correct option: (A) -80

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$a_n = a + (n-1)d$

where a is the first term of the AP and d is the common difference.

Given, the 11th term of an AP is 80 and the 16th term is 40.

Let's first find the first term and the common difference of the AP.

Using the fact that 11th term of an AP is 80, we get:

$80 = a + (11-1)d$

$80 = a + 10d$

Also, since 16th term is 40, we also have:

$40 = a + (16-1)d$

$40 = a + 15d$

Now, subtracting these two equations, we get:

$80 - 40 = a + 10d - (a + 15d)$

$40 = -5d$

$d = -8$

Substituting $d = -8$ into $80 = a + 10d$, we get:

$80 = a + 10(-8)$

$80 = a - 80$

$a = 160$

Hence, the first term of the AP is 160 and the common difference is -8.

Now, the 31st term of the AP is:

$a_n = a + (n-1)d$

$a_{31} = 160 + (31-1)(-8)$

$a_{31} = 160 - 240$

$a_{31} = -80$

Therefore, if the 11th term of an AP is 80 and the 16th term is 40, then the 31st term of the AP is -80.

3. Correct option: (C) -1

Solution: Given, $a_1 = a_2 = 2, a_n = a_{n-1} - 1 (n > 2)$

Now, the next term in the sequence is :

$a_3 = a_2 - 1$

$a_3 = 2 - 1$

$a_3 = 1$

Similarly, the next term in the sequence is :

$a_4 = a_3 - 1$

$a_4 = 1 - 1$

$a_4 = 0$

Similarly, the next term in the sequence is :

$a_5 = a_4 - 1$

$a_5 = 0 - 1$

$a_5 = -1$

Therefore, if $a_1 = a_2 = 2, a_n = a_{n-1} - 1 (n > 2)$, then $a_5 = -1$.

4. Correct option: (D) 0

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$a_n = a + (n-1)d$

where a is the first term of the AP and d is the common difference.

Given, in a certain A.P., 5 times the 5th term is equal to 8 times the 8th term.

Let's first find a relation between the first term and the common difference of the AP.

Using the fact that 5 times the 5th term is equal to 8 times the 8th term, we get:

$5a_5 = 8a_8$

$5(a + 4d) = 8(a + 7d)$

$-3a = 36d$

$a = -12d$

Now, the 13rd term of the AP is:

$a_n = a + (n-1)d$

$a_{13} = -12d + 12d$

$a_{13} = 0$

Therefore, if 5 times the 5th term is equal to 8 times the 8th term, then the 13rd term of the AP is 0.

5. Correct option: (B) $\frac{-3}{b^2} + \frac{4}{ac}$

Solution: We know, when three numbers a, b, c are in AP, then they satisfy:

$2b = a + c$

Given, $\frac{1}{a},\frac{1}{b},\frac{1}{c}$ are in AP.

Hence, we have, $\frac{2}{b}=\frac{1}{a}+\frac{1}{c}$.

Now, $\left(\frac{1}{a}+\frac{1}{b}-\frac{1}{c}\right)\left(\frac{1}{b}+\frac{1}{c}-\frac{1}{a}\right)$ can be simplified as:

$$\left(\frac{1}{a}+\frac{1}{b}-\frac{1}{c}\right)\left(\frac{1}{b}+\frac{1}{c}-\frac{1}{a}\right)=\frac{1}{b^2}-\left(\frac{1}{a}-\frac{1}{c}\right)^2$$

$$\left(\frac{1}{a}+\frac{1}{b}-\frac{1}{c}\right)\left(\frac{1}{b}+\frac{1}{c}-\frac{1}{a}\right)=\frac{1}{b^2}-\left(\frac{1}{a^2}+\frac{1}{c^2}-\frac{2}{ac}\right)$$

$$\left(\frac{1}{a}+\frac{1}{b}-\frac{1}{c}\right)\left(\frac{1}{b}+\frac{1}{c}-\frac{1}{a}\right)=\frac{1}{b^2}-\left(\frac{4}{b^2}-\frac{2}{ac}\right)+\frac{2}{ac}$$

$$\left(\frac{1}{a}+\frac{1}{b}-\frac{1}{c}\right)\left(\frac{1}{b}+\frac{1}{c}-\frac{1}{a}\right)=\frac{-3}{b^2}+\frac{4}{ac}$$

Therefore, if $\frac{1}{a},\frac{1}{b},\frac{1}{c}$ are in AP, then $\left(\frac{1}{a}+\frac{1}{b}-\frac{1}{c}\right)\left(\frac{1}{b}+\frac{1}{c}-\frac{1}{a}\right)=\frac{-3}{b^2}+\frac{4}{ac}$.

6. Correct option: (C)16

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$a_n = a + (n-1)d$

where a is the first term of the AP and d is the common difference.

Now, sequence of two-digit numbers divisible by 6 is: 6, 12, 18, …, 96

We see this form an AP with first term 6 and common difference 6.

Therefore, the number of two-digit numbers divisible by 6 are:

$a_n = a+(n-1)d$

$96 = 6+(n-1)6$

$96 = 6n$

$n = 16$

Therefore, the number of two-digit numbers divisible by 6 are 16.

7. Correct option: (C) $\frac{(c-2a+b)}{2(b-a)}(a+c)$

Solution: We know, the sum of first n terms of an arithmetic progression (AP) is given by:

$S_n = \frac{n}{2}(a+l)$

where l is the last term of the AP.

We also know, the n th term in an Arithmetic Progression (AP) is given by:

$a_n = a+(n-1)d$

where a is the first term of the AP and d is the common difference.

Given, the first, second and last term of an A.P. are a,b,c respectively.

Hence, the common difference is $d = b-a$.

Now, the number of terms is the AP is:

$c = a+(n-1)(b-a)$

$c = a+n(b-a)-b+a$

$n = \frac{c-2a+b}{b-a}$

Therefore, the sum of the AP is:

$$S_n = \frac{n}{2}(a+l)$$

$$S_n = \frac{(c-2a+b)}{2(b-a)}(a+c)$$

8. Correct option: (C) $\frac{n(n+1)}{\sqrt{2}}$

Solution: We know, the sum of first n natural number is:

$$S_n = \frac{n}{2}(n+1)$$

Given series:

$\sqrt{2}+\sqrt{8}+\sqrt{18}+\sqrt{32}+...$

We can rewrite the given series as:

$\sqrt{2}(1+2+3+4+...)$

Therefore, sum of first n terms of the series is:

$$\sqrt{2}(1+2+3+4+...) = \sqrt{2}\frac{n}{2}(n+1)$$

$$\sqrt{2}(1+2+3+4+...) = \frac{n(n+1)}{\sqrt{2}}$$

9. Correct option: (C) 38

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$$a_n = a+(n-1)d$$

where a is the first term of the AP and d is the common difference.

Now, list of numbers lying between 100 and 500 that are divisible by 7 is:

$105,112,...,497$

We see this form an AP with first term 105 and common difference 7.

Therefore, number of numbers lying between 100 and 500 that are divisible by 7 is:

$a_n = a+(n-1)d$

$497 = 105+(n-1)7$

$399 = 7n$

$n = 57$

Now, list of numbers lying between 100 and 500 that are divisible by 21 is:

$105,126,...,483$

We see this form an AP with first term 105 and common difference 21.

Therefore, number of numbers lying between 100 and 500 that are divisible by 21 is:

$a_n = a+(n-1)d$

$483 = 105+(n-1)21$

$399 = 21n$

$n = 19$

Therefore, the number of numbers lying between 100 and 500 that are divisible by 7 but not by 21 is $57-19=38$.

10. Correct option: (B) 1

Solution: We know, the arithmetic mean AM between a and b is given by:

$$\text{AM} = \frac{a+b}{2}$$

Given, $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$ is the AM between a and b.

Hence, we have:

$$\frac{a^n + b^n}{a^{n-1} + b^{n-1}} = \frac{a+b}{2}$$

$$\frac{a^1 + b^1}{a^{1-1} + b^{1-1}} = \frac{a+b}{2}$$

Therefore, the value of n is 1.

11. Correct option: (A) $\frac{n(n+1)(n+2)(n+3)}{4}$

Solution: We know, the sum of first n natural number is $\frac{n(n+1)}{2}$

Also, the sum of square of first n natural number is $\frac{n(n+1)(2n+1)}{6}$

And the sum of cube of first n natural numbers is $\left[\frac{n(n+1)}{2}\right]^2$

Given series: $1.2.3 + 2.3.4 + ...$

Now, the general term of the series is:

$$a_n = n(n+1)(n+2)$$

$$a_n = n(n^2 + 3n + 2)$$

$$a_n = n^3 + 3n^2 + 2n$$

Now, the sum of the series is:

$$S_n = \left[\frac{n(n+1)}{2}\right]^2 + 3\frac{n(n+1)(2n+1)}{6} + 2\frac{n(n+1)}{2}$$

$$S_n = \frac{n(n+1)}{2}\left[\frac{n(n+1)}{2} + \frac{3(2n+1)}{3} + 2\right]$$

$$S_n = \frac{n(n+1)}{2}\left[\frac{n(n+1)}{2} + (2n+3)\right]$$

$$S_n = \frac{n(n+1)}{2}\left[\frac{n^2 + 5n + 6}{2}\right]$$

$$S_n = \frac{n(n+1)(n+2)(n+3)}{4}$$

Therefore, the sum of the series $1.2.3 + 2.3.4 + ...$ is $\frac{n(n+1)(n+2)(n+3)}{4}$

12. Correct option: (D) 13 terms

Solution: We know, the n th term of the Geometric Progression (GP) is:

$$a_n = ar^{n-1}$$

where a is the first term of the GP and r is the common ration of the GP.

Given, a GP with terms: $2, 2\sqrt{2}, 4, ..., 128$

We see the first term of the GP is 2 and the common ratio is $\sqrt{2}$.
Therefore, the number of terms in the GP is:

$a_n = ar^{n-1}$

$128 = 2(\sqrt{2})^{n-1}$

$2^7 = 2^{\frac{n+1}{2}}$

$7 = \frac{n+1}{2}$

$n = 13$

Therefore, the number of terms in the GP $2, 2\sqrt{2}, 4, ..., 128$ is 13.

13. Correct option: (C) $\sqrt{pq}$

Solution: We know, the n th term of the Geometric Progression (GP) is: $a_n = ar^{n-1}$
where a is the first term of the GP and r is the common ration of the GP.
Given, in a G.P., if $(m+n)^{th}$ term is p and $(m-n)^{th}$ term is q.
Using the fact that $(m+n)^{th}$ term of GP is p, we get: $p = ar^{m+n-1}$
Similarly, using the fact that $(m-n)^{th}$ term of GP is q, we have: $q = ar^{m-n-1}$
Multiplying these two equations, we get:

$pq = a^2 r^{2m-2}$

$pq = (ar^{m-1})^2$

$pq = (a_m)^2$

$a_m = \sqrt{pq}$

Therefore, if in a G.P., if $(m+n)^{th}$ term is p and $(m-n)^{th}$ term is q, then its m th term is $\sqrt{pq}$.

14. Correct option: (B) GP

Solution: We know, the n th term of the Geometric Progression (GP) is: $a_n = ar^{n-1}$
where a is the first term of the GP and r is the common ration of the GP.
We also know, the n th term in an Arithmetic Progression (AP) is given by: $a_n = a + (n-1)d$
where a is the first term of the AP and d is the common difference.
Given, x, y, z are in A.P.
Hence,

$y = x + d$

$z = x + 2d$

where d is the common difference.
Now,

$e^{-y} = e^{-x-d} = e^{-x}e^{-d}$

$e^{-z} = e^{-x-2d} = e^{-x}(e^{-d})^2$

This is a GP with first term e^{-x} and common ratio e^{-d}.
Therefore, if x, y, z are in A.P, then e^{-x}, e^{-y}, e^{-z} are in GP.

15. Correct option: (B) ± 3

Solution: We know, the n th term of the Geometric Progression (GP) is: $a_n = ar^{n-1}$

where a is the first term of the GP and r is the common ration of the GP.

Given, the first term of a G.P. is 1 and the sum of the third and fifth terms is 90.

Therefore,

$a_3 + a_5 = ar^2 + ar^4$

$90 = r^2 + r^4$

Factorising this:

$(r^2 + 10)(r^2 - 9) = 0$

$r = 3, -3$

In last step we restrict to real common ratio.

Therefore, if the first term of a G.P. is 1 and the sum of the third and fifth terms is 90, the common ratio is ± 3

16. Correct option: (D) 6

Solution: We know, the n th term of the Geometric Progression (GP) is:

$a_n = ar^{n-1}$

where a is the first term of the GP and r is the common ration of the GP.

Also, the sum of first n terms of the GP is:

$$S_n = a\frac{r^n - 1}{r - 1}$$

where a is the first term of the GP and r is the common ration of the GP.

Given, in a G.P $a_1 = 3, a_n = 96, S_n = 189$.

Now, using $a_1 = 3$, we have

$a_1 = a$

$3 = a$

Also, using $a_n = 96$, we get

$a_n = ar^{n-1}$

$96 = 3r^{n-1}$

$32 = r^{n-1}$

Further, using $S_n = 189$, we have

$$189 = 3\frac{r^n - 1}{r - 1}$$

$$63 = \frac{r^n - 1}{r - 1}$$

$$63 = \frac{r^{n-1} - 1/r}{1 - 1/r}$$

$$63 = \frac{32 - 1/r}{1 - 1/r}$$

$$63 - \frac{63}{r} = 32 - \frac{1}{r}$$

$$31 = \frac{1}{r}(63 - 1)$$

$$31 = \frac{62}{r}$$

$r = 2$

Now, substituting $r = 2$ in $32 = r^{n-1}$, we get:

$32 = 2^{n-1}$

$2^5 = 2^{n-1}$

$5 = n - 1$

$n = 6$

Therefore, if in a G.P $a_1 = 3, a_n = 96, S_n = 189$, then $n = 6$.

17. Correct option: (B) $\frac{7}{15}$

Solution: We know, the sum of infinite terms of a GP is:

$$S_n = \frac{a}{1-r}$$

where a is the first term of the GP and r is the common ration of the GP with magnitude less than 1.

Given, $3 + 3x + 3x^2 + ...\infty = \frac{45}{8}$

We can rewrite the given series as:

$3(1 + x + x^2 + ...\infty) = \frac{45}{8}$

We see the term in bracket is a GP, with first term 1 and common ratio x. Hence,

$3\frac{1}{1-x} = \frac{45}{8}$

$24 = 45 - 45x$

$45x = 21$

$x = \frac{7}{15}$

Therefore, if $3 + 3x + 3x^2 + ...\infty = \frac{45}{8}$, then $x = \frac{7}{15}$.

18. Correct option: (C) x

Solution: We know, the sum of infinite terms of a GP is: $S_n = \frac{a}{1-r}$

where a is the first term of the GP and r is the common ration of the GP with magnitude less than 1.

Given, $x^{\frac{1}{2}}.x^{\frac{1}{4}}.x^{\frac{1}{8}}.x^{\frac{1}{16}}...\infty$

We can rewrite the given series as:

$x^{\left(\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+...\right)}$

We see the term in bracket is a GP, with first term $\frac{1}{2}$ and common ratio $\frac{1}{2}$. Hence,

$x^{\frac{1}{2}}.x^{\frac{1}{4}}.x^{\frac{1}{8}}.x^{\frac{1}{16}}...\infty = x^{\frac{1/2}{1-1/2}}$

$x^{\frac{1}{2}}.x^{\frac{1}{4}}.x^{\frac{1}{8}}.x^{\frac{1}{16}}...\infty = x$

Therefore, $x^{\frac{1}{2}}.x^{\frac{1}{4}}.x^{\frac{1}{8}}.x^{\frac{1}{16}}...\infty = x$

19. Correct option: (A) 12

Solution: We know, the geometric mean GM between two numbers a and b is: $GM = \sqrt{ab}$
Given, two numbers -9 and -16.
Therefore, geometric mean between -9 and -16 is:

$GM = \sqrt{(-9)(-16)}$

$GM = \sqrt{144}$

$GM = 12$

Therefore, geometric mean between -9 and -16 is 12.

20. Correct option: (B) 64

Solution: We know, the n th term of the Geometric Progression (GP) is:

$a_n = ar^{n-1}$

where a is the first term of the GP and r is the common ration of the GP.
We know, if three numbers a, b, c are in GP, then

$b = \sqrt{ac}$

Given, $2, a, 2^3, b, 2^5$ are in GP.
Hence, a is given by:

$a = \sqrt{2 \times 2^3}$

$a = 2^2$

$a = 4$

Similarly, b is given by:

$b = \sqrt{2^3 \times 2^5}$

$b = 2^4$

$b = 16$

Hence the product ab is $16(4) = 64$

LEVEL-II

1. Correct option: (A) 34

Solution**:** We know, the n th term in an Arithmetic Progression (AP) is given by:

$a_n = a + (n-1)d$

where a is the first term of the AP and d is the common difference.
Given, an AP has terms 7, 11, 15, … ,139
We see, the first term of the AP is $a = 7$ and the common difference is

$d = 11 - 7$

$d = 4$

Now, the number of terms in the AP is:

$a_n = a + (n-1)d$

$139 = 7 + (n-1)4$

$132 = (n-1)4$

$33 = n - 1$

$n = 34$

Therefore, if the terms of the AP are 7, 11, 15, … ,139, then the number of terms in the AP is 34.

2. Correct option: (A) $2n-8-3p$

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$a_n = a+(n-1)d$

where a is the first term of the AP and d is the common difference.

Given, $2n-5, 2n-8, 2n-11$ are in A.P.

We see, the first term of the AP is $a = 2n-5$ and the common difference is

$d = 2n-8-(2n-5)$

$d = -3$

Now, the $(p+2)$ th term of the AP is:

$a_{p+2} = a+(p+2-1)d$

$a_{p+2} = 2n-5+(p+2-1)(-3)$

$a_{p+2} = 2n-8-3p$

Therefore, if $2n-5, 2n-8, 2n-11$ are in A.P, then the $(p+2)$ th term of the AP is $2n-8-3p$.

3. Correct option: (C) 191

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$a_n = a+(n-1)d$

where a is the first term of the AP and d is the common difference.

Given, two series $3+7+11+15+19+23+27+31...$ and $1+6+11+16+21+26+31...$

We see, the first series is that of an AP with first term 3 and common difference 7-3=4.

Similarly, the second series is also an AP with first term 1 and common difference 6-1=5

Now, the first common term is 11 and the next common term is 31. The series of common terms will be again an AP with first term 11 and common difference 31-11=20.

Then the 10th common term is:

$a_n = 11+(n-1)(20)$

$a_{10} = 11+(10-1)(20)$

$a_{10} = 11+180$

$a_{10} = 191$

4. Correct option: (C) 900

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$a_n = a+(n-1)d$

where a is the first term of the AP and d is the common difference.

We know, the sum of first n terms of an arithmetic progression (AP) is given by:

$$S_n = \frac{n}{2}\left(2a+(n-1)d\right)$$

where d is the common difference of the AP.

Given, $a_1, a_2, a_3, a_4, ...$ are in AP such that $a_1+a_5+a_{10}+a_{15}+a_{20}+a_{24} = 225$.

We can rewrite this as:

$a+a+4d+a+9d+a+14d+a+19d+a+23d = 225$

$6a+69d = 225$

$3(2a+23d) = 225$

$2a+23d = 75$

Now, the sum of first 24 terms of the series is:

$$a_1 + a_2 + a_3 + ... + a_{23} + a_{24} = \frac{24}{2}(2a + 23d)$$

$$a_1 + a_2 + a_3 + ... + a_{23} + a_{24} = 12(2a + 23d)$$

$$a_1 + a_2 + a_3 + ... + a_{23} + a_{24} = 12(75)$$

$$a_1 + a_2 + a_3 + ... + a_{23} + a_{24} = 900$$

Therefore, if $a_1, a_2, a_3, a_4, ...$ are in AP such that $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$, then $a_1 + a_2 + a_3 + ... + a_{23} + a_{24} = 900$.

5. Correct option: (A) 1

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$$a_n = a + (n-1)d$$

where a is the first term of the AP and d is the common difference.

We know, the n th term of the Geometric Progression (GP) is:

$$a_n = ar^{n-1}$$

where a is the first term of the GP and r is the common ration of the GP.

Given, p, q, r are in AP and x, y, z are in GP.

Then, we have:

$$q = p + d$$

$$r = p + 2d$$

And,

$$y = xr$$

$$z = xr^2$$

Therefore, $x^{q-r} y^{r-p} z^{p-q}$ can be simplified as:

$$x^{q-r} y^{r-p} z^{p-q} = x^{p+d-p-2d} (xr)^{p+2d-p} (xr^2)^{p-p-d}$$

$$x^{q-r} y^{r-p} z^{p-q} = x^{-d} (xr)^{2d} (xr^2)^{-d}$$

$$x^{q-r} y^{r-p} z^{p-q} = x^{-d+2d-d} r^{2d-2d}$$

$$x^{q-r} y^{r-p} z^{p-q} = 1$$

Therefore, if p, q, r are in AP and x, y, z are in GP, then $x^{q-r} y^{r-p} z^{p-q} = 1$.

6. Correct option: (A) 6, 18, 54, 162

Solution: We know, we can insert n geometric means between two numbers a, b such that $a, m_1, m_2, m_3, ..., m_n, b$ forms a GP where $m_n = ar^n$ and $r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$

We want to insert four G.M.'s between 2 and 486.

Now, the common ratio is:

$$r = \left(\frac{486}{2}\right)^{\frac{1}{5}}$$

$$r = \left(\frac{486}{2}\right)^{\frac{1}{5}}$$

$$r = 3$$

Therefore, the four geometric means are:

$m_1 = (2)(3) = 6$

$m_2 = (2)(3)^2 = 18$

$m_3 = (2)(3)^3 = 54$

$m_4 = (2)(3)^4 = 162$

Therefore, four GM's between 2 and 486 are 6,18,54,162

7. Correct option: (B) 3

Solution: We know, the sum of infinite terms of a GP is:

$$S_n = \frac{a}{1-r}$$

where a is the first term of the GP and r is the common ration of the GP with magnitude less than 1.

Given, the sum of an infinite number of terms in G.P. is 20 and the sum of their squares is 100.

Using the fact that the sum of an infinite number of terms in G.P. is 20, we get:

$$20 = \frac{a}{1-r}$$

Now, if we have a GP $a, ar, ar^2, \ldots$ then the sequence of square of each term $a^2, a^2r^2, a^2r^4, \ldots$ is also a GP with first term a^2 and common ratio r^2.

Therefore, using the fact that the sum of square of infinite number of terms in G.P. is 100, we get:

$$100 = \frac{a^2}{1-r^2}$$

Squaring the first equation and dividing by second, we get:

$$\frac{100}{400} = \frac{a^2}{1-r^2} \times \frac{(1-r)^2}{a^2}$$

$$\frac{1}{4} = \frac{1-r}{1+r}$$

$$1 + r = 4 - 4r$$

$$5r = 3$$

$$r = 3/5$$

Therefore, five-times of the common ratio of G.P. is $5\frac{3}{5} = 3$

8. Correct option: (D) 24.2

Solution: Also, the sum of first n terms of the GP is:

$$S_n = a\frac{r^n - 1}{r - 1}$$

where a is the first term of the GP and r is the common ration of the GP.

Given sequence: 16.2, 5.4, 1.8,…

We see, this is a GP with first term 16.2 and common ratio $5.4/16.2 = 0.33$

Therefore, sum of 16.2, 5.4, 1.8,… to 5 terms is:

$$S_5 = 16.2\frac{0.33^5 - 1}{0.33 - 1}$$

$$S_5 = 24.08$$

Therefore, sum of 16.2, 5.4, 1.8,… to 5 terms is 24.2 approximately.

9. Correct option: (D) a

Solution: We know, the sum of infinite terms of a GP is:

$$S_n = \frac{a}{1-r}$$

where a is the first term of the GP and r is the common ration of the GP with magnitude less than 1.

Given, $\sum_{n=1}^{\infty}\left(\frac{a}{a+1}\right)^n$

Writing a first few terms of the series, we get:

$$\frac{a}{a+1}+\left(\frac{a}{a+1}\right)^2+\left(\frac{a}{a+1}\right)^3+\ldots$$

We see, this is an infinite GP with first term $\frac{a}{a+1}$ and common ratio $\frac{a}{a+1}$.

Therefore,

$$\sum_{n=1}^{\infty}\left(\frac{a}{a+1}\right)^n = \frac{\frac{a}{a+1}}{1-\frac{a}{a+1}}$$

$$\sum_{n=1}^{\infty}\left(\frac{a}{a+1}\right)^n = \frac{a}{a+1-a}$$

$$\sum_{n=1}^{\infty}\left(\frac{a}{a+1}\right)^n = a$$

10. Correct option: (A) $\frac{n(n+1)(n+2)}{6}$

Solution: We know, the sum of first n natural number is $\frac{n(n+1)}{2}$

Also, the sum of square of first n natural number is $\frac{n(n+1)(2n+1)}{6}$

Given, $\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j}1$

We see the first summation is j times 1, hence:

$$\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j}1 = \sum_{i=1}^{n}\sum_{j=1}^{i} j$$

The next summation is sum of first i natural numbers, thus:

$$\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j} 1 = \sum_{i=1}^{n}\frac{i(i+1)}{2}$$

$$\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j} 1 = \sum_{i=1}^{n}\frac{i^2+i}{2}$$

$$\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j} 1 = \frac{1}{2}\sum_{i=1}^{n} i^2 + \frac{1}{2}\sum_{i=1}^{n} i$$

$$\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j}1=\frac{1}{2}\frac{n(n+1)(2n+1)}{6}+\frac{1}{2}\frac{n(n+1)}{2}$$

$$\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j}1=\frac{1}{2}\frac{n(n+1)}{2}\left(\frac{2n+1}{3}+1\right)$$

$$\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j}1=\frac{n(n+1)}{4}\left(\frac{2n+4}{3}\right)$$

$$\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j}1=\frac{n(n+1)(n+2)}{6}$$

Therefore, $\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j}1=\frac{n(n+1)(n+2)}{6}$.

11. Correct option: (B) n^2**, D)** $6n-n^2$

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$a_n=a+(n-1)d$

where a is the first term of the AP and d is the common difference.

We know, the sum of first n terms of an arithmetic progression (AP) is given by:

$$S_n=\frac{n}{2}\left(2a+(n-1)d\right)$$

where d is the common difference of the AP.

Given, the sum of the first three consecutive terms of an A.P. is 9 and the sum of their squares is 35.

Let the consecutive terms of the AP be $a-d,a,a+d$

Therefore,

$a-d+a+a+d=9$

$(a-d)^2+a^2+(a+d)^2=35$

Or simplifying this, we get:

$a=3$

$(3-d)^2+3^2+(3+d)^2=35$

$9+d^2-6d+9+9+d^2+6d=35$

$27+2d^2=35$

$2d^2=8$

$d=2,-2$

Therefore, the sum to n terms of the series is:

$$S_n=\frac{n}{2}\left(2(1)+(n-1)2\right);S_n=\frac{n}{2}\left(2(5)+(n-1)(-2)\right)$$

$$S_n=\left(n+n^2-n\right);S_n=\left(5n-n^2+n\right)$$

$$S_n=n^2;S_n=\left(6n-n^2\right)$$

12. Correct options: (A)9*5-10, B)7*4+7, C)8*4+3 , D)2*17+1

Solution: We know, the arithmetic mean between a,b is given by:

$$AM = \frac{a+b}{2}$$

Given, $3k-7$ is the arithmetic mean of $3k+4$ and $k+12$.
Therefore,

$$3k-7 = \frac{3k+4+k+12}{2}$$

$$3k-7 = \frac{4k+16}{2}$$

$$3k-7 = 2k+8$$

$$k = 15$$

Therefore, the value of $2k+5$ is:

$$2k+5 = 2(15)+5$$

$$2k+5 = 35$$

13. Correct options: (A) $a = 32, b = 8$ **, D)** $a = 8, b = 32$

Solution: We know, the arithmetic mean AM between a and b is given by:

$$\text{AM} = \frac{a+b}{2}$$

We know, the geometric mean GM between two numbers a and b is:

$$GM = \sqrt{ab}$$

Given, the A.M. and G.M. of two terms a and b are 20 and 16. Hence,

$$20 = \frac{a+b}{2}$$

$$40 = a+b$$

$$16 = \sqrt{ab}$$

$$256 = ab$$

Substituting $b = 40-a$ in $256 = ab$ and simplifying, we get:

$$256 = \text{a}(40-\text{a})$$

$$\text{a}^2 - 40\text{a} + 256 = 0$$

$$\text{a} = \frac{40 \pm \sqrt{40^2 - 4(1)(256)}}{2\text{a}}$$

$$\text{a} = 32,8$$

Hence using $b = 40-a$, we get:

$$b = 8, 32$$

14. Correct option: (B) $\frac{n}{2}(13-3n)$

Solution: Given, a sequence 5,2,-1, …
We see, this is an AP with first term 5 and common difference 2-5=-3
Therefore, sum to n terms is:

$$S_n = \frac{n}{2}\left(2a+(n-1)d\right)$$

$$S_n = \frac{n}{2}\left(2(5)+(n-1)(-3)\right)$$

$$S_n = \frac{n}{2}(13-3n)$$

15. Correct option: (A) $\frac{2S_n-(n^2-n)d}{2n}$

Solution: Given, the sum of first n terms of an arithmetic progression (AP) is given by:

$$S_n=\frac{n}{2}\left(2a+(n-1)d\right)$$

where d is the common difference of the AP.

Now, solving for a, we get:

$$2S_n=\left(2an+(n^2-n)d\right)$$

$$2S_n-(n^2-n)d=2an$$

$$a=\frac{2S_n-(n^2-n)d}{2n}$$

16. Correct option: (B) $\frac{8}{3}$

Solution: Given, $S_n=400, n=16, a=5$

We know, the sum of first n terms of an arithmetic progression (AP) is given by:

$$S_n=\frac{n}{2}\left(2a+(n-1)d\right)$$

where d is the common difference of the AP.

Therefore,

$$400=\frac{16}{2}\left(2(5)+(16-1)d\right)$$

$$400=8\left(10+15d\right)$$

$$50=10+15d$$

$$d=\frac{8}{3}$$

17. Correct Matchin is: a) ->Q) , b) ->P) , c) ->S), d) -> R)

Solution: a)We know, the geometric mean GM between two numbers a and b is:

$$GM=\sqrt{ab}$$

Therefore, geometric mean between 25 and 16 is:

$$GM=\sqrt{(25)(16)}$$

$$GM=20$$

b) We know, the n th term of the Geometric Progression (GP) is:

$$a_n=ar^{n-1}$$

where a is the first term of the GP and r is the common ration of the GP.

Now, if $x, 2x+2, 3x+3$ are the first three terms of the GP, we should have:

$$\frac{2x+2}{x}=\frac{3x+3}{2x+2}$$

$$\frac{2x+2}{x}=\frac{3}{2}$$

$$4x+4=3x$$

$$x=-4$$

Therefore, the first three terms of the GP are $-4, -6, -9$.

We see the first term of the GP is -4 and the common ratio is 3/2.
Therefore, the fourth term of the GP is:

$a_4 = (-4)(3/2)^3$

$a_4 = (-4)(27/8)$

$a_4 = \frac{-27}{2}$

c) We know, the sum of first n terms of an arithmetic progression (AP) is given by:

$S_n = \frac{n}{2}\left(2a + (n-1)d\right)$

where d is the common difference of the AP.
Given series, $a - (a+d) + (a+2d) - (a+3d) + ...$
We can rewrite the given series as:
$a - (a+d) + (a+2d) - (a+3d) + ... = a + (a+2d) + ... - (a+d) - (a+3d) + ...$
We see, the series break down into two series, each of which is an AP. The first AP has first term a and common ratio $2d$. The second AP has first term $-a-d$ and common ratio $-2d$.
Therefore, the sum of first $2n+1$ terms is:

$S_n = \frac{n+1}{2}\left(2a + (n+1-1)2d\right) + \frac{n}{2}\left(2(-a-d) + (n-1)(-2d)\right)$

$S_n = \frac{n}{2}(-2a - 2d - 2nd + 2d + 2a + 2nd) + \frac{1}{2}(2a + 2nd)$

$S_n = a + nd$

d) We know, the sum of first n terms of an arithmetic progression (AP) is given by:

$S_n = \frac{n}{2}\left(2a + (n-1)d\right)$

where d is the common difference of the AP.
Given, the sum of the series $2 + 5 + 8 + 11 + ...$ n terms is 60100.
We see, this forms an AP with first term 2 and common difference is 3.
Therefore, n is:

$60100 = \frac{n}{2}\left(2(2) + (n-1)3\right)$

$120200 = n + 3n^2$

$n = \frac{-1 \pm \sqrt{1^2 - 4(3)(-120200)}}{2(3)}$

$n = 200$

In last step, we restrict n to just the positive value.
Therefore, the correct match is: a) ->Q) , b) ->P) , c) ->S), d) -> R)

18. Correct Answer is: 1

Solution: We know, the n th term of the Geometric Progression (GP) is:

$a_n = ar^{n-1}$

where a is the first term of the GP and r is the common ration of the GP.
Also, the sum of first n terms of the GP is:

$S_n = a\frac{r^n - 1}{r - 1}$

where a is the first term of the GP and r is the common ratio of the GP.
Given, the n th term of the GP is 128 and the sum of its n terms is 255. The common ratio is 2.
Now,

$$128 = a(2^{n-1})$$

$$256 = a(2^n)$$

$$255 = a\frac{2^n - 1}{2 - 1}$$

$$255 = 256 - a$$

$$a = 1$$

Therefore, if the n th term of the GP is 128 and the sum of its n terms is 255 and the common ratio is 2, then the first term is 1.

19. Correct Answer is: 5

Solution: We know, the arithmetic mean AM between a and b is given by:

$$\text{AM} = \frac{a+b}{2}$$

Therefore, the arithmetic mean of $3+\sqrt{2}$ and $7-\sqrt{2}$ is:

$$\frac{(3+\sqrt{2}+7-\sqrt{2})}{2} = 5$$

20. Correct Answer is: 7

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$$a_n = a + (n-1)d$$

where a is the first term of the AP and d is the common difference.
Given, two A.P.'s have the same common difference, the first term of one of these is –1 and that of the other is –8.
Therefore, the general term of the two AP is:

$$a_n = -1 + (n-1)d$$

$$b_n = -8 + (n-1)d$$

Now, difference between their 4th term is:

$$a_4 - b_4 = [-1 + (4-1)d] - [-8 + (4-1)d]$$

$$a_4 - b_4 = 7$$

Therefore, if two A.P.'s have the same common difference, the first term of one of these is –1 and that of the other is –8, the difference between their 4th term is 7.

STUDENT PRACTICE QUESTIONS

1. Correct option: (C) 89

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$$a_n = a + (n-1)d$$

where a is the first term of the AP and d is the common difference.
Given, the 7th term of an AP is 34 and the 13th term is 64.
Let's first find the first term and the common difference of the AP.
Using the fact that 7th term of an AP is 34, we get:

$34 = a + (7-1)d$

$34 = a + 6d$

Also, since 13th term is 64, we also have:

$64 = a + (13-1)d$

$64 = a + 12d$

Now, subtracting these two equations, we get:

$64 - 34 = a + 12d - (a + 6d)$

$30 = 6d$

$d = 5$

Substituting $d = 5$ into $64 = a + 12d$, we get:

$64 = a + 12(5)$

$64 = a + 60$

$a = 4$

Hence, the first term of the AP is 4 and the common difference is 5.

Now, the 18th term of the AP is:

$a_n = a + (n-1)d$

$a_{18} = 4 + (18-1)(5)$

$a_{18} = 4 + 17(5)$

$a_{18} = 4 + 85$

$a_{18} = 89$

Therefore, if the 7th term of an AP is 34 and the 13th term is 64, then the 18th term of the AP is 89.

2. Correct option: (C) $-13 - 4n$

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$a_n = a + (n-1)d$

where a is the first term of the AP and d is the common difference.

Given, -17, -21, -25 are in AP.

Let's first find the first term and the common difference of the AP.

For the common difference we have:

$d = -21 - (-17)$

$d = -4$

The first term of the AP is -17.

Therefore, the n th term is:

$a_n = -17 + (n-1)(-4)$

$a_n = -13 - 4n$

Therefore, if -17, -21, -25 are in AP, then the n th term is $-13 - 4n$.

3. Correct option: (D) $q + p - r$

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$a_n = a + (n-1)d$

where a is the first term of the AP and d is the common difference.

Given, the pth term of an AP is q and the qth term is p.

Let's first find the first term and the common difference of the AP.

Using the fact that pth term of an AP is q, we get:

$q = a + (p-1)d$

Also, since qth term is p, we also have:

$p = a + (q-1)d$

Now, subtracting these two equations, we get:

$q - p = a + (p-1)d - (a + (q-1)d)$

$q - p = d(p - 1 - q + 1)$

$d = \frac{q-p}{p-q}$

$d = -1$

Substituting $d = -1$ into $q = a + (p-1)d$, we get:

$q = a + (p-1)(-1)$

$a = q + p - 1$

Hence, the first term of the AP is $q + p - 1$ and the common difference is -1.

Now, the rth term of the AP is:

$a_r = a + (r-1)d$

$a_r = q + p - 1 + (r-1)(-1)$

$a_r = q + p - r$

Therefore, if the pth term of an AP is q and the qth term is p, then the rth term of the AP is $q + p - r$.

4. Correct option: (C)0

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$a_n = a + (n-1)d$

where a is the first term of the AP and d is the common difference.

Given, a, b, c, d, e are in AP.

Hence, we have:

$b = a + D$

$c = a + 2D$

$d = a + 3D$

$e = a + 4D$

where D is the common difference of the sequence.

Now, $a - 4b + 6c - 4d + e$ can be simplified as:

$a - 4b + 6c - 4d + e = a - 4(a + D) + 6(a + 2D) - 4(a + 3D) + a + 4D$

$a - 4b + 6c - 4d + e = a - 4a - 4D + 6a + 12D - 4a - 12D + a + 4D$

$a - 4b + 6c - 4d + e = 0$

Therefore, if a, b, c, d, e are in AP, then $a - 4b + 6c - 4d + e = 0$

5. Correct option: (D) $\frac{10}{3}$

Solution: We know, the sum of first n terms of an arithmetic progression (AP) is given by:

$S_n = \frac{n}{2}(2a + (n-1)d)$

where d is the common difference of the AP.

Given, $S_n = 200, n = 10, a = 5$

Therefore, the common difference is:

$S_n = \frac{n}{2}\left(2a + (n-1)d\right)$

$200 = \frac{10}{2}\left(2(5) + 9d\right)$

$40 = 10 + 9d$

$30 = 9d$

$d = \frac{10}{3}$

Therefore, if $S_n = 200, n = 10, a = 5$, then d = $\frac{10}{3}$.

6. Correct option: (C) $\frac{n}{n+1}$

Solution: Given series is:

$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \ldots$

We can rewrite this as:

$\frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \ldots + \frac{1}{n} - \frac{1}{n+1}$

We see, many of the terms cancel.

The sum of n terms of the series is:

$1 - \frac{1}{n+1} = \frac{n}{n+1}$

7. Correct option: (C) $pr = q^2$

Solution: We know, the n th term of the Geometric Progression (GP) is:

$a_n = ar^{n-1}$

where a is the first term of the GP and r is the common ration of the GP.

Given, 4th ,7th and 10th terms of a G.P. are p, q and r respectively.

Now,

$p = ar^3$

$q = ar^6$

$r = ar^9$

Multiplying the first and last term, we get:

$pr = (ar^3)(ar^9)$

$pr = a^2 r^{12}$

$pr = (ar^6)^2$

$pr = q^2$

Therefore, if 4th ,7th and 10th terms of a G.P. are p, q and r respectively, then $pr = q^2$.

8. Correct option: (D) $\frac{1}{9}\left(10^{n+1} - 9n - 10\right)$

Solution: We know, the sum of first n terms of the GP is:

$S_n = a\frac{r^n - 1}{r - 1}$

where a is the first term of the GP and r is the common ration of the GP.

Given series: $9 + 99 + 999 + \ldots$

We can rewrite the given series as:

$10-1+100-1+1000-1+...=10+100+1000+...-1-1-1-...$

We see, the first group of terms on the right is a GP with first term 10 and common ration 10. Therefore, the sum of first n terms of the series is:

$$S_n = 10\frac{10^n-1}{10-1}-n$$

$$S_n = \frac{1}{9}\left(10^{n+1}-9n-10\right)$$

Therefore, sum of the series $9+99+999+...$ up to n terms is $\frac{1}{9}\left(10^{n+1}-9n-10\right)$.

9. Correct option: (A) $\frac{3}{2},1,\frac{2}{3}$

Solution: We know, we can insert n geometric means between two numbers a,b such that $a,m_1,m_2,m_3,...,m_n,b$ forms a GP where $m_n = ar^n$ and $r=\left(\frac{b}{a}\right)^{\frac{1}{n+1}}$

We want to insert three G.M.'s between $2\frac{1}{4}$ and $\frac{4}{9}$.

Now, the common ratio is:

$$r=\left(\frac{4/9}{9/4}\right)^{\frac{1}{4}}$$

$$r=\left(\frac{16}{81}\right)^{\frac{1}{4}}$$

$$r=\frac{2}{3}$$

Therefore, the three geometric means are:

$$m_1=\frac{9}{4}.\frac{2}{3}=\frac{3}{2}$$

$$m_2=\frac{9}{4}.\left(\frac{2}{3}\right)^2=1$$

$$m_3=\frac{9}{4}.\left(\frac{2}{3}\right)^3=\frac{2}{3}$$

Therefore, three GM's between $2\frac{1}{4}$ and $\frac{4}{9}$ are $\frac{3}{2},1,\frac{2}{3}$

10. Correct option: (D) $2(3)^{n-1}$

Solution: We know, the n th term of the Geometric Progression (GP) is:

$a_n = ar^{n-1}$

where a is the first term of the GP and r is the common ration of the GP.

Given, 4th and 9th terms of GP are 54 and 13122.

Let's find the common ratio and the first term of the GP.

Using the fact that 4th term of the GP is 54 and 9th term is 13122, we get:

$54 = ar^3$

$13122 = ar^8$

Dividing these two equations, we get:

$$\frac{13122}{54} = \frac{ar^8}{ar^3}$$

$$243 = r^5$$

$$3^5 = r^5$$

$$r = 3$$

Substituting $r = 3$ in $54 = ar^3$, we get:

$$54 = a3^3$$

$$54 = a(27)$$

$$a = 2$$

Therefore, the first term of the GP is 2 and the common ration is 2.

Therefore, the n th term of the GP is:

$$a_n = 2(3)^{n-1}$$

11. Correct option: (A)1 , C)5

Solution: We know, the sum of infinite terms of a GP is:

$$S_n = \frac{a}{1-r}$$

where a is the first term of the GP and r is the common ration of the GP with magnitude less than 1.

Given, the sum of the first two terms of a G.P. is 5/3 and the sum of an infinity of the series is 3

Hence,

$$3 = \frac{a}{1-r}$$

$$a + ar = \frac{5}{3}$$

$$a(1+r) = \frac{5}{3}$$

Substituting $3(1-r) = a$ in the last expression, we get:

$$3(1-r)(1+r) = \frac{5}{3}$$

$$1 - r^2 = \frac{5}{9}$$

$$\frac{4}{9} = r^2$$

$$r = \frac{2}{3}, -\frac{2}{3}$$

Now, using $3(1-r) = a$, we get:

$$a = 3(1-\frac{2}{3}); a = 3(1+\frac{2}{3})$$

$$a = 1; a = 5$$

Therefore, if the sum of the first two terms of a G.P. is 5/3 and the sum of an infinity of the series is 3, then the first term is either 1 or 5.

12. Correct options: (B) 19, C) 18

Solution: We know, the sum of first n terms of an arithmetic progression (AP) is given by:

$$S_n = \frac{n}{2}\left(2a + (n-1)d\right)$$

where d is the common difference of the AP.

Given sequence 54,51,48,…

We see this is an AP with first term 54 and common difference $51-54=-3$

Hence, numbers of terms of the sequence taken so that their sum is 513 is:

$$513 = \frac{n}{2}\left(2(54) + (n-1)(-3)\right)$$

$$1026 = n(111-3n)$$

$$3n^2 - 111n + 1026 = 0$$

Using the quadratic formula, we get:

$$n = \frac{111 \pm \sqrt{111^2 - 4(3)(1026)}}{2(3)}$$

$$n = 19,18$$

13. Correct option: (B) 9

Solution: We know, we can insert n geometric means between two numbers a,b such that $a, m_1, m_2, m_3, ..., m_n, b$ forms a GP where $m_n = ar^n$ and $r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$

Further, the product of n geometric mean between two numbers a,b is the n th power of their geometric mean:

$$m_1(m_2)(m_3)...(m_n) = (\sqrt{ab})^n$$

Given, product of geometric mean between 1 and 1024 is 2^{45} .Hence,

$$2^{45} = (\sqrt{1024})^n$$

$$2^{45} = 2^{5n}$$

$$45 = 5n$$

$$n = 9$$

Therefore, the value of n is 9.

14. Correct option: 8763750

Solution: We know, the sum of n arithmetic mean between two numbers a,b is: $\frac{n(a+b)}{2}$

Given, sum of arithmetic means between -2 and 102 is 1025×171.

Hence,

$$\frac{m(-2+102)}{2} = 1025 \times 171$$

$$m = 1025 \times 171 \times 50$$

$$m = 8763750$$

15. Correct option: (D) 1022

Solution: We know, we can insert n geometric means between two numbers a,b such that $a, m_1, m_2, m_3, ..., m_n, b$ forms a GP where $m_n = ar^n$ and $r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$

Also, the sum of first n terms of the GP is: $S_n = a\frac{r^n - 1}{r-1}$

where a is the first term of the GP and r is the common ration of the GP.

Given, $G_1, G_2, G_3, ..., G_n$ are the geometric means between 1 and 1024.

Therefore, the common ratio is:

$r = (1024)^{\frac{1}{9+1}}$

$r = 2$

Therefore, the product of geometric means is:

$S_9 = 2\dfrac{2^9 - 1}{2 - 1}$

$S_9 = 1022$

16. Correct Matching is: a) ->R), b) ->P), c) ->S), d) -> Q)

Solution: a) We know, when three numbers a, b, c are in GP, they satisfy: $b = \sqrt{ac}$

Given, $\dfrac{1}{3}, x_1, x_2, 9$ are in GP.

Therefore,

$x_1^{\,2} = \dfrac{x_2}{3}$

$x_2^{\,2} = 9x_1$

Eliminating x_1 from these two equations, we get:

$x_1^{\,2} = \dfrac{x_2}{3}$

$x_2^{\,4} = 81\dfrac{x_2}{3}$

$x_2^{\,3} = 27$

$x_2 = 3$

b) We know, the n th term of the Geometric Progression (GP) is: $a_n = ar^{n-1}$

where a is the first term of the GP and r is the common ration of the GP.

Given, the third term of a G.P. is 4.

Now, the product of its five terms is:

$a(ar)(ar^2)(ar^3)(ar^4) = a^5 r^{10}$

$a(ar)(ar^2)(ar^3)(ar^4) = (ar^2)^5$

$a(ar)(ar^2)(ar^3)(ar^4) = (4)^5$

$a(ar)(ar^2)(ar^3)(ar^4) = 1024$

c) We know, the sum of infinite terms of a GP is: $S_n = \dfrac{a}{1 - r}$

where a is the first term of the GP and r is the common ration of the GP with magnitude less than 1.

Given, $\dfrac{1}{3} + \dfrac{1}{9} + \dfrac{1}{27} + ...\infty$

We see this is a GP, with first term $\dfrac{1}{3}$ and common ratio $\dfrac{1}{3}$. Hence,

$$\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+...\infty=\frac{1/3}{1-1/3}$$

$$\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+...\infty=\frac{1}{2}$$

d) We know, the sum of first n terms of the GP is: $S_n=a\frac{r^n-1}{r-1}$

where a is the first term of the GP and r is the common ration of the GP.
Given, sum of $1+3+9+27+...$ is 364.
We see, the this is a GP with first term 1 and common ration 3. Therefore,

$$364=\frac{3^n-1}{3-1}$$

$$729=3^n$$

$$3^6=3^n$$

$$n=6$$

Therefore, the correct match is: a) ->R), b) ->P), c) ->S), d) -> Q)

17. Correct Answer is: 1

Solution: We know, the n th term in an Arithmetic Progression (AP) is given by:

$$a_n=a+(n-1)d$$

where a is the first term of the AP and d is the common difference.
Given, a,b,c are three terms of an AP. Hence,

$$b=a+d$$

$$c=a+2d$$

Now,

$$\frac{b-c}{a-b}=\frac{a+d-a-2d}{a-a-d}$$

$$\frac{b-c}{a-b}=\frac{-d}{-d}$$

$$\frac{b-c}{a-b}=1$$

Therefore, if a,b,c are three terms of an AP such that $a\neq b$, then $\frac{b-c}{a-b}=1$.

18. Correct Answer is: 2

Solution: We know, the sum of first n terms of an arithmetic progression (AP) is given by:

$$S_n=\frac{n}{2}\left(2a+(n-1)d\right)$$

where d is the common difference of the AP.
Given, the sum of n terms of an A.P. is $3n^2-n$ and its common difference is 6.
Now,

$$3n^2-n=\frac{n}{2}\left(2a+(n-1)6\right)$$

$$6n^2-2n=2an+6n^2-6n$$

$$4n=2an$$

$$2=a$$

Therefore, if the sum of n terms of an A.P. is $3n^2-n$ and its common difference is 6, then its first term is 2.

6. Sets
Solutions

LEVEL – I

1. Correct option: (B)

The given set in roster form is $\{1, 2, 3, 4, 5, 6\}$.

2. Correct option: (C)

(i) A = {–18, –15, –12, –9, –6, –3, 0, 3, 6, 9, 12, 15, 18}

(ii) B = {1, 2, 3}

3. Correct option: (A)

Let B = {2}, $\therefore$ A = {1, 2}

$\therefore$ P(A) = {ϕ, {1}, {B}, {1, B}} = {ϕ, {1}, {{2}}, {1, {2}}}

4. Correct option: (D)

Let B = {2, 3}, $\therefore$ A = {1, B}

5. Correct option: (B)

{3, 6, 15, 18, 21}

6. Correct option: (B)

{3, 15, 18, 21}

7. Correct option: (A)

{5, 7, 9}

8. Correct option: (D)

{1, 2, 3, 4}

9. Correct option: (C)

We have $A' \cap B' = (A \cup B)'$

$\therefore$ $n(A' \cap B') = n((A \cup B)') = n(U) - n(A \cup B)$

$= n(U) - [n(A) + n(B) - n(A \cap B)]$

$= 700 - (200 + 300 - 100) = 300.$

10. Correct option: (D)

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$\Rightarrow 42 = 20 + n(B) - 4$

$\Rightarrow n(B) = 42 - 16 = 26$

11. Correct option: (A)

$n(A - B) = n(A) - n(A \cap B) = 20 - 4 = 16$

12. Correct option: (B)

$n(B - A) = n(B) - n(A \cap B) = 26 - 4 = 22$

13. Correct option: (C)

Since $n(A \cup B) \geq n(A)$ $(\because A \cup B \supseteq A, A \cup B \supseteq B)$

$n(A \cup B) \geq n(B)$

$\therefore$ minimum number of elements in $A \cup B = 6$

14. Correct option: (B)

$(A \times B) \cap (B \times A) = (A \cap B) \times (B \cap A)$

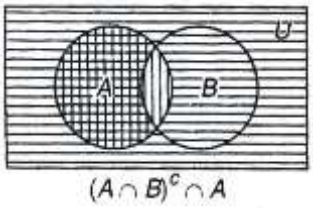
$= (2,3) \times (2 \times 3)$

$= \{(2,2), (2 \times 3), (3,2), (3,3)\}$

$\therefore$ number of elements = 4

15. Correct option: (B)

From Venn diagram

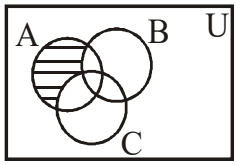


$(A \cap B)^c$ = Portion exterior to $A \cap B$

$(A \cap B)^c \cap A$ = Portion showing both shadings = A–B.

16. Correct option: (C)

$\because A-(B \cup C)=(A-B)\cap(A-C)$



17. Correct option: (B)

If $A \subseteq B$ then $A-B=\phi$

$A \Delta B=(A-B)\cup(B-A)$

$=\phi\cup(B-A)=(B-A)$

18. Correct option: (D)

$(A \cup B)=\{1,2,3,4\}$ and $A \cap B=\{3\}$

$\therefore\ (A \cup B)\times(A \cap B)=\{(1,3),(2,3),(3,3),(4,3)\}$

19. Correct option: (C)

Given that n (A) = 8, $n(A \cap B)=2$

$$n(A-B)=n(A)-n(A \cap B)=8-2=6$$

LEVEL-II

1. Correct option: (A)

x is a natural number less than 8

2. Correct option: (A)

A∪B = {x : x ∈ A or x ∈ B}

3. Correct option: (B)

A∩B = {x : x ∈ A or x ∈ B}

4. Correct option: (D)

A′ = ∪ – A

5. Correct option: (A)

Let U be the universal set. Then,

(i) A∪(A∩B) = (A∩B)∪(A∩B) [A = A∩U]

= A∩B(U∪B) [by distributive law]

= A∩U [U∪B = U]

= A [A∩U = A]

∴ A∪(A∩B) = A.

6. Correct option: (B)

1, 2 ∈ N and are factors of 6 and 8 both.

7. Correct option: (C)

We have $(A')' = U - A'$

$= \{1, 2, 3, 4, 5, 6, 7, 8\} - \{1, 3, 5, 7\}$

$= \{2, 4, 6, 8\} = A$

8. Correct option: (A)

The collection of all the days of a week is well defined.

9. Correct option: (D)

$B \cup C = \{7, 9, 11, 13, 15\}$

$A \cap (B \cup C) = \{7, 9, 11\}$

10. Correct option: (B)

$B \cap C = \{e, g\}$

$A - \{B \cap C\} = \{a, b, c, d\}$

11. Correct option: (C)

$A \cap B$ has no element

12. Correct option: (D)

Number of students enrolled in mathematics alone = a = 60.

Physics alone = 35.

Chemistry alone = g = 13.

13. Correct option: (B)

Since the two finite sets have m and n elements.

$\therefore$ no. of subsets of these sets will be 2^m and 2^n respectively.

$\therefore \quad 2^m - 2^n = 56$

Putting m = 6, n = 3, we get

$2^6 - 2^3 = 64 - 8 = 56$.

Hence m = 6, n = 3

14. Correct option: (B)

Number of subsets of $A = {}^nC_0 + {}^nC_1 + \ldots + {}^nC_n = 2^n$.

15. Correct option: (A)

$A \cap B = \{2, 3, 4, 8, 10\} \cap \{3, 4, 5, 10, 12\}$

$= \{3, 4, 10\}$

$A \cup C = \{4\} \therefore (A \cap B) \cup (A \cap BC) = \{3, 4, 10\}$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

16. Correct option: (B, C, D)

Number of elements is countable in the set A only

17. Correct option: (A, D)

No such natural number exists which is less than 5 and greater than 8.

18. Correct option: (B, C)

The sets C and D have no common element.

19. Correct option: (A, D)

Number of non-empty subsets of $\{1, 2, 3, 4\} = 2^4 - 1 = 15$.

20. Correct option: (A, C)

The number of proper subsets $= 2^n - 1 = 2^5 - 1 = 31$.

COMPREHENSION TYPE

21. Correct option: (A)

M = {13, 18, 23, 28},

N = {10, 12, 21, 23}

22. Correct option: (D) 12

23. Correct option: (A) 15

24. Correct option: (C) 550

25. Correct option: (B) 450

26. Correct option: (A) 350

MATRIX MATCHING TYPE

27. Correct Matching is: (A-q), (B-r), (C-s), (D-p)

(A) If $A = \{1,\{2,3\}\}$

then $A = \{1, B\}$ where B = {2,3}

Hence, $P(A) = \{\phi, \{1\}, \{B\}, \{1, B\}\}$

(B) $A \cup B = \{1,2,3,5,7\}$

(C) If A = {1,3,5, 7, 9} and B = {2,3,5,7,11},

then, $(A - B) = (1,9)$ and $(B - A) = \{2,11\}$

Hence, $A\Delta B = (A - B) \cap (B - A) = \{1,2,9,11\}$

(D) $A \cap B = \{1,3\}$

28. Correct Matching is: (A-r), (B-p), (C-q), (D-s)

(A) $A \cup B = \{2,3,4,5,8,10,12\}$ and $A \cup C = \{2,3,4,5,6,8,10,12,14\}$

$\therefore \quad (A \cup B) \cap (A \cup C) = \{2,3,4,5,8,10,12\}$

(B) Since, $\frac{A}{B} = \{3,4,5,6\}$ {by definition}

$$\Rightarrow \frac{A}{\left(\frac{A}{B}\right)} = \{1,2\} = A \cap B$$

(C) Since, $A \cup \{1,2\} = \{1,2,3,5,9\}$ {given}

$\Rightarrow A = \{3,5,9\}$ at least

(D) $A\Delta B = \left\{x : x \in \frac{A}{B} \text{ or } x \in \frac{B}{A}\right\}$

INTEGER ANSWERS TYPE

29. Correct Answer is:

Total number of subsets = 24 = 16

30. Correct Answer is:

Number of subsets of a set containing n elements is 2n.

31. Correct Answer is:

Power set of a set is the set of all its subsets.

32. Correct Answer is:

Total number of subsets = 22 = 4

33. Correct Answer is:

Number of subsets is $2^0 = 1$.

7. Trigonometry
Solutions

LEVEL – I

1. Correct option: (C)

To find the value of required expression first of all we need to calculate the value of $\tan\theta, \sec\theta, \cos\theta$ using $\sin\theta$

To find $\cos\theta$, we can use the Pythagorean trigonometric identity:

$$\sin^2\theta + \cos^2\theta = 1$$

Putting the values

$$\cos^2\theta = 1 - \left(\frac{3}{5}\right)^2 = \frac{16}{25} = \left(\frac{4}{5}\right)^2 \Rightarrow \cos\theta = \frac{4}{5}\ as(\theta < 90°)$$

To find the $\tan\theta$ we can use the definition of same as

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3}{4}$$

Now, we need to find $\sec\theta$, which is the reciprocal of $\cos\theta$:

$$\sec\theta = \frac{1}{\cos\theta} = \frac{5}{4}$$

Therefore the given expression

$$\frac{\tan\theta - 2\cos\theta}{3\sin\theta + \sec\theta} = \frac{\frac{3}{4} - 2\times\frac{4}{5}}{3\times\frac{3}{5} + \frac{5}{4}} = \frac{\left(\frac{3}{4} - \frac{8}{5}\right)}{\frac{9}{5} + \frac{5}{4}} = \frac{15 - 32}{36 + 25} = -\frac{17}{61}$$

Then the correct option is C)

2. Correct option: (B)

Given $7\tan\theta = 24$

We have to find the value of $\sin\theta + \cos\theta$

Using the definition of $\tan\theta$

$$\tan\theta = \frac{opposite\ side}{Adjacent\ side} = \frac{7}{24} \Rightarrow hypotenuse = \sqrt{7^2 + 24^2} = 25$$

So we can write

$$\theta < 90° \Rightarrow \sin\theta = \frac{opposite\ side}{hypotenuse} = \frac{7}{25}, \cos\theta = \frac{Adjacent\ side}{hypotenuse} = \frac{24}{25}$$

Therefore

$$\sin\theta + \cos\theta = \frac{7}{25} + \frac{24}{25} = \frac{31}{25}$$

Then the correct option is B)

3. Correct option: (C)

Given $\sin\theta = \frac{1}{3}$

We have to find the value of $2\cot^2\theta + 2$

Using the definition of $\sin\theta$

$$\sin\theta = \frac{opposite\ side}{hypotenuse} = \frac{1}{3} \Rightarrow Adjacent\ side = \sqrt{3^2 - 1} = \sqrt{8}$$

So we can write

$$\theta < 90° \Rightarrow \cot\theta = \frac{Adjacent\ side}{opposite\ side} = \frac{\sqrt{8}}{1}$$

Therefore

$$2\cot^2\theta + 2 = 2\times 8 + 2 = 18$$

Then the correct option is C)

4. Correct option: (B)

Given $\cos ec\theta = \frac{13}{12}$

We have to find the value of

$$\frac{2\sin\theta - 3\cos\theta}{4\sin\theta - 9\cos\theta}$$

Using the definition of $\cos ec\theta$

$$\cos ec\theta = \frac{hypotenuse}{opposite\ side} = \frac{13}{12} => Adjacent\ side = \sqrt{13^2 - 12^2} = 5$$

So we can write

$$\theta < 90° => \sin\theta = \frac{opposite\ side}{hypotenuse} = \frac{12}{13}, \cos\theta = \frac{Adjacent\ side}{hypotenuse} = \frac{5}{13}$$

Therefore

$$\frac{2\sin\theta - 3\cos\theta}{4\sin\theta - 9\cos\theta} = \frac{\left(2\times\frac{12}{13} - 3\times\frac{5}{13}\right)}{4\times\frac{12}{13} - 9\times\frac{5}{13}} = \frac{24-15}{48-45} = \frac{9}{3} = 3$$

Then the correct option is B)

5. Correct option: (B)

We have to find the value of

$$\frac{\tan 45°}{\sin 30° + \cos 60°}$$

We know that

$$\tan 45° = 1, \sin 30° = \frac{1}{2}, \cos 60° = \frac{1}{2}$$

So we can write

$$\frac{\tan 45°}{\sin 30° + \cos 60°} = \frac{1}{\frac{1}{2} + \frac{1}{2}} = 1$$

Then the correct option is B)

6. Correct option: (A)

We have to find the value of

$$(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2$$

Expanding the terms

$$\sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta + \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta$$
$$= 2(\sin^2\theta + \cos^2\theta)$$

We know that $\sin^2\theta + \cos^2\theta = 1$

So we can write

$$(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2 = 2(\sin^2\theta + \cos^2\theta) = 2$$

Then the correct option is A)

7. Correct option: (A)

We have to find the value of

$$\frac{\cos A}{1+\sin A} + \frac{1+\sin A}{\cos A}$$

Expanding and solving the terms

$$\frac{\cos A}{1+\sin A} + \frac{1+\sin A}{\cos A} = \frac{\cos^2 A + (1+\sin A)^2}{1+\sin A\cos A} = \frac{\cos^2 A + (1+\sin A)^2}{(1+\sin A)\cos A}$$
$$= \frac{\cos^2 A + \sin^2 A + 1 + 2\sin A}{(1+\sin A)\cos A} = \frac{2(1+\sin A)}{(1+\sin A)\cos A} = \frac{2}{\cos A} = 2\sec A$$

Then the correct option is A)

8. Correct option: (A)

We are given that

$$7\sin^2\theta + 3\cos^2\theta = 4$$

We have to find the value of

$$\tan\theta$$

Using trigonometric identity and given expression

$$7\sin^2\theta + 3\cos^2\theta = 4\sin^2\theta + 3\sin^2\theta + 3\cos^2\theta = 4\sin^2\theta + 3$$

Therefore

$$4\sin^2\theta + 3 = 4 => 4\sin^2\theta = 1 => \sin^2\theta = \frac{1}{4} => \sin\theta = \pm\frac{1}{2}$$

Therefore

$$\theta < 90° => \tan\theta = \frac{1}{\sqrt{3}}$$

Then the correct option is A)

9. Correct option: (C)

We are given that

$$\sec\theta + \tan\theta = k$$

We have to find the value of

$$\sin\theta$$

Using trigonometric identity and given expression

$$\sec\theta + \tan\theta = \frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta} = \frac{1+\sin\theta}{\cos\theta} = k$$

Squaring both side and adding 1

$$\left(\frac{1+\sin\theta}{\cos\theta}\right)^2 + 1 = k^2 + 1 => \frac{1+\sin^2\theta+2\sin\theta+\cos^2\theta}{\cos^2\theta} = k^2 + 1$$

$$= \frac{2+2\sin\theta}{\cos^2\theta} = \frac{2(1+\sin\theta)}{1-\sin^2\theta} = \frac{2(1+\sin\theta)}{(1+\sin\theta)(1-\sin\theta)} = \frac{2}{1-\sin\theta}$$

Therefore

$$\frac{2}{1-\sin\theta} = k^2 + 1 => \frac{2}{k^2+1} = 1 - \sin\theta => \sin\theta = 1 - \frac{2}{k^2+1}$$

$$\sin\theta = \frac{k^2-1}{k^2+1}$$

Then the correct option is C)

10. Correct option: (D)

We are given that

$$\cos ec\theta + \cot\theta = 5$$

We have to find the value of

$$\cos\theta$$

Using trigonometric identity and given expression

$$\cos ec\theta + \cot\theta = \frac{1}{\sin\theta} + \frac{\cos\theta}{\sin\theta} = \frac{1+\cos\theta}{\sin\theta} = k$$

Squaring both side and adding 1

$$\left(\frac{1+\cos\theta}{\sin\theta}\right)^2 + 1 = 5^2 + 1 => \frac{1+\cos^2\theta+2\cos\theta+\sin^2\theta}{\sin^2\theta} = 26$$

$$= \frac{2+2\cos\theta}{\sin^2\theta} = \frac{2(1+\cos\theta)}{1-\cos^2\theta} = \frac{2(1+\cos\theta)}{(1+\cos\theta)(1-\cos\theta)} = \frac{2}{1-\cos\theta}$$

Therefore

$$\frac{2}{1-\cos\theta} = 26 => \frac{2}{26} = 1 - \cos\theta => \cos\theta = 1 - \frac{1}{13} = \frac{12}{13}$$

Then the correct option is D)

11. Correct option: (C)

We have to find the value of

$$\frac{1+\tan^2\theta}{1+\cot^2\theta}$$

Using trigonometric identity and given expression

$$\frac{1+\tan^2\theta}{1+\cot^2\theta}=\frac{\sec^2\theta}{\cos ec^2\theta}=\left(\frac{\sec\theta}{\cos ec\theta}\right)^2=\left(\frac{\sec\theta}{\cos ec\theta}\times\frac{\sin\theta-\cos\theta}{\sin\theta-\cos\theta}\right)^2$$

$$=\left(\frac{\sec\theta}{\cos ec\theta}\times\frac{\sin\theta-\cos\theta}{\sin\theta-\cos\theta}\right)^2=\left(\frac{\frac{(\sin\theta-\cos\theta)}{\cos\theta}}{\frac{(\sin\theta-\cos\theta)}{\sin\theta}}\right)^2=\left(\frac{\tan\theta-1}{1-\cot\theta}\right)^2=\left(\frac{1-\tan\theta}{1-\cot\theta}\right)^2$$

Therefore

$$\frac{1+\tan^2\theta}{1+\cot^2\theta}=\left(\frac{1-\tan\theta}{1-\cot\theta}\right)^2$$

Then the correct option is C)

12. Correct option: (B)

We have to find the value of $(1+\cos A)(1-\cos A)(1+\cot^2 A)$

Using trigonometric identity and given expression

$$(1+\cos A)(1-\cos A)(1+\cot^2 A)=(1-\cos^2 A)(1+\cot^2 A)$$

$$\sin^2 A\cos ec^2 A=\frac{\sin^2 A}{\sin^2 A}=1$$

Therefore $(1+\cos A)(1-\cos A)(1+\cot^2 A)=1$

Then the correct option is B)

13. Correct option: (B)

Given $x=r\cos A\sin B, y=r\cos A.\cos B, z=r\sin A$

We have to find the value of $x^2+y^2+z^2$

Using trigonometric identity and given expression

$$x^2+y^2+z^2=(r\cos A\sin B)^2+(r\cos A.\cos B)^2+(r\sin A)^2$$

$$=r^2(\cos^2 A\sin^2 B+\cos^2 A.\cos^2 B+\sin^2 A)=r^2(\cos^2 A\,(\sin^2 B+\cos^2 B)+\sin^2 A)$$

$$=r^2(\cos^2 A+\sin^2 A)=r^2$$

Therefore $x^2+y^2+z^2=r^2$

Then the correct option is B)

14. Correct option: (D)

Given $m=\tan\theta+\sin\theta, n=\tan\theta-\sin\theta$

We have to find the value of m^2-n^2

Using trigonometric identity and given expression

$$m^2-n^2=(m-n)(m+n)$$

$$=(\tan\theta+\sin\theta)^2-(\tan\theta-\sin\theta)^2=4\tan\theta\sin\theta$$

Find the value of mn

$$mn=(\tan\theta+\sin\theta)(\tan\theta-\sin\theta)=\tan^2\theta-\sin^2\theta=\sin^2\theta\left(\frac{1}{\cos^2\theta}-1\right)=\sin^2\theta\tan^2\theta$$

Therefore $4\tan\theta\sin\theta=4\sqrt{mn}\Rightarrow m^2-n^2=4\sqrt{mn}$

Then the correct option is D)

15. Correct option: (A)

Given $2\sin\theta=2-\cos\theta$

We have to find the value of $\sin\theta$

Using trigonometric identity and given expression

$$2\sin\theta=2-\cos\theta$$

$$2-2\sin\theta=\cos\theta$$

Squaring both side

$$4 + 4\sin^2\theta - 8\sin\theta = \cos^2\theta = 1 - \sin^2\theta$$
$$5\sin^2\theta - 8\sin\theta + 3 = 0$$

Using quadratic equation

$$\sin\theta = \frac{-(-8) \pm \sqrt{64 - 60}}{10} = \frac{8 \pm 2}{10} = 1, \frac{6}{10}$$

Therefore

$$\sin\theta = 1, \frac{3}{5}$$

Then the correct option is A)

16. Correct option: (A)

Given

$$\tan^2\theta + \sec\theta = 5$$

We have to find the value of

$$\cos\theta$$

Using trigonometric identity and given expression

$$\tan^2\theta + \sec\theta = 5$$
$$=> \frac{1 - \cos^2\theta}{\cos^2\theta} + \frac{1}{\cos\theta} = 5$$
$$=> (1 - \cos^2\theta + \cos\theta) = 5\cos^2\theta$$
$$=> 6\cos^2\theta - \cos\theta - 1 = 0$$

Using quadratic equation

$$\cos\theta = \frac{-(-1) \pm \sqrt{1 - (-24)}}{12} = \frac{1 \pm 5}{12} = \frac{1}{2}, -\frac{1}{3}$$

Therefore

$$\cos\theta = \frac{1}{2}, -\frac{1}{3}$$

Then the correct option is A)

17. Correct option: (C)

Given

$$\sec^2\theta = 2 + 2\tan\theta$$

We have to find the value of

$$\tan\theta$$

Using trigonometric identity and given expression

$$1 + \tan^2\theta = 2 + 2\tan\theta$$
$$=> \tan^2\theta - 1 - 2\tan\theta = 0$$

Using quadratic equation

$$\tan\theta = \frac{-(-2) \pm \sqrt{8}}{2} = 1 \pm \sqrt{2}$$

Therefore

$$\tan\theta = 1 \pm \sqrt{2}$$

Then the correct option is C)

18. Correct option: (A)

We have to find the value of

$$(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A)$$

Using trigonometric identity and given expression

$$(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = \left(1 + \frac{\cos A}{\sin A} - \frac{1}{\sin A}\right) + \left(1 + \frac{\sin A}{\cos A} + \frac{1}{\cos A}\right)$$

$$= \left(\frac{\sin A + \cos A - 1}{\sin A}\right) + \left(\frac{\cos A + \sin A + 1}{\cos A}\right)$$

$$= \frac{(\sin A + \cos A)^2 - 1}{\sin A \cos A} = \frac{\sin^2 A + \cos^2 A + 2\sin A \cos A - 1}{\sin A \cos A} = \frac{2\sin A \cos A}{\sin A \cos A} = 2$$

Therefore $(1 + \cot A - \cos ecA)(1 + \tan A + \sec A) = 2$

Then the correct option is A)

19. Correct option: (A)

Given $x = a(\cos ec\theta + \cot\theta), y = b(\cos ec\theta - \cot\theta)$

Using option and given expression

$$xy = a(\cos ec\theta + \cot\theta) \times b(\cos ec\theta - \cot\theta)$$

$$= ab(\cos ec^2\theta - \cot^2\theta)$$

Using $\cos ec^2\theta - \cot^2\theta = 1$ we can write $xy = ab$

Then the correct option is A)

20. Correct option: (C)

Given $x = a(\cos\theta)^2, y = b(\sin\theta)^2$

Given expressions

$$\frac{x}{a} = \cos^2\theta, \frac{y}{b} = \sin^2\theta$$

Using $\cos^2\theta + \sin^2\theta = 1$ we can write

$$\frac{x}{a} + \frac{y}{b} = 1$$

Then the correct option is C)

LEVEL – II

1. Correct option: (C)

Given $\sqrt{3}\sin\theta - \cos\theta = 0$

We have to find the value of

$$\frac{\tan^2\theta + 3\tan\theta}{1 + \tan^2\theta}$$

From given expression we can write

$$\sqrt{3}\sin\theta - \cos\theta = 0 \Rightarrow \tan\theta = \frac{1}{\sqrt{3}}$$

Therefore

$$\frac{\tan^2\theta + 3\tan\theta}{1 + \tan^2\theta} = \frac{\left(\frac{1}{3} + 3 \times \frac{1}{\sqrt{3}}\right)}{1 + \frac{1}{3}} = \frac{1 + 3\sqrt{3}}{4}$$

Then the correct option is C)

2. Correct option: (*A)

Given $9\sin\theta + 4\cos\theta = 41$

We have to find the value of

$$41\cos\theta$$

From given expression we can write

$$9\sin\theta = 41 - 4\cos\theta$$

$$81\sin^2\theta = (41 - 4\cos\theta)^2$$

$$81\sin^2\theta = 1681 + 16\cos^2\theta - 328\cos\theta$$

$$\Rightarrow 0 = 97\cos^2\theta - 328\cos\theta + 1600$$

Therefore

$$D = 107584 - 620800 < 0$$

So we can say that no real solution for $\cos\theta$

3. Correct option: (D)

Given $\cot\theta = \frac{q}{p}$

We have to find the value of

$$\frac{p\sin\theta - q\cos\theta}{p\sin\theta + q\cos\theta}$$

From given expression we can write

$$\cot\theta = \frac{q}{p} => \frac{\cos\theta}{\sin\theta} = \frac{q}{p} => \frac{p - \frac{q\cos\theta}{\sin\theta}}{p + \frac{q\cos\theta}{\sin\theta}} = \frac{p - q\left(\frac{q}{p}\right)}{p + q\left(\frac{q}{p}\right)} = \frac{(p^2 - q^2)}{p^2 + q^2}$$

Therefore

$$\frac{p\sin\theta - q\cos\theta}{p\sin\theta + q\cos\theta} = \frac{p^2 - q^2}{p^2 + q^2}$$

Then the correct option is D)

4. Correct option: (A)

Given

$$15\cos A - 8\sin A = 0$$

We have to find the value of

$$\frac{\sin A + \cos A}{2\cos A - \sin A}$$

From given expression we can write

$$\frac{\cos A}{\sin A} = \frac{8}{15} => \frac{\sin A + \cos A}{2\cos A - \sin A} = \frac{1 + \frac{\cos A}{\sin A}}{\frac{2\cos A}{\sin A} - 1}$$

$$= \frac{1 + \frac{\cos A}{\sin A}}{\frac{2\cos A}{\sin A} - 1} = \frac{1 + \frac{8}{15}}{\frac{16}{15} - 1} = \frac{23}{15} \times \frac{15}{1} = 23$$

Therefore

$$\frac{\sin A + \cos A}{2\cos A - \sin A} = 23$$

Then the correct option is A)

5. Correct option: (A)

Given

$$\tan\theta + \cot\theta = 2$$

We have to find the value of θ as $\theta < 90°$

From given expression we can write

$$\tan\theta + \cot\theta = 2$$

$$\tan\theta + \frac{1}{\tan\theta} = 2$$

$$=> \tan^2\theta - 2\tan\theta + 1 = 0 => (\tan\theta - 1)^2 = 0$$

$$(\tan\theta - 1)^2 = 0 => \tan\theta = 1$$

Therefore

$$\theta < 90°, \tan\theta = 1 => \theta = 45°$$

Then the correct option is A)

6. Correct option: (A)

Given $x = a\sin\theta\,, y = b\cos\theta$

We have to find the value of

$$\frac{a^2}{x^2} - \frac{b^2}{y^2}$$

Putting the given values to find the required expression value

$$\frac{a^2}{x^2} - \frac{b^2}{y^2} = \frac{a^2}{(a\sin\theta)^2} - \frac{b^2}{(b\cos\theta)^2}$$

$$=> \frac{1}{\sin^2\theta} - \frac{1}{\cos^2\theta} = \cos ec^2\theta - \cot^2\theta$$

Using the identity $\cos ec^2\theta - \cot^2\theta = 1$

Therefore

$$\frac{a^2}{x^2} - \frac{b^2}{y^2} = \cos ec^2\theta - \cot^2\theta = 1$$

Then the correct option is A)

7. Correct option: (*B)

Given $a\cos\theta + b\sin\theta = 1, a\cos\theta - b\sin\theta = 1$

We have to find the value of $a^2 + b^2$

Solving both of the given equations

$$=> a\cos\theta + b\sin\theta + a\cos\theta - b\sin\theta = 2 => a\cos\theta = 1$$

$$=> a\cos\theta + b\sin\theta - a\cos\theta + b\sin\theta = 0 => b\sin\theta = 0$$

Therefore

$$a^2 + b^2 = \left(\frac{1}{\cos\theta}\right)^2 + b^2$$

So the given expression has many solutions.

8. Correct option: (A, B, D)

We have to find the value of $\sin 0° \sin 1° \ldots . \sin 90°$

We know that $\sin 0° = 0$

$$\sin 0° = 0 => \sin 0° \sin 1° \ldots . \sin 90° = 0$$

Therefore

$$\sin 0° \sin 1° \ldots . \sin 90° = 0 = \cos 90° = \tan 0°$$

Then the correct option is A, B, D)

9. Correct option: (A, B, C, D)

Given $\sin A + \cos A = \sqrt{2}$

Then we can write

$$\sin A = \sqrt{2} - \cos A$$

Squaring both of the side

$$\sin^2 A = \left(\sqrt{2} - \cos A\right)^2 = 2 + \cos^2 A - 2\sqrt{2}\cos A$$

$$1 - \cos^2 A = 2 + \cos^2 A - 2\sqrt{2}\cos A => 2\cos^2 A - 2\sqrt{2}\cos A + 1 = 0$$

$$\left(\sqrt{2}\cos A - 1\right)^2 = 0 => \cos A = \frac{1}{\sqrt{2}} => \sin A = \frac{1}{\sqrt{2}}$$

$$\cos A = \frac{1}{\sqrt{2}} => \sin A = \frac{1}{\sqrt{2}} => \tan A = 1, \cos A \sin A = \frac{1}{2}$$

Then the correct option is A, B, C, D)

10. Correct option: (B)

Given $(\sin\theta + \cos ec\theta)^2 + (\cos\theta + \sec\theta)^2 = k + \tan^2\theta + \cot^2\theta$

We have to find the value of k

Solving the left side of the equation

$$\sin^2\theta + \cos ec^2\theta + 2 + \cos^2\theta + \sec^2\theta + 2 = \sin^2\theta + \cos^2\theta + 4 + \sec^2\theta + \cos ec^2\theta$$

Using trigonometric identity

$$=> 5 + 1 + 1 + \tan^2\theta + \cot^2\theta = 7 + \tan^2\theta + \cot^2\theta$$

Therefore the value of k is 7

Then the correct option is B)

11. Correct option: (B, C)

Given $\tan\theta = p - \frac{1}{4p}$

We have to find the value of $\sec\theta - \tan\theta$

Squaring both side of the given equation and use the identity

$$\tan^2\theta = p^2 + \frac{1}{16p^2} - \frac{1}{2} \Rightarrow \sec^2\theta - 1 = p^2 + \frac{1}{16p^2} - \frac{1}{2}$$

$$\sec^2\theta = \left(p + \frac{1}{4p}\right)^2 \Rightarrow \sec\theta = \pm\left(p + \frac{1}{4p}\right)$$

Therefore

$$\sec\theta - \tan\theta = \left(p + \frac{1}{4p}\right) - \left(p - \frac{1}{4p}\right) = \frac{1}{2p} \text{ or } -\left(p + \frac{1}{4p}\right) - \left(p - \frac{1}{4p}\right) = -2p$$

Then the correct option is B, C)

12. Correct option: (C)

Given $\triangle ABC, \angle B = 90°, AB = x, BC = y$

We have to find the value of $\sin A$

Using the basic definition of the trigonometric functions

$$\sin A = \frac{opposite\ side}{hypotenuse} = \frac{y}{\sqrt{x^2 + y^2}}$$

Then the correct option is C)

13. Correct option: (B)

Given $\triangle ABC, \angle B = 90°, AB = x, BC = y$

We have to find the value of $\sin C$

Using the basic definition of the trigonometric functions

$$\sin C = \frac{opposite\ side}{hypotenuse} = \frac{x}{\sqrt{x^2 + y^2}}$$

Then the correct option is B)

14. Correct option: (D)

Given $\triangle ABC, \angle B = 90°, AB = x, BC = y$

We have to find the value of $\sec^2\theta - \tan^2\theta$

Above required expression is an identity so we can simply write

$$\sec^2\theta - \tan^2\theta = 1\ (\theta \neq (2n+1)90°)$$

Then the correct option is D)

15. Correct Matching is: a-R, b-S, c-P, T, d-Q

Given $m = \cot\theta + \cos\theta, n = \cot\theta - \cos\theta$

a) $\frac{m+n}{m-n}$

Putting the value of m and n

$$\frac{m+n}{m-n} = \frac{\cot\theta + \cos\theta + \cot\theta - \cos\theta}{\cot\theta + \cos\theta - \cot\theta + \cos\theta} = \frac{\cot\theta}{\cos\theta} = \frac{1}{\sin\theta} = \cos ec\theta$$

a) -→ R)

b) $m^2 - n^2$

Putting the value of m and n

$$m^2 - n^2 = (\cot\theta + \cos\theta)^2 - (\cot\theta - \cos\theta)^2 = 4\cot\theta\cos\theta$$

b) -→ S)

c) $2(m^2 + n^2)$

Putting the value of m and n

$$2(m^2 + n^2) = 2((\cot\theta + \cos\theta)^2 + (\cot\theta - \cos\theta)^2) = 4(\cot^2\theta + \cos^2\theta)$$

c) -→ P,T)

d) $m^2 + n^2 + 2mn = (m+n)^2$

Putting the value of m and n

$$m^2 + n^2 + 2mn = (m+n)^2 = (\cot\theta + \cos\theta + \cot\theta - \cos\theta)^2 = (2\cot\theta)^2 = 4\cot^2\theta$$

d) -→ Q)

16. Correct Answer is: (2)

Given $\frac{\cos A}{1+\sin A} + \frac{\cos A}{1-\sin A} = k\sec A$

We have to find the value of k

Solving the given expression

$$\frac{\cos A\,(1-\sin A) + \cos A\,(1+\sin A)}{(1+\sin A)(1-\sin A)} = \frac{2\cos A}{1-\sin^2 A} = \frac{2\cos A}{\cos^2 A} = \frac{2}{\cos A} = 2\sec A$$

Comparing the given $k\sec A = 2\sec A \Rightarrow k = 2$

Therefore the value of k is 2.

17. Correct Answer is: (2)

We have to find the value of $(\sin A - \cos A)^2 + (\sin A + \cos A)^2$

Solving the given expression

$$(\sin A - \cos A)^2 + (\sin A + \cos A)^2 = \sin^2 A + \cos^2 A - 2\sin A\cos A + \sin^2 A + \cos^2 A + 2\sin A\cos A$$

$$\Rightarrow 2(\sin^2 A + \cos^2 A) = 2$$

Therefore the value of the given expression is 2.

18. Correct Answer is: (5)

Given $\cos\alpha + \cos^2\alpha = 1$

We have to find the value of $2\sin^4\alpha + 2\sin^2\alpha + 3$

Solving the given expression $\cos\alpha + \cos^2\alpha = 1 \Rightarrow \cos\alpha = 1 - \cos^2\alpha = \sin^2\alpha$

Putting the above value in required expression we get

$$2\sin^4\alpha + 2\sin^2\alpha + 3 = 2\cos^2\alpha + 2\cos\alpha + 3 = 2(\cos^2\alpha + \cos\alpha) + 3$$

$$= 2 + 3$$

Therefore the value of the given expression is 5.

19. Correct Answer is: (2)

We have to find the value of

$$(\sin^4\theta - \cos^4\theta + 1)\cos ec^2\theta$$

Solving the given expression

$$(\sin^4\theta - \cos^4\theta + 1)\cos ec^2\theta = \frac{(\sin^2\theta + \cos^2\theta)(\sin^2\theta - \cos^2\theta) + 1}{\sin^2\theta}$$

$$\Rightarrow \frac{(\sin^2\theta - \cos^2\theta) + 1}{\sin^2\theta} = \frac{\left(\sin^2\theta - (1 - \sin^2\theta)\right) + 1}{\sin^2\theta} = \frac{2\sin^2\theta}{\sin^2\theta} = 2$$

Therefore the value of the given expression is 2.

20. Correct Answer is: (1)

We have to find the value of

$$(\sin^6\theta + \cos^6\theta + 3\sin^2\theta\,.\cos^2\theta)$$

Solving the given expression

$$(\sin^6\theta + \cos^6\theta + 3\sin^2\theta\,.\cos^2\theta) = (\sin^2\theta)^3 + (\cos^2\theta)^3 + 3\sin^2\theta\,.\cos^2\theta\,(\sin^2\theta + \cos^2\theta)$$

$$= (\sin^2\theta + \cos^2\theta)^3 = 1^3$$

Therefore the value of the given expression is 1.

STUDENT PRACTICE QUESTIONS

1. Correct option: (A)

Given $\tan\theta = \frac{1}{\sqrt{7}}$

We have to find the value of

$$\frac{\cos ec^2\theta - \sec^2\theta}{\cos ec^2\theta + \sec^2\theta}$$

Using the definition of $\tan\theta$

$$\tan\theta = \frac{opposite\ side}{Adjacent\ side} = \frac{1}{\sqrt{7}} => hypotenuse = \sqrt{8}$$

So we can write

$$\theta < 90° => \cos ec\theta = \frac{hypotenuse}{opposite\ side} = \frac{\sqrt{8}}{1}, \sec\theta = \frac{hypotenuse}{Adjacent\ side} = \frac{\sqrt{8}}{\sqrt{7}}$$

Therefore

$$\frac{\cos ec^2\theta - \sec^2\theta}{\cos ec^2\theta + \sec^2\theta} = \frac{\left(8 - \frac{8}{7}\right)}{8 + \frac{8}{7}} = \frac{48}{64} = \frac{3}{4}$$

Then the correct option is A)

2. Correct option: (C)

Given $\sin\theta = \frac{1}{2}$

We have to find the value of $\cos\theta \,.\, \tan\theta \,.\, \sec\theta \,.\, \cos ec\theta \,.\, \cot\theta$

Using the relationship between trigonometry function we can write

$$\cos\theta \,.\, \tan\theta \,.\, \sec\theta \,.\, \cos ec\theta \,.\, \cot\theta = \cos\theta \,.\, \tan\theta \,.\, \frac{1}{\cos\theta} \,.\, \cos ec\theta \,.\, \frac{1}{\tan\theta}$$

$$= \cos ec\theta$$

So we can write

$$\cos\theta \,.\, \tan\theta \,.\, \sec\theta \,.\, \cos ec\theta \,.\, \cot\theta = \cos ec\theta = \frac{1}{\sin\theta} = 2$$

Then the correct option is C)

3. Correct option: (C)

We have to find the value of

$$\frac{1}{\cos\theta} - \cos\theta$$

Expanding and solving the terms

$$\frac{1}{\cos\theta} - \cos\theta = \frac{1 - \cos^2\theta}{\cos\theta}$$

We know that

$$\sin^2\theta + \cos^2\theta = 1 => \sin^2\theta = 1 - \cos^2\theta$$

Therefore

$$= \frac{1 - \cos^2\theta}{\cos\theta} = \frac{\sin^2\theta}{\cos\theta} = \sin\theta \,.\, \tan\theta$$

Then the correct option is C)

4. Correct option: (A, C)

Given

$$8\sin\theta = 4 + \cos\theta$$

We have to find the value of

$$\sin\theta$$

Using trigonometric identity and given expression

$$8\sin\theta - 4 = \cos\theta$$

Squaring both side

$$64\sin^2\theta + 16 - 64\sin\theta = \cos^2\theta = 1 - \sin^2\theta$$
$$65\sin^2\theta + 15 - 64\sin\theta$$

Using quadratic equation

$$\sin\theta = \frac{-(-64) \pm \sqrt{4096 - 3900}}{130} = \frac{64 \pm 14}{130} = \frac{50}{130}, \frac{78}{130}$$

Therefore

$$\sin\theta = \frac{5}{13}, \frac{3}{5}$$

Then the correct option is A, C)

5. Correct option: (A)

Given $x = a(\cos ec\theta + \cot\theta)^2, y = b(\cos ec\theta - \cot\theta)^2$

We have to find the value of x^2y^2

Given expression

$$x^2y^2 = (a(\cos ec\theta + \cot\theta)^2)^2 \times (b(\cos ec\theta - \cot\theta)^2)^2$$
$$= a^2b^2(\cos ec^2\theta - \cot^2\theta)^4$$

Using $\cos ec^2\theta - \cot^2\theta = 1$ we can write $x^2y^2 = a^2b^2$

Then the correct option is A)

6. Correct option: (B)

Given

$$5\cos\theta + 12\sin\theta = 13$$

We have to find the value of

$$\sin\theta$$

We can write the given equation as

$$5\cos\theta = 13 - 12\sin\theta$$

Squaring both side

$$25\cos^2\theta = 169 + 144\sin^2\theta - 312\sin\theta$$
$$25(1 - \sin^2\theta) = 169 + 144\sin^2\theta - 312\sin\theta$$
$$=> 169\sin^2\theta - 312\sin\theta + 144 = 0$$

Using quadratic formula

$$\sin\theta = \frac{312 \pm \sqrt{97344 - 97344}}{2 \times 169} = \frac{312}{2 \times 169} = \frac{12}{13}$$

Therefore

$$5\cos\theta + 12\sin\theta = 1 => \sin\theta = \frac{12}{13}$$

Then the correct option is B)

8. Mensuration

Solutions

LEVEL – I

1. Correct option: (A)

Given the ratio of sides 3:5:7

Let's our side 3x, 5x and 7x with given perimeter as 300m.

Then we can write $3x + 5x + 7x = 300 \Rightarrow 15x = 300$

$$x = 20$$

So length of the sides is 60m, 100m and 140 m

To find the area of the triangle we can use Heron's formula

$$A = \sqrt{[s(s-a)(s-b)(s-c)]}$$

Where A is area and a, b, c are the length of the sides of a triangle with semi perimeter s.

Then

$$A = \sqrt{[150(150-60)(150-100)(150-140)]}$$

$$A = \sqrt{150(90)(50)(10)} = 1500\sqrt{3}$$

Therefore the correct option is A)

2. Correct option: (A)

Perimeter of an equilateral triangle (P) = 3 × a

Given area (A) = $24\sqrt{3}$

Side length (a) = $\sqrt{A \times \left(\frac{4}{\sqrt{3}}\right)}$

$$a = \sqrt{24\sqrt{3} \times \left(\frac{4}{\sqrt{3}}\right)}$$

$$a = \sqrt{96}$$

$$a = 4\sqrt{6}$$

Then

$$P = 12\sqrt{6}$$

Therefore the correct option is A)

3. Correct option: (D)

Area (A) = 39 cm²

Height (h) = 13 cm

$$Base\ (b) = \frac{2 \times A}{h}$$

$$b = \frac{2 \times 39\ cm^2}{13\ cm} cm$$

$$b = \frac{78\ cm^2}{13\ cm} cm$$

$$b = 6\ cm$$

So, the base of the triangle is 6 cm.

Therefore the correct option is D)

4. Correct option: (C)

Area (A) of a triangle can be calculated using the formula:

$$A = \left(\frac{1}{2}\right) \times base \times height$$

In this case, you are given:

BC = 8 cm (base)

Altitude from A to BC = 4 cm (height)
Now, we can calculate the area:

$$A = \left(\frac{1}{2}\right) \times 8\,cm \times 4\,cm$$

$$A = \left(\frac{1}{2}\right) \times 32\,cm^2$$

$$A = 16\,cm^2$$

So, the area of triangle ABC is 16 square centimeters.
Therefore the correct option is C)

5. Correct option: (B)

The formula to calculate the area (A) of a trapezium is:

$$A = \left(\frac{1}{2}\right) \times (a + b) \times (h)$$

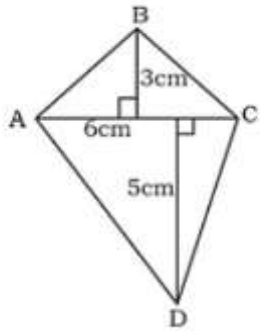
Where a and b are length of the parallel sides and h is the height.
Here a = 7, b = 9 and h = 8

Then area $= \frac{1}{2} \times (7 + 9) \times 8 = 64$

Therefore the correct option is B)

6. Correct option: (A)

We can find the area of the quadrilateral as sum of both the triangles ABC and ACD



For triangle ABC $area = \frac{1}{2} \times 3 \times 6 = 9\,cm^2$

For triangle ACD $area = \frac{1}{2} \times 5 \times 6 = 15\,cm^2$

So area of the quadrilateral ABCD is $9cm^2 + 15cm^2 = 24cm^2$
Therefore the correct option is A)

7. Correct option: (C)

To find the area of the parallelogram:
1. Calculate the semi-perimeter (s) of one of the triangles formed by the diagonal:

$$s = \frac{5\,cm + 3.5\,cm + 6.5\,cm}{2}$$

$$s = 7.5\,cm$$

2. Use Heron's formula to find the area (A) of one triangle:

$$A = \sqrt{[s(s - 5\,cm)(s - 3.5\,cm)(s - 6.5\,cm)]}$$

$$A = \sqrt{[7.5\,cm \times 2.5\,cm \times 4\,cm \times 1\,cm]}$$

$$A = 5\sqrt{3}\,cm^2$$

3. Double the area of one triangle to find the area of the parallelogram:

$$Area\ of\ parallelogram = 2 \times A$$

$$Area\ of\ parallelogram = 2 \times 5\sqrt{3}\,cm^2$$

$$Area\ of\ parallelogram = 10\sqrt{3}\,cm^2$$

Therefore the correct option is C)

8. Correct option: (D)

Given:

AB = 16 cm

AE = 8 cm

CF = 10 cm

In parallelogram ABCD, opposite sides are equal in length

So CD = AB = 16 cm.

To find the length of AD, we can use the formula for the area of a parallelogram:

Area of a parallelogram = Base × Corresponding altitude

In this case, CD and AD act as the base, and AE ⊥ CD and CF ⊥ AD are the corresponding altitudes.

Then

Area of parallelogram ABCD = CD × AE

Area of parallelogram ABCD = AD × CF

Now, solve for AD:

$$AD = \frac{CD \times AE}{CF}$$

$$AD = \frac{16\ cm \times 8\ cm}{10\ cm}$$

$$AD = \frac{128\ cm^2}{10\ cm}$$

$$AD = 12.8\ cm$$

Therefore the correct option is D)

9. Correct option: (B)

Given: ∠CDB = ∠ABD = 90°.

Also, DC = AB = 2.5 cm

Now, we know that a quadrilateral is a parallelogram if a pair of opposite sides is parallel and is of equal length. Therefore,

Quadrilateral ABCD is a parallelogram.

Now, to find the area of the parallelogram ABCD, we can use the formula:

$$Area = Base \times Corresponding\ Altitude$$

Therefore,

$$Area\ of\ parallelogram\ ABCD = AB \times BD$$

$$Area\ of\ parallelogram\ ABCD = 2.5\ cm \times 4\ cm$$

$$Area\ of\ parallelogram\ ABCD = 10\ square\ cm$$

Therefore the correct option is B)

10. Correct option: (C)

Let the rectangle have length l and breadth b.

Then we can write as

$$2(l + b) = 28 => l + b = 14$$

$$l \times b = 48$$

Using above equations we can write

$$(l - b)^2 = (l + b)^2 - 4lb = 196 - 192 = 4 = 2^2$$

$$=> (l - b) = \pm 2$$

So values are

$$l + b = 14, l - b = 2 => l = 8, b = 6$$

$$l + b = 14, l - b = -2 => l = 6, b = 8$$

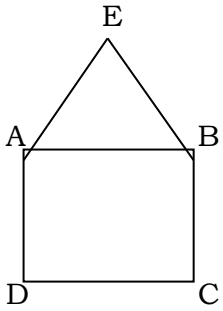
Then diagonal can be find using the pythagorean theorem

$$diagonal = \sqrt{6^2 + 8^2} = \sqrt{10^2} = 10$$

Therefore the correct option is C)

11. Correct option: (D)

Area of figure can be calculated as: $A = area\ of\ square\ ABCD + area\ of\ triangle\ AEB$



Area of square ABCD = $8^2 cm^2 = 64\ cm^2$

For triangle AEB

Side AE = BE and perimeter is 18 cm.

Then sides are AB = 8 , $AE = BE = \frac{18-8}{2} = 5\ cm$

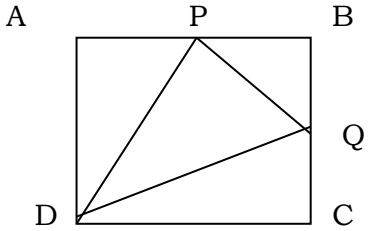
So area of triangle AEB

$$A' = \sqrt{9(9-5)(9-5)(9-8)} = \sqrt{9 \times 4 \times 4 \times 1} = 3 \times 4 = 12\ cm^2$$

Then $A = 64\ cm^2 + A' = 64\ cm^2 + 12\ cm^2 = 76\ cm^2$

Therefore the correct option is D)

12. Correct option: (A)



To find the area of the triangle DPQ we can subtract the area of all three right angle triangle PAD, PBD and QCD from area of square ABCD

Area of square is 1 m^2 , so all sides of square will be of 1 m.

Then

Area of triangle PBQ = $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$

Area of triangle APD = $\frac{1}{2} \times \frac{1}{2} \times 1 = \frac{1}{4}$

Area of triangle QCD = $\frac{1}{2} \times \frac{1}{2} \times 1 = \frac{1}{4}$

Then area of triangle DPQ

$$= 1 - \frac{1}{8} - \frac{1}{4} - \frac{1}{4} = 1 - \frac{5}{8} = \frac{3}{8}$$

Therefore the correct option is A)

13. Correct option: (B)

Given that area of a parallelogram is $x^2 - 4$ with base $x + 4$ and altitude to the base is $x - 3$

Then we can write the equation as: $x^2 - 4 = (x + 4)(x - 3)$

Because area of the parallelgram can be calculated as product of base and altitude to the base.

$$x^2 - 4 = x^2 + 4x - 3x - 12$$

$$12 - 4 = x \Rightarrow x = 8$$

Then actual area of the parallelogram is

$$x^2 - 4 = 64 - 4 = 60\ sq.units$$

Therefore the correct option is B)

14. Correct option: (B)

The area (A) of a rhombus can be calculated using the formula:

$$A = \left(\frac{1}{2}\right) \times d_1 \times d_2$$

In this case, we are given:

A = 25 sq. cm (Area of the rhombus)

d_1 = 10 cm (Length of one diagonal)

Now, let's solve for d_2 using the formula:

$$25\ sq.cm = \left(\frac{1}{2}\right) \times 10\ cm \times d_2$$

$$d_2 = 5\ cm$$

So, the length of the other diagonal d_2 is 5 cm.

In a rhombus, all four sides are equal in length. Since we know the lengths of both diagonals, we can use the properties of a rhombus to find the side length (s) using the Pythagorean theorem.

$$s^2 = (0.5 \times 10\ cm)^2 + (0.5 \times 5\ cm)^2$$

$$s^2 = (5\ cm)^2 + (2.5\ cm)^2$$

$$s^2 = 25\ cm^2 + 6.25\ cm^2$$

$$s^2 = 31.25\ cm^2$$

$$s = \sqrt{31.25}\ cm \approx 5.59\ cm \approx 5.6\ cm$$

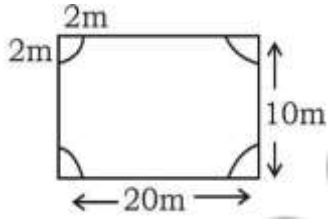
Therefore the correct option is B)

15. Correct option: (C)

Given:

Four quadrants form the corners of the shape.

Each quadrant has a radius of 2 meters.



1. Calculate the perimeter of the four quadrants:

$$Perimeter\ of\ 4\ quadrants = 4 \times \left(\frac{1}{4}\right) \times 2 \times 2\pi = 4\pi\ meters$$

2. Calculate the horizontal portions' perimeter:

$$Horizontal\ portions'\ perimeter = 2 \times (20 - 4) = 32\ meters$$

3. Calculate the vertical portions' perimeter:

$$Vertical\ portions'\ perimeter = 2 \times (10 - 4) = 12\ meters$$

4. Find the total perimeter:

$$Total\ perimeter = 4\pi + 32 + 12\ meters$$

$$= 12.57 + 44 = 56.57 \approx 56.6$$

Therefore the correct option is C)

16. Correct option: (D)

Let A be the area of the circle and B be the area of the square. According to the given information, A = B.

Since A = B, we have:

$$\pi r^2 = s^2$$

Now, let's find the ratio of their perimeters:

Perimeter of the circle = 2πr

Perimeter of the square = 4s

Now, calculate the ratio of their perimeters:

$$Ratio\ of\ perimeters\ = = \frac{2\pi r}{4s}$$

We can simplify this ratio further by using the relationship A = B:

$$s\ = \sqrt{\pi r^2}$$

$$s\ = r\sqrt{\pi}$$

Now, substitute this value of s into the ratio of perimeters:

$$Ratio\ of\ perimeters\ = \frac{2\pi r}{4s}\ = \frac{2\pi r}{\left[4\left(r\sqrt{\pi}\right)\right]} = \frac{\sqrt{\pi}}{2}$$

Therefore the correct option is D)

17. Correct option: (C)

To find the area of a sector, we can use the formula:

$$Area\ of\ Sector\ = \left(\frac{\theta}{360}\right) \times\ \pi\ \times\ r^2$$

In this case, we have:

$$\theta\ =\ 144°$$

$$r\ =\ 28\ cm$$

Now, plug these values into the formula:

$$Area\ of\ Sector\ = \left(\frac{144}{360}\right) \times \pi\ \times (28\ cm)^2$$

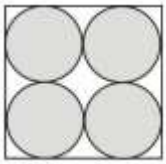
$$Area\ of\ Sector\ = \left(\frac{2}{5}\right) \times \pi\ \times (784\ cm^2) = \frac{2}{5} \times \frac{22}{7} \times 784\ cm^2$$

$$= 985.6\ cm^2$$

Therefore the correct option is C)

18. Correct option: (C)

To find the area of region which is not coverd by the coins we can subtract the area of all four coins from the area of the square.



Given that $\frac{A_c}{P_c} = \frac{P_c}{A_c}$

From above we can write

$$\frac{\pi r^2}{2\pi r} = \frac{2\pi r}{\pi r^2} => r^4 = 4r^2 => r = 0, 2$$

Then side of the square is

$$s = 4r = 8$$

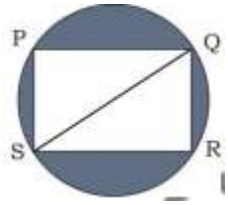
Then area of the required region

$$= 8^2 - 4 \times \pi \times r^2 = 64 - 16\pi = 16(4 - \pi)$$

Therefore the correct option is C)

19. Correct option: (A)

To find the area of shaded region we can subtract the area of the rectangle from the area of the circle.



Form above we can conclude that SQ is diameter of the circle and to find the value of SQ we can use the right angle triangle SPQ as

$$SQ^2 = 6^2 + 8^2 = 10^2 => SQ = 10$$

Radius of the circle is 5 cm.

Area of rectangle is $6 \times 8 = 48$

Then area of the shaded region will be

$$A = \pi \times r^2 - 48 = 25\pi - 48 = 78.57 - 48 = 30.57\ cm^2$$

Therefore the correct option is A)

20. Correct option: (C)

Given: Radius of the sector = 7 cm

Area of the sector = 70 sq.cm

The formula for the perimeter of a sector is:

$$P = \left(\frac{\theta}{360}\right) \times 2\ \pi r$$

Now, let's solve for the unknown:

$$70 = \left(\frac{\theta}{360}\right) \times\ \pi \times r^2 = \left(\frac{\theta}{360}\right) \times \frac{22}{7} \times 7 \times 7$$

$$\frac{10}{22} = \frac{\theta}{360}$$

Then P is

$$P = \frac{10}{22} \times 2 \times \frac{22}{7} \times 7 = 20\ cm$$

Therefore the perimeter of the sector

$$P + 2r = 20 + 14 = 34\ cm$$

Therefore the correct option is C)

LEVEL-II

1. Correct option: (C)

Given:

Radius of the sector = 7 cm

Area of the sector = 70 sq.cm

The formula for the perimeter of a sector is:

$$P = \left(\frac{\theta}{360}\right) \times 2\ \pi r$$

Now, let's solve for the unknown:

$$70 = \left(\frac{\theta}{360}\right) \times\ \pi \times r^2 = \left(\frac{\theta}{360}\right) \times \frac{22}{7} \times 7 \times 7$$

$$\frac{10}{22} = \frac{\theta}{360}$$

Then P is

$$P = \frac{10}{22} \times 2 \times \frac{22}{7} \times 7 = 20\ cm$$

Therefore the perimeter of the sector

$$P + 2r = 20 + 14 = 34\ cm$$

Therefore the correct option is C)

2. Correct option: (C)

Given: The diagonal of the square = $12\sqrt{2}$ cm.

Let's denote: The side length of the square as "s" cm.

The side length of the equilateral triangle as "a" cm.

The perimeter of both the square and the triangle as "P" cm.

According to the given information, the perimeters are equal:

$$P_s = P_t$$

Now, we know that the diagonal of the square divides it into two congruent right triangles. Using one right angle triangle we can write

$$s^2 = \left(\frac{1}{2}\right) \times \left(12\sqrt{2}\right)^2 => s = 12\ cm$$

Now that we know the side length of the square, we can find the perimeter of the square:

$$P_s = 48\ cm$$

Now, we can equate the perimeters again:

$$48\ cm = 3a$$

$$a = 16\ cm$$

$$Area_{triangle} = \left(\frac{\sqrt{3}}{4}\right) \times a^2$$

$$Area_{triangle} = \left(\frac{\sqrt{3}}{4}\right) \times 256 = 64\sqrt{3}$$

Therefore the correct option is C)

3. Correct option: (B)

Given:

Sides of the rectangle: 10m and 12m.

Base of the parallelogram: 20m.

Areas of the rectangle and parallelogram are equal.

First, let's find the area of the rectangle:

$$Area\ of\ rectangle = 10m \times 12m = 120\ m^2$$

Now, since the areas of the rectangle and parallelogram are equal, the area of the parallelogram is also 120 square meters.

The formula for the area of a parallelogram is:

$$Area\ of\ parallelogram = base \times height$$

Now, solve for the height: $120 = 20m \times height$

Divide both sides by 20m to find the height:

$$height = 6\ meters$$

So, the altitude of the parallelogram is 6 meters.

Therefore the correct option is B)

4. Correct option: (C)

Given: Area of the rhombus = 128 cm^2

Perimeter of the rhombus = 32 cm

Let's denote the side length of the rhombus as "s" cm.

$$P = 4 \times s$$

Now, let's find the perimeter in terms of the side length:

$$P = 4 \times s = 32$$

$$4 \times s = 32$$

$$s = \frac{32}{4}$$

$$s = 8$$

Now, let's find the altitude (h) of the rhombus using the side length (s) and the area. The altitude (h) can be found using the formula: $h = \frac{A}{s}$

$$h = \frac{128}{8} = 16\ cm$$

So, the altitude of the rhombus is 16 cm.

Therefore the correct option is C)

5. Correct option: (A)

Given: Perimeter of the circle = Circumference of circle = 88 cm

Then

$$2\pi r = 88 => r = \frac{88}{2} \times \frac{7}{22} = 14$$

Now area of the circle

$$\pi r^2 = \frac{22}{7} \times 14 \times 14 = 616\ sq.cm$$

Therefore the correct option is A)

6. Correct option: (B)

Let's the radius of the smaller circle is a and bigger circle is b.

Given:

$$b - a = 7$$
$$\pi b^2 - \pi a^2 = 1078$$

Then

$$\pi b^2 - \pi a^2 = 1078 => (b^2 - a^2) = \frac{1078}{\pi} = 1078 \times \frac{7}{22} = 49 \times 7$$

Now solving the equations

$$(b - a)(b + a) = 7^3 => (b + a) = 7^2$$

Then

$$b - a = 7, b + a = 49 => b = 28, a = 21$$

a = 7 which is value of radius of smaller circle.

Therefore the correct option is B)

7. Correct option: (D)

Let's the radius of the smaller (inner) circle is a and bigger(outer) circle is b.

Given:

$$a = 41, b = 55$$

Then area of ring

$$A = \pi b^2 - \pi a^2 = \frac{22}{7}(55^2 - 41^2) = \frac{22}{7} \times 14 \times 96 = 4224$$

Therefore the correct option is D)

8. Correct option: (B)

Given:

$$radius = 105, angle = 72°$$

Then perimeter of paddy growing which is a sector of angle 72° will be sum of perimeter of sector and double of its radius.

$$P = \left(\frac{\theta}{360}\right) \times 2\ \pi r + 2r = P = \left(\frac{72}{360}\right) \times 2 \times \frac{22}{7} \times 105 + 210 = 2 \times 22 \times 3 + 210 = 132 + 210 = 342$$

Therefore the correct option is B)

9. Correct option: (A, C)

Given: $h = 2b, area = 400\ cm^2$

Area of the triangle in form of base and height

$$= \frac{1}{2} \times b \times h = \frac{bh}{2}$$

Equating the equations

$$\frac{bh}{2} = \frac{b \times 2b}{2} = b^2 = 400 => b = 20$$

Then $h = 2b = 40 = \sqrt{1600}$

Therefore the correct option is A, C)

10. Correct option: (A)

Given: $l = 20, b = 15\ (for\ rectangle), base = 30\ (for\ triangle)$

Area of the triangle in form of base and height

$$= \frac{1}{2} \times b \times h = 15h$$

We are given that area of both body is same

Then we can write

$$20 \times 15 = 15h => h = 20$$

The height of the triangle whose area is equal to the given rectangle is 20cm.

Therefore the correct option is A)

11. Correct option: (A, B)

Given: $r = 7\ cm, angle = 240°$

Then length of the bonding arc will be calculated using the given formula where θ is angle of arc.

$$P = \left(\frac{\theta}{360}\right) \times 2\,\pi r = P = \left(\frac{240}{360}\right) \times 2 \times \frac{22}{7} \times 7 = \frac{88}{3} cm$$

$$= \frac{88}{3} = 29\frac{1}{3}$$

Therefore the correct option is A, B)

12. Correct option: (A)

Let's take AE = AF = x

Then

$$x^2 + x^2 = \left(\sqrt{2}\right)^2 = 2 => x^2 = 1 => x = 1\ m$$

So value of AF = 1 m

Therefore the correct option is A)

13. Correct option: (D)

Let's take AE = AF = x

Then

$$x^2 + x^2 = \left(\sqrt{2}\right)^2 = 2 => x^2 = 1 => x = 1\ m$$

So value of AF = 1 m

Similarly CH = CG = 1 m

To find the value of FG use right angle triangle FBG

$$FG^2 = FB^2 + BG^2 = 7^2 + 7^2 = 2 \times 7^2$$

$$FG = \sqrt{2 \times 7^2} = 7\sqrt{2}$$

Therefore the correct option is D)

14. Correct option: (B)

Area of the rectangle EFGH

$$= FG \times AF$$

As we know the value of FG and AF then area

$$= 7\sqrt{2} \times \sqrt{2} = 14\ m^2$$

Therefore the correct option is B)

15. Correct Matching is: a-S, b-P, c-Q, d-R

Here's the matching of the polygons (Column I) with the corresponding formulas for area (Column II):

a) Square → S) a^2

b) Rhombus → P) $\frac{1}{2}\ d_1 d_2$ (This formula calculates the area of a rhombus using its diagonals.)

c) Trapezium → Q) $\frac{1}{2}(h_1 + h_2)d$ (This formula calculates the area of a trapezium using its heights and one base.)
d) Parallelogram → R) bh (This formula calculates the area of a Parallelogram using its base and height).

16. Correct Matching is: a-S, b-T, c-P, d-S, e-R

a) Side of a square is 12 cm, then its area is
- Match: S) 144 sq. cm
- Explanation: The area of a square is calculated using the formula $Area = Side^2$. In this case, Area = 12 cm × 12 cm = 144 sq. cm.

b) h = 12 cm, b = 6 cm, then the area of the parallelogram is
- Match: T) 72 sq. cm
- Explanation: The area of a parallelogram is calculated using the formula Area = $base \times height$. In this case, $Area = 6\,cm \times 12\,cm = 72\,sq.cm$.

c) Perimeter of a square is 88 cm, then the side is
- Match: P) 22 cm
- Explanation: The perimeter of a square is calculated as $Perimeter = 4 \times Side$.

In this case, 88 cm = 4 × Side, so $Side = \frac{88\,cm}{4} = 22\,cm$.

d) The perimeter of a rhombus whose side is a cm is
- Match: S) 4a
- Explanation: The perimeter of a rhombus is calculated as Perimeter = 4 × Side. In this case, the side is represented as "a," so the perimeter is 4a.

e) The perimeter of a parallelogram if sides are 3 cm and 2 cm
- Match: R) 10 cm
- Explanation: $P = 2(3 + 2) = 10$

17. Correct Answer is: (5)

Let's denote the base of the parallelogram as "b" and the height as "h." According to the given information, the ratio of the base to the height is 1:5, which can be written as:

$$\frac{b}{h} = \frac{1}{5}$$

We're also given that the area of the parallelogram is 125 cm². The formula for the area (A) of a parallelogram is: $125\,cm^2 = b \times h$

$$b = \left(\frac{1}{5}\right)h$$

$$125\,cm^2 = \left[\left(\frac{1}{5}\right)h\right] \times h$$

$$125\,cm^2 = \left(\frac{1}{5}\right)h^2$$

$$h^2 = 125\,cm^2 \times 5$$

$$h^2 = 625\,cm^2$$

Now, take the square root of both sides to find h: $h = \sqrt{625\,cm^2}$

$$h = 25\,cm$$

So, the height of the parallelogram is 25 cm.

Then $b = \left(\frac{1}{5}\right)h$

$$b = \left(\frac{1}{5}\right) \times 25\,cm$$

$$b = 5\,cm$$

Therefore, the base of the parallelogram is 5 cm.

18. Correct Answer is: (9)

We can find the breadth (width) of the rectangle by using the formula for the area of a rectangle, which is: $Area = Length \times Breadth$

In this case, you're given the area (144 cm²) and the length (16 cm). We need to solve for the breadth (B): $144\ cm^2 = 16\ cm \times B$

$$B = \frac{144\ cm^2}{16\ cm}$$

$$B = 9\ cm$$

So, the breadth of the rectangle is 9 cm.

19. Correct Answer is: (4)

To find the side length of a square when we know the perimeter, we can use the formula for the perimeter of a square, which is: $Perimeter = 4 \times Side$

In this case, we're given the perimeter as 16 cm, so we can set up the equation:

$$16\ cm = 4 \times Side$$

$$Side = \frac{16\ cm}{4}$$

$$Side = 4\ cm$$

So, the side of the square is 4 cm.

20. Correct Answer is: (6)

The sides of a triangle is x, x+1, 2x-1 and its area is $x\sqrt{10}\ sq.units$

To find the value of x we can use the heron's formula as

Value of semi perimeter

$$s = \frac{x + x + 1 + 2x - 1}{2} = 2x$$

$$A = \sqrt{2x(2x - x)(2x - x - 1)(2x - 2x + 1)} = \sqrt{2x^2(x - 1)}$$

The value of area is given as $x\sqrt{10}$

Equating the areas as

$$\sqrt{2x^2(x - 1)} = x\sqrt{10} => 2x^2(x - 1) = x^2 \times 10$$

$$(x - 1) = 5 => x = 6$$

Then the value of x is 6 units.

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)

Given:

Side a = 6 cm

Side b = 12 cm

Perimeter (p) = 26 cm

Then c:

c = p - (a + b)

c = 8 cm

Now, use Heron's formula to find the area (A):

Semi-perimeter (s) = 13

$$A = \sqrt{[13(13 - 6)(13 - 12)(13 - 8)]}$$

$$A = \sqrt{[13(7)(1)(5)]}$$

$$A = \sqrt{455}$$

Therefore the correct option is C)

2. Correct option: (B)

In a right isosceles triangle, the two legs are congruent, and the hypotenuse is $\sqrt{2}$ times the length of one of the legs.

Given the hypotenuse: $16\sqrt{2}$ cm

Then

$$a = \frac{16\sqrt{2}}{\sqrt{2}}$$

$$a = 16\,cm$$

$$A = \left(\frac{1}{2}\right) \times base \times height$$

In this case, the base and height are both equal to the length of one leg, which is 16 cm:

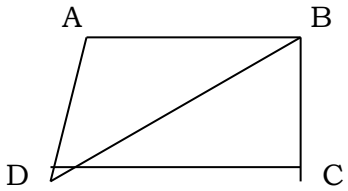
$$A = \left(\frac{1}{2}\right) \times 16\,cm \times 16\,cm$$

$$A = \left(\frac{1}{2}\right) \times 256\,cm^2$$

$$A = 128\,cm^2$$

Therefore the correct option is B)

3. Correct option: (B)



To find the area of park we can add area of the triangle ABD and BCD

To find the area of triangle BCD we can use the formula

$$= \frac{1}{2} \times base \times height = \frac{1}{2} \times DC \times BC = \frac{1}{2} \times 12 \times 5 = 30$$

as triangle BCD is a right angle triangle we can find the length of BD as

$$BD = \sqrt{12^2 + 5^2} = \sqrt{169} = 13$$

To find the area of triangle ABD we can use Heron's formula

$$s = \frac{9 + 13 + 8}{2} = 15$$

$$A = \sqrt{15(15-9)(15-8)(15-13)} = \sqrt{15 \times 6 \times 7 \times 2} = 6\sqrt{35}$$

$$= 35.46$$

Then area of park is $= 30 + 35.46 = 65.46 \approx 65.5\,m^2$

Therefore the correct option is B)

4. Correct option: (A)

The area (A) of a parallelogram can be calculated using the formula:

$$A = base \times altitude$$

In this case, we are given:

Base = 8 meters

Altitude = 4.5 meters

Then

$$A = 8\,meters \times 4.5\,meters$$

$$A = 36\,m^2$$

Therefore the correct option is A)

5. Correct option: (A)

Let the original square have side length "s."

The area of the original square = s^2

From the given conditions, the dimensions of the new rectangle are 0.9s and 1.1s.
The area of the new rectangle = $(0.9s) \times (1.1s)$
Area of the new rectangle = $0.99s^2$
Then

$$\Delta = (1 - 0.99)s^2 = 0.01s^2$$

So change in area is

$$\frac{0.01}{1} \times 100\ \% = 1\%$$

Therefore the correct option is A)

6. Correct option: (B)

Let the rectangle have length l and breadth b.
Then area of all four square given as

$$l^2 + b^2 + l^2 + b^2 = 68 => l^2 + b^2 = 34$$

From the given perimeter we can also write

$$2(l + b) = 16 => (l + b) = 8$$

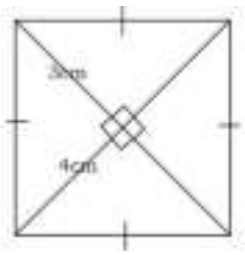
We can solve 2 equation to find the value of lb as

$$(l + b)^2 = l^2 + b^2 + 2lb = 64 => 2lb = 30$$
$$=> lb = 15$$

For a rectangle the area is equal to lb which is 15.
Therefore the correct option is B)

7. Correct option: (B)

Given figure

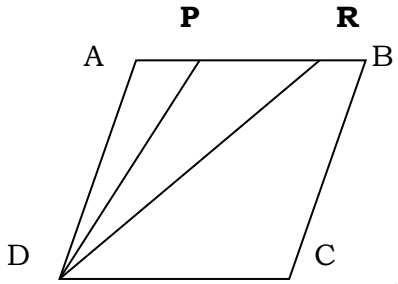


The area of given figure can be calculated as 4 × area of one triangle
So area of the given figure is

$$4 \times \frac{1}{2} \times 3 \times 4 = 24\ sq.cm$$

Therefore the correct option is B)

8. Correct option: (C)



Let's denote the area of parallelogram ABCD as "⊿" square units and the area of triangle DPR as "B" square units.
Given:
Area of ABCD = 8 times the area of DPR
PR = 5 cm
Then
⊿ = 8B

Now, we need to find the length of CD. The base of triangle DPR is PR, which is given as 5 cm. To find the length of CD, we need to determine the height of triangle DPR, which is also the height of parallelogram ABCD.

Δ = CD × height

$$height = \frac{\Delta}{CD}$$

Substitute the relationship between Δ and B: $height = \frac{8B}{CD}$

Now, we know the area of a triangle can be calculated as: $B = \left(\frac{1}{2}\right) \times base \times height$

Substitute the known values: $B = \left(\frac{1}{2}\right) \times 5\,cm \times \left(\frac{8B}{CD}\right)$

$$CD = \left(\frac{1}{2}\right) \times 5\,cm \times \left(\frac{8B}{B}\right)$$

$$CD = \left(\frac{1}{2}\right) \times 5\,cm \times 8$$

$$CD = 20\,cm$$

Therefore the correct option is C)

9. Correct option: (D)

To determine the length of the rope by which a cow must be tethered in order to graze over an area of 616 square meters, we need to consider the situation as a circular grazing area. When a cow is tethered with a rope, it can only move in a circular path.

We want to find the radius (r) given the area (616 square meters).

$$r^2 = \frac{Area}{\pi}$$

$$r^2 = \frac{616}{\frac{22}{7}} = 616 \times \frac{7}{22} = 28 \times 7 = 196$$

$$r^2 = 196 \Rightarrow r = 14$$

So, the length of the rope by which the cow must be tethered in order to graze over an area of 616 square meters is 14 meters.

Therefore the correct option is D)

10. Correct option: (D)

To find the area of shaded region we can subtract the area of all four

Sector from the area of square

In this case, we have: $A = (12)^2 - 4 \times \frac{1}{4} \times \frac{22}{7} \times 6 \times 6\,cm^2$

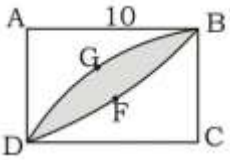
$$= 144 - 36\pi = 36(4 - \pi)cm^2$$

Therefore the correct option is D)

11. Correct option: (C)

To find the area of shaded region, firstly we find the area of the DGBC and then subtract the area of triangle DCB.

As given figure is square and to find the area of shaded region we can double the area found in first step as both area will same.



In this case, we have: $A = (area\ ABFD - area\ of \Delta ABD) + (area\ BGDC - area\ of \Delta BCD)$

$$A = 2\left(\frac{1}{4} \times \pi \times 10^2 - \frac{1}{2} \times 10^2\right) = 50\pi - 100$$

Therefore the correct option is C)

12. Correct option: (D)

Given: Perimeter of the semicircle = 18 cm

The formula for the perimeter of a semicircle is:

$P = \left(\frac{1}{2}\right) \times \pi \times d + d$, where d is the diameter.

Now, let's solve for the diameter (d):

$$18\ cm = \left(\frac{1}{2}\pi + 1\right) d = \left(\frac{\pi + 2}{2}\right) d$$

$$18 = \frac{36}{14} d => d = 7\ cm$$

So, the diameter of the semicircle is 7cm

Therefore the correct option is D)

13. Correct option: (D)

Let's the radius of the smaller (inner) circle is a and bigger(outer) circle is b.

Given: $a = 64.5, b = 68$

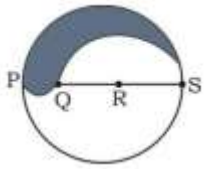
Then area of ring $A = \pi b^2 - \pi a^2 = \frac{22}{7}(68^2 - (64.5)^2) = \frac{22}{7} \times 3.5 \times 132.5 = 1457.5$

Therefore the correct option is D)

14. Correct option: (A)

Let's the radius of the circle is r then diameter is 2r.

Given: PQRS is diameter of circle with PQ = QR = RS



To find the perimeter of the shaded region we can first find the perimeter of circle with diameter as QS and cirlce with diameter with PQ.

From above we can write that

$$QS = \frac{2}{3}PS = \frac{2}{3}(2r) = \frac{4}{3}r \text{ and } PQ = \frac{1}{3}PS = \frac{2}{3}r$$

Then perimeter of the shaded region

$$Perimeter = \frac{1}{2}(\pi \times PS + \pi \times QS + \pi \times PS) = \frac{1}{2}\left(\pi \times 2r + \pi \times \frac{4}{3}r + \pi \times \frac{2}{3}r\right)$$

$$= \frac{1}{2}\pi(2r + 2r) = 2\pi r$$

Therefore the correct option is A)

15. Correct option: (C)

We are given that an equilateral triangle is described on the diagonal of the square then all side of the triangle will be equal to the diagonal of the square.

Let's the side of the square is s then using the right angle property of the triangle made by the diagonal we can write

$$diagonal = \sqrt{2}s$$

and area of the square is s^2

Area of the equilateral triangle

$$= \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4}\left(\sqrt{2}s\right)^2 = \frac{\sqrt{3}}{4} \times 2s^2 = \frac{\sqrt{3}}{2}s^2$$

Then the ratio of the areas

$$= \frac{\sqrt{3}}{2}s^2 : s^2 = \sqrt{3} : 2$$

Therefore the correct option is C)

16. Correct option: (A)

To find the value of a+b we can use above equations

Putting the value of b from equation 1 to equation 2

$$\frac{23}{22}a - a = 5 \Rightarrow a = 110$$

Then b

$$b = a \times \frac{23}{22} = 115$$

So a + b

$$= 110 + 115 = 225\, m$$

Therefore the correct option is A)

17. Correct option: (A)

To find the area of the path we can subtract the area of circle with smaller radius from area of the circle with bigger radius

$$\pi(b^2 - a^2) = \pi(b + a)(b - a) = \frac{22}{7} \times 5 \times 225 = 3535.71\, m^2$$

Therefore the correct option is A)

18. Correct option: (C)

To find the sum of perimeter of inner and outer circle we use

$$2\pi(b + a) = 2\pi \times 225 = 450\pi$$

Therefore the correct option is C)

19. Correct Matching is: a-R, b-S, c-P, d-Q

a) Area of sector **(R)** = $\left(\frac{x}{360}\right) \times \pi r^2$

Explanation: The area of a sector is calculated as a fraction of the total circle's area based on the central angle 'x' out of 360 degrees.

b) Length of an arc of a sector **(S)** = $\left(\frac{x}{360}\right) \times 2\pi r$

Explanation: The length of an arc of a sector is a fraction of the total circumference of the circle based on the central angle 'x' out of 360 degrees.

c) Perimeter of a sector **(P)** = $l + 2r$

Explanation: The perimeter of a sector is the sum of the length of the arc (l) and twice the radius (2r).

d) If 'l' is the length of arc, then area is **(Q)** = $\left(\frac{lr}{2}\right)$

Explanation: The area of a sector can be found by multiplying the length of the arc (l) by half the radius (r) of the sector.

20. Correct Matching is: a-Q, b-S, c-P, d-R

a) πr^2 = **(Q)** Area of a circle

Explanation: The area of a full circle is given by πr^2.

b) $\frac{\pi r^2}{2}$ **(S)** = Area of a semi-circle

Explanation: The formula for the area of a semicircle is half the area of a full circle, which is πr^2 divided by 2.

c) πr **(P)** = Circumference of a semi-circle

Explanation: The formula for the circumference of a semi arc only is half the circumference of a full circle.

d) 2r **(R)** = Diameter of a circle

Explanation: The diameter of a circle is twice the radius.

21. Correct Answer is: (7)

To find the length of the second diagonal of a rhombus when we know its area and one diagonal, we can use the following formula:

$$Area = \left(\frac{1}{2}\right) \times Product\ of\ Diagonals$$

In this case, we're given the area as 112 sq. cm and one diagonal as 32 cm. Let's denote the length of the second diagonal as "d_2." The formula becomes:

$$112\ sq.cm = \left(\frac{1}{2}\right) \times 32\ cm \times d_2$$

$$112\ sq.cm = 16\ cm \times d_2$$

$$d_2 = \frac{112\ sq.cm}{16\ cm}$$

$$d_2 = 7\ cm$$

So, the length of the second diagonal of the rhombus is 7 cm.

22. Correct Answer is: (8)

In an isosceles right-angled triangle, the two legs are congruent (equal) and the angles at the base are each 45°. To find the length of a side in such a triangle, we can use the following formula for the area: $Area = \left(\frac{1}{2}\right) \times Leg^2$

In this case, we're given the area as 32 sq. cm. Let's denote the length of each leg as "L." The formula becomes:

$$32\ sq.cm = \left(\frac{1}{2}\right) \times L^2$$

Now, solve for L:

$$32\ sq.cm = \left(\frac{1}{2}\right) \times L^2$$

$$64\ sq.cm = L^2$$

$$L = \sqrt{64\ sq.cm}$$

$$L = 8\ cm$$

So, the length of each side of the isosceles right-angled triangle is 8 cm.

PRISM, CUBE AND CUBOID

LEVEL-I

1. Correct option: (B)

To find the total length of the sides of a triangular prism, we need to calculate the perimeter of the two triangular bases and then add 3 times of height

Perimeter of triangular base = 12 cm + 16 cm + 20 cm = 48 cm.

Since there are two triangular bases, the total perimeter for the bases is:

Total perimeter of bases = 2 × 48 cm = 96 cm.

Now total perimeter of the prism is

$$= 96 + 3 \times 15 = 96 + 45 = 141\ cm$$

Then the correct option is B)

2. Correct option: (A)

To find the volume of a prism with an equilateral triangle as its base, we can use the formula:

$$Volume\ (V) = (Area\ of\ Base) \times Height$$

In this case, the base is an equilateral triangle with side length 8 cm and height 12 cm. Let's find the area of the equilateral triangle first:

$$A_t = \frac{\sqrt{3}}{4}(8^2) = 16\sqrt{3}\ cm^2$$

Now that we have the area of the equilateral triangle, we can calculate the volume of the prism:

$$V = 16\sqrt{3} \times 12\ cm^3 = 192\sqrt{3}\ cm^3$$

Then the correct option is A)

3. Correct option: (B)

To find the total surface area of a cube, we can use the formula:

$$Total\ Surface\ Area\ (TSA)\ =\ 6\ \times\ (Edge\ Length)^2$$

In this case, the edge length of the cube is given as 8 cm, so:

$$TSA\ =\ 6\ \times\ (8\ cm)^2$$
$$TSA\ =\ 6\ \times\ 64\ cm^2$$
$$TSA\ =\ 384\ cm^2$$

So, the total surface area of the cube is 384 square centimeters.

Then the correct option is B)

4. Correct option: ($300m^2$.**)**

Given the dimensions of the room:

- Length = 12 meters
- Width = 18 meters
- Height = 5 meters

Total Area of Four Walls = 2 × (Area of one long wall + Area of one short wall)

Total Area of Four Walls = 2 × ($18 \times 5\ m^2$+ $12 \times 5m^2$)

$$Total\ Area\ of\ Four\ Walls\ =\ 2\ \times\ 150\ m^2$$
$$Total\ Area\ of\ Four\ Walls\ =\ 300\ m^2$$

So, the total area of the four walls of the room is $300m^2$.

5. Correct option: (D)

Given:

Surface area of a cube = 600 square units

To find the length of the cube's diagonal, we can use the formula for the diagonal of a cube in terms of its edge length (a):

$$Diagonal\ (d)\ =\ a\sqrt{3}$$

First, let's find the edge length (a) using the given surface area:

$$Surface\ Area\ (SA)\ =\ 6a^2$$
$$6a^2\ =\ 600$$
$$a^2\ = \frac{600}{6}$$
$$a^2\ =\ 100 => a = 10$$

Now

$$Diagonal\ (d)\ =\ a\sqrt{3}$$
$$d\ =\ 10\ units\ \times\ \sqrt{3}$$

Then the correct option is D)

6. Correct option: (D)

If the edge of a cube is doubled, the new volume of the cube will be 8 times the original volume.

Let the original edge length be "a" units, and the original volume be "V" cubic units.

Original Volume: $V = a^3$

When the edge length is doubled, the new edge length becomes "2a" units.

New Volume: $V' = (2a)^3 = 8a^3$

So, the new volume is 8 times the original volume:

Then the correct option is D)

7. Correct option: (C)

Given:

The volumes of two cubes are in the ratio 1:27.

Let V_1 be the volume of the first cube and V_2be the volume of the second cube.

According to the given ratio:

$$\frac{V_1}{V_2} = \frac{1}{27}$$

Now, let's find the ratio of the lengths of their sides, as volume is directly proportional to the cube of the side length:

$$\left(\frac{V_1}{V_2}\right)^{\frac{1}{3}} = \left(\frac{1}{27}\right)^{\frac{1}{3}} = \frac{1}{3}$$

$$\frac{S_1}{S_2} = \left(\frac{1}{3}\right)$$

Now, let's find the ratio of the areas of one face of the first cube to one face of the second cube:

$$\frac{Area\ of\ one\ face\ of\ the\ first\ cube}{Area\ of\ one\ face\ of\ the\ second\ cube} = \left(\frac{S_1}{S_2}\right)^2$$

Plugging in the ratio of side lengths we found earlier:

$$= \left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

So, the ratio of the area of one face of the first cube to one face of the second cube is 1:9.

Then the correct option is C)

8. Correct option: (D)

Given:

- Cost of painting the whole surface area of a cube = Rs. 343.98
- Rate of painting per square centimeter = 13 paise per sq. cm

First, let's find the total surface area of the cube.

Let "s" represent the side length of the cube.

$$TSA = 6 \times (s^2)$$

We know that the cost of painting the whole surface area is Rs. 343.98, and the rate is 13 paise per sq. cm, which is equal to Rs. 0.13 per sq. cm.

$$Cost = Rate \times Area$$

$$343.98 = 0.13 \times TSA$$

$$TSA = \frac{343.98}{0.13}$$

$$TSA = 2646\ sq.cm$$

Now that we have the total surface area, we can calculate the side length "s" of the cube:

$$6 \times (s^2) = 2646$$

$$s^2 = \frac{2646}{6}$$

$$s^2 = 441$$

$$s = \sqrt{441}$$

$$s = 21\ cm$$

Now, we can calculate the volume (V) of the cube using the side length "s":

$$V = s\text{^}3$$

$$V = (21\ cm)^3$$

$$V = 9261\ cubic\ cm$$

So, the volume of the cube is 9261 cubic cm.

Then the correct option is D)

9. Correct option: (B)

Given: Total surface area of cube = 96 cm^2

To find the side length (s) of the cube, we can use the formula for the total surface area of a cube:

$$Total\ Surface\ Area\ (TSA) = 6 \times (Side\ Length)^2$$
$$96 = 6 \times (s)^2$$
$$16 = s^2$$
$$s = \sqrt{16}$$
$$s = 4\ cm$$

So, the side length of the cube is 4 cm.

Then the correct option is B)

10. Correct option: (A)

Given:

- Side lengths of the triangular base: a = 15 cm, b = 20 cm, c = 25 cm
- Height of the triangular prism: h = 40 cm

To find the lateral surface area (LSA) of a triangular prism, we can use the formula:

$$LSA = Perimeter\ of\ the\ base \times Height$$

First, calculate the perimeter of the triangular base:

$$Perimeter = a + b + c$$
$$Perimeter = 15\ cm + 20\ cm + 25\ cm$$
$$Perimeter = 60\ cm$$

Now, use the formula to find the lateral surface area:

$$LSA = Perimeter \times Height$$
$$LSA = 60\ cm \times 40\ cm$$
$$LSA = 2400\ square\ cm$$

So, the lateral surface area of the triangular prism is 2400 square centimeters.

Then the correct option is A)

11. Correct option: (D)

Given:

- Total Surface Area (TSA) of the cuboid = 40 m²
- Lateral Surface Area (LSA) of the cuboid = 26 m²

As we know

$$TSA = 2 \times (Area\ of\ Base) + LSA$$

Let's rearrange the formula to solve for Base area:

$$B = \frac{TSA - LSA}{2}$$

Now, substitute the given values:

$$B = \frac{40\ m^2 - 26\ m^2}{2}$$
$$B = \frac{14\ m^2}{2}$$
$$B = 7\ m^2$$

So, the area of the base of the cuboid is 7 square meters.

Then the correct option is D)

12. Correct option: (A)

Given:

- Sum of the length, breadth, and depth of the cuboid is 19 cm.
- Length of its diagonal is 11 cm.

Let's denote the length, breadth, and depth of the cuboid as "L," "B," and "D," respectively.

We are given the sum of these dimensions:
L + B + D = 19 cm
We are also given the length of the diagonal (d):
d = 11 cm
Now, we can relate these dimensions to the diagonal using the Pythagorean theorem. In a cuboid, the diagonal, length, breadth, and depth form a right-angled triangle.
Using the Pythagorean theorem: $d^2 = L^2 + B^2 + D^2$
Substitute the value of d:

$$(11\,cm)^2 = L^2 + B^2 + D^2$$
$$121\,cm^2 = L^2 + B^2 + D^2$$

Now, let's find an expression for the surface area (SA) of the cuboid.
The surface area of a cuboid is given by: $SA = 2(LB + BD + DL)$
We'll use the following identity to simplify the surface area formula:

$$2(LB + BD + DL) = (L + B + D)^2 - (L^2 + B^2 + D^2)$$

$$2(LB + BD + DL) = (19\,cm)^2 - 121\,cm^2$$
$$2(LB + BD + DL) = 361\,cm^2 - 121\,cm^2$$
$$2(LB + BD + DL) = 240\,cm^2$$

So, the surface area of the cuboid is 240 square centimeters.
Then the correct option is A)

13. Correct option: (C)

The volume of a cube is calculated using the formula: $Volume\,(V) = Side\,Length^3$
In this case, the side length of the cube is given as 7 cm, so we can simply substitute this value into the formula:

$$V = (7\,cm)^3$$
$$V = 7\text{^}3\,cm^3$$
$$V = 343\,cm^3$$

So, the volume of the cube is 343 cubic centimeters.
Then the correct option is C)

14. Correct option: (C)

The product of the areas of three adjacent faces of a cuboid, which share a common vertex, is equal to the square of the volume of the cuboid. This relationship is expressed as:

$$(Product\ of\ areas) = (Length \times Breadth) \times (Breadth \times Height) \times (Height \times Length)$$
$$= (l \times b) \times (b \times h) \times (h \times l) = (l \times b \times h)^2 = (Volume)^2$$

So, the product of the areas of these three adjacent faces is equal to the square of the volume of the cuboid.
Then the correct option is C)

15. Correct option: (B)

Given:
The areas of three adjacent faces of a rectangular block are in the ratio 2:3:4.
The volume of the rectangular block is 9000 cubic centimeters.
Let's denote the areas of the three adjacent faces as A_1, A_2, and A_3, and the sides of the rectangular block as L, B, and H (length, breadth, and height).
We are using the relationship $V^2 = (A_1 \times A_1 \times A_1)$ to solve for the sides of the rectangular block.

$$A_1 : A_2 : A_3 = 2x : 3x : 4x$$

Then

$$(9000)^2 = (2x \times 3x \times 4x) = 24x^3 \Rightarrow x^3 = 3375000$$
$$x = 150$$

As maximum area comes from the larger sides then we can write

$$maximum\ area = 4x = 600$$

Then the smallest side is $(smallest\ side) \times (maximum\ area) = 9000$

$$smallest\ side = \frac{9000}{600} = 15$$

So, the length of the shortest side is 15 cm.

Then the correct option is B)

16. Correct option: (D)

Given dimensions of the iron cuboid:

- Length (L) = 270 cm
- Breadth (B) = 100 cm
- Height (H) = 64 cm

To find the surface area (SA) of the resulting cube when the cuboid is melted and reformed into a cube, we need to consider that the volume of the cuboid will be equal to the volume of the cube.

$$Volume\ of\ cuboid\ = L \times B \times H$$

$$Volume\ of\ cube\ = (side\ length)^3$$

$$L\ x\ B\ x\ H\ = (side\ length)^3$$

We can find the side length of the cube:

$$(side\ length)^3\ = 270\ cm \times 100\ cm \times 64\ cm$$

$$(side\ length)^3\ = 1728000\ cm^3$$

$$side\ length = 120\ cm$$

$$Surface\ Area\ of\ cube\ = 6 \times (120\ cm)^2$$

$$Surface\ Area\ of\ cube\ = 6 \times 14400\ cm^2$$

$$Surface\ Area\ of\ cube\ = 86400\ cm^2$$

So, the surface area of the cube formed from the melted cuboid is 86400 square centimeters.

Then the correct option is D)

17. Correct option: (C)

Given dimensions of the rectangular box:

- Length (L) = 1.6 m
- Breadth (B) = 1 m
- Depth (D) = 60 cm = 0.6 m

$$Volume\ of\ the\ box\ = L \times B \times D$$

$$Volume\ of\ the\ box\ = 1.6\ m \times 1\ m \times 0.6\ m\ = 0.96\ cubic\ meters$$

Now, let's calculate the volume of a single cubical block:

$$Volume\ of\ a\ cubical\ block\ = E^3$$

$$Volume\ of\ a\ cubical\ block\ = (0.2\ m)^3\ = 0.008\ cubic\ meters$$

Now, to find the number of cubical blocks that can be packed inside the box, we'll divide the volume of the box by the volume of a single cubical block:

$$Number\ of\ blocks\ = \frac{Volume\ of\ the\ box}{Volume\ of\ a\ single\ block}$$

$$Number\ of\ blocks\ = \frac{0.96\ cubic\ meters}{0.008\ cubic\ meters\ per\ block}$$

$$Number\ of\ blocks\ = 120$$

So, we can pack 120 cubical blocks, each with an edge length of 20 cm, inside the rectangular box.

Then the correct option is C)

18. Correct option: (B)

Given: Dimensions of the wooden block: L = 20 cm, B = 10 cm, H = 5 cm

1. To construct a solid wooden cube of minimum size, use the L.C.M. of the dimensions:
   L.C.M. of 5, 10, and 20 = 20 cm (Edge length of the cube)
2. Calculate the volume of a single wooden block:

$$Volume\ of\ a\ single\ block\ =\ L\ \times\ B\ \times\ H$$
$$=\ 20\ cm\ \times 10\ cm\ \times\ 5\ cm\ =\ 1000\ (cm^3)$$

3. Calculate the volume of the cube:

$$Volume\ of\ the\ cube\ =\ (Edge\ length)^3$$
$$=\ (20\ cm)^3\ =\ 8000\ (cm^3)$$

4. Find the number of blocks required:

$$Number\ of\ blocks\ = \frac{Volume\ of\ the\ cube}{Volume\ of\ a\ single\ block}$$
$$= \frac{8000\ cm^3}{1000\ cm^3 per\ block}\ =\ 8$$

Then the correct option is B)

19. Correct option: (B)

Given: - Length (L) = 2 times the breadth (B)
- Height (H) = 3 meters
- Area of its four walls = 108 m²

1. We know L = 2B, so the area equation becomes: 4BH + 2BH = 108 m².
2. Simplify: 6BH = 108 m².
3. Solve for B: B = 6 meters.
4. Find L: $L\ =\ 2\ \times\ B\ =\ 12$ meters.
5. Calculate volume: Volume = $L\ \times\ B\ \times\ H\ =\ 216\ m^3$

The volume of the cold storage is 216 cubic meters.
Then the correct option is B)

20. Correct option: (C)

Given: - Side length of the square base (S) = 10 cm
- Height of the pyramid (H) = 12 cm

To find the lateral surface area of a right pyramid, we can use the formula:

$$Lateral\ Surface\ Area\ (LSA) = \frac{(Perimeter\ of\ the\ base) \times\ (Slant\ height)}{2}$$

First, let's find the slant height (l) of the pyramid using the Pythagorean theorem:

$$l^2\ =\ H^2\ +\ \left(\frac{S}{2}\right)^2$$
$$l^2\ =\ (12\ cm)^2\ +\ (5\ cm)^2$$
$$l^2\ =\ 144\ cm^2\ +\ 25\ cm^2$$
$$l^2\ =\ 169\ cm^2$$
$$l\ =\ \sqrt{169}\ cm$$
$$l\ =\ 13\ cm$$
$$Perimeter\ of\ the\ base\ =\ 40\ cm$$

Now, use the formula for Lateral Surface Area:

$$LSA\ = \frac{(40\ cm) \times\ (13\ cm)}{2}$$
$$LSA\ = \frac{520\ cm^2}{2}$$
$$LSA\ =\ 260\ cm^2$$

So, the lateral surface area of the right pyramid is 260 square centimeters.
Then the correct option is C)

LEVEL-II

1. Correct option: (A)

When three cubes of metal are melted to form a new cube, the volume of the new cube will be equal to the sum of the volumes of the three original cubes.

Given the edge lengths of the three cubes:

1. Cube 1: Edge length (a) = 3 cm
2. Cube 2: Edge length (b) = 4 cm
3. Cube 3: Edge length (c) = 5 cm

The volume of each cube is given by V = a^3, b^3, and c^3, respectively.

Now, calculate the volumes of the three cubes:

1. Volume of Cube 1 = (3 cm)³ = 27 cubic centimeters (cm^3)
2. Volume of Cube 2 = (4 cm)³ = 64 cubic centimeters (cm^3)
3. Volume of Cube 3 = (5 cm)³ = 125 cubic centimeters (cm^3)

The total volume of the new cube is the sum of these volumes:

$$Total\ Volume\ =\ 27\ cm^3\ +\ 64\ cm^3\ +\ 125\ cm^3\ =\ 216\ cm^3$$

To find the edge length (L) of the new cube, take the cube root of the total volume:

$$L\ =\ \sqrt[3]{216\ cm^3}\ =\ 6\ cm$$

Now that we have the edge length of the new cube (6 cm), we can calculate its surface area using the formula for the surface area of a cube:

$$Surface\ Area\ (SA)\ =\ 6L^2$$
$$SA\ =\ 6\ \times\ (6\ cm)^2$$
$$SA\ =\ 6\ \times\ 36\ cm^2$$
$$SA\ =\ 216\ cm^2$$

So, the surface area of the new cube is 216 square centimeters.

Then the correct option is A)

2. Correct option: (B)

If we know the areas of three adjacent sides of a cuboid (let's call them x, y, and z), then we can find the volume of the cuboid using the formula:

$$Volume\ (V)\ =\ \sqrt{x\ \times\ y\ \times\ z}$$

So, the volume of the cuboid is the square root of the product of the areas of these three adjacent sides.

Then the correct option is B)

3. Correct option: (C)

In a cube, the length of the diagonal (l) can be related to its side (s) using the following formula:

$$l\ =\ s\sqrt{3}$$

Then volume of cube is related with diagonal as

$$V = s^3 \Rightarrow V = \left(\frac{l}{\sqrt{3}}\right)^3 = \frac{l^3}{3\sqrt{3}} \Rightarrow 3\sqrt{3}V = l^3$$

This formula relates the length of the diagonal to the cube's volume.

Then the correct option is C)

4. Correct option: (C)

Given:

Sum of all edges of a cube = 36 cm

To find the volume of the cube, we can use the formula for the sum of all edges of a cube:

$$Sum\ of\ all\ edges\ =\ 12\times\ Edge\ length$$
$$Edge\ length\ =\frac{36\ cm}{12}\ =\ 3\ cm$$

Now that we know the edge length (E) is 3 cm, we can find the volume of the cube using the formula:

$$Volume\ (V)\ =\ Edge\ length^3$$
$$Volume\ =\ (3\ cm)^3\ =\ 27\ (cm^3)$$

So, the volume of the cube is 27 cm³.
Then the correct option is C)

5. Correct option: (B)

Given: Area of the floor of the room $=\ 15\ m^2\ = 15 \times 10 \times 10\ dm^2$
Height of the room $=\ 4\ m = 40\ dm$
Volume (V) = Area of the floor × Height

$$V\ =\ 1500\ dm^2\ \times\ 40\ dm\ =\ 60000\ dm^3$$

So, the volume of the air contained in the room is 60000 cubic decimeters (dm³).
Then the correct option is B)

6. Correct option: (B)

The volume of a cube with side length 's' is given by $V_{cube}\ =\ s^3$
The volume of the rectangular sheet is given by

$$V_{sheet} = (length)\ \times\ (width)\ \times\ (thickness).$$

Given that the sides of the sheet are in the ratio 1:5, let the length be 'x' and the width be '5x'. The thickness is 0.5 cm.

$$V_{cube} = V_{sheet}$$
$$10^3\ =\ x\ \times\ 5x\ \times\ 0.5$$
$$1000\ =\ 2.5x^2$$
$$x^2\ =\ 400$$
$$x\ =\ 20\ cm$$

Therefore, the dimensions of the rectangular sheet are:

$$Length\ =\ x\ =\ 20\ cm$$
$$Width\ =\ 5x\ =\ 5\ \times\ 20\ cm\ =\ 100\ cm$$

Then the correct option is B)

7. Correct option: (C)

The volume of a cube is directly proportional to the cube of its edge length (side length).
Let the edge length of the first cube be 'a' and the edge length of the second cube be 'b'.
We are given that the volumes are in the ratio 27:1, which means:

$$\frac{a^3}{b^3} = \frac{27}{1}$$

Now, simplify the equation:

$$a^3\ =\ 27b^3$$
$$a\ =\ 3b$$
$$a{:}\,b\ =\ 3{:}\,1$$

Therefore, the correct answer is: C) 3:1

8. Correct option: (A)

To find the edge length of the new cube formed by melting three smaller cubes, we need to find the cube root of the total volume of the smaller cubes.

$$Volume\ of\ the\ first\ cube\ =\ 125\ cm^3$$
$$Volume\ of\ the\ second\ cube\ =\ 64\ cm^3$$
$$Volume\ of\ the\ third\ cube\ =\ 27\ cm^3$$
$$Total\ volume\ =\ 125\ cm^3\ +\ 64\ cm^3\ +\ 27\ cm^3\ =\ 216\ cm^3$$

Now, take the cube root of the total volume to find the edge length of the new cube:

$$Edge\ length\ of\ the\ new\ cube\ =\ \sqrt[3]{216\ cm^3}\ =\ \sqrt[3]{6^3 cm^3}\ =\ 6\ cm$$

Therefore, the correct option is A)

9. Correct option: (B)

When two cubes with side lengths of 12 cm each are joined end to end, they form a cuboid. To find the surface area of this resulting cuboid, we can use the formula for the surface area of a cuboid, which is:

$$Surface\ Area\ =\ 2(lw\ +\ lh\ +\ wh)$$

In this case, the length (l) of the cuboid is the sum of the side lengths of the two cubes, which is $12\ cm\ +\ 12\ cm\ =\ 24\ cm.$

The width (w) and height (h) of the cuboid are each equal to the side length of one of the cubes, which is 12 cm.

$$Surface\ Area\ =\ 2(24\ cm \times 12\ cm + 24\ cm \times 12\ cm + 12\ cm \times 12\ cm)$$
$$Surface\ Area\ =\ 2(288\ cm^2\ +\ 288\ cm^2\ +\ 144\ cm^2)$$
$$Surface\ Area\ =\ 2(720\ cm^2)$$
$$Surface\ Area\ =\ 1440\ cm^2$$

So, the surface area of the resulting cuboid is 1440 square centimeters.

Therefore, the correct option is B)

10. Correct option: (B)

Let the side lengths of the three cubes be 3x, 4x, and 5x, where x is a positive constant.

$$Volume\ of\ the\ first\ cube\ =\ (3x)^3\ =\ 27x^3 cm^3$$
$$Volume\ of\ the\ second\ cube\ =\ (4x)^3\ =\ 64x^3 cm^3$$
$$Volume\ of\ the\ third\ cube\ =\ (5x)^3\ =\ 125x^3 cm^3$$

Side of the single cube where $12\sqrt{3}$ is length of the diagonal

$$a\sqrt{3} = 12\sqrt{3} => a = 12$$
$$Volume\ of\ the\ single\ cube\ =\ 12^3\ cm^3$$

Now, since the volumes of the three smaller cubes add up to the volume of the single cube, we can write:

$$27x^3\ +\ 64x^3\ +\ 125x^3\ =\ 1728$$
$$216x^3 = 1728 => x^3 = 8 => x = 2$$

So, the side lengths of the three cubes are:

$$First\ cube\colon 3x = 6\ cm$$
$$Second\ cube\colon 4x = 8\ cm$$
$$Third\ cube\colon 5x = 10\ cm$$

Therefore, the sides of the cubes are 6 cm, 8 cm, and 10 cm.

Therefore, the correct option is B)

11. Correct option: (B)

The volume of a cube is directly proportional to the cube of its side length (edge length).

$$\frac{a^3}{b^3} = \frac{8}{27} => \frac{a}{b} = \frac{2}{3}$$

Now, to find the ratio of their surface areas, we can use the formula for the surface area of a cube, which is:

$$\frac{6a^2}{6b^2} = \left(\frac{a}{b}\right)^2$$

Now, square both sides to find the ratio of their surface areas:

$$\left(\frac{a^2}{b^2}\right) = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$$

Therefore, the correct option is B)

12. Correct option: (B)

To find the ratio of the total surface area of one of the small cubes to that of the large cube, we need to first calculate the total surface area of both cubes.

1. Large Cube:
   Edge length (a) = 5 cm
   Total surface area of the large cube = $6 \times 5^2 = 150$
2. Small Cube:
   Edge length (a) = 1 cm
   Total surface area of each small cube = $6 \times 1^2 = 6$

$$Ratio = \frac{6\ cm^2}{150\ cm^2}$$

$$Ratio = \left(\frac{6}{150}\right) = \left(\frac{1}{25}\right)$$

Therefore, the correct option is B)

13. Correct option: (A)

Given: Sum of length, breadth, and depth: L + B + H = 19 cm
Length of the diagonal: 11 cm
First, we need to calculate the squares of the length, breadth, and height:

$$L^2 + B^2 + H^2 = 121\ cm^2$$

Now, we can find the surface area of the cuboid using the formula:

$$Surface\ Area = 2(LB + LH + BH) = (L + B + H)^2 - (L^2 + B^2 + H^2)$$

Plug in the given values:

$$Surface\ Area = 2(LB + LH + BH) = (19\ cm)^2 - 121\ cm^2$$

$$Surface\ Area = 361\ cm^2 - 121\ cm^2$$

$$Surface\ Area = 240\ cm^2$$

So, the surface area of the cuboid is 240 square centimeters.
Therefore, the correct option is A)

14. Correct option: (A)

To find the slant height (l) of the right pyramid:
Given:

$$Side\ of\ the\ base\ (s) = 10\ cm$$

$$Height\ (h) = 12\ cm$$

We can use the Pythagorean theorem to calculate the slant height (l), which is the hypotenuse of a right triangle with one side being half of the base and the other side being the height of the pyramid:

$$l^2 = \left(\frac{1}{2} \times s\right)^2 + h^2$$

$$l^2 = \left(\frac{1}{2} \times 10\ cm\right)^2 + (12\ cm)^2$$

$$l^2 = (5\ cm)^2 + (12\ cm)^2$$

$$l^2 = 25\ cm^2 + 144\ cm^2$$

$$l^2 = 169\ cm^2 => l = 13\ cm$$

So, the slant height of the pyramid is 13 cm, and the correct option is A) 13 cm.

15. Correct option: (B)

To find the lateral surface area (LSA) of the right pyramid, you can use the formula:

$$LSA = \left(\frac{1}{2}\right) \times Perimeter\ of\ the\ base \times Slant\ Height$$

Since the base is a square with side length 10 cm, its perimeter (P) is:

$$P = 4 \times side = 4 \times 10\ cm = 40\ cm$$

Then

$$LSA = \left(\frac{1}{2}\right) \times 40\ cm \times 13\ cm$$

$$LSA = 20\ cm \times 13\ cm$$

$$LSA = 260\ cm^2$$

So, the lateral surface area of the right pyramid is 260 square centimeters.
Therefore, the correct option is B)

16. Correct option: (400 c.c)

To find the volume (V) of the right pyramid, we can use the formula:

$$V = \left(\frac{1}{3}\right) \times Base\ Area \times Height$$

First, calculate the base area (A) of the square base:

$$A = s^2 = (10\ cm)^2 = 100\ cm^2$$

Now, plug the values into the formula for volume:

$$V = \left(\frac{1}{3}\right) \times 100\ cm^2 \times 12\ cm$$

$$V = \left(\frac{1}{3}\right) \times 1200\ cm^3$$

$$V = 400\ cm^3$$

So, the volume of the right pyramid is 400 cubic centimeters.

17. Correct Matching is: a-Q, b-R, c-S, d-P

Given that there are a cubiod of length , breadth b, height h, then

a) Total surface area of the cuboid:-→(Q)

$$Total\ Surface\ Area = 2(lb + lh + bh)$$

b) Lateral surface area of the cuboid:-→(R)

$$Lateral\ Surface\ Area = 2h(l + b)$$

c) Length of all edges of the cuboid:-→(S)

$$Length\ of\ all\ edges = 4(l + b + h)$$

d) Diagonal of the cuboid:-→(P)

$$Diagonal = \sqrt{l^2 + b^2 + h^2}$$

18. Correct Answer is: (6)

Let's the breadth of the box is b.

Given:

Length (L) is double the breadth, so L = 2b.

Height (H) is 3 meters.

The total surface area of the four walls (including doors) is 108 square meters.

Then

$$108 = 2(L \times H + b \times H)$$

$$108 = 2(2b \times 3 + b \times 3)$$

$$108 = 2(6b + 3b)$$

$$108 = 2(9b) = 18b$$

$$b = \frac{108}{18} = 6$$

So, the breadth (b) of the cold storage cuboid is 6 meters.

19. Correct Answer is: (8)

Given:

Sum of all edges of the cube = 24 cm

A cube has 12 edges, and each edge has the same length. Let's denote the length of each edge as "a."

Then we can write

$$12a = 24$$

$$a = \frac{24}{12}$$

$$a = 2\ cm$$

Then volume is

$$V = a^3$$

$$V = (2\ cm)^3 = 8\ cm^3$$

So, the volume of the cube is 8 cubic centimeters.

20. Correct Answer is: (4)

Given: $Total\ surface\ area\ of\ the\ cube\ =\ 96\ cm^2$

In a cube, there are 6 identical square faces, and the total surface area is the sum of the areas of these squares.

Let "a" be the side length of the cube.

$$Total\ Surface\ Area\ (TSA)\ =\ 6a^2$$

We are given that TSA is 96 cm², so: $6a^2\ =\ 96\ cm^2$

$$a^2\ =\ \frac{96\ cm^2}{6}\ =\ 16\ cm^2$$

$$a\ =\ \sqrt{16\ cm^2}\ =\ 4\ cm$$

So, the side length of the cube is 4 cm.

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

To find the volume of a prism with a rhombus as its base, we can use the formula:

$$Volume\ (V)\ =\ (Area\ of\ Base)\ \times\ Height$$

In this case, the base is a rhombus with diagonals of 28 cm and 96 cm, and the height of the prism is 32 cm.

First, let's find the area of the rhombus. The formula for the area of a rhombus is:

$$A\ =\ \left(\frac{1}{2}\right)\ \times\ (28\ cm\ \times\ 96\ cm)$$

$$A\ =\ (1/2)\ \times\ 2688\ cm^2$$

$$A\ =\ 1344\ cm^2$$

Now that we have the area of the rhombus, we can calculate the volume of the prism:

$$V\ =\ 1344\ cm^2\ \times\ 32\ cm$$

$$V\ =\ 43{,}008\ cm^3$$

So, the volume of the prism is 43,008 cubic centimeters.

Then the correct option is B)

2. Correct option: (C)

When the edge of a cube is increased by 100%, it means the new edge length is twice the original edge length.

Let the original edge length be "a" units. The new edge length is "2a" units.

The surface area of a cube is given by:

Original Surface Area (OSA) = $6a^2$

New Surface Area (NSA) = $6(2a)^2 = 6(4a^2) = 24a^2$

Now, let's find the percentage increase in surface area:

$$Percentage\ Increase\ =\ \left[\frac{New\ Surface\ Area\ -\ Original\ Surface\ Area}{Original\ Surface\ Area}\right]\ \times\ 100\%$$

$$Percentage\ Increase\ =\ \left[\frac{24a^2\ -\ 6a^2}{6a^2}\right]\ \times\ 100\%$$

$$Percentage\ Increase\ =\ \left[\frac{18a^2}{6a^2}\right]\ \times\ 100\%$$

$$Percentage\ Increase\ =\ \left(\frac{18}{6}\right)\times\ 100\%$$

$$Percentage\ Increase\ =\ 3\ \times\ 100\%$$

$$Percentage\ Increase\ =\ 300\%$$

So, when the edge of a cube is increased by 100%, the surface area of the cube is increased by 300%.

Then the correct option is C)

3. Correct option: (D)

Given: Diagonal of the cube = $6\sqrt{3}$ m

The diagonal of a cube is related to its edge length (s) using the Pythagorean Theorem. The formula is: $Diagonal\ (d) = s\sqrt{3}$

We are given the diagonal (d), so we can solve for the edge length (s):

$$s\sqrt{3} = 6\sqrt{3}\,m$$

Now, divide both sides by $\sqrt{3}$ to find the edge length (s):

$$s = 6\,m$$

Now that we know the edge length (s) is 6 meters, we can find the volume (V) of the cube:

$$Volume\ (V) = s^3$$
$$Volume\ (V) = (6\,m)^3$$
$$Volume\ (V) = 216\ cubic\ meters$$

So, the volume of the cube is 216 cubic meters.

Then the correct option is D)

4. Correct option: (D)

When the volume and surface area of a cube are equal, we can set up an equation to find the edge length (s) in terms of the unit measurement.

Let s be the edge length of the cube.

The formula for the volume (V) of a cube is:

$$V = s^3$$

The formula for the surface area (SA) of a cube is:

$$SA = 6s^2$$

Given that the volume and surface area are equal:

$$V = SA$$
$$s^3 = 6s^2$$
$$s = 6$$

The edge length of the cube in terms of the unit measurement is 6 times the unit measurement.

Then the correct option is D)

5. Correct option: (C)

To find the total surface area (TSA) of a rectangular prism (chalk box), we can use the formula:

$$TSA = 2 \times (Length \times Breadth + Length \times Height + Breadth \times Height)$$

Given the dimensions of the chalk box:

- Length (L) = 16 cm
- Breadth (B) = 8 cm
- Height (H) = 6 cm

Now, plug these values into the formula:

$$TSA = 2 \times (16\,cm \times 8\,cm + 16\,cm \times 6\,cm + 8\,cm \times 6\,cm)$$
$$TSA = 2 \times (128\,cm^2 + 96\,cm^2 + 48\,cm^2)$$
$$TSA = 2 \times 272\,cm^2$$
$$TSA = 544\,cm^2$$

So, the total surface area of the chalk box is 544 square centimeters.

Then the correct option is C)

6. Correct option: (B)

Given:

- Sum of the length, breadth, and depth of the cuboid = 19 cm
- Length of its diagonal = $5\sqrt{5}$ cm

Let's denote the length, breadth, and depth of the cuboid as "L," "B," and "D," respectively.
We are given the sum of these dimensions:
L + B + D = 19 cm
We are also given the length of the diagonal (d):

$$d = 5\sqrt{5}\,cm$$

As diagonal is related with L, B, D as

$$d^2 = L^2 + B^2 + D^2$$
$$(5\sqrt{5}\,cm)^2 = L^2 + B^2 + D^2$$
$$25 \times 5\,cm^2 = L^2 + B^2 + D^2$$
$$125\,cm^2 = L^2 + B^2 + D^2$$

We know

$$SA = 2(LB + BD + DL)$$
$$SA = 2(LB + BD + DL) = (L + B + D)^2 - (L^2 + B^2 + D^2)$$
$$SA = (19\,cm)^2 - 125\,cm^2$$
$$SA = 361\,cm^2 - 125\,cm^2$$
$$SA = 236\,cm^2$$

So, the surface area of the cuboid is 236 square centimeters.
Then the correct option is B)

7. Correct option: (A)

Given the areas of three adjacent faces of a cubical box:
1. Area 1 = 120 cm²
2. Area 2 = 72 cm²
3. Area 3 = 60 cm²

We can use the relationship, where the square of the volume (V²) is equal to the product of the areas of these three adjacent faces:

$$V^2 = Area\,1 \times Area\,2 \times Area\,3$$
$$V^2 = 120\,cm^2 \times 72\,cm^2 \times 60\,cm^2$$
$$V^2 = 12^2 \times 60^2\,cm^6$$
$$V = 720\,cm^3$$

So, the volume of the cubical box is 720 cubic centimeters
Then the correct option is A)

8. Correct option: (B)

To find the volume of a cuboid, we can simply multiply its three dimensions together.
Given dimensions of the cuboid:
- Length (L) = 12 meters
- Breadth (B) = 8 meters
- Height (H) = 3 meters

$$Volume\ of\ the\ cuboid = L \times B \times H$$
$$Volume = 12\,m \times 8\,m \times 3\,m = 288\,cubic\ meters$$

So, the volume of the cuboid is 288 cubic meters.
Then the correct option is B)

9. Correct option: (B)

Given room dimensions:
- Length (L) = 12 meters
- Breadth (B) = 8 meters
- Height (H) = 9 meters

The length of the longest pole (P) is given by:

$$P = \sqrt{L^2 + B^2 + H^2}$$
$$P = \sqrt{12^2 + 8^2 + 9^2}$$

$$P = \sqrt{144 + 64 + 81}$$
$$P = \sqrt{289}$$
$$P = 17\ meters$$

So, the length of the longest pole that can be placed in the room is 17 meters.
Then the correct option is B)

10. Correct option: (B)

Given:
- Side length of each cube (S) = 5 cm

1. When three cubes are joined end to end, they form a cuboid. The resulting cuboid will have the following dimensions:
   - Length (L) = 3S (since there are three cubes joined together)
   - Breadth (B) = S (as each cube's breadth becomes the cuboid's breadth)
   - Height (H) = S (as each cube's height becomes the cuboid's height)
2. Calculate the surface area (SA) of the resulting cuboid using the formula: $SA = 2(LB + BH + HL)$
3. Substitute the dimensions into the formula:

$SA = 2(3S \times S + S \times S + S \times 3S)$

$$SA = 2(3S^2 + S^2 + 3S^2)$$
$$SA = 2(7S^2)$$
$$SA = 14S^2$$
$$SA = 14(5\ cm)^2$$
$$SA = 14(25\ cm^2)$$
$$SA = 350\ cm^2$$

So, the surface area of the resulting cuboid is 350 square centimeters.
Then the correct option is B)

11. Correct option: (A)

Volume = Length × Breadth × Depth
Given:
The breadth is x cm.
The length is thrice the breadth, which means length = 3 x.
The depth is 25 cm.
The tank holds 3000 liters of water.
We need to convert the volume to cubic centimeters because the other dimensions are in centimeters. Also, we need to account for the conversion from liters to cubic centimeters:

$$1\ liter = 1000\ cubic\ centimeters\ (cm^3)$$
$$Volume = 3000\ liters \times 1000\ cm^3/liter = 3{,}000{,}000\ cm^3$$

Now, we can use the formula to find the base area:

$$Volume = Length \times Breadth \times Depth$$
$$3{,}000{,}000\ cm^3 = 75x^2$$
$$x^2 = \frac{3000000}{75} = 40000 \Rightarrow x = 200$$

Then area of base is $3x^2 = 1200\ cm^2$
Therefore, the correct option is A)

12. Correct option: (B)

1. Calculate the surface area of the inside of the cistern.
   Length (L) = 12.5 meters
   Breadth (B) = 8.5 meters
   Height (H) = 4 meters

$$2.\ Surface\ area\ of\ bottom\ (L \times B) = 12.5m \times 8.5m = 106.25\ sq.m$$

$$3. Surface\ area\ of\ two\ sides\ (2 \times L \times H) = 2 \times (12.5m \times 4m) = 100\ sq.m$$
$$4. Surface\ area\ of\ base\ 2(B \times H) = 2 \times 8.5m \times 4m = 68\ sq.m$$
$$Total\ Surface\ Area = 274.25\ sq.m$$

4. Calculate the cost at Rs. 24 per sq.m:

$$Cost = 274.25\ sq.m \times Rs.24/sq.m = Rs.6582$$

So, the cost of cementing the inside of the cistern is Rs. 6582.
Therefore, the correct option is B)

13. Correct option: (B)

To find the diagonal (D) of a rectangular room with dimensions 12m, 7m, and 5m, we can use the 3D Pythagoras' theorem. The formula is:

$$D = \sqrt{L^2 + W^2 + H^2}$$
$$D = \sqrt{12^2 + 7^2 + 5^2}$$
$$D = \sqrt{144 + 49 + 25}$$
$$D = \sqrt{218}$$

So, the diagonal of the room is $\sqrt{218}$ meters.
Therefore the correct option is B)

14. Correct option: (B)

To find the cost of flooring, we need to calculate the total floor area and then multiply it by the rate per square meter.
First, calculate the floor area (A) of the room:

$$A = L \times W = 12m \times 7m = 84\ (m^2)$$

Now, multiply the floor area by the rate per square meter:

$$Cost = A \times Rate = 84\ m^2 \times Rs.2/m^2 = Rs.168$$

So, the cost of flooring at the rate of Rs. 2 per square meter is Rs. 168.
Therefore the correct option is B)

15. Correct option: (822)

The room has 4 walls (two with length L and two with length W) and a ceiling (with length L and width W).
1. Calculate the total surface area of the walls and ceiling:

$$Surface\ Area = 2 \times (L \times H + W \times H) + (L \times W)$$
$$Surface\ Area = 2 \times (12m \times 5m + 7m \times 5m) + (12m \times 7m)$$
$$Surface\ Area = 190m^2 + 84m^2$$
$$Surface\ Area = 274m^2$$

2. Now, multiply the surface area by the rate per square meter to find the cost of whitewashing:

$$Cost = Surface\ Area \times Rate = 274\ m^2 \times Rs.3/m^2 = Rs.822$$

So, the cost of whitewashing the room (excluding the floor) at the rate of Rs. 3 per square meter is Rs. 822.

16. Correct Matching is: a-T, b-P, c-Q, d-R

a) Lateral Surface Area of a Prism:-→(T)
The lateral surface area (LSA) of a prism can be calculated using the formula:

$$LSA = Perimeter\ of\ the\ base \times Height$$

b) Volume of a Prism:-→(P)
The volume (V) of a prism can be calculated using the formula:

$$V = Area\ of\ the\ base \times Height$$

c) Lateral Surface Area of a Pyramid:→(Q)

The lateral surface area (LSA) of a pyramid can be calculated using the formula:

$$LSA = \frac{Perimeter\ of\ the\ base \times Slant\ height}{2}$$

d) Volume of a Pyramid:

The volume (V) of a pyramid can be calculated using the formula:

$$V = \left(\frac{1}{3}\right) \times Area\ of\ the\ base \times Height$$

17. Correct Answer is: (6)

To find the sum of the digits of the volume of the matchbox, we can simply multiply the three dimensions together and then add the digits of the result.

Given the dimensions of the matchbox:

Length (L) = 4 cm

Width (W) = 2.5 cm

Height (H) = 1.5 cm

$$Volume\ (V) = L \times W \times H$$

$$V = 4\ cm \times 2.5\ cm \times 1.5\ cm = 15\ cm^3$$

Now, find the sum of the digits of the volume:

$$1 + 5 = 6$$

So, the sum of the digits of the volume of the matchbox is 6.

18. Correct Answer is: (9)

1. Find the total volume of the three metal cubes:

$$Volume\ of\ the\ first\ cube = 27\ cm^3$$

$$Volume\ of\ the\ second\ cube = 64\ cm^3$$

$$Volume\ of\ the\ third\ cube = 125\ cm^3$$

$$Total\ volume = 27\ cm^3 + 64\ cm^3 + 125\ cm^3 = 216\ cm^3$$

2. Since a new cube is formed, let's find the side length of this cube. The volume of a cube is given by:

$$Volume\ (V) = (a)^3$$

We can write as

$$a = \sqrt[3]{216} = 6\ cm$$

3. Calculate the lateral surface area of the new cube. In a cube, all six faces are identical squares, and the lateral surface area is the sum of the areas of these squares.

$$Lateral\ Surface\ Area\ (LSA) = 4a^2$$

$$LSA = 4(6\ cm)^2 = 4(36\ cm^2) = 144\ cm^2$$

4. Find the sum of the digits of the lateral surface area:

$$1 + 4 + 4 = 9$$

So, the sum of the digits of the lateral surface area of the new cube is 9.

CYLINDER, CONE AND SPHERE

LEVEL-I

1. Correct option: (B)

To find the curved surface area (CSA) of a cylinder, we can use the following formula:

CSA = 2πrh

Given:

Radius (r) = 0.7 m

Height (h) = 1 m

$$CSA = 2 \times \left(\frac{22}{7}\right) \times 0.7\,m \times 1\,m$$
$$CSA = \left(\frac{44}{7}\right) \times 0.7\,m^2$$
$$CSA = \left(\frac{44}{10}\right) m^2$$
$$CSA = 4.4\,m^2$$

So, the curved surface area of the cylinder is 4.4 square meters.
Then the correct option is B)

2. Correct option: (A)

Given:
Height of the pillar (h) = 15 m
Diameter of the base (d) = 350 cm = 3.5 m
Cost of painting per square meter = Rs. 25/m²
First, find the radius (r) of the base by dividing the diameter by 2:

$$r = \frac{d}{2} = \frac{3.5}{2} = 1.75\,m$$

Now, calculate the curved surface area (CSA) of the cylinder using the formula:

$$CSA = 2\pi rh$$
$$CSA = 2 \times \left(\frac{22}{7}\right) \times 1.75\,m \times 15\,m$$
$$CSA = \left(\frac{44}{7}\right) \times 1.75\,m \times 15\,m$$
$$CSA = 165\,m^2$$

Now, find the cost of painting the curved surface area at the given rate:

$$Cost = CSA \times Cost\ per\ square\ meter$$
$$Cost = 165\,m^2 \times Rs.25/m^2$$

So, the cost of painting the curved surface area of the pillar is approximately Rs. 4125.
Then the correct option is A)

3. Correct option: (C)

Given:

$$Lateral\ Surface\ Area\ (LSA) = 176\,cm^2$$
$$Base\ Area = 38.5\,cm^2$$

First, we need to find the radius (r) of the cylinder using the formula for the base area:

$$Base\ Area = \pi r^2$$
$$38.5\,cm^2 = \left(\frac{22}{7}\right) \times r^2$$
$$r^2 = \frac{38.5 \times 7}{22}$$
$$r^2 = 12.25$$

Taking the square root of both sides:

$$r = \sqrt{12.25}$$
$$r = 3.5\,cm$$

Now that we have the radius, we can calculate the volume of the cylinder using the formula:

$$Volume = \frac{Base\ Area \times LSA}{Base\ Perimeter}$$

First, let's find the base perimeter (circumference):

$$Base\ Perimeter = 2\pi r$$
$$Base\ Perimeter = 2 \times \left(\frac{22}{7}\right) \times 3.5\,cm$$

$$Base\ Perimeter\ =\ 22\ cm$$

$$Volume\ =\frac{38.5\ cm^2\ \times\ 176\ cm^2}{22\ cm}=308\ cm^3$$

So, the volume of the cylinder is 308 cubic centimeters.
Then the correct option is C)

4. Correct option: (A)

1. Length of the sheet becomes the perimeter of the base:
Perimeter of the base (2πr) = Length of the sheet (22 cm)

$$2\pi r\ =\ 22\ cm$$

2. Solve for the radius (r):

$$r\ =\ 22\frac{cm}{2\pi}$$

$$r=\frac{7}{2}\ cm$$

3. Now, we can calculate the volume of the cylinder using the formula as width of the sheet will be the height of the cylinder:

$$Volume\ =\ \pi r^2 h$$

$$Volume\ =\ \pi\ \times\ \left(\frac{7}{2}cm\right)^2\ \times\ 10\ cm$$

$$Volume\ =\ \left(\frac{22}{7}\right)\times\left(\frac{49}{4}\right)\times\ 10\ cm^3$$

$$Volume\ =\ 385\ cm^3$$

So, the volume of the cylinder is 385 cubic centimeters.
Then the correct option is A)

5. Correct option: (C)

Given dimensions of the cuboidal metal:

$$Length\ (l)\ =\ 44\ cm$$

$$Width\ (w)\ =\ 30\ cm$$

$$Height\ (h)\ =\ 15\ cm$$

$$Volume\ of\ the\ cuboidal\ metal\ =\ l\ \times\ w\ \times\ h$$

$$Volume\ of\ the\ cuboidal\ metal\ =\ 44\ cm\ \times\ 30\ cm\ \times\ 15\ cm$$

$$Volume\ of\ the\ cuboidal\ metal\ =\ 19800\ cm^3$$

Now, this volume is equal to the volume of the cylinder formed:

$$Volume\ of\ the\ cylinder\ =\ \pi\ \times\ r^2\ \times\ h$$

We know that the height (h) of the cylinder is 28 cm.

$$\pi r^2\ \times 28\ cm\ =\ 19800\ cm^3$$

$$r^2\ =\frac{19800\ cm^3}{\pi\ \times\ 28\ cm}$$

$$r^2=225\ =>\ r=15$$

So, the radius of the cylinder formed is approximately 15 cm.
Then the correct option is C)

6. Correct option: (C)

Given that

$$r_x\ =\frac{1}{2}r_y, h_x=h_y=h$$

So volume of both of the cylinders

$$V_x=\pi(r_x)^2h_x=\pi(r_x)^2h, V_y=\pi\left(r_y\right)^2h_y=\pi\left(r_y\right)^2h$$

After doubling the height of cylinder x

$$h_x=2h$$

Now, let's compare the new volume of x to the volume of y:
New volume of the cylinder x

$$V_{x'} = \pi(r_x)^2 \times 2h = \pi\left(\frac{r_y}{2}\right)^2 \times 2h = \frac{\pi(r_y)^2 h}{2}$$

As we can see, the new volume of x is half the volume of y
Then the correct option is C)

7. Correct option: (D)

Since the volume remains the same, let's denote the original radius as "r" and the original length as "h."
The original volume (V) is:

$$V = \pi \times r^2 \times h$$

Now, the new radius is one-third of the original radius, so it's $\left(\frac{1}{3}\right)r$. Let's denote the new length as "h'"
The new volume (V') is:

$$V' = \pi \times \left(\frac{1}{3r}\right)^2 \times h'$$

Since the volume remains the same, V = V', we can set up the equation:

$$\pi \times r^2 \times h = \pi \times \left(\frac{1}{3r}\right)^2 \times h'$$

$$r\text{^}2 \times h = \left(\frac{1}{9}\right) \times r^2 \times h'$$

$$h = \left(\frac{1}{9}\right) \times h'$$

$$h' = 9h$$

So, the new length (h') is 9 times the original length (h).
Then the correct option is D)

8. Correct option: (B)

1. Calculate the volume of the earth removed from the well:

$$Volume\ of\ earth\ removed = \pi \times (7\,m)^2 \times 10\,m$$

$$Volume\ of\ earth\ removed = 1540\ cubic\ meters$$

2. Now, calculate the volume of the embankment created by spreading this earth evenly around the well:

Volume of embankment = Volume of earth removed

3. The volume of a cylindrical embankment can be calculated using the formula:

$$Volume\ of\ cylindrical\ embankment = \pi\left((outer\ radius)^2 - (inner\ radius)^2\right) \times height$$

Outer radius = 7 m + 21 m = 28 m

So, the volume of the embankment is:

$$Volume\ of\ embankment = \pi(28^2 - 7^2)h\ cubic\ meters = 2310h$$

$$2310h = 1540 \Rightarrow h = \frac{1540}{2310} = \frac{2}{3}\,m$$

So, the height of the embankment is $\frac{2}{3}$ meters.
Then the correct option is B)

9. Correct option: (C)

Given:
Circumference of the base = 12 meters
Height = π meters
First, let's find the radius (r) of the cylinder's base using the formula for the circumference of a circle:
Circumference (C) = 2πr

$$12\ meters\ =\ 2\pi r$$
$$r\ =\frac{12\ meters}{2\pi}$$
$$r\ =\frac{6}{\pi}\ meters$$

Now, we have the values of the radius (r) and height (h), so we can calculate the volume (V):

$$Volume\ (V)\ =\ \pi\ \times\ \left(\frac{6}{\pi}meters\right)^2\ \times\ \pi\ meters$$
$$Volume\ (V)\ =\ 36\ meters^3$$

So, the volume of the cylinder is 36 cubic meters.
Then the correct option is C)

10. Correct option: (C)

Given:
Diameter (d) = 14 cm
Slant height (l) = 10 cm
First, find the radius (r):

$$r\ =\frac{d}{2}=7\ cm$$

Now, plug the values into the formula:

$$CSA\ =\ \pi\ \times\ 7\ cm\ \times\ 10\ cm$$
$$CSA\ =\ 70\pi\ cm^2=220\ cm^2$$

So, the curved surface area of the cone is 220 square centimeters.
Then the correct option is C)

11. Correct option: (*1963cm²)

Given:
Cost of painting the curved surface area = Rs. 35.20
Cost per square meter (m²) = 5 paise
Slant height (l) = 25 cm
First, calculate the curved surface area (CSA) of the cone using the Cost:

$$CSA=\frac{35.20\times100}{5}=704\ m^2$$

From the CSA we can calculate the r:

$$CSA=\pi\times r\times l$$

To find the r we can use

$$704=\pi\times r\times25\ =>\ r=8.96$$

With the radius (r) and slant height (l), we can now find the volume (V) of the cone using the formula:

$$V\ =\ \left(\frac{1}{3}\right)\times\ \pi\ \times\ r^2\times\ h$$

Where:

h is the height of the cone, which can be found using the Pythagorean theorem: $h\ =\ \sqrt{l^2-\ r^2}=\sqrt{625-80.2816}=\sqrt{544.7184}=23.34\ m$

$$V=\frac{1}{3}\times\frac{22}{7}\times(8.96)^2\times(23.34)=1962.99\approx1963$$

So, the volume of the cone is approximately 1963 cubic centimeters.

12. Correct option: (C)

Given:

For the cone:

Radius of the base (r_{cone}) = 3 cm

Height (h_{cone}) = 5 cm

For the cylinder:

Radius of the base ($r_{cylinder}$)) = 2 cm

Height ($h_{cylinder}$) = 4 cm

Now, calculate the volumes:

$$\frac{V_{cone}}{V_{cylinder}} = \frac{\left(\frac{1}{3}\,\pi(r_{cone})^2 h_{cone}\right)}{\pi\left(r_{cylinder}\right)^2 h_{cylinder}} = \frac{1}{3} \times \left(\frac{(r_{cone})^2 h_{cone}}{\left(r_{cylinder}\right)^2 h_{cylinder}}\right) = \frac{1}{3}\frac{(3^2 \times 5)}{2^2 \times 4}$$

$$= \frac{15}{16}$$

Therefore, the correct option is (C).

13. Correct option: (C)

First, identify the relevant measurements:

Given:

Radius (r) = 4 cm

Height (h) = 3 cm

Volume of the cone (V)

$$V = \left(\frac{1}{3}\right) \times \pi \times (4\,cm)^2 \times 3\,cm$$

$$V = \left(\frac{1}{3}\right) \times \pi \times 16\,cm^2 \times 3\,cm$$

$$V = 16\pi\,cm^3$$

So, the volume of the cone is 16π cubic centimeters.

Therefore, the correct option is (C).

14. Correct option: (D)

First, calculate the volume of the cylinder using the formula for the volume of a cylinder:

$$(V_{cylinder}) = \pi \times V_{cylinder}^2 \times h_{cylinder}$$

$$V_{cylinder} = \pi \times (8\,cm)^2 \times 2\,cm = 128\pi\ cubic\ cm$$

Now, since the volume of the cylinder is equal to the volume of the cone, we can set up the equation:

$$V_{cone} = \frac{1}{3}\,\pi \times r_{cone}^2 \times h_{cone}$$

$$\frac{1}{3} \times \pi \times r_{cone}^2 \times 6 = 128\pi$$

$$r_{cone}^2 = 64 => r_{cone} = 8\,cm$$

So, the radius of the cone formed is 8 cm.

Therefore, the correct option is (D).

15. Correct option: (B)

The formula for the surface area (A) of a sphere with radius (r) is given by:

$$A = 4\pi r^2$$

Given the radius (r) as 14 cm

$$A = 4 \times \left(\frac{22}{7}\right) \times (14\,cm)^2$$

$$A = 2464\,cm^2$$

So, the surface area of the sphere is 2464 square centimeters.

Therefore, the correct option is (B).

16. Correct option: (D)

Let's denote the initial radius of the sphere as "r" and the increased radius as "r + 2."
If the radius is increased by 2 cm, the new surface area will be:

$$New\ Surface\ Area\ =\ 4\pi(r\ +\ 2)^2$$

According to the problem, the increase in surface area is 352 cm². So, we can set up the equation:
New Surface Area - Initial Surface Area = 352 cm²

$$4\pi(r\ +\ 2)^2\ -\ 4\pi r^2\ =\ 352\ cm^2$$
$$4\pi[(r\ +\ 2)^2\ -\ r^2]\ =\ 352\ cm^2$$
$$4\pi(4r\ +\ 4)\ =\ 352\ cm^2$$
$$4r\ +\ 4\ =\frac{352\ cm^2}{4\pi}=\frac{352\times 7}{88}=28$$
$$r+1=7 \Rightarrow r=6$$

So, the initial radius of the sphere before the increase was 6 cm.
Therefore, the correct option is (D).

17. Correct option: (D)

The formula for the surface area (A) of a sphere with radius (r) is: $A\ =\ 4\pi r^2$
So ratio of the surface area of the sphere given

$$\frac{S_A}{S_B}=\frac{4\pi r_A^2}{4\pi r_B^2}=\frac{r_A^2}{r_B^2}=\frac{40^2}{10^2}=\frac{16}{1}$$
$$Ratio\ =\ 16:1$$

So, the ratio of the surface area of sphere A to the surface area of sphere B is 16:1.
Therefore, the correct option is (D).

18. Correct option: (B)

The volume (V) of a sphere of radius "r" is obtained by multiplying its surface area (A) by the radius (r) and dividing by 3:

$$V\ =\left(\frac{1}{3}\right)\times\ A\ \times\ r$$

In mathematical notation, we can express it as: $V\ =\left(\frac{1}{3}\right)\times(4\pi r^2)\times\ r$

Therefore, the correct option is (B).

19. Correct option: ($\frac{6}{\pi}$)

Let "s" represent the side length of the cube and "d" represent the diameter of the sphere.
Given that the side of the cube is equal to the diameter of the sphere:
s = d
The formula for the volume (V) of a cube with side length "s" is: $V_{cube}\ =\ s^3$
The formula for the volume (V) of a sphere with diameter "d" is:

$$V_{sphere}\ =\left(\frac{4}{3}\right)\times\ \pi\ \times\left(\frac{d}{2}\right)^3$$

Since s = d, we can substitute s for d in the formula for the sphere's volume:

$$V_{sphere}\ =\left(\frac{4}{3}\right)\times\ \pi\ \times\left(\frac{d}{2}\right)^3=\left(\frac{4}{3}\right)\times\ \pi\ \times\left(\frac{s}{2}\right)^3$$
$$\frac{V_{cube}}{V_{sphere}}=\frac{s^3}{\left[\left(\frac{4}{3}\right)\times\ \pi\ \times\left(\frac{s^3}{8}\right)\right]}$$
$$\frac{V_{cube}}{V_{sphere}}=\frac{6}{\pi}$$

So, the ratio of the volume of the cube to the volume of the sphere is $\frac{6}{\pi}$.

20. Correct option: (A)

Let's denote the height of the cone, hemisphere, and cylinder as "h" and the radius of each as "r."

Volume of the cone (V_{cone}) = $\left(\frac{1}{3}\right) \times \pi \times r^2 \times h$

Volume of the hemisphere ($V_{\text{hemispher}}$) = $\left(\frac{2}{3}\right) \times \pi \times r^3$

Volume of the cylinder (V_{cylinder}) = $\pi \times r^2 \times h$

Now, we're given that they have equal bases and the same height:

$$Ratio\ of\ volumes\ (cone : hemisphere : cylinder) = (V_{cone} : V_{\text{hemispher}} : V_{\text{cylinder}})$$

$$Ratio = \left[\left(\frac{1}{3}\right) \times \pi \times r^2 \times r : \left(\frac{2}{3}\right) \times \pi \times r^3 : \pi \times r^2 \times r\right] = \left[\frac{1}{3} : \frac{2}{3} : 1\right]$$

Let's cancel out common factors: $Ratio = [1 : 2 : 3]$

So, the ratio of the volumes of the cone, hemisphere, and cylinder is 1:2:3.

Then the correct option is A)

LEVEL-II

1. Correct option: (B)

1. First, calculate the radius and volume of the original sphere:

$$Radius\ of\ sphere\ (r) = \frac{Diameter}{2} = 4\ cm$$

$$Volume\ of\ sphere = \left(\frac{4}{3}\right) \times \pi \times r^3$$

$$= \left(\frac{4}{3}\right) \times \pi \times (4\ cm)^3$$

$$= \left(\frac{256}{3}\right) \pi\ cm^3$$

$$Volume\ of\ wire = \pi \times (radius\ of\ wire)^2 \times length\ of\ wire$$

$$\left(\frac{256}{3}\right) \times \pi\ cm^3 = \pi \times (0.15\ cm)^2 \times length\ of\ wire$$

$$=> l = \frac{256 \times 100 \times 100}{3 \times 15 \times 15} = 3792\ cm = 37.9\ m$$

Then the correct option is B)

2. Correct option: (C)

When a solid object is submerged in a liquid, it displaces an amount of liquid equal to its own volume. In this case, a solid sphere with a radius of 3 cm is submerged in the water inside a cylindrical vessel with a radius of 4 cm.

$$Volume\ of\ sphere = \left(\frac{4}{3}\right) \times \pi \times r^3$$

$$V_{sphere} = 36\pi\ cm^3$$

Now, this volume of water is displaced when the sphere is submerged. Therefore, the water level in the vessel will rise by this volume. To find the rise in water level, we need to calculate the height (h) by which the water level will rise.

The volume of water displaced is equal to the volume of the sphere:

$$36\pi\ cm^3 = \pi \times (4\ cm)^2 \times h$$

$$36\pi\ cm^3 = \pi \times 16\ cm^2 \times h$$

$$h = \frac{36\pi\ cm^3}{\pi \times 16\ cm^2}$$

$$h = \left(\frac{36}{16}\right) cm$$

$$h = \frac{9}{4}\ cm$$

So, the water level in the vessel will rise by 2.25 cm when the sphere is completely immersed.

Then the correct option is C)

3. Correct option: (B)

We're given that the height (h) is 14 cm and the curved surface area (CSA) is 704 sq.cm. We can use this information to calculate the radius (r) of the cylinder:

$$CSA = 2\pi rh$$
$$704\ sq.cm = 2\pi r \times 14\ cm$$
$$2\pi r = \frac{704\ sq.cm}{14\ cm}$$
$$r = 8\ cm$$

Now that we have the radius (r) and height (h), we can find the volume (V) of the cylinder using the formula for the volume of a cylinder:

$$V = \pi r^2 h$$
$$V = \pi \times (8\ cm)^2 \times 14\ cm$$
$$V = \pi \times 64\ cm^2 \times 14\ cm$$
$$V = 2816\ cm^3$$

So, the volume of the cylinder is 2816cubic centimeters.

Then the correct option is B)

4. Correct option: (C)

We are given that the CSA is 2640 cm² and the circumference is 66 cm. We can use these formulas to find the values of "r" and "h."

Let's solve for "r" first: $2\pi r = 66\ cm \Rightarrow r = 10.5$

Now that we have the radius "r," we can use the CSA formula to find "h":

$$2640\ cm^2 = 2\pi \times 10.5\ cm \times h \Rightarrow h = 40$$

Now volume of cylinder

$$V = \pi r^2 h$$
$$V = \pi \times (10.5\ cm)^2 \times 40\ cm = 13860\ cm^3$$

Then the correct option is C)

5. Correct option: (14000*)

1. Volume of the cuboid:

$$Volume = 198\ cm^3$$

2. Volume of an individual lead shot:

$$Volume = \left(\frac{4}{3}\right) \times \pi \times r^3$$
$$Volume = (4/3) \times \pi \times (0.15\ cm)^3$$
$$Volume = \frac{4}{3} \times \frac{22}{7} \times 0.15 \times 0.15 \times 0.15 = \frac{0.099}{7}\ cm^3$$
$$Number\ of\ shots = \frac{Volume\ of\ cuboid}{Volume\ of\ one\ shot}$$
$$Number\ of\ shots = \frac{198\ cm^3 \times 7}{0.099\ cm^3} = 14000$$

6. Correct option: (C)

1. Calculate the volume of the iron ball using its diameter:

$$Volume\ of\ the\ iron\ ball = \left(\frac{4}{3}\right) \times \pi \times (radius\ of\ the\ ball)^3$$

$$Volume\ of\ the\ iron\ ball = (4/3) \times \pi \times (7\ cm)^3$$

2. Calculate the volume of the solid cylinder:

$$Volume\ of\ the\ cylinder\ =\ \pi\ \times (radius\ of\ the\ base)^2 \times\ (height\ of\ the\ cylinder)$$

$$Volume\ of\ the\ cylinder\ =\ \pi\ \times \left[\left(\frac{d}{2}\right)^2\right] \times \left(\frac{7}{3} cm\right)$$

$$\left(\frac{4}{3}\right) \times\ \pi\ \times\ (7\ cm)^3\ =\ \pi\ \times \left[\left(\frac{d}{2}\right)^2\right] \times \left(\frac{7}{3} cm\right)$$

$$\left(\frac{d}{2}\right)^2 = 2^2 \times 7^2 => \frac{d}{2} = 14 => d = 28$$

So, the diameter of the base of the cylinder is 28 cm.

Then the correct option is C)

7. Correct option: (A)

1. Calculate the surface area of the box:

The box consists of the lateral surface area of the cylinder, as well as the top and bottom areas.

$$Lateral\ Surface\ Area\ (LSA)\ of\ the\ cylinder\ =\ 2\pi rh$$

$$LSA\ =\ 2\ \times \left(\frac{22}{7}\right) \times\ 0.35\ m\ \times\ 1.25\ m$$

$$LSA =\ 2.75\ m^2$$

$$Top\ Area\ (circular) =\ Bottom\ Area\ =\ \pi r^2$$

$$Top\ Area\ =\ \left(\frac{22}{7}\right) \times\ (0.35\ m)^2$$

$$Top\ Area =\ 0.385\ m^2$$

$$Total\ Surface\ Area\ =\ LSA\ +\ Top\ Area\ +\ Bottom\ Area$$

$$Total\ Surface\ Area =\ 2.75\ m^2\ +\ 0.385\ m^2\ +\ 0.385\ m^2$$

$$Total\ Surface\ Area =\ 3.52\ m^2$$

2. Calculate the cost of the material using the cost per square meter:

$$Cost\ =\ Total\ Surface\ Area\ \times\ Cost\ per\ square\ meter$$

$$Cost\ =\ 3.52\ m^2\ \times\ Rs.80/m^2$$

$$Cost\ =\ Rs.281.60$$

So, the cost of the material used in the closed metallic cylindrical box is Rs. 281.60.

Then the correct option is A)

8. Correct option: (D)

To find the volume of the earth dug out when a circular well with a diameter of 2 meters is dug to a depth of 14 meters, we can use the formula for the volume of a cylinder:

$$Volume\ =\ \pi\ \times\ (radius)^2 \times\ height$$

For earth dug out we can take the first 14 m high cylinder volume

$$V = \frac{22}{7} \times 1 \times 1 \times 14 = 44\ m^3$$

Then the correct option is D)

9. Correct option: (C)

Given that curved surface area is CSA, and base area of the cone is BA.

$$CSA = \pi rl, BA = \pi r^2$$

Then r is $\sqrt{\frac{BA}{\pi}} = r => l = \frac{CSA}{\pi r} => l = \frac{CSA}{\sqrt{\pi BA}}$

To find the value of h

$$h = \sqrt{l^2 - r^2} = \sqrt{\left(\frac{CSA}{\sqrt{\pi BA}}\right)^2 - \left(\sqrt{\frac{BA}{\pi}}\right)^2}$$

then the volume of the cone is

$$V = \frac{1}{3}\pi\left(\sqrt{\frac{BA}{\pi}}\right)^2\left(\sqrt{\left(\frac{CSA}{\sqrt{\pi BA}}\right)^2 - \left(\sqrt{\frac{BA}{\pi}}\right)^2}\right)$$

10. Correct option: (B)

Let's denote the radius of the base of the cylinder as "r" meters and the height as "h" meters. We are given that the sum of the radius and height is 37 meters, which can be written as:

$$r + h = 37$$

We are also given that the total surface area of the cylinder is 1628 square meters. The formula for the total surface area (TSA) of a cylinder is:

$$TSA = 2\pi rh + 2\pi r^2 = 2\pi r(h + r) = 1628 = 2\pi r \times 37 =>$$
$$r = 7$$

Then the value of h is

$$h = 37 - 7 = 30$$

So volume of the cylinder is

$$V = \pi r^2 h = \frac{22}{7} \times 7 \times 7 \times 30 = 4620\ m^3$$

Then the correct option is B)

11. Correct option: (B)

Let the original radius be "R" and the original height be "H."

1. Original Cylinder Volume $V_1 = \pi \times R^2 \times H$
2. Reduced Cylinder Radius $\left(\frac{R}{2}\right)$ and Height (H) remain the same.
3. Reduced Cylinder Volume $V_2 = \pi \times \left(\frac{R}{2}\right)^2 \times H$

$$Ratio\ \left(\frac{V_2}{V_1}\right) = \frac{\pi \times \left(\frac{R}{2}\right)^2 \times H}{\pi \times R^2 \times H}$$
$$Ratio = \frac{\frac{R^2}{4}}{R^2}$$
$$Ratio = \left(\frac{1}{4}\right)$$

Then the correct option is B)

12. Correct option: (BD)

The volume of the sphere is equal to the volume of the cone.

$$1.Volume\ of\ the\ sphere\ (V_S) = \left(\frac{4}{3}\right) \times \pi \times \left(r_{sphere}\right)^3$$
$$V_S = \left(\frac{4}{3}\right) \times \pi \times (3\ cm)^3 = 36\pi$$
$$2.Volume\ of\ the\ cone\ (V_C) = (1/3) \times \pi \times (r_{cone})^2 \times h_{cone}$$
$$V_S = V_C$$
$$\left(\frac{4}{3}\right) \times \pi \times (3\ cm)^3 = \left(\frac{1}{3}\right) \times \pi \times (4\ cm)^2 \times h_{cone}$$
$$3^3 = 4h_{cone} => h_{cone} = 6.75\ cm$$

So, the height of the cone is 6.75 cm.

13. Correct option: (A, D)

Volume of the water raised

$$V = \pi r^2\left(h_f - h_i\right) = \pi \times 12 \times 12 \times (6.75) = 972\pi\ cm^3$$

Raised volume will be equal to the volume of sphere ball

$$972\pi = \frac{4}{3}\pi R^3 => R^3 = 729 => R = 9$$

So radius of the ball will be 9 cm

Then the correct option is A, D)

14. Correct option: (A)

The combined volume of lead balls will be equal the raised volume

So raised volume $V_r = \pi r^2 h = 4\pi \times 0.8\, cm^3 = 3.2\pi\, cm^3$

Combined volume of lead ball

$$V_c = 300(volume\ of\ one\ ball) = 300 \times \frac{4}{3} \times \pi \times R^3$$

Now we can find the radius of each ball R

$$3.2\pi = 300 \times \frac{4}{3} \times \pi \times R^3 => \frac{8}{1000} = R^3 => R = \frac{2}{10} = 0.2\, cm$$

Then the correct option is A)

15. Correct option: (B)

The raised volume $V_r = \pi r^2 h = 4\pi \times 0.8\, cm^3 = 3.2\pi\, cm^3 = 10.05\, cm^3$

Then the correct option is B)

16. Correct option: (D)

The combined volume 300 ball will be same as the raised volume

$$V_r = \pi r^2 h = 4\pi \times 0.8\, cm^3 = 3.2\pi\, cm^3 = 10.05\, cm^3$$

Then the correct option is D)

17. Correct Matching is: a-T, b-R, c-S, d-P

a) If the base radii of two cylinders are equal, the height ratio (H:h) will determine the curved surface area ratio ($CSA_1 : CSA_2$).

$$\frac{CSA_1}{CSA_2} = \frac{2\pi r \times h}{2\pi R \times H} = \frac{h}{H}$$

Therefore, a) ---> T) h:H

b) If the curved surface areas are equal, the volume ratio $\left(\frac{V1}{V2}\right)$ will be determined by the base radius ratio (r:R).

From CSA we can conclude that

$$\frac{rh}{RH} = 1$$

$$\frac{V_1}{V_2} = \frac{\pi r^2 h}{\pi R^2 H} = \frac{r}{R}$$

Therefore, b) ---> R) r:R

c) If the volumes are equal, the curved surface area ratio ($CSA_1 : CSA_2$) will be determined by the base radius ratio (R:r).

$$\frac{V_1}{V_2} = \frac{\pi r^2 h}{\pi R^2 H} = 1 => \frac{r^2 h}{R^2 H} => \frac{rh}{RH} = \frac{R}{r}$$

So CSA ratio

$$\frac{CSA_1}{CSA_2} = \frac{2\pi r \times h}{2\pi R \times H} = \frac{rh}{RH} = \frac{R}{r}$$

Therefore, c) ---> S) R:r

d) If the heights are equal, the ratio of their base radii squared (r^2:R^2) will determine the volume.

Given $h = H$

$$\frac{V_1}{V_2} = \frac{\pi r^2 h}{\pi R^2 H} => \frac{r^2 h}{R^2 H} => \frac{r^2}{R^2}$$

Therefore, d) ---> P) $r^2 : R^2$

18. Correct Answer is: (3)

Find the area of the sector:

$$Area\ of\ Sector\ =\ \left(\frac{216}{360}\right)\times\ \pi\ \times\ (15\ cm)^2$$

$$Area\ of\ Sector\ =\ 135\pi\ cm^2$$

Since the sector is used to create the curved surface of the cone, the curved surface area (CSA) of the cone is will remain same with slant height equal to radius of circle.

$135\pi\ =\ \pi\ \times\ 15\ cm\ \times\ r$

$$r\ =\ 9\ cm$$

To find the height (h) of the cone, we can use the Pythagorean theorem:

$$h = \sqrt{l^2 - R^2} = \sqrt{225 - 81} = \sqrt{144} = 12\ cm$$

so one fourth of the height is 3.

19. Correct Answer is: (6)

To find the sum of the digits of the radius of the cone, we can use the Pythagorean theorem because we have the height (h) and slant height (l) of the cone.

According to the Pythagorean theorem:

$$l^2\ =\ r^2\ +\ h^2$$

$$r\ =\ \sqrt{l^2\ -\ h^2}$$

$$r\ =\ \sqrt{25^2\ -\ 7^2}$$

$$r\ =\ \sqrt{625\ -\ 49}$$

$$r\ =\ \sqrt{576}$$

$$r\ =\ 24\ cm$$

The radius of the cone is 24 cm, and the sum of its digits is 2 + 4 = 6.

So, the sum of the digits of its radius is 6.

20. Correct Answer is: (5)

The volume of the both body remain same

So we can write

$$V_c = \pi r^2 h = \pi \times \frac{5}{2} \times \frac{5}{2} \times \frac{10}{3} = \frac{250}{12}\pi$$

The volume of sphere is

$$V_s = \frac{4}{3} \times \pi \times R^3$$

Therefore

$$\frac{250}{12}\pi = \frac{4}{3}\pi R^3 \Rightarrow \frac{125}{8} = R^3 \Rightarrow R = \frac{5}{2}$$

Then diameter of the sphere is 5 cm.

STUDENT PRACTICE QUESTIONS

1. Correct option: (D)

1. Lateral Surface Area of the Cylinder:

The lateral surface area of a cylinder can be calculated using the formula:

$$Lateral\ Surface\ Area\ =\ 2\pi rh,$$

$$Lateral\ Surface\ Area\ =\ 440\ cm^2$$

2. Area of Each Circular End:

The area of a circle can be calculated using the formula:

$$Area\ of\ a\ Circle\ =\ \pi r^2$$

$$Area\ of\ Each\ Circular\ End\ =\ 154\ cm^2$$

3. Total Metal Sheet Area Required:

$$Total\ Metal\ Sheet\ Area\ =\ Lateral\ Surface\ Area\ +\ 2\ \times\ Area\ of\ Each\ Circular\ End$$

$$Total\ Metal\ Sheet\ Area\ =\ 440\ cm^2\ +\ 308\ cm^2$$

$$Total\ Metal\ Sheet\ Area\ =\ 748\ cm^2$$

So, the area of the metal sheet required to make the closed cylindrical container is 748 square centimeters.

Then the correct option is D)

2. Correct option: (B)

Given:

Total Surface Area (TSA) = 220 cm²

Height (h) = 6.5 cm

1. First, find the radius (r) of the cylinder. Since we have the total surface area, we can use the formula for the total surface area of a cylinder:

$$TSA\ =\ 2\pi rh\ +\ 2\pi r^2$$

$$220\ cm^2\ =\ 2\pi rh\ +\ 2\pi r^2$$

$$220\ cm^2\ =\ 2\pi r(6.5\ cm)+\ 2\pi r^2\ =>\ 2\pi(6.5r+r^2)$$

$$\frac{220}{2\pi}=110\times\frac{7}{22}=35=6.5r+r^2\ =>\ r^2+6.5r-35=0$$

$$r=\frac{-6.5\pm\sqrt{(13.5)^2}}{2}=\frac{7}{2},-10\ =>\ r=\frac{7}{2}$$

Now that we have the radius, we can calculate the volume of the cylinder using the formula:

$$Volume\ =\ \pi r^2h$$

$$Volume\ =\ \pi(3.5\ cm)^2(6.5\ cm)=22\times0.5\times3.5\times6.5=250.25\ cm^3$$

So, the volume of the cylinder is 250.25 cubic centimeters.

Then the correct option is B)

3. Correct option: (B)

Given:

$$Area\ of\ the\ base\ (A)\ =\ 616\ cm^2$$

$$Height\ (h)\ =\ 48\ cm$$

First, let's find the radius (r) of the cone's base using the formula for the area of a circle (base):

$$A\ =\ \pi r^2$$

$$616\ cm^2\ =\ \pi r^2\ =>\ r^2=196\ =>\ r=14\ cm$$

The curved surface area (CSA):

$$CSA\ =\ \pi\ \times\ r\ \times\ l$$

To find the slant height (l), we can use the Pythagorean theorem, as we have the height (h) and the radius (r):

$$l^2\ =\ r^2\ +\ h^2$$

$$l^2\ =\ (14\ cm)^2\ +\ (48\ cm)^2$$

$$l^2\ =\ 196\ cm^2\ +\ 2304\ cm^2$$

$$l^2\ =\ 2500\ cm^2\ =>\ l=50\ cm$$

$$CSA\ =\ \pi\ \times\ 14\ cm\ \times\ 50\ cm$$

$$CSA\ =\ 2200\ cm^2$$

$$TSA\ =\ A\ +\ CSA=616+2200=2816\ cm^2$$

So, the total surface area of the cone is 2816 square centimeters.

Then the correct option is B)

4. Correct option: (D)

Given information:

Height of the cylindrical part ($h_{cylinder}$) = 4.8 m

Diameter of the base of the cylinder (d) = 9 m

Total height of the tent (h_{total}) = 10.8 m

The radius (r) of the cylinder is half of the diameter (d):

$$r = \frac{d}{2} = \frac{9}{2} = 4.5\,m$$

The formula for the curved surface area of a cylinder is 2πrh, where r is the radius and h is the height:

$$S_{cylinder} = 2\pi r h_{cylinder} = 135.77\,m^2$$

To find the slant height (l) of the cone

$$h_{\text{cone}} = h_{\text{total}} - h_{cylinder} = 10.8\,m - 4.8\,m = 6\,m$$

Now, we can calculate the slant height (l) using the Pythagorean Theorem:

$$l = \sqrt{h_{\text{cone}}^2 + r^2} = \sqrt{(6\,m)^2 + (4.5\,m)^2} = \sqrt{36\,m^2 + 20.25\,m^2} = 7.5\,m$$

Calculate the curved surface area of the cone (S_{cone}):

$$S_{cone} = \pi r l = \pi \times 4.5\,m \times 7.5\,m = 106.07\,m^2$$

Calculate the total area of the canvas:

$$Total\ Area = S_{cylinder} + S_{cone} = (135.77 + 106.07)m^2 = 241.84\,m^2$$

So, the total area of canvas required for the tent is $241.84\,m^2$

Therefore, the correct option is (D).

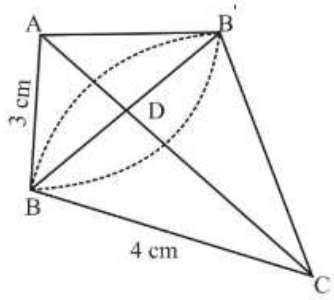
5. Correct option: (B)

Calculate the length of the hypotenuse AC of right-angled triangle ABC using the Pythagorean Theorem:

$$AC = \sqrt{AB^2 + BC^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5\,cm$$

Determine the area of triangle ABC using the formula for the area of a triangle:

$$Area\,(ABC) = \left(\frac{1}{2}\right) \times AB \times AC = \left(\frac{1}{2}\right) \times 4\,cm \times 3\,cm = 6\,cm^2$$



Calculate the length of side DB of triangle ABC using area: $6 = \frac{1}{2} \times AC \times BD \Rightarrow BD = 2.4\,cm$

Now, we have the dimensions of both cones:

Radius (r) = 2.4 cm (DB)

Height (h) = 5 cm (the total height of the double cone)

Calculate the volume of the double cone using the formula for the volume of a cone:

$$Volume = \left(\frac{1}{3}\right) \times \pi \times r^2 \times h_1 + \left(\frac{1}{3}\right) \times \pi \times r^2 \times h_2$$

$$= \frac{1}{3}\pi \times r^2(h_1 + h_2) = \frac{1}{3} \times 3.14 \times 2.4 \times 2.4 \times 5 \approx 30.14\,cm^3$$

So, the volume of the double cone is approximately 30.14 cubic centimeters.

Therefore, the correct option is (B).

6. Correct option: (D)

The formula for the volume (V) of a right circular cone is:

$$V = \left(\frac{1}{3}\right) \times \pi \times r^2 \times h$$

If both the radius (r) and height (h) are doubled, we get:

$$V_{new} = \left(\frac{1}{3}\right) \times \pi \times (2r)^2 \times (2h)$$

$$V_{new} = \left(\frac{1}{3}\right) \times \pi \times 4 \times r^2 \times 2 \times h$$

$$V_{new} = \left(\frac{8}{3}\right) \times \pi \times r^2 \times h$$

$$V_{new} = 8 \times \left[\left(\frac{1}{3}\right) \times \pi \times r^2 \times h\right]$$

$$V_{new} = 8 \times V$$

So, when both the radius and height of the cone are doubled, the new volume becomes 8 times the original volume.

Therefore, the correct option is (D).

7. Correct option: (B)

Let's denote the radii of the two cones as r_1 and r_2, and their respective heights as h_1 and h_2. Given that the radii are in the ratio 2:1, we can write:

$$\frac{r_1}{r_2} = \frac{2}{1}$$

Since the volumes of the two cones are equal, we can equate their volume formulas:

$$(1/3) \times \pi \times {r_1}^2 \times h_1 = (1/3) \times \pi \times {r_2}^2 \times h_2$$

$${r_1}^2 \times h_1 = {r_2}^2 \times h_2$$

$$(2r_2)^2 \times h_1 = {r_2}^2 \times h_2$$

$$4 \times {r_2}^2 \times h_1 = {r_2}^2 \times h_2$$

$$4 \times h_1 = h_2$$

$$\frac{h_1}{h_2} = \frac{1}{4}$$

So, the ratio of their heights is 1:4.

Therefore, the correct option is (B).

8. Correct option: (A)

1. First, find the radius (r) of the sphere using the given circumference (C).

$$Circumference\ (C) = 2\pi r$$

$$88\ cm = 2\pi r$$

$$r = \frac{88\ cm}{2\pi}$$

2. With the radius (r) determined, we can now calculate the surface area (A) of the sphere using the formula:

$$Surface\ Area\ (A) = 4\pi r^2$$

$$A = 4\pi(14\ cm)^2$$

$$A = 4\pi(196\ cm^2)$$

$$A = 784\pi\ cm^2$$

$$A = 784 \times \left(\frac{22}{7}\right) cm^2$$

$$A = 2464\ cm^2$$

So, the surface area of the sphere is 2464 square centimeters.

Therefore, the correct option is (A).

9. Correct option: (A)

$$Volume = \left(\frac{4}{3}\right) \times \pi \times (radius)^3$$

Given the original sphere has a diameter of 3 cm, its radius is 1.5 cm.

$$Volume\ of\ original\ sphere = \left(\frac{4}{3}\right) \times \pi \times (1.5\ cm)^3 = \frac{9}{2}\pi\ cm^3$$

Now, this volume is divided into three equal parts for the new spheres. Let the radius of the third ball be "r."

Then we can write that

$$\frac{9}{2}\pi = \frac{4}{3}\pi\ ((0.75)^3 + (1)^3 + r^3)$$

$$\frac{27}{8} - 1 - (0.75)^3 = 1.953125 = r^3 => r = 1.25$$

Then the diameter is 2.5 cm

Therefore, the correct option is (A).

10. Correct option: (B)

Let's calculate the volumes of the three original spheres:

1. For the sphere with a radius of 6 cm:

$$V_1 = \left(\frac{4}{3}\right) \times \pi \times (6\ cm)^3$$

2. For the sphere with a radius of 8 cm:

$$V_2 = \left(\frac{4}{3}\right) \times \pi \times (8\ cm)^3$$

3. For the sphere with a radius of 10 cm:

$$V_2 = \left(\frac{4}{3}\right) \times \pi \times (10\ cm)^3$$

Now, find the total volume of the three spheres:

$$Total\ Volume = V_1 + V_2 + V_3$$

$$Total\ Volume = \left(\frac{4}{3}\right) \times \pi \times R^3$$

$$\left(\frac{4}{3}\right) \times \pi \times R^3 = \frac{4}{3}\pi(6^3 + 8^3 + 10^3) => R^3 = 1728 => R = 12$$

So, the diameter of the new sphere will be twice the radius, which is:

$$Diameter = 2 \times R = 2 \times 12\ cm = 24\ cm$$

Then the correct option is B)

11. Correct option: (B, C)

Given the diameter of the hemisphere is 14 cm

1. Curved Surface Area of Hemisphere (CSA):

$$CSA = \left(\frac{1}{2}\right) \times 4\pi r^2$$

$$CSA = 2 \times \pi \times (7\ cm)^2 = 308\ cm^2$$

2. Area of the Circular Base (A):

$$A = \pi r^2$$

$$A = \pi \times (7\ cm)^2 = 154\ cm^2$$

$$TSA = CSA + A = 462\ cm^2$$

So, the total surface area of the solid hemisphere is 462 cm^2.

Then the correct option is B, C)

12. Correct option: (A)

The volume of the hemisphere is equal to the volume of the cone:

$$1. Volume\ of\ Hemisphere\ = \left(\frac{2}{3}\right) \times \pi \times R^3$$

$$2. Volume\ of\ Cone\ = \left(\frac{1}{3}\right) \times \pi \times R^2 \times H$$

Since the volumes are equal, we can set them equal to each other:

$$\left(\frac{2}{3}\right) \times \pi \times R^3 = \left(\frac{1}{3}\right) \times \pi \times R^2 \times H$$

$$2 \times R = H$$

So, the height (H) of the cone is equal to twice the radius (R) of the hemisphere.

Then the correct option is A)

13. Correct option: (A)

Find the area of the sector:

$$Area\ of\ Sector\ = \left(\frac{216}{360}\right) \times \pi \times (15\ cm)^2$$

$$Area\ of\ Sector\ = 135\pi\ cm^2$$

Since the sector is used to create the curved surface of the cone, the curved surface area (CSA) of the cone is will remain same with slant height equal to radius of circle.

$135\pi = \pi \times 15\ cm \times r$

$$r = 9\ cm$$

To find the height (h) of the cone, we can use the Pythagorean theorem:

$$h = \sqrt{l^2 - R^2} = \sqrt{225 - 81} = \sqrt{144} = 12\ cm$$

Finally, calculate the volume of the cone:

$$Volume\ of\ Cone\ = \left(\frac{1}{3}\right) \times \pi \times R^2 \times h = \frac{1}{3} \times \frac{22}{7} \times 9 \times 9 \times 12 = 1018.28$$

So, the volume of the cone is approximately 1018.28 cubic centimeters.

Then the correct option is A)

14. Correct option: (D)

Find the area of the sector which will be equal to cone CSA

$$Area\ of\ Sector\ = \left(\frac{216}{360}\right) \times \pi \times (15\ cm)^2$$

$$Area\ of\ Sector\ = 135\pi\ cm^2$$

The base area of the cone

$$= \pi R^2 = 81\pi\ cm^2$$

So total surface area = $216\pi \approx 678.58\ cm^2$

Then the correct option is D)

15. Correct option: (D)

The slant height of the cone will remain equal to radius of the circle as we fold it on radius

So slant height is 15 cm.

Then the correct option is B)

16. Correct Matching is: a-Q, b-P, c-T, d-R

a) Curved surface area of cylinder:

$$a)\ ---> Q)\ 2\pi rh$$

b) Volume of cylinder:

$$b)\ ---> P)\ \pi r^2 h$$

c) Volume of cone:

$$c)\ ---> T)\ \left(\frac{1}{3}\right)\pi r^2 h$$

d) Total surface area of sphere: $d)\ ---> R)\ 4\pi r^2$

17. Correct Answer is: (7)

Given:

Length of the rectangle (l) = 44 cm

Now, we know that the circumference of a cylinder (C) is related to its radius (r) by the formula:

$$C = 2\pi r$$

In this case, the length of the rectangle (l) becomes the circumference of the cylinder:

$$44 = 2\pi r => r = 7$$

so radius of the cylinder is 7 cm.

18. Correct Answer is: (9)

Given:

Cost of painting the CSA of the cone = Rs. 176

Slant height (l) = 25 cm

From above information

$$CSA \times Cost\ per\ square\ centimeter = Total\ Cost$$

$$(\pi rl) \times \left(0.25 \frac{Rs}{cm^2}\right) = 176\ Rs$$

$$\pi \times r \times 25\ cm = \frac{176}{0.25}$$

$$r = 8.96$$

So, the radius of the base of the cone is approximately 9 cm.

19. Correct Answer is: (9)

The volume of the sphere and the cylindrical wire will be same

So we can write

$$V_S = \frac{4}{3}\pi r^3 = 36\pi$$

For cylindrical wire we can find the volume as

$$V_c = \pi R^2 h = \frac{\pi}{100} h$$

Then

$$36\pi = \frac{\pi}{100} h => h = 3600$$

So sum of the digit of height

$$3 + 6 + 0 + 0 = 9$$

20. Correct Answer is: (9)

The base area of the cone is

$$= 38.5\ cm^2$$

Volume of the cone is

$$= 77\ cm^3$$

We know that volume formula in form of base area and height of the cone

$$V = \frac{1}{3}\pi r^2 h = \frac{Base\ area \times h}{3} => h = \frac{3V}{Base\ area} = \frac{77 \times 3}{38.5} = 6\ cm$$

Therefore the height of the cone is 6 cm.